

FINAL TRIM AND STABILITY BOOKLET OF KB-26 (YARD NO. CHISTI/015)

AS AMENDED PAGE:2,23,25-39, 65-67, 70,71,114

SEE LETTER E-116437-156477



Rev.	Description			Date	Prepared	Review	
<u>Issued for</u> : <u>Approval</u>							
Client: K.B.SHIPPING & CO., JAMNAGAR Builder: CHISTI MARINE ENGINEERING WORKS				Project: PONTOON (YARD NO. CHISTI/015)			
Doc #				Document Title:		Prepared	OM
CM20-V-111-0207				FINAL		Checked	MG
Scale	Size	Sheet	Date	TRIM AND STABILITY BOOKLET		Review 1	
NTS	A4	1 of 174	01-01-2021			Review 2	
 Cybermarine www.cybermarine.net info@cybermarine.net							
10, Bukit Batok Crescent, #12-01, The Spire Building, Singapore-658079.		501, Technocity IT Premises Co-op. Soc. Ltd, Plot No. X-5/3, Mahape, Maharashtra- 400710, India.		22503, Katy Freeway #7, Katy, Texas, USA 77450.			

CONTENTS

General

Page No

1.General Particulars	03
2.Stability Criteria	04
3.Notes to the Master	08
4.Draft & Trim Calculation	09
5.Worked Out Example	12
6.Definitions, Symbols and Sign Conventions	20
7.Metric Conversion Table	21

Ship's Data

1. Angle of Down flooding	22
2. Capacity Plan and Table	23
3. Sounding Tables	25
4. Table of Hydrostatic Particulars	40
5. Definition of KN	51
6. Volume used For KN Computation	52
7. Displacement Vs KN tables	53
8. Deadweight Table	64
9. Table of Maximum Permissible KG _{SHIP}	65

Summary of Conditions

70

Loading Conditions

Conditions with 150T Crane

1. Cond#0- Lightship	NOT A SAILING CONDITION	72
2. Cond#1- Lightship + Crane + Ballast	NOT A SAILING CONDITION	76
3. Cond#2- Lightship + Load at Aft + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Aft		80
4. Cond#3- Lightship + Load at Stbd + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Port		91
5. Cond#4- Lightship + Load at Fwd + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Fwd		102
6. * Cond#5- Lightship + Crane + Ballast + Deck Cargo		113
7. * Cond#6- Lightship + Max. Deck Cargo	* TOWING CONDITION	117

Conditions with 250T Crane

8. Cond#7- Lightship + Load at Aft + Ballast Crane Lifting 36.6T @18m Radius Aft	121
9. Cond#8- Lightship + Load at Aft + Ballast Crane Lifting 36.6T @18m Radius Stbd	132
10. Cond#9- Lightship + Load at Aft + Ballast Crane Lifting 36.6T @18m Radius Fwd	143

Appendix

I. Lightship Particulars	154
II. Windage area calculation	155

GENERAL PARTICULARS

Name of the Vessel	KB-26
Client	K.B. SHIPPING & CO., JAMNAGAR
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/015
Type of Vessel	PONTOON
Class Notation	HSU-PONTOON, "INDIAN COASTAL SERVICE" DESCRIPTIVE NOTE "FOR CRANE OPERATION" "DECK STRENGTHENED FOR {10T/M ² }

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	15.000 M
Depth (Mld.)	03.000 M
Draft (Mld.)	02.200 M
Draft (Ext.)	02.212 M
Displacement(Ext.)	1569.15 T
Lightship	375.10 T
Deadweight	1194.05 T
Gross Tonnage	564
Net Tonnage	170

Stability Criteria

REGULATION

Intact Stability Criteria in accordance **IS CODE 2008**

- 1) The area under a righting lever curve up to the angle of maximum righting lever should not be less than 0.080 meter-radian.
- 2) The static angle of heel due to a uniformly distributed wind load of 0.54 kPa (wind speed 30 m/s) should not exceed an angle corresponding to half freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.
- 3) The minimum range of stability should be:
For $L \leq 100\text{m}$: 20°
For $L \geq 150\text{m}$: 15°
For intermediate length by interpolation

Intact stability criteria during cargo lifting

The following intact stability criteria are to be complied with:

- $\theta_C \leq 15^\circ$
- $GZC \leq 0,6 GZ_{MAX}$
- $A1 \geq 0.4 ATOT$

where:

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 1)

GZC , GZ_{MAX} : Defined in Fig 1

$A1$: Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle equal to the lesser of:

- heeling angle θ_R of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 1)
- heeling angle θ_F , corresponding to flooding of unprotected openings

$ATOT$: Total area, in m.rad, below the righting lever curve.

In the above formula, the heeling arm, corresponding to the cargo lifting, is to be obtained, in m, from the following formula:

$$b = (Pd - Zz) / \Delta$$

where:

P : Cargo lifting mass, in t

d : Transversal distance, in m, of lifting cargo to the longitudinal plane (see Fig 1)

Z : Mass, in t, of ballast used for righting the pontoon, if applicable (see Fig 1)

z : Transversal distance, in m, of the centre of gravity of Z to the longitudinal plane (see Fig 1)

Δ : Displacement, in t, at the loading condition

The above check is to be carried out considering the most unfavourable situations of cargo lifting combined with the lesser initial metacentric height GM . The vertical position of the centre of gravity of cargo lifting is to be assumed in correspondence of the suspension point.

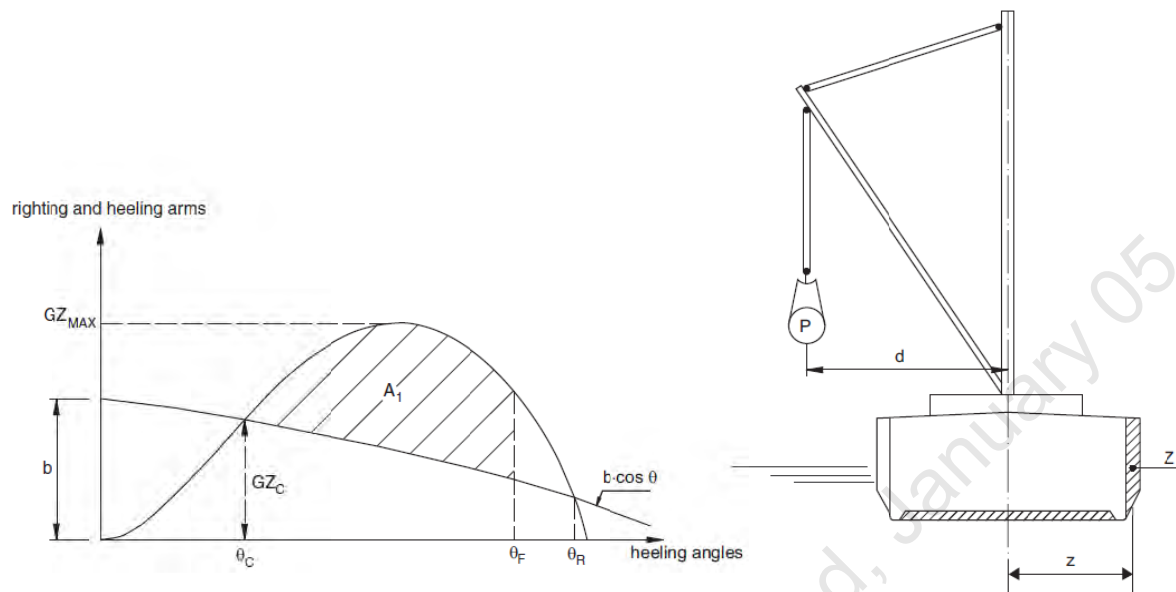


FIGURE 1: CARGO LIFTING

Intact stability criteria in the event of sudden loss of cargo during lifting

This additional requirement is compulsory when counterweights or ballasting of the ship are necessary or when deemed necessary by the Society taking into account the ship dimensions and the weights lifted.

The case of a hypothetical loss of cargo during lifting due to a break of the lifting cable is to be considered.

In this case, the following intact stability criteria are to be compiled with

- $A_2 / A_1 \geq 1$
- $\theta_2 - \theta_3 \geq 20^\circ$

where:

A_1 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_1 to the heeling angle θ_C (see Fig 2)

A_2 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_2 (see Fig 2)

A_3 : Area, in m.rad, contained between the righting lever and the heeling arm curves, measured from the heeling angle θ_C to the heeling angle θ_3 (see Fig 2)

θ_1 : Heeling angle of equilibrium during lifting (see Fig 2)

θ_2 : Heeling angle corresponding to the lesser of θ_R and θ_F

θ_C : Heeling angle of equilibrium, corresponding to the first intersection between heeling and righting arms (see Fig 2)

θ_3 : Maximum heeling angle due to roll, at which $A_3 = A_1$, to be taken not greater than 30° (angle in correspondence of which the loaded cargo on deck is assumed to shift (see Fig 2)

θ_R : Heeling angle of loss of stability, corresponding to the second intersection between heeling and righting arms (see Fig 2).

θ_F : Heeling angle at which progressive flooding may occur (see Fig 2)

In the above formulae, the heeling arm, induced on the ship by the cargo loss, is to be obtained, in m, from the following formula

$$b = Zz \cos\theta/\Delta$$

HEELING AND RIGHTING ARMS

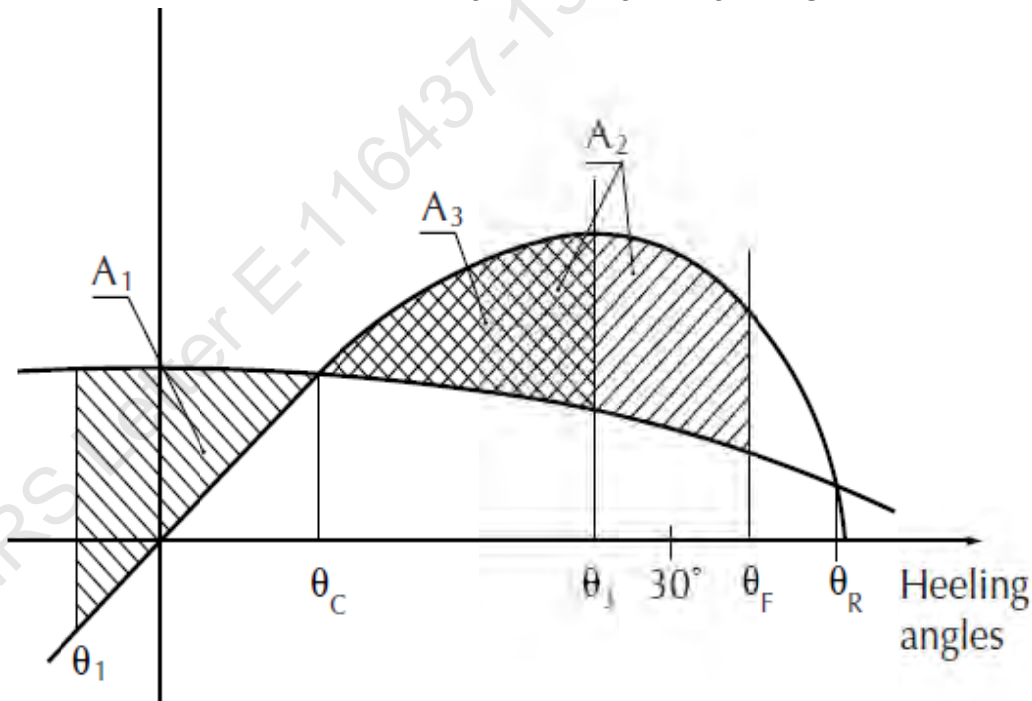


FIGURE 2: CARGO LOSS

Notes to the Master

General precautions against capsizing

1. Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season of the year, weather forecasts, the navigational zone, and should take appropriate action as to speed and course warranted by the prevailing circumstances.
2. Loading conditions shown in this booklet are typical of those which may be encountered in service and are worked out to be used for guidance only. Further the booklet contains sufficient data to enable the ship's staff to compute the ship's stability in various loading conditions.
3. Excessive trim by forward is to be avoided in all cases as this may lead to difficulties of maneuverability and course keeping as also infringe the minimum Bow-height requirements.
4. All longitudinal levers, namely LCB, LCG, and LCF are computed with reference to the A.P of Ship.
5. In this booklet hydrostatic particulars and cross curves are given at following trims.

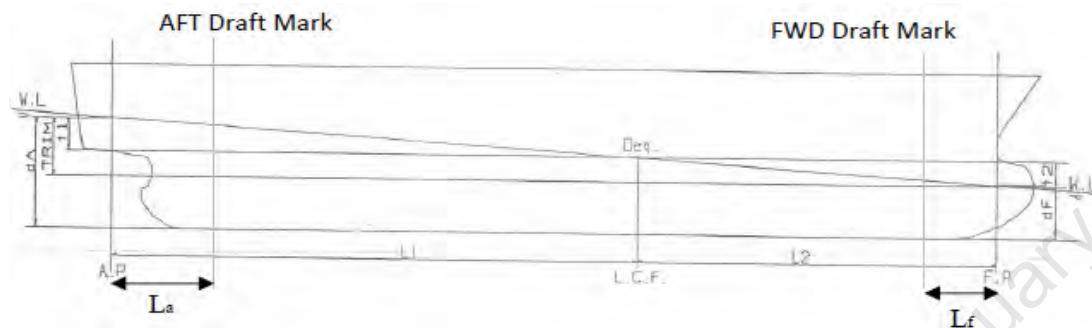
-0.50m, -0.25m, 0.00m, 0.25m, 0.50m

These trims are the expected operational trims.

Notes on Use of Cross Curves of Stability

The purpose of the Cross-Curves of Stability is to enable Statical Stability curves to be drawn for the ship in any loading condition. Intact stability is then determined on the basis of this Curve. WORKED OUT EXAMPLE demonstrates the use of Cross Curves.

Draft and Trim Calculation



The following data are required for ascertaining trim and draft:

- a) Longitudinal Centre of gravity of the ship calculated as follows:

$$\text{LCG} = \frac{\text{Summation of longitudinal weight moments}}{\text{Displacement}}$$

- b) Longitudinal centre of buoyancy, LCB
- c) Displacement, Δ
- d) Moment to change trim, MCT1cm
- e) Longitudinal centre of floatation, LCF

LCB, LCF and MCT1cm are taken from the hydrostatic tables (trim = 0 m) corresponding to the actual displacement. The trim is calculated according to the following formula:

$$\text{Trim} = \frac{\Delta * (\text{LCB} - \text{LCG})}{\text{MCT1cm} * 100}$$

The Extreme drafts at AP (TAP) and at FP (TFP) are calculated according to the following formulae:

$$T_{AP} = \frac{\text{LCF} * \text{Trim}}{\text{LBP}} + T (\text{extreme})$$

$$T_{FP} = T_{AP} - \text{Trim}$$

Where T is the Extreme draft taken from the hydrostatic tables (at trim = 0) corresponding to the actual displacement.

The draft at the Aft and Forward reading marks are as follows:

$$T_a = T_{AP} - \frac{\text{Trim} * L_a}{LBP}$$

$$T_f = T_{FP} + \frac{\text{Trim} * L_f}{LBP}$$

Where,

T_a = Draft at Aft Draft Mark

T_f = Draft at Fwd Draft Mark

L_a = Distance of Aft Draft Mark from AP and it is consider positive when Aft Draft Mark is Fwd of AP

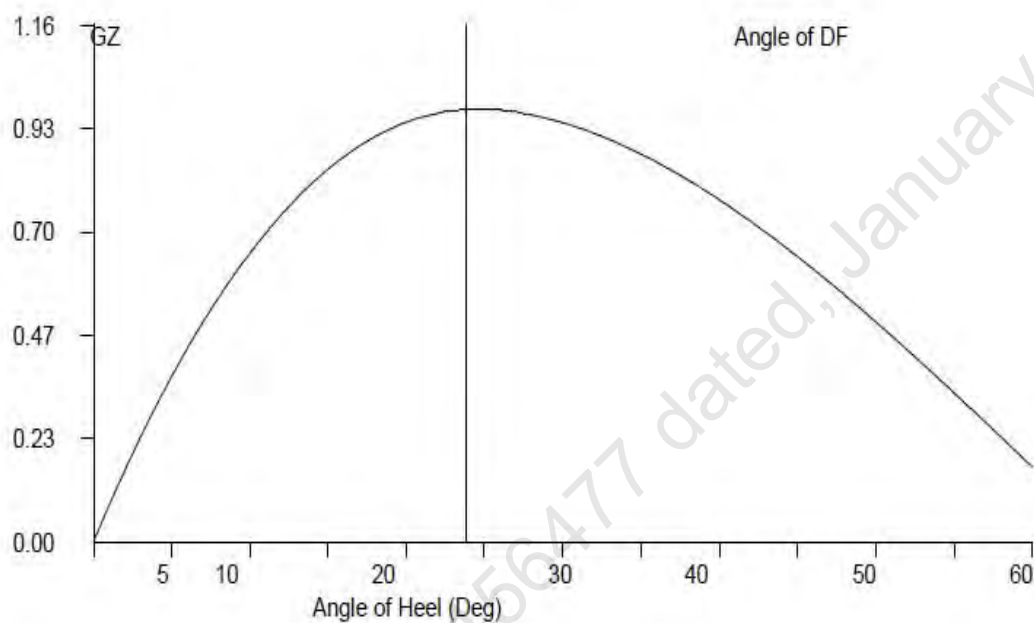
L_f = Distance of Fwd Draft Mark from FP and it is consider positive when Fwd Draft Mark is Aft of FP

GZ Calculation:

- 1) $GZ = KN - KGo \sin (\theta)$
- 2) Area under 0-10 = $10 * 0.01745 * [GZ0 + GZ10] / 2$
- 3) Area under 10-20 = $10 * 0.01745 * [GZ10 + GZ20] / 2$
- 4) Area under 20-30 = $10 * 0.01745 * [GZ20 + GZ30] / 2$
- 5) Area under 30-40 = $10 * 0.01745 * [GZ30 + GZ40] / 2$
- 6) Area under 40-50 = $10 * 0.01745 * [GZ40 + GZ50] / 2$
- 7) Area under 50-60 = $10 * 0.01745 * [GZ50 + GZ60] / 2$
- 8) Area upto 40 deg / Angle of DF = sum of 2+3+4+5
- 9) Max Gz = Read from Gz Curve

Plot GZ values against the angles of heel as above. At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin. This is the Tangent to the GZ Curve at the Origin. Find the

Maxima of GZ curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding & between 30 degrees to 40 degrees/angle of flooding. If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Examples**1 Loading the Vessel**

List out the weights, LCG, VCG, TCG and work out longitudinal moment and vertical moment for each load and finally sum up the entire loads to get Displacement, LCG and VCG of the vessel.

For this worked out Example,

Condition 1: Lightship + Crane + Ballast

Displacement	645.67	Tonnes
LCG	24.469	M Fwd of AP
VCG (KG)	3.522	M above Base Line
TCG	0.000	M Off centerline

2 Free Surface Correction

A tank which is partially filled, adversely affects the stability of the vessel. This is known as Free Surface Effect and is expressed as 'Loss of GM' or 'Virtual Rise in KG' and is calculated as follows:

$$\text{Free Surface Correction (M)} = \frac{\text{Sum of Free Surface Moments of all Tanks}}{\text{Displacement of Vessel}}$$

$$\text{Free Surface moment for a Tank} = \frac{\text{Moment of Inertia (M}^4\text{) for a Tank} \times \text{Density of Liquid in Tank}}{12}$$

$$\text{KGo (Fluid)} = \text{KG (Solid)} + \text{Free Surface Correction.}$$

Hydrostatic Particulars

The trim of a particular loading condition is obtained by

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where,

Trim* = Trim at which hydrostatic particulars have been computed.

CONDITION 1 - LIGHTSHIP + CRANE + BALLAST

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
WBT6.P	39.01	1.025	39.990	47.660	1905.923	-4.967	-198.630	1.915	76.581	41.850
WBT6.C	39.61	1.025	40.600	47.664	1935.158	0.000	0.000	1.904	77.302	41.850
WBT6.S	39.01	1.025	39.990	47.660	1905.923	4.967	198.630	1.915	76.581	41.850
TOTAL TANK			120.580	47.661	5747.005	0.000	0.000	1.911	230.464	125.550

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			375.100	25.000	9,377.500	0.000	0.000	3.000	1125.300
CRANE			150.000	4.500	675.000	0.000	0.000	6.120	918.000
DECK CARGO			0.000	0.000	0.000	0.000	0.000	0.000	0.000
LOAD LIFTED			0.000	0.000	0.000	0.000	0.000	0.000	0.000
TOTAL FIXED WEIGHT			525.100	19.144	10052.500	0.000	0.000	3.891	2043.30

ITEM			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
TOTAL TANK			120.580	47.661	5747.005	0.000	0.000	1.911	230.464
TOTAL FIXED WEIGHT			525.100	19.144	10052.500	0.000	0.000	3.891	2043.300
TOTAL WEIGHT			645.680	24.470	15799.505	0.000	0.000	3.522	2273.764

CONDITION 1 - LIGHTSHIP + CRANE + BALLAST

Free Surface Correction for VCG = FSM/Displacement

	0.194	m
VCGc	3.716	m

From the MaxVCG table,

Displacement	645.68	T
Allowable Max VCG	13.488	m

VCGc < Maxvcg

PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement =	645.680	T
Draft ,Tm @ 25.00f =	0.969	m
LCB =	25.000	m
LCF =	25.000	m
MCT =	24.886	T-m/cm
KMT =	21.061	m

The vessels trim is to be determined as follow,

Trim=(LCB-LCG)xDisplacement /(MCTX100)
0.138 m,(+) is aft trim,(-) is fwd trim

Determining the Forward perpendicular,Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.000 (MS),Tmid	0.969	m
Draft @ 50.0f (FP),Tfwd	=Tm - Trim/2	
	0.900	m
Draft@0(AP),Taft	=Tm + Trim/2	
	1.038	m

CONDITION 1 - LIGHTSHIP + CRANE + BALLAST**From HYDROSTATICS**

Hydrostatic Particulars			Drafts & Trims	
MCT	24.886	T-m	Draft (Mean)	0.969 m
LCB	25.000	m	Draft AP (USK)	1.038 m
			Draft FP (USK)	0.900 m
Metacentric Height				
KMt	21.061	m		
KG	3.522	m		
KG0	3.716	m		
GMt	17.345	m	Trim	0.138 m

The condition shown above is first worked using Even Keel hydrostatic particulars Program then interpolates the hydrostatic particulars from the values available for the displacement

From HYDROSTATICS

Hydrostatic Particulars			Drafts & Trims	
MCT	24.430	T-m	Draft (Mean)	0.969 m
LCB	24.461	m	Draft AP (USK)	1.039 m
			Draft FP (USK)	0.889 m
Metacentric Height				
KMt	21.010	m		
KG	3.522	m		
KG0	3.716	m		
GMt	17.294	m	Trim	0.140 m

Trim is calculated as follows:

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{displacement} / (\text{MCT} \times 100) + \text{Trim}^*$$

Where Trim* is trim at which hydrostatics were calculated.

CONDITION 1 - LIGHTSHIP + CRANE + BALLAST**Intact Stability (Primary Criterion)**

Displacement	645.68	T	
KG0	3.716	M	
GMt	17.294	M	0.150 M

Residual Arm Curve

Angle (Deg)	0	10	20	30	40	50
KN (M)	-0.012	3.393	4.540	4.953	4.794	4.412
GZ (M)	-0.012	2.750	3.271	3.097	2.407	1.567

	Computed values	Minimum Required
Residual Area from abs 0 deg to MAXRA	0.8339 m-rad	0.080 m-rad
Angle from Equilibrium to RAZero or flood	Large m-rad	20.00 Deg.
Angle from Equilibrium to 50% DK Imm Angle	8.51 m-rad	0.00 Deg.

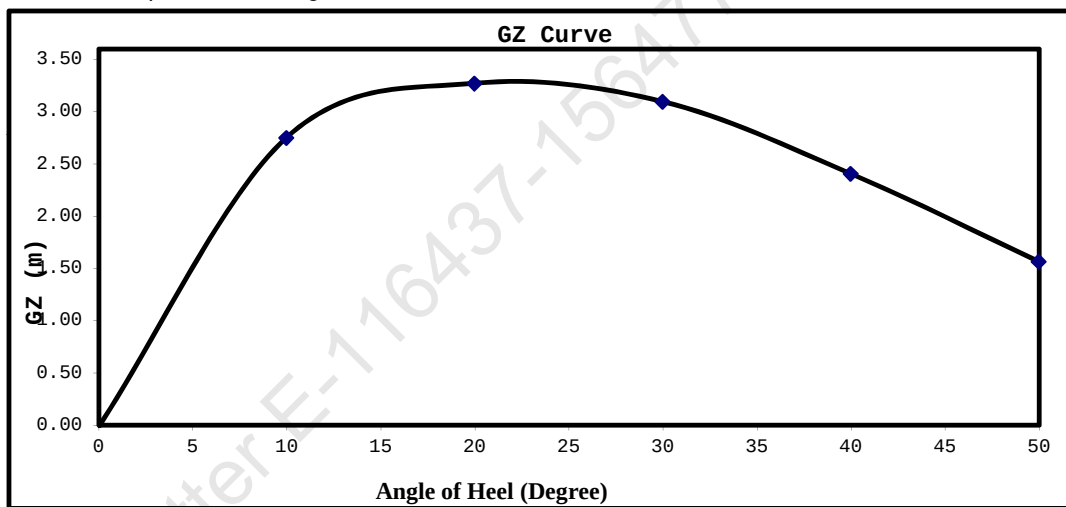
Plot GZ values against the angles of heel as above.

At 57.3 degree erect an ordinate equal to GM fluid on the same scale as GZ ordinates. Join the point to the origin.

This is the Tangent to the GZ Curve at the Origin. Find the Maxima of GZ

curve, ordinates under GZ curve upto 30 degrees, 40 degrees/angle of flooding

If maximum GZ occurs at an angle less than 30 degrees then area upto this angle must be calculated.



Worked Out Example Contd.

Mean Draft	0.969	M
LCF from AP	24.874	M
LCB from AP	24.461	M
MCT	24.430	T-M/Cm
Trim	0.140	M
Draft (AP)	1.039	M
Draft (FP)	0.899	M
KG	3.522	M
GoMt	17.294	M

LOADLINE DRAFT: Check that Mean Draft (Midship) does not exceed the Summer Draft (MLD) for the corresponding season, as per the assigned loadline.

BLANK CALCULATION SHEET

TANK NAME	VOL(Cu.M.)	SpGr	Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT	FSM
				METER	T-M	METER	T-M	METER	T-M	T-M
WBTk1.P		1.025								
WBTk1.C		1.025								
WBTk1.S		1.025								
WBTk2.C		1.025								
WBTk3.P		1.025								
WBTk3.C		1.025								
WBTk3.S		1.025								
WBTk4.P		1.025								
WBTk4.C		1.025								
WBTk4.S		1.025								
WBTk5.P		1.025								
WBTk5.C		1.025								
WBTk5.S		1.025								
WBTk6.P		1.025								
WBTk6.C		1.025								
WBTk6.S		1.025								
TOTAL TANK										

FIXED WEIGHT			Weight(MT)	LCG	LCG MOMENT	TCG	TCG MOMENT	VCG	VCG MOMENT
				METER	T-M	METER	T-M	METER	T-M
LIGHT SHIP			375.100	25.000	9,206.750	0.000	0.000	3.000	1104.810
CRANE									
DECK CARGO									
LOAD LIFTED									
TOTAL FIXED WEIGHT									

Free Surface Correction for VCG = FSM/Displacement

	m
VCGc	m

From the MaxVCG table,

Displacement	T
Allowable Max VCG	m

Allowable Max VCG	m
-------------------	---

$$\text{VCGc} < \text{Maxvcg}$$

PASS

Note : The Master should always check the obtained VCG value against Maximum limiting VCG curve for intact
Refer to the Hydrostatic Table for even trim & Even Heel Condition of Stability Booklet.

Displacement = T

Draft , Tm @ 25.00f = m

LCB = _____ m

LCF = _____ m

MCT = T-m/cm

KMT =

The vessels trim is to be determined as follow,

$$\text{Trim} = (\text{LCB} - \text{LCG}) \times \text{Displacement} / (\text{MCTX} \times 100)$$

m, (+) is aft trim, (-) is fwd trim

Determining the Forward perpendicular, Midship and aft Perpendicular Draft

LBP = 50m

Draft @ 25.000 (MS), Tmid m

$$\text{Draft @ 50.0 f (FP), } T_{fwd} = \frac{T_m - T_{rim}}{2}$$
$$\text{Draft}_{00}(\text{AP}), \text{Taft} = T_m + T_{\text{Trim}}/2$$

Definitions, Symbols and Sign Conventions:

AP	Fr.0
FP	Fr.100
Midship	Fr.50
Frame Spacing	500mm throughout

List of Symbols

Symbol	Unit	Description
LOA	M	Length Overall
LBP	M	Length between perpendiculars
B (MLD)	M	Moulded Breadth
D (MLD)	M	Moulded Depth
TPCm	T/Cm	Tonnes per 1 Cm immersion
MCTCm	T-M/Cm	Moment to change Trim by 1 Cm
Draft AP (USK)	M	Draft at AP wrt Keel line
Draft FP (USK)	M	Draft at FP wrt Keel line
Draft (Mean)	M	Draft wrt Keel line at Midship
Draft (Mean)	M	[Draft AP(USK) + Draft FP(USK)] x 0.5
VCG, KG	M	Vertical Center of Gravity above Base Line before correcting for free surface
KGo	M	Vertical Center of Gravity above Base Line after correcting for free surface
KB	M	Vertical Center of Buoyancy from Base Line
KMt	M	Transverse Metacenter above Base Line
GoMt	M	Transverse Metacenter above KGo
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
sp.gr.		Specific Gravity
LCG	M	Longitudinal Center of Gravity
LCB	M	Longitudinal Center of Buoyancy
LCF	M	Longitudinal Center of Floataction

Sign Convention

LCB, LCG, LCF	Fwd of AP is denoted by f and aft of AP is denoted by a
TCG	port is negative and stbd is positive
Trim by Aft	Positive
Trim by Ford	Negative

Metric Conversion Table

<u>MULTIPLY BY</u>	<u>TO CONVERT FROM</u>	<u>TO OBTAIN</u>	
0.03937	millimeters	Inches	25.400
0.39370	Centimeters	Inches	2.5400
3.28080	Meters	Feet	0.3050
2.20460	kilograms	Pounds	0.4540
0.00098	kilogram's	tons(2240 lb.)	1016.1
0.98420	Metric tonnes (i.e. tonnes of 1000 kg.)	tons(2240 lb.)	1.0160
2.49980	Metric tonnes per centimeter of immersion	tons per inch (immersion)	0.4000
8.20140	moment to change trim one centimeter one (tonne meter units)	moment to change trim inch(foot ton units)	0.1220
187.976	Meter Radians	Feet degrees	0.0053
	<u>TO OBTAIN</u>	<u>TO CONVERT FROM</u>	<u>MULTIPLY</u>

Relation between Weight and volume

1000mm cubed	1 Cubic Centimeters
1 Cubic Centimeter of Fresh Water (S.G.1.0)	1 Gram
1000 cubic centimeter of fresh Water (S.G.1.0)	1 Kilogram (1000 Gm.)
1 Cubic Meter of fresh water (S.G.1.0)	1 Tonne (1000 Kg.)
1 Cubic Meter of fresh water (S.G.1.0)	1 tonne
1 Cubic Meter of Salt water (S.G. 1.025)	1.025 Tonnes
1 Tonne of Salt Water (S.G.1.025)	0.975 Cubic Meters

Angle of Downflooding

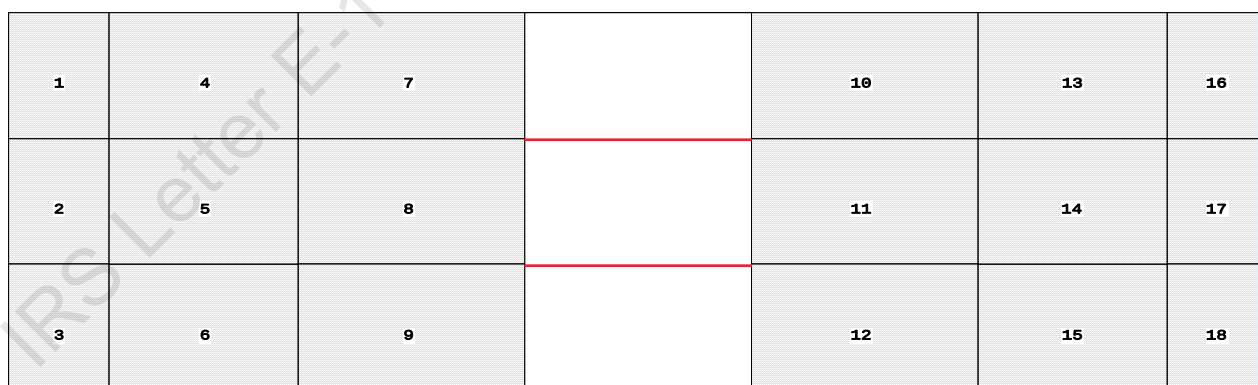
1. The angle of flooding is the angle of immersion of the sill or coaming of the opening on or above the freeboard deck through which progressive downflooding can occur as the vessel heels.
2. If this opening has a weathertight closing device which cannot be kept closed and secured at all times while the vessel is underway (e.g. engine room air supply trunk), or has no weathertight closing device, progressive downflooding will occur when the sill of the opening immerses. This angle is to be marked on the GZ curve if it comes within the apparent positive range of the GZ curve, and the GZ curve terminates at this angle: i.e., the actual angle of flooding becomes the range of the GZ curve.
3. If an opening is fitted with a weathertight closing device which can be kept closed and secured at all times while the vessel is underway, progressive downflooding can only occur through the opening if its closing device is not closed and secured. The angle of immersion of the lower sill of this opening is a 'potential' angle of flooding. If the closing appliance is not closed and secured, downflooding will occur when the lower sill of this opening reaches the water level.
If the lower sill of the opening through which downflooding can occur immerses at 40 degree or smaller angles of heel, its closing appliances must be kept closed and secured at all times when the vessel is underway or at sea where this opening is used for access in the normal working of the vessel, an alternate access should be used in heavy weather. In moderate weather, the opening may be used for access, but the closing appliances must be closed and secured again immediately after use.

As there is no any unprotected opening exists above freeboard deck, hence the downflooding angle is considered at 50 deg.

Profile View



Plan View



Tanks

1 WBTk1.P	3 WBTk1.S	6 VOID2.S	9 WBTk3.S	12 WBTk4.S	15 WBTk5.S	18 WBTk6.S
2 WBTk1.C	4 VOID.P	7 WBTk3.P	10 WBTk4.P	13 WBTk5.P	16 WBTk6.P	
	5 WBTk2.C	8 WBTk3.C	11 WBTk4.C	14 WBTk5.C	17 WBTk6.C	

24-07-19 15:30:04 Cybermarine Technologies PTE. Ltd.
GHS 16.68C

TANK STATUS

Trim: zero, Heel: zero

Part-----	Cu.M.-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	39.0	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.C	39.6	1.025	40.60	2.336f	0.000	1.904	41.85*
WBTk1.S	39.0	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk2.C	110.3	1.025	113.02	7.750f	0.000	1.500	78.48*
WBTk3.P	131.2	1.025	134.48	16.000f	4.980p	1.511	94.17*
WBTk3.C	132.3	1.025	135.61	16.000f	0.000	1.500	94.17*
WBTk3.S	131.2	1.025	134.48	16.000f	4.980s	1.511	94.17*
WBTk4.P	131.2	1.025	134.48	34.000f	4.980p	1.511	94.17*
WBTk4.C	132.3	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk4.S	131.2	1.025	134.48	34.000f	4.980s	1.511	94.17*
WBTk5.P	109.3	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTk5.C	110.3	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk5.S	109.3	1.025	112.08	42.251f	4.980s	1.511	78.49*
WBTk6.P	39.0	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	39.6	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTk6.S	39.0	1.025	39.99	47.660f	4.967s	1.915	41.84*

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk1.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.001	0.054	0.05	3.920f	4.536p	0.056	0.033	0.502
200	.005	0.201	0.21	3.847f	4.720p	0.133	0.208	3.954
300	.012	0.469	0.48	3.766f	4.784p	0.202	0.244	5.469
400	.021	0.829	0.85	3.698f	4.808p	0.266	0.318	7.650
500	.033	1.280	1.31	3.635f	4.825p	0.332	0.358	9.400
600	.047	1.830	1.88	3.571f	4.845p	0.398	0.413	11.320
700	.063	2.475	2.54	3.507f	4.861p	0.464	0.496	13.933
800	.082	3.215	3.30	3.444f	4.876p	0.530	0.553	15.913
900	.104	4.050	4.15	3.381f	4.887p	0.596	0.615	17.878
1000	.128	4.980	5.10	3.318f	4.896p	0.663	0.677	19.939
1100	.154	6.005	6.16	3.254f	4.904p	0.729	0.738	21.934
1200	.183	7.126	7.30	3.190f	4.911p	0.795	0.801	24.042
1300	.214	8.332	8.54	3.128f	4.917p	0.861	0.775	25.381
1400	.246	9.593	9.83	3.072f	4.920p	0.926	0.825	27.288
1500	.281	10.979	11.25	3.010f	4.925p	0.992	0.995	30.123
1600	.320	12.486	12.80	2.943f	4.929p	1.059	1.055	31.968
1700	.361	14.087	14.44	2.876f	4.934p	1.126	1.010	33.008
1800	.405	15.780	16.17	2.811f	4.935p	1.194	1.166	35.784
1900	.451	17.576	18.02	2.744f	4.938p	1.261	1.262	38.398
2000	.499	19.466	19.95	2.678f	4.941p	1.328	1.307	40.474
2100	.549	21.406	21.94	2.618f	4.944p	1.393	1.198	40.879
2200	.599	23.352	23.94	2.568f	4.948p	1.456	1.106	41.198
2300	.649	25.304	25.94	2.524f	4.951p	1.517	1.026	41.482
2400	.699	27.259	27.94	2.487f	4.954p	1.577	0.955	41.627
2500	.749	29.216	29.95	2.454f	4.957p	1.636	0.893	41.737
2600	.799	31.175	31.95	2.426f	4.959p	1.693	0.837	41.760
2700	.849	33.133	33.96	2.401f	4.962p	1.750	0.788	41.783
2800	.900	35.092	35.97	2.378f	4.964p	1.805	0.744	41.807
2900	.950	37.052	37.98	2.358f	4.966p	1.861	0.705	41.831
3000	1.000	39.012	39.99	2.340f	4.967p	1.915		

Soundings in mm.-----Other distances in METERS.-----

WBTk1.P Reference Point: Long.= 2.000f Trans.= 5.005p Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk1.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.002	0.068	0.07	3.908f	0.000	0.056	0.042	0.739
200	.006	0.245	0.25	3.836f	0.000	0.130	0.188	5.060
300	.013	0.532	0.55	3.765f	0.000	0.197	0.248	7.182
400	.023	0.916	0.94	3.699f	0.000	0.261	0.305	9.219
500	.035	1.393	1.43	3.634f	0.000	0.327	0.362	11.227
600	.050	1.967	2.02	3.570f	0.000	0.392	0.425	13.283
700	.067	2.635	2.70	3.506f	0.000	0.458	0.484	15.298
800	.086	3.399	3.48	3.441f	0.000	0.524	0.546	17.330
900	.108	4.258	4.36	3.377f	0.000	0.590	0.609	19.374
1000	.132	5.213	5.34	3.312f	0.000	0.656	0.672	21.417
1100	.158	6.264	6.42	3.248f	0.000	0.722	0.735	23.461
1200	.187	7.411	7.60	3.183f	0.000	0.788	0.798	25.505
1300	.218	8.646	8.86	3.120f	0.000	0.855	0.833	26.842
1400	.252	9.974	10.22	3.056f	0.000	0.921	0.926	29.592
1500	.288	11.400	11.69	2.992f	0.000	0.987	0.991	31.646
1600	.326	12.931	13.25	2.927f	0.000	1.054	1.058	33.731
1700	.368	14.557	14.92	2.861f	0.000	1.120	1.119	35.750
1800	.411	16.278	16.69	2.796f	0.000	1.187	1.186	37.839
1900	.457	18.097	18.55	2.730f	0.000	1.254	1.251	39.897
2000	.505	20.010	20.51	2.665f	0.000	1.320	1.306	41.854
2100	.555	21.970	22.52	2.606f	0.000	1.385	1.190	41.854
2200	.604	23.930	24.53	2.556f	0.000	1.448	1.092	41.854
2300	.654	25.890	26.54	2.514f	0.000	1.509	1.009	41.854
2400	.703	27.850	28.55	2.478f	0.000	1.568	0.938	41.854
2500	.753	29.810	30.56	2.446f	0.000	1.626	0.877	41.854
2600	.802	31.770	32.56	2.419f	0.000	1.683	0.823	41.854
2700	.852	33.730	34.57	2.394f	0.000	1.739	0.775	41.854
2800	.901	35.690	36.58	2.373f	0.000	1.795	0.732	41.854
2900	.951	37.650	38.59	2.353f	0.000	1.850	0.694	41.854
3000	1.000	39.610	40.60	2.336f	0.000	1.904		

Soundings in mm.-----Other distances in METERS.-----

WBTk1.C Reference Point: Long.= 2.000f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: WBTk1.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.001	0.054	0.05	3.920f	4.536s	0.056	0.033	0.502
200	.005	0.201	0.21	3.847f	4.720s	0.133	0.208	3.954
300	.012	0.469	0.48	3.766f	4.784s	0.202	0.244	5.469
400	.021	0.829	0.85	3.698f	4.808s	0.266	0.318	7.650
500	.033	1.280	1.31	3.635f	4.825s	0.332	0.358	9.400
600	.047	1.830	1.88	3.571f	4.845s	0.398	0.413	11.320
700	.063	2.475	2.54	3.507f	4.861s	0.464	0.496	13.933
800	.082	3.215	3.30	3.444f	4.876s	0.530	0.553	15.913
900	.104	4.050	4.15	3.381f	4.887s	0.596	0.615	17.878
1000	.128	4.980	5.10	3.318f	4.896s	0.663	0.677	19.939
1100	.154	6.005	6.16	3.254f	4.904s	0.729	0.738	21.934
1200	.183	7.126	7.30	3.190f	4.911s	0.795	0.801	24.042
1300	.214	8.332	8.54	3.128f	4.917s	0.861	0.775	25.381
1400	.246	9.593	9.83	3.072f	4.920s	0.926	0.825	27.288
1500	.281	10.979	11.25	3.010f	4.925s	0.992	0.995	30.123
1600	.320	12.486	12.80	2.943f	4.929s	1.059	1.055	31.968
1700	.361	14.087	14.44	2.876f	4.934s	1.126	1.010	33.008
1800	.405	15.780	16.17	2.811f	4.935s	1.194	1.166	35.784
1900	.451	17.576	18.02	2.744f	4.938s	1.261	1.262	38.398
2000	.499	19.466	19.95	2.678f	4.941s	1.328	1.307	40.474
2100	.549	21.406	21.94	2.618f	4.944s	1.393	1.198	40.879
2200	.599	23.352	23.94	2.568f	4.948s	1.456	1.106	41.198
2300	.649	25.304	25.94	2.524f	4.951s	1.517	1.026	41.482
2400	.699	27.259	27.94	2.487f	4.954s	1.577	0.955	41.627
2500	.749	29.216	29.95	2.454f	4.957s	1.636	0.893	41.737
2600	.799	31.175	31.95	2.426f	4.959s	1.693	0.837	41.760
2700	.849	33.133	33.96	2.401f	4.962s	1.750	0.788	41.783
2800	.900	35.092	35.97	2.378f	4.964s	1.805	0.744	41.807
2900	.950	37.052	37.98	2.358f	4.966s	1.861	0.705	41.831
3000	1.000	39.012	39.99	2.340f	4.967s	1.915		

Soundings in mm.-----Other distances in METERS.-----

WBTk1.S Reference Point: Long.= 2.000f Trans.= 5.005s Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL**TANK CHARACTERISTICS**

No Trim, No Heel

Tank: WBTk3.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume		Weight		Center of Gravity			FSM m.-MT
		CU. METERS	METRIC TON			LCG	TCG	VCG	
0	.000	0.000	0.00						
100	.031	4.013	4.11	16.000f	4.775p	0.050	68.243	73.332	
200	.062	8.114	8.32	16.000f	4.800p	0.101	34.484	78.219	
300	.094	12.304	12.61	16.000f	4.826p	0.152	23.226	83.319	
400	.126	16.582	17.00	16.000f	4.851p	0.203	17.593	88.636	
500	.160	20.948	21.47	16.000f	4.877p	0.254	14.210	94.165	
600	.193	25.358	25.99	16.000f	4.899p	0.306	11.739	94.173	
700	.227	29.768	30.51	16.000f	4.914p	0.357	10.000	94.173	
800	.261	34.178	35.03	16.000f	4.925p	0.408	8.710	94.173	
900	.294	38.588	39.55	16.000f	4.933p	0.458	7.714	94.173	
1000	.328	42.998	44.07	16.000f	4.940p	0.509	6.923	94.173	
1100	.361	47.408	48.59	16.000f	4.946p	0.559	6.279	94.173	
1200	.395	51.818	53.11	16.000f	4.950p	0.609	5.745	94.173	
1300	.429	56.228	57.63	16.000f	4.954p	0.659	5.294	94.173	
1400	.462	60.638	62.15	16.000f	4.958p	0.710	4.909	94.173	
1500	.496	65.048	66.67	16.000f	4.960p	0.760	4.576	94.173	
1600	.529	69.458	71.19	16.000f	4.963p	0.810	4.286	94.173	
1700	.563	73.868	75.71	16.000f	4.965p	0.860	4.030	94.173	
1800	.597	78.278	80.23	16.000f	4.967p	0.910	3.803	94.173	
1900	.630	82.688	84.75	16.000f	4.969p	0.960	3.600	94.173	
2000	.664	87.098	89.28	16.000f	4.970p	1.011	3.418	94.173	
2100	.697	91.508	93.80	16.000f	4.972p	1.061	3.253	94.173	
2200	.731	95.918	98.32	16.000f	4.973p	1.111	3.103	94.173	
2300	.765	100.328	102.84	16.000f	4.974p	1.161	2.967	94.173	
2400	.798	104.738	107.36	16.000f	4.975p	1.211	2.842	94.173	
2500	.832	109.148	111.88	16.000f	4.976p	1.261	2.727	94.173	
2600	.866	113.558	116.40	16.000f	4.977p	1.311	2.621	94.173	
2700	.899	117.968	120.92	16.000f	4.978p	1.361	2.523	94.173	
2800	.933	122.378	125.44	16.000f	4.979p	1.411	2.432	94.173	
2900	.966	126.788	129.96	16.000f	4.980p	1.461	2.348	94.173	
3000	1.000	131.197	134.48	16.000f	4.980p	1.511			

Soundings in mm.-----Other distances in METERS.-----

WBTk3.P Reference Point: Long.= 16.000f Trans.= 5.000p Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk3.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume		Weight		Center of Gravity			FSM m.-MT
		CU.	METERS	METRIC	TON	LCG	TCG	VCG	GML
0	.000	0.000		0.00					
100	.033	4.410		4.52		16.000f	0.000	0.050	67.499
200	.067	8.820		9.04		16.000f	0.000	0.100	33.749
300	.100	13.230		13.56		16.000f	0.000	0.150	22.500
400	.133	17.640		18.08		16.000f	0.000	0.200	16.875
500	.167	22.050		22.60		16.000f	0.000	0.250	13.500
600	.200	26.460		27.12		16.000f	0.000	0.300	11.250
700	.233	30.870		31.64		16.000f	0.000	0.350	9.643
800	.267	35.280		36.16		16.000f	0.000	0.400	8.437
900	.300	39.690		40.68		16.000f	0.000	0.450	7.500
1000	.333	44.100		45.20		16.000f	0.000	0.500	6.750
1100	.367	48.510		49.72		16.000f	0.000	0.550	6.136
1200	.400	52.920		54.24		16.000f	0.000	0.600	5.625
1300	.433	57.330		58.76		16.000f	0.000	0.650	5.192
1400	.467	61.740		63.28		16.000f	0.000	0.700	4.821
1500	.500	66.150		67.80		16.000f	0.000	0.750	4.500
1600	.533	70.560		72.32		16.000f	0.000	0.800	4.219
1700	.567	74.970		76.84		16.000f	0.000	0.850	3.971
1800	.600	79.380		81.36		16.000f	0.000	0.900	3.750
1900	.633	83.790		85.88		16.000f	0.000	0.950	3.553
2000	.667	88.200		90.40		16.000f	0.000	1.000	3.375
2100	.700	92.610		94.92		16.000f	0.000	1.050	3.214
2200	.733	97.020		99.45		16.000f	0.000	1.100	3.068
2300	.767	101.430		103.97		16.000f	0.000	1.150	2.935
2400	.800	105.840		108.49		16.000f	0.000	1.200	2.812
2500	.833	110.250		113.01		16.000f	0.000	1.250	2.700
2600	.867	114.660		117.53		16.000f	0.000	1.300	2.596
2700	.900	119.070		122.05		16.000f	0.000	1.350	2.500
2800	.933	123.479		126.57		16.000f	0.000	1.400	2.411
2900	.967	127.889		131.09		16.000f	0.000	1.450	2.328
3000	1.000	132.299		135.61		16.000f	0.000	1.500	

Soundings in mm.-----Other distances in METERS.-----

WBTk3.C Reference Point: Long.= 16.000f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk3.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume		Weight		Center of Gravity			FSM m.-MT
		CU. METERS	METRIC TON			LCG	TCG	VCG	
0	.000	0.000	0.00						
100	.031	4.013	4.11	16.000f	4.775s	0.050	68.243	73.332	
200	.062	8.114	8.32	16.000f	4.800s	0.101	34.484	78.219	
300	.094	12.304	12.61	16.000f	4.826s	0.152	23.226	83.319	
400	.126	16.582	17.00	16.000f	4.851s	0.203	17.593	88.636	
500	.160	20.948	21.47	16.000f	4.877s	0.254	14.210	94.165	
600	.193	25.358	25.99	16.000f	4.899s	0.306	11.739	94.173	
700	.227	29.768	30.51	16.000f	4.914s	0.357	10.000	94.173	
800	.261	34.178	35.03	16.000f	4.925s	0.408	8.710	94.173	
900	.294	38.588	39.55	16.000f	4.933s	0.458	7.714	94.173	
1000	.328	42.998	44.07	16.000f	4.940s	0.509	6.923	94.173	
1100	.361	47.408	48.59	16.000f	4.946s	0.559	6.279	94.173	
1200	.395	51.818	53.11	16.000f	4.950s	0.609	5.745	94.173	
1300	.429	56.228	57.63	16.000f	4.954s	0.659	5.294	94.173	
1400	.462	60.638	62.15	16.000f	4.958s	0.710	4.909	94.173	
1500	.496	65.048	66.67	16.000f	4.960s	0.760	4.576	94.173	
1600	.529	69.458	71.19	16.000f	4.963s	0.810	4.286	94.173	
1700	.563	73.868	75.71	16.000f	4.965s	0.860	4.030	94.173	
1800	.597	78.278	80.23	16.000f	4.967s	0.910	3.803	94.173	
1900	.630	82.688	84.75	16.000f	4.969s	0.960	3.600	94.173	
2000	.664	87.098	89.28	16.000f	4.970s	1.011	3.418	94.173	
2100	.697	91.508	93.80	16.000f	4.972s	1.061	3.253	94.173	
2200	.731	95.918	98.32	16.000f	4.973s	1.111	3.103	94.173	
2300	.765	100.328	102.84	16.000f	4.974s	1.161	2.967	94.173	
2400	.798	104.738	107.36	16.000f	4.975s	1.211	2.842	94.173	
2500	.832	109.148	111.88	16.000f	4.976s	1.261	2.727	94.173	
2600	.866	113.558	116.40	16.000f	4.977s	1.311	2.621	94.173	
2700	.899	117.968	120.92	16.000f	4.978s	1.361	2.523	94.173	
2800	.933	122.378	125.44	16.000f	4.979s	1.411	2.432	94.173	
2900	.966	126.788	129.96	16.000f	4.980s	1.461	2.348	94.173	
3000	1.000	131.197	134.48	16.000f	4.980s	1.511			

Soundings in mm.-----Other distances in METERS.-----

WBTk3.S Reference Point: Long.= 16.000f Trans.= 5.000s Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk4.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.013	4.11	34.000f	4.775p	0.050	68.237	73.332
200	.062	8.114	8.32	34.000f	4.800p	0.101	34.480	78.219
300	.094	12.304	12.61	34.000f	4.826p	0.152	23.227	83.318
400	.126	16.582	17.00	34.000f	4.851p	0.203	17.594	88.635
500	.160	20.948	21.47	34.000f	4.877p	0.254	14.209	94.165
600	.193	25.358	25.99	34.000f	4.899p	0.306	11.741	94.172
700	.227	29.768	30.51	34.000f	4.914p	0.357	10.001	94.172
800	.261	34.178	35.03	34.000f	4.925p	0.408	8.711	94.172
900	.294	38.588	39.55	34.000f	4.933p	0.458	7.715	94.172
1000	.328	42.998	44.07	34.000f	4.940p	0.509	6.924	94.172
1100	.361	47.408	48.59	34.000f	4.946p	0.559	6.280	94.172
1200	.395	51.818	53.11	34.000f	4.950p	0.609	5.745	94.172
1300	.429	56.228	57.63	34.000f	4.954p	0.659	5.295	94.172
1400	.462	60.638	62.15	34.000f	4.958p	0.710	4.910	94.172
1500	.496	65.048	66.67	34.000f	4.960p	0.760	4.577	94.172
1600	.529	69.458	71.19	34.000f	4.963p	0.810	4.286	94.172
1700	.563	73.868	75.71	34.000f	4.965p	0.860	4.030	94.172
1800	.597	78.278	80.23	34.000f	4.967p	0.910	3.803	94.172
1900	.630	82.688	84.75	34.000f	4.969p	0.960	3.600	94.172
2000	.664	87.098	89.27	34.000f	4.970p	1.011	3.418	94.172
2100	.697	91.508	93.80	34.000f	4.972p	1.061	3.253	94.172
2200	.731	95.918	98.32	34.000f	4.973p	1.111	3.104	94.172
2300	.765	100.328	102.84	34.000f	4.974p	1.161	2.967	94.172
2400	.798	104.738	107.36	34.000f	4.975p	1.211	2.842	94.172
2500	.832	109.148	111.88	34.000f	4.976p	1.261	2.728	94.172
2600	.866	113.558	116.40	34.000f	4.977p	1.311	2.622	94.172
2700	.899	117.968	120.92	34.000f	4.978p	1.361	2.524	94.172
2800	.933	122.378	125.44	34.000f	4.979p	1.411	2.433	94.172
2900	.966	126.788	129.96	34.000f	4.980p	1.461	2.348	94.172
3000	1.000	131.197	134.48	34.000f	4.980p	1.511		

Soundings in mm.-----Other distances in METERS.-----

WBTk4.P Reference Point: Long.= 34.000f Trans.= 5.000p Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk4.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m. -MT
				LCG	TCG	VCg		
0	.000	0.000	0.00					
100	.033	4.410	4.52	34.000f	0.000	0.050	67.502	94.171
200	.067	8.820	9.04	34.000f	0.000	0.100	33.751	94.171
300	.100	13.230	13.56	34.000f	0.000	0.150	22.501	94.171
400	.133	17.640	18.08	34.000f	0.000	0.200	16.876	94.171
500	.167	22.050	22.60	34.000f	0.000	0.250	13.500	94.171
600	.200	26.460	27.12	34.000f	0.000	0.300	11.250	94.171
700	.233	30.870	31.64	34.000f	0.000	0.350	9.643	94.171
800	.267	35.280	36.16	34.000f	0.000	0.400	8.438	94.171
900	.300	39.690	40.68	34.000f	0.000	0.450	7.500	94.171
1000	.333	44.100	45.20	34.000f	0.000	0.500	6.750	94.171
1100	.367	48.510	49.72	34.000f	0.000	0.550	6.137	94.171
1200	.400	52.920	54.24	34.000f	0.000	0.600	5.625	94.171
1300	.433	57.330	58.76	34.000f	0.000	0.650	5.192	94.171
1400	.467	61.740	63.28	34.000f	0.000	0.700	4.822	94.171
1500	.500	66.150	67.80	34.000f	0.000	0.750	4.500	94.171
1600	.533	70.560	72.32	34.000f	0.000	0.800	4.219	94.171
1700	.567	74.970	76.84	34.000f	0.000	0.850	3.971	94.171
1800	.600	79.380	81.36	34.000f	0.000	0.900	3.750	94.171
1900	.633	83.790	85.88	34.000f	0.000	0.950	3.553	94.171
2000	.667	88.200	90.40	34.000f	0.000	1.000	3.375	94.171
2100	.700	92.610	94.92	34.000f	0.000	1.050	3.214	94.171
2200	.733	97.020	99.45	34.000f	0.000	1.100	3.068	94.171
2300	.767	101.430	103.97	34.000f	0.000	1.150	2.935	94.171
2400	.800	105.840	108.49	34.000f	0.000	1.200	2.813	94.171
2500	.833	110.249	113.01	34.000f	0.000	1.250	2.700	94.171
2600	.867	114.659	117.53	34.000f	0.000	1.300	2.596	94.171
2700	.900	119.069	122.05	34.000f	0.000	1.350	2.500	94.171
2800	.933	123.479	126.57	34.000f	0.000	1.400	2.411	94.171
2900	.967	127.889	131.09	34.000f	0.000	1.450	2.328	94.171
3000	1.000	132.299	135.61	34.000f	0.000	1.500		

Soundings in mm.-----Other distances in METERS.-----

WBTk4.C Reference Point: Long.= 34.000f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk4.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.031	4.013	4.11	34.000f	4.775s	0.050	68.237	73.332
200	.062	8.114	8.32	34.000f	4.800s	0.101	34.480	78.219
300	.094	12.304	12.61	34.000f	4.826s	0.152	23.227	83.318
400	.126	16.582	17.00	34.000f	4.851s	0.203	17.594	88.635
500	.160	20.948	21.47	34.000f	4.877s	0.254	14.209	94.165
600	.193	25.358	25.99	34.000f	4.899s	0.306	11.741	94.172
700	.227	29.768	30.51	34.000f	4.914s	0.357	10.001	94.172
800	.261	34.178	35.03	34.000f	4.925s	0.408	8.711	94.172
900	.294	38.588	39.55	34.000f	4.933s	0.458	7.715	94.172
1000	.328	42.998	44.07	34.000f	4.940s	0.509	6.924	94.172
1100	.361	47.408	48.59	34.000f	4.946s	0.559	6.280	94.172
1200	.395	51.818	53.11	34.000f	4.950s	0.609	5.745	94.172
1300	.429	56.228	57.63	34.000f	4.954s	0.659	5.295	94.172
1400	.462	60.638	62.15	34.000f	4.958s	0.710	4.910	94.172
1500	.496	65.048	66.67	34.000f	4.960s	0.760	4.577	94.172
1600	.529	69.458	71.19	34.000f	4.963s	0.810	4.286	94.172
1700	.563	73.868	75.71	34.000f	4.965s	0.860	4.030	94.172
1800	.597	78.278	80.23	34.000f	4.967s	0.910	3.803	94.172
1900	.630	82.688	84.75	34.000f	4.969s	0.960	3.600	94.172
2000	.664	87.098	89.27	34.000f	4.970s	1.011	3.418	94.172
2100	.697	91.508	93.80	34.000f	4.972s	1.061	3.253	94.172
2200	.731	95.918	98.32	34.000f	4.973s	1.111	3.104	94.172
2300	.765	100.328	102.84	34.000f	4.974s	1.161	2.967	94.172
2400	.798	104.738	107.36	34.000f	4.975s	1.211	2.842	94.172
2500	.832	109.148	111.88	34.000f	4.976s	1.261	2.728	94.172
2600	.866	113.558	116.40	34.000f	4.977s	1.311	2.622	94.172
2700	.899	117.968	120.92	34.000f	4.978s	1.361	2.524	94.172
2800	.933	122.378	125.44	34.000f	4.979s	1.411	2.433	94.172
2900	.966	126.788	129.96	34.000f	4.980s	1.461	2.348	94.172
3000	1.000	131.197	134.48	34.000f	4.980s	1.511		

Soundings in mm.-----Other distances in METERS.-----

WBTk4.S Reference Point: Long.= 34.000f Trans.= 5.000s Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTK5.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.027	2.964	3.04	42.266f	4.772p	0.044	53.336	60.667
200	.058	6.374	6.53	42.257f	4.798p	0.095	25.348	64.715
300	.090	9.857	10.10	42.255f	4.823p	0.145	16.739	68.945
400	.123	13.413	13.75	42.254f	4.849p	0.197	12.559	73.355
500	.156	17.043	17.47	42.253f	4.874p	0.248	10.087	77.949
600	.189	20.718	21.24	42.252f	4.897p	0.300	8.316	78.486
700	.223	24.393	25.00	42.252f	4.912p	0.351	7.062	78.486
800	.257	28.068	28.77	42.252f	4.924p	0.401	6.138	78.485
900	.290	31.743	32.54	42.252f	4.933p	0.452	5.428	78.485
1000	.324	35.419	36.30	42.251f	4.940p	0.502	4.865	78.485
1100	.358	39.094	40.07	42.251f	4.945p	0.553	4.407	78.484
1200	.391	42.769	43.84	42.251f	4.950p	0.603	4.029	78.484
1300	.425	46.444	47.61	42.251f	4.954p	0.653	3.710	78.484
1400	.458	50.119	51.37	42.251f	4.957p	0.704	3.437	78.483
1500	.492	53.794	55.14	42.251f	4.960p	0.754	3.203	78.483
1600	.526	57.469	58.91	42.251f	4.963p	0.804	2.998	78.482
1700	.559	61.144	62.67	42.251f	4.965p	0.854	2.818	78.482
1800	.593	64.819	66.44	42.251f	4.967p	0.904	2.658	78.482
1900	.626	68.494	70.21	42.251f	4.969p	0.954	2.515	78.481
2000	.660	72.170	73.97	42.251f	4.970p	1.005	2.387	78.481
2100	.694	75.845	77.74	42.251f	4.972p	1.055	2.271	78.480
2200	.727	79.520	81.51	42.251f	4.973p	1.105	2.167	78.480
2300	.761	83.195	85.27	42.251f	4.974p	1.155	2.071	78.480
2400	.794	86.870	89.04	42.251f	4.975p	1.205	1.983	78.479
2500	.828	90.545	92.81	42.251f	4.976p	1.255	1.903	78.479
2600	.862	94.220	96.58	42.251f	4.977p	1.305	1.828	78.479
2700	.895	97.895	100.34	42.251f	4.978p	1.355	1.760	78.478
2800	.929	101.570	104.11	42.251f	4.979p	1.405	1.696	78.478
2900	.962	105.245	107.88	42.251f	4.980p	1.455	1.637	78.477
3000	.996	108.920	111.64	42.251f	4.980p	1.505	1.582	78.477
3012	1.000	109.348	112.08	42.251f	4.980p	1.511		

Soundings in mm.-----Other distances in METERS.-----

WBTK5.P Reference Point: Long.= 42.250f Trans.= 5.003p Vert.= -0.012

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTK5.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.030	3.261	3.34	42.265f	0.000	0.044	52.821	78.476
200	.063	6.936	7.11	42.257f	0.000	0.094	24.836	78.476
300	.096	10.611	10.88	42.255f	0.000	0.144	16.234	78.476
400	.130	14.286	14.64	42.253f	0.000	0.194	12.058	78.476
500	.163	17.961	18.41	42.253f	0.000	0.244	9.591	78.476
600	.196	21.636	22.18	42.252f	0.000	0.294	7.962	78.476
700	.230	25.311	25.94	42.252f	0.000	0.344	6.806	78.476
800	.263	28.986	29.71	42.252f	0.000	0.394	5.943	78.476
900	.296	32.661	33.48	42.252f	0.000	0.444	5.274	78.476
1000	.330	36.336	37.24	42.251f	0.000	0.494	4.741	78.476
1100	.363	40.011	41.01	42.251f	0.000	0.544	4.306	78.476
1200	.396	43.686	44.78	42.251f	0.000	0.594	3.943	78.476
1300	.430	47.361	48.55	42.251f	0.000	0.644	3.637	78.476
1400	.463	51.036	52.31	42.251f	0.000	0.694	3.375	78.476
1500	.496	54.711	56.08	42.251f	0.000	0.744	3.149	78.476
1600	.530	58.386	59.85	42.251f	0.000	0.794	2.951	78.476
1700	.563	62.061	63.61	42.251f	0.000	0.844	2.776	78.476
1800	.596	65.736	67.38	42.251f	0.000	0.894	2.621	78.476
1900	.630	69.411	71.15	42.251f	0.000	0.944	2.482	78.476
2000	.663	73.086	74.91	42.251f	0.000	0.994	2.357	78.476
2100	.696	76.761	78.68	42.251f	0.000	1.044	2.244	78.476
2200	.729	80.436	82.45	42.251f	0.000	1.094	2.142	78.476
2300	.763	84.111	86.21	42.251f	0.000	1.144	2.048	78.476
2400	.796	87.786	89.98	42.251f	0.000	1.194	1.962	78.476
2500	.829	91.461	93.75	42.251f	0.000	1.244	1.884	78.476
2600	.863	95.136	97.51	42.251f	0.000	1.294	1.811	78.476
2700	.896	98.811	101.28	42.250f	0.000	1.344	1.743	78.476
2800	.929	102.486	105.05	42.250f	0.000	1.394	1.681	78.476
2900	.963	106.161	108.82	42.250f	0.000	1.444	1.623	78.476
3000	.996	109.836	112.58	42.250f	0.000	1.494	1.568	78.476
3012	1.000	110.264	113.02	42.250f	0.000	1.500		

Soundings in mm.-----Other distances in METERS.-----

WBTK5.C Reference Point: Long.= 42.250f Trans.= 0.000 Vert.= -0.012

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk5.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.027	2.964	3.04	42.266f	4.772s	0.044	53.336	60.667
200	.058	6.374	6.53	42.257f	4.798s	0.095	25.348	64.715
300	.090	9.857	10.10	42.255f	4.823s	0.145	16.739	68.945
400	.123	13.413	13.75	42.254f	4.849s	0.197	12.559	73.355
500	.156	17.043	17.47	42.253f	4.874s	0.248	10.087	77.949
600	.189	20.718	21.24	42.252f	4.897s	0.300	8.316	78.486
700	.223	24.393	25.00	42.252f	4.912s	0.351	7.062	78.486
800	.257	28.068	28.77	42.252f	4.924s	0.401	6.138	78.485
900	.290	31.743	32.54	42.252f	4.933s	0.452	5.428	78.485
1000	.324	35.419	36.30	42.251f	4.940s	0.502	4.865	78.485
1100	.358	39.094	40.07	42.251f	4.945s	0.553	4.407	78.484
1200	.391	42.769	43.84	42.251f	4.950s	0.603	4.029	78.484
1300	.425	46.444	47.61	42.251f	4.954s	0.653	3.710	78.484
1400	.458	50.119	51.37	42.251f	4.957s	0.704	3.437	78.483
1500	.492	53.794	55.14	42.251f	4.960s	0.754	3.203	78.483
1600	.526	57.469	58.91	42.251f	4.963s	0.804	2.998	78.482
1700	.559	61.144	62.67	42.251f	4.965s	0.854	2.818	78.482
1800	.593	64.819	66.44	42.251f	4.967s	0.904	2.658	78.482
1900	.626	68.494	70.21	42.251f	4.969s	0.954	2.515	78.481
2000	.660	72.170	73.97	42.251f	4.970s	1.005	2.387	78.481
2100	.694	75.845	77.74	42.251f	4.972s	1.055	2.271	78.480
2200	.727	79.520	81.51	42.251f	4.973s	1.105	2.167	78.480
2300	.761	83.195	85.27	42.251f	4.974s	1.155	2.071	78.480
2400	.794	86.870	89.04	42.251f	4.975s	1.205	1.983	78.479
2500	.828	90.545	92.81	42.251f	4.976s	1.255	1.903	78.479
2600	.862	94.220	96.58	42.251f	4.977s	1.305	1.828	78.479
2700	.895	97.895	100.34	42.251f	4.978s	1.355	1.760	78.478
2800	.929	101.570	104.11	42.251f	4.979s	1.405	1.696	78.478
2900	.962	105.245	107.88	42.251f	4.980s	1.455	1.637	78.477
3000	.996	108.920	111.64	42.251f	4.980s	1.505	1.582	78.477
3012	1.000	109.348	112.08	42.251f	4.980s	1.511		

Soundings in mm.-----Other distances in METERS.-----

WBTk5.S Reference Point: Long.= 42.250f Trans.= 5.003s Vert.= -0.012

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk6.P, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.001	0.054	0.05	46.080f	4.536p	0.056	0.033	0.486
200	.005	0.201	0.21	46.153f	4.720p	0.133	0.203	3.952
300	.012	0.469	0.48	46.234f	4.784p	0.202	0.245	5.459
400	.021	0.829	0.85	46.302f	4.808p	0.266	0.316	7.643
500	.033	1.280	1.31	46.365f	4.825p	0.332	0.358	9.389
600	.047	1.830	1.88	46.429f	4.845p	0.398	0.414	11.307
700	.063	2.475	2.54	46.493f	4.861p	0.464	0.497	13.927
800	.082	3.215	3.30	46.556f	4.876p	0.530	0.553	15.907
900	.104	4.050	4.15	46.619f	4.887p	0.596	0.614	17.872
1000	.128	4.980	5.10	46.682f	4.896p	0.663	0.676	19.932
1100	.154	6.005	6.16	46.746f	4.904p	0.729	0.736	21.927
1200	.183	7.126	7.30	46.810f	4.911p	0.795	0.801	24.036
1300	.214	8.332	8.54	46.872f	4.917p	0.861	0.776	25.371
1400	.246	9.593	9.83	46.928f	4.920p	0.926	0.825	27.268
1500	.281	10.979	11.25	46.990f	4.925p	0.992	0.995	30.116
1600	.320	12.486	12.80	47.057f	4.929p	1.059	1.054	31.960
1700	.361	14.087	14.44	47.124f	4.934p	1.126	1.009	32.987
1800	.405	15.780	16.17	47.189f	4.935p	1.194	1.166	35.771
1900	.451	17.576	18.02	47.256f	4.938p	1.261	1.262	38.392
2000	.499	19.466	19.95	47.322f	4.941p	1.328	1.306	40.468
2100	.549	21.406	21.94	47.382f	4.944p	1.393	1.198	40.873
2200	.599	23.352	23.94	47.432f	4.948p	1.456	1.105	41.193
2300	.649	25.304	25.94	47.476f	4.951p	1.517	1.027	41.478
2400	.699	27.259	27.94	47.513f	4.954p	1.577	0.955	41.624
2500	.749	29.216	29.95	47.546f	4.957p	1.636	0.894	41.736
2600	.799	31.175	31.95	47.574f	4.959p	1.693	0.838	41.760
2700	.849	33.133	33.96	47.599f	4.962p	1.750	0.789	41.783
2800	.900	35.092	35.97	47.622f	4.964p	1.805	0.744	41.807
2900	.950	37.052	37.98	47.642f	4.966p	1.861	0.706	41.831
3000	1.000	39.012	39.99	47.660f	4.967p	1.915		

Soundings in mm.-----Other distances in METERS.-----

WBTk6.P Reference Point: Long.= 48.000f Trans.= 5.005p Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL
TANK CHARACTERISTICS
No Trim, No Heel

Tank: WBTk6.C, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.002	0.068	0.07	46.092f	0.000	0.056	0.039	0.715
200	.006	0.245	0.25	46.164f	0.000	0.130	0.184	5.060
300	.013	0.532	0.55	46.235f	0.000	0.197	0.248	7.182
400	.023	0.916	0.94	46.301f	0.000	0.261	0.306	9.219
500	.035	1.393	1.43	46.366f	0.000	0.327	0.362	11.227
600	.050	1.967	2.02	46.430f	0.000	0.392	0.424	13.283
700	.067	2.635	2.70	46.494f	0.000	0.458	0.486	15.298
800	.086	3.399	3.48	46.559f	0.000	0.524	0.546	17.330
900	.108	4.258	4.36	46.623f	0.000	0.590	0.609	19.374
1000	.132	5.213	5.34	46.688f	0.000	0.656	0.673	21.417
1100	.158	6.264	6.42	46.752f	0.000	0.722	0.736	23.461
1200	.187	7.411	7.60	46.817f	0.000	0.788	0.798	25.505
1300	.218	8.646	8.86	46.881f	0.000	0.855	0.833	26.837
1400	.252	9.974	10.22	46.944f	0.000	0.921	0.927	29.593
1500	.288	11.400	11.69	47.008f	0.000	0.987	0.992	31.646
1600	.326	12.931	13.25	47.073f	0.000	1.054	1.059	33.731
1700	.368	14.557	14.92	47.139f	0.000	1.120	1.120	35.750
1800	.411	16.278	16.69	47.204f	0.000	1.187	1.187	37.839
1900	.457	18.097	18.55	47.270f	0.000	1.254	1.251	39.897
2000	.505	20.010	20.51	47.335f	0.000	1.320	1.306	41.854
2100	.555	21.970	22.52	47.394f	0.000	1.385	1.190	41.854
2200	.604	23.930	24.53	47.444f	0.000	1.448	1.093	41.854
2300	.654	25.890	26.54	47.486f	0.000	1.509	1.010	41.854
2400	.703	27.850	28.55	47.522f	0.000	1.568	0.939	41.854
2500	.753	29.810	30.56	47.554f	0.000	1.626	0.877	41.854
2600	.802	31.770	32.56	47.581f	0.000	1.683	0.823	41.854
2700	.852	33.730	34.57	47.606f	0.000	1.739	0.775	41.854
2800	.901	35.690	36.58	47.627f	0.000	1.795	0.733	41.854
2900	.951	37.650	38.59	47.647f	0.000	1.850	0.694	41.854
3000	1.000	39.610	40.60	47.664f	0.000	1.904		

Soundings in mm.-----Other distances in METERS.-----

WBTk6.C Reference Point: Long.= 48.000f Trans.= 0.000 Vert.= 0.000

(Zero Sounding is at the Reference Point.)

15-02-19 13:23:11 Cybermarine Technologies PTE. Ltd.
GHS 16.50

NOT PART OF THIS APPROVAL

TANK CHARACTERISTICS

No Trim, No Heel

Tank: WBTk6.S, Contents: SALT WATER at 1.025 Specific Gravity

Snding	Load	Volume CU. METERS	Weight METRIC TON	Center of Gravity			GML	FSM m.-MT
				LCG	TCG	VCG		
0	.000	0.000	0.00					
100	.001	0.054	0.05	46.080f	4.536s	0.056	0.033	0.486
200	.005	0.201	0.21	46.153f	4.720s	0.133	0.203	3.952
300	.012	0.469	0.48	46.234f	4.784s	0.202	0.241	5.459
400	.021	0.829	0.85	46.302f	4.808s	0.266	0.316	7.643
500	.033	1.280	1.31	46.365f	4.825s	0.332	0.358	9.389
600	.047	1.830	1.88	46.429f	4.845s	0.398	0.414	11.307
700	.063	2.475	2.54	46.493f	4.861s	0.464	0.497	13.927
800	.082	3.215	3.30	46.556f	4.876s	0.530	0.553	15.907
900	.104	4.050	4.15	46.619f	4.887s	0.596	0.614	17.872
1000	.128	4.980	5.10	46.682f	4.896s	0.663	0.677	19.932
1100	.154	6.005	6.16	46.746f	4.904s	0.729	0.736	21.927
1200	.183	7.126	7.30	46.810f	4.911s	0.795	0.801	24.036
1300	.214	8.332	8.54	46.872f	4.917s	0.861	0.776	25.371
1400	.246	9.593	9.83	46.928f	4.920s	0.926	0.825	27.268
1500	.281	10.979	11.25	46.990f	4.925s	0.992	0.995	30.116
1600	.320	12.486	12.80	47.057f	4.929s	1.059	1.054	31.960
1700	.361	14.087	14.44	47.124f	4.934s	1.126	1.009	32.987
1800	.405	15.780	16.17	47.189f	4.935s	1.194	1.166	35.771
1900	.451	17.576	18.02	47.256f	4.938s	1.261	1.262	38.392
2000	.499	19.466	19.95	47.322f	4.941s	1.328	1.306	40.468
2100	.549	21.406	21.94	47.382f	4.944s	1.393	1.198	40.873
2200	.599	23.352	23.94	47.432f	4.948s	1.456	1.105	41.193
2300	.649	25.304	25.94	47.476f	4.951s	1.517	1.027	41.478
2400	.699	27.259	27.94	47.513f	4.954s	1.577	0.955	41.624
2500	.749	29.216	29.95	47.546f	4.957s	1.636	0.894	41.736
2600	.799	31.175	31.95	47.574f	4.959s	1.693	0.838	41.760
2700	.849	33.133	33.96	47.599f	4.962s	1.750	0.789	41.783
2800	.900	35.092	35.97	47.622f	4.964s	1.805	0.744	41.807
2900	.950	37.052	37.98	47.642f	4.966s	1.861	0.706	41.831
3000	1.000	39.012	39.99	47.660f	4.967s	1.915		

Soundings in mm.-----Other distances in METERS.-----

WBTk6.S Reference Point: Long.= 48.000f Trans.= 5.005s Vert.= 0.000

(Zero Sounding is at the Reference Point.)

HYDROSTATIC PARTICULARS

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Fwd 0.500/50.000, No Heel, Fixed VCG = 0.000

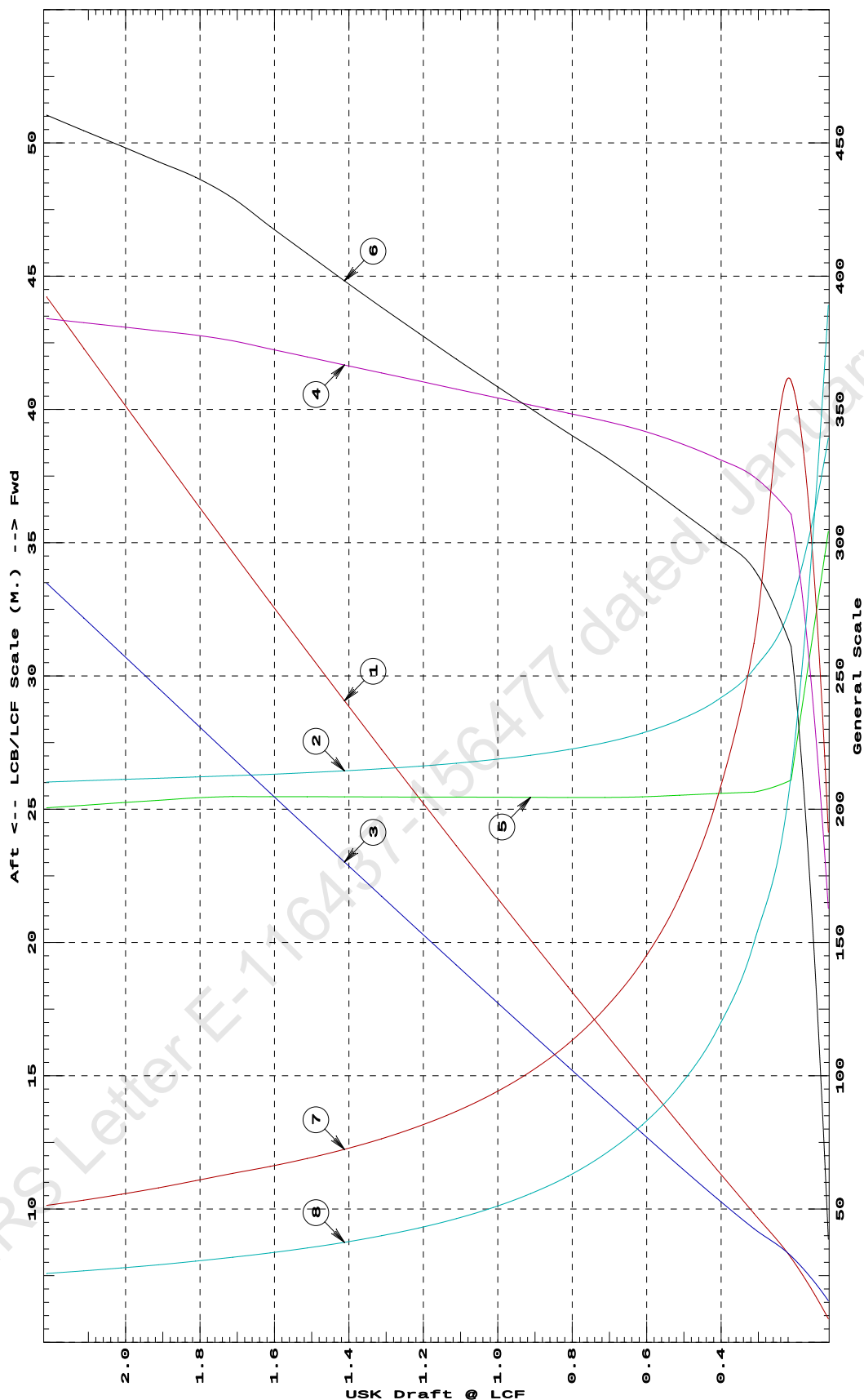
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT-----	
0.112	35.31	38.933f	0.062	3.26	35.422f	2.70	382.72	155.640	
0.212	126.67	32.680f	0.129	6.21	26.091f	18.28	721.37	85.218	
0.312	193.60	30.258f	0.171	6.49	25.644f	20.31	524.43	59.944	
0.412	259.63	29.079f	0.216	6.63	25.587f	21.12	406.76	46.782	
0.512	326.98	28.354f	0.264	6.75	25.525f	21.85	334.10	38.458	
0.612	395.34	27.859f	0.314	6.84	25.465f	22.58	285.53	32.636	
0.712	464.24	27.501f	0.364	6.91	25.439f	23.27	250.56	28.275	
0.812	533.64	27.233f	0.414	6.97	25.443f	23.89	223.80	24.906	
0.912	603.65	27.026f	0.465	7.03	25.446f	24.52	203.11	22.305	
1.012	674.25	26.861f	0.515	7.09	25.451f	25.17	186.64	20.238	
1.112	745.46	26.726f	0.566	7.15	25.454f	25.83	173.21	18.554	
1.212	817.28	26.614f	0.618	7.21	25.457f	26.50	162.12	17.166	
1.312	889.68	26.520f	0.669	7.27	25.462f	27.18	152.75	15.996	
1.412	962.69	26.440f	0.721	7.33	25.465f	27.88	144.78	15.003	
1.512	1,036.38	26.371f	0.773	7.39	25.470f	28.59	137.93	14.149	
1.612	1,110.58	26.311f	0.825	7.45	25.473f	29.31	131.94	13.408	
1.712	1,185.40	26.258f	0.877	7.51	25.476f	30.05	126.74	12.763	
1.812	1,261.19	26.210f	0.929	7.56	25.427f	30.60	121.30	12.168	
1.912	1,337.62	26.163f	0.982	7.59	25.335f	31.00	115.89	11.627	
2.012	1,414.37	26.115f	1.034	7.62	25.243f	31.42	111.05	11.147	
2.112	1,491.44	26.068f	1.087	7.65	25.148f	31.82	106.66	10.716	
2.212	1,568.83	26.020f	1.139	7.68	25.053f	32.23	102.71	10.332	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HYDROSTATIC PROPERTIES at 0.5 M. FWD TRIM



- ① Displacement 1=4 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.004 M.
- ④ Immersion 1=.02 MT/cm
- ④ WPA 1=1.95 Sq.M.
- ⑤ LCF (use top scale)
- ⑥ Moment/Trim 1=.07 M.-MT/cm
- ⑦ KML 1=2 M.
- ⑧ KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Fwd 0.250/50.000, No Heel, Fixed VCG = 0.000

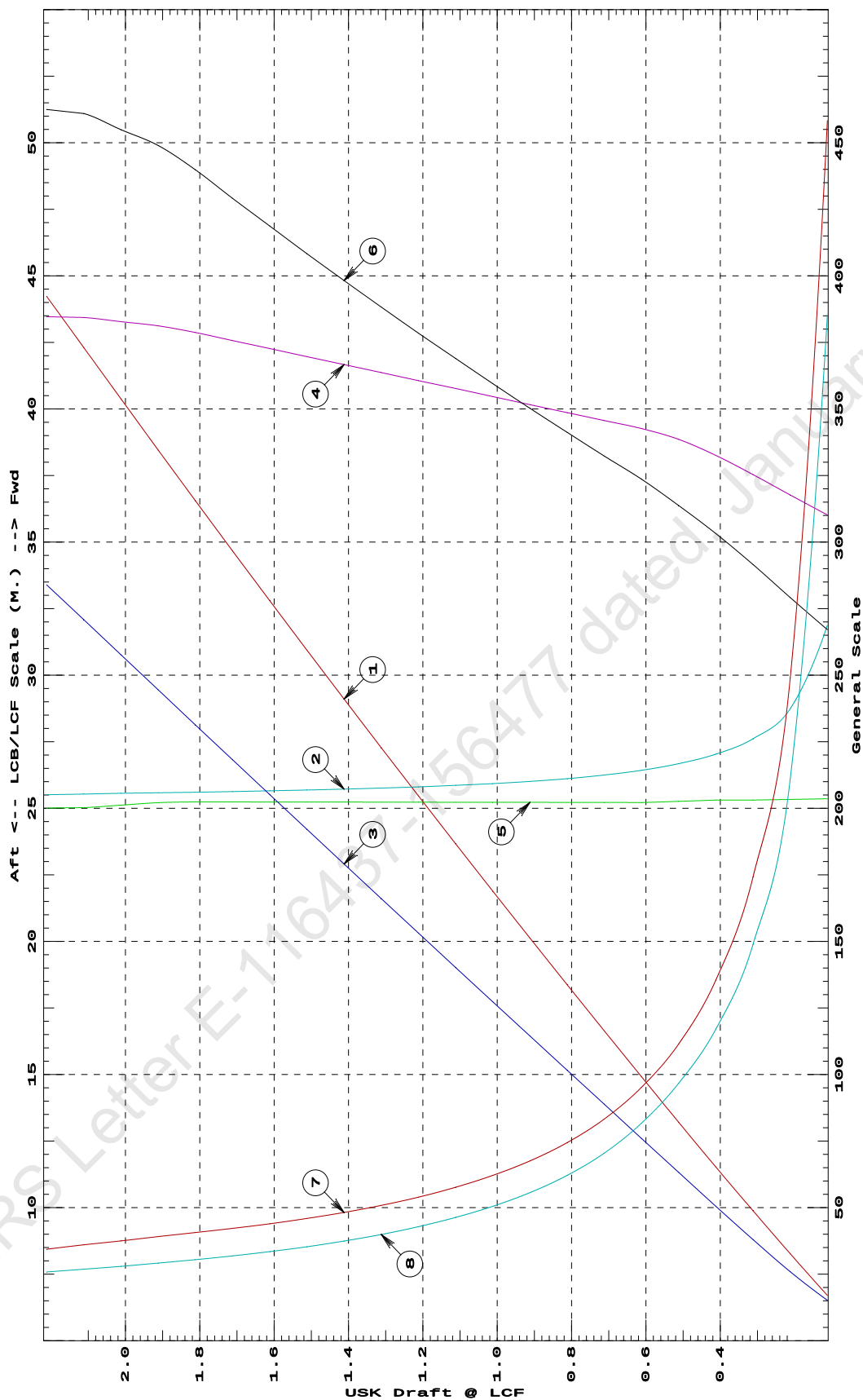
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT-----	
0.112	67.99	31.878f	0.061	6.20	25.355f	18.70	1375.02	154.077	
0.212	130.85	28.739f	0.104	6.35	25.333f	19.54	746.69	84.364	
0.312	195.25	27.612f	0.152	6.51	25.309f	20.41	522.70	59.562	
0.412	261.01	27.032f	0.202	6.65	25.309f	21.22	406.53	46.823	
0.512	328.33	26.675f	0.253	6.77	25.258f	21.96	334.41	38.687	
0.612	396.62	26.426f	0.304	6.85	25.218f	22.65	285.59	32.693	
0.712	465.44	26.248f	0.355	6.91	25.219f	23.26	249.88	28.193	
0.812	534.86	26.114f	0.406	6.97	25.221f	23.89	223.30	24.847	
0.912	604.89	26.011f	0.458	7.03	25.223f	24.52	202.66	22.253	
1.012	675.51	25.929f	0.509	7.09	25.225f	25.16	186.26	20.193	
1.112	746.74	25.862f	0.561	7.15	25.227f	25.83	172.93	18.521	
1.212	818.57	25.806f	0.613	7.21	25.229f	26.49	161.82	17.132	
1.312	891.00	25.759f	0.664	7.27	25.230f	27.18	152.52	15.970	
1.412	964.03	25.719f	0.716	7.33	25.233f	27.88	144.57	14.979	
1.512	1,037.66	25.685f	0.768	7.39	25.234f	28.58	137.73	14.127	
1.612	1,111.90	25.655f	0.821	7.45	25.237f	29.31	131.80	13.391	
1.712	1,186.73	25.629f	0.873	7.51	25.238f	30.04	126.56	12.744	
1.812	1,262.18	25.605f	0.925	7.57	25.240f	30.79	121.97	12.177	
1.912	1,338.33	25.584f	0.978	7.62	25.209f	31.43	117.41	11.662	
2.012	1,415.08	25.561f	1.031	7.66	25.117f	31.84	112.52	11.180	
2.112	1,492.21	25.536f	1.083	7.69	25.024f	32.27	108.13	10.753	
2.212	1,569.14	25.510f	1.136	7.69	25.009f	32.37	103.16	10.342	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HYDROSTATIC PROPERTIES at 0.25 M. FWD TRIM



- ① Displacement 1=4 MT
 ② LCB (use top scale)
 ③ VCB (KB) 1=.004 M.
 ④ Immersion 1=.02 MT/cm
 ⑤ WPA 1=1.95 Sq.M.
 ⑥ LCF (use top scale)
 ⑦ Moment/Trim 1=.07 M.-MT/cm
 ⑧ KML 1=3 M.
 ⑨ KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
Trim is per 50 M. "K" = BPL

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES
No Trim, No Heel, Fixed VCG = 0.000

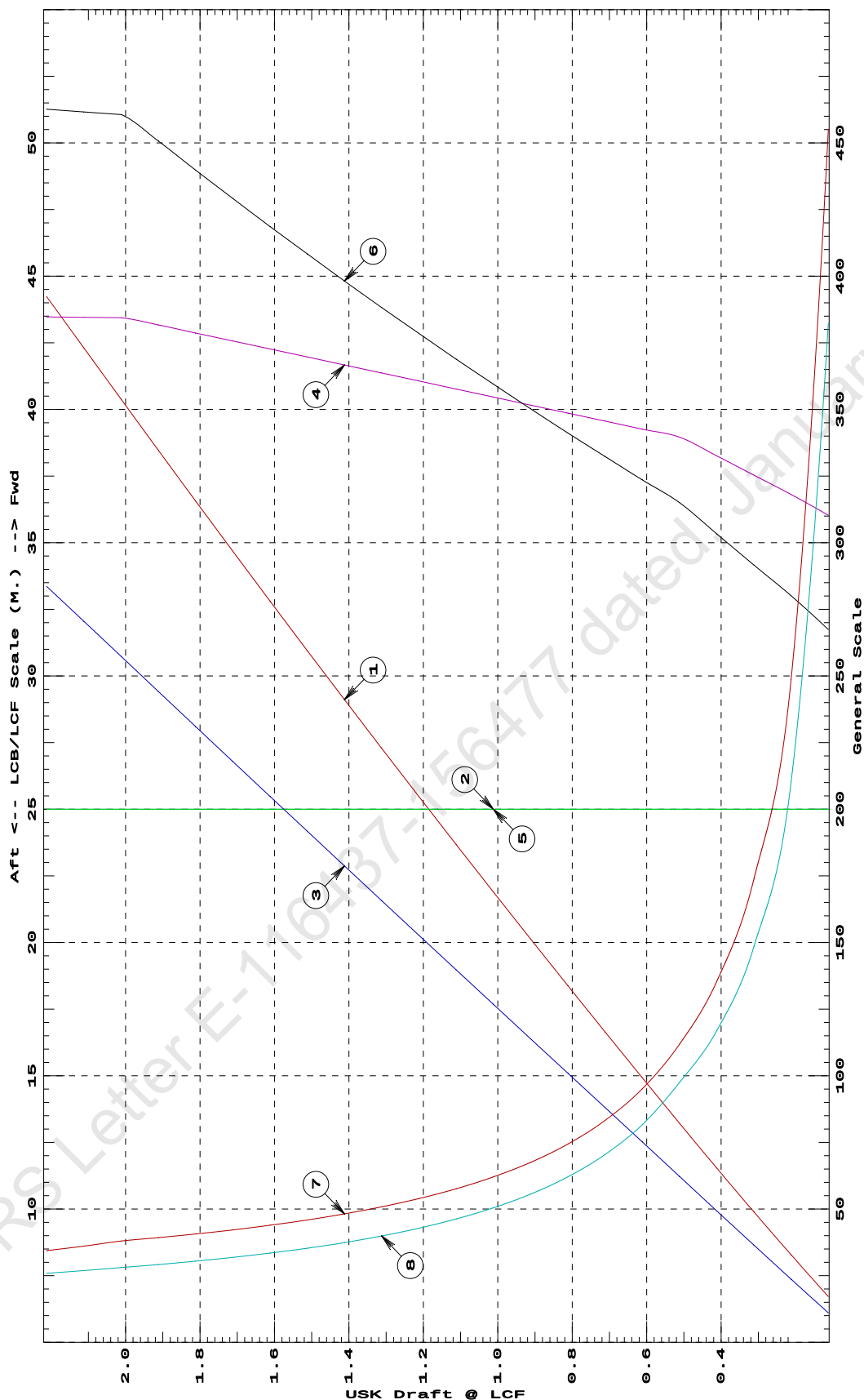
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT	
0.112	68.53	25.000f	0.044	6.21	25.000f	18.72	1365.62	152.943	
0.212	131.38	25.000f	0.095	6.36	25.000f	19.61	746.26	84.110	
0.312	195.72	25.000f	0.146	6.51	25.000f	20.41	521.46	59.400	
0.412	261.52	25.000f	0.197	6.65	25.000f	21.24	406.00	46.750	
0.512	328.77	25.000f	0.249	6.79	25.000f	22.05	335.40	38.975	
0.612	396.99	25.000f	0.301	6.85	25.000f	22.65	285.27	32.654	
0.712	465.83	25.000f	0.352	6.91	25.000f	23.27	249.73	28.175	
0.812	535.26	25.000f	0.404	6.97	25.000f	23.88	223.09	24.822	
0.912	605.29	25.000f	0.456	7.03	25.000f	24.52	202.53	22.237	
1.012	675.92	25.000f	0.507	7.09	25.000f	25.16	186.15	20.181	
1.112	747.15	25.000f	0.559	7.15	25.000f	25.82	172.79	18.506	
1.212	818.99	25.000f	0.611	7.21	25.000f	26.50	161.77	17.125	
1.312	891.43	25.000f	0.663	7.27	25.000f	27.18	152.43	15.959	
1.412	964.46	25.000f	0.715	7.33	25.000f	27.87	144.50	14.970	
1.512	1,038.11	25.000f	0.767	7.39	25.000f	28.59	137.68	14.122	
1.612	1,112.35	25.000f	0.819	7.45	25.000f	29.30	131.72	13.382	
1.712	1,187.19	25.000f	0.872	7.51	25.000f	30.04	126.53	12.740	
1.812	1,262.64	25.000f	0.924	7.57	25.000f	30.79	121.92	12.171	
1.912	1,338.68	25.000f	0.977	7.63	25.000f	31.55	117.84	11.669	
2.012	1,415.33	25.000f	1.029	7.69	25.000f	32.24	113.91	11.217	
2.112	1,492.22	25.000f	1.082	7.69	25.000f	32.31	108.27	10.755	
2.212	1,569.15	25.000f	1.134	7.69	25.000f	32.38	103.19	10.344	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HYDROSTATIC PROPERTIES at LEVEL TRIM



- ① Displacement 1=4 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.004 M.
- ④ Immersion 1=.02 MT/cm
- ⑤ WPA 1=1.95 Sq.M.
- ⑥ LCF (use top scale)
- ⑦ Moment/Trim 1=.07 M.-MT/cm
- ⑧ KML 1=3 M.
- ⑨ KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Aft 0.250/50.000, No Heel, Fixed VCG = 0.000

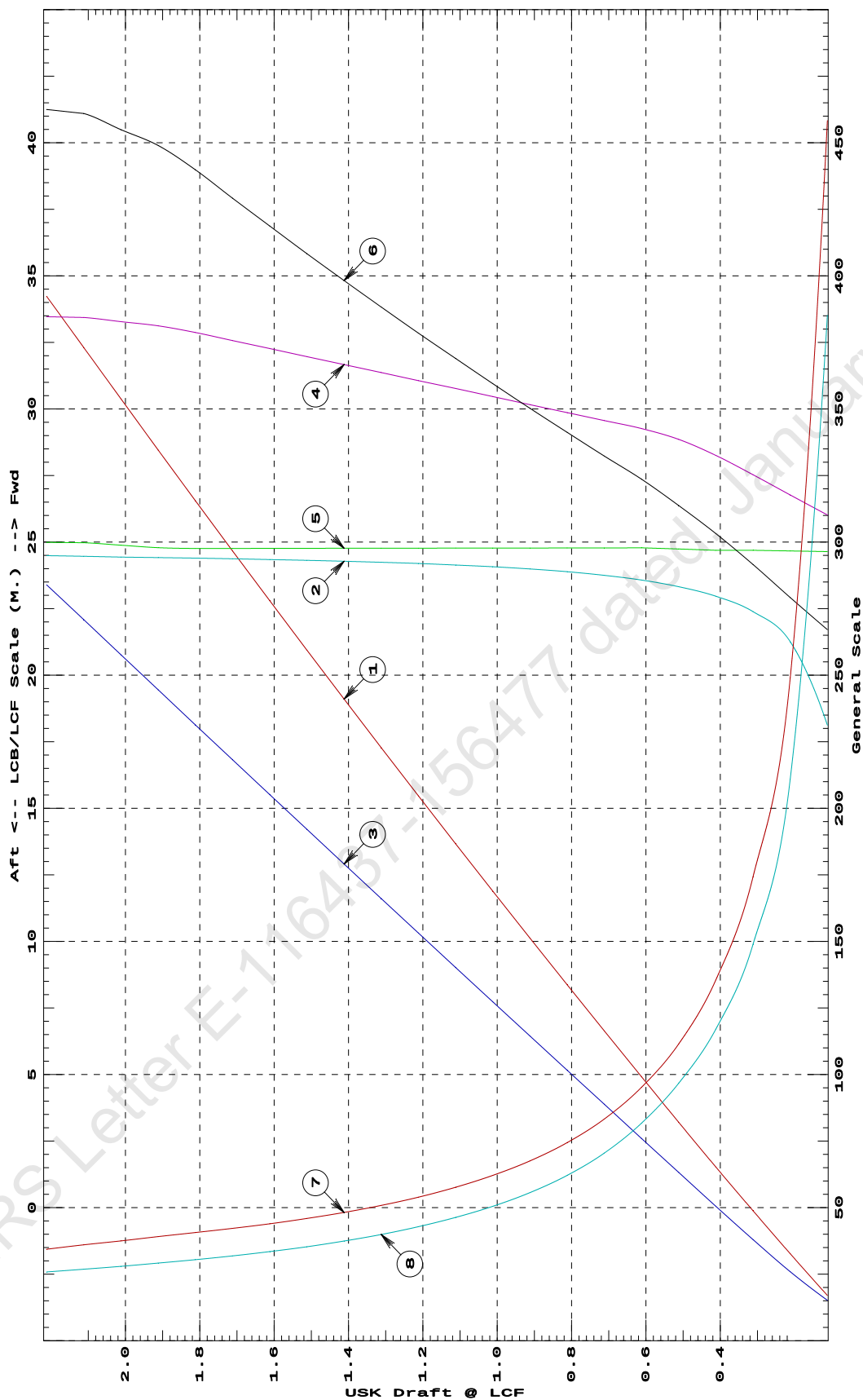
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT-----	
0.112	67.99	18.121f	0.061	6.20	24.645f	18.70	1375.05	154.081	
0.212	130.84	21.260f	0.104	6.35	24.667f	19.54	746.72	84.368	
0.312	195.22	22.387f	0.152	6.51	24.691f	20.41	522.75	59.569	
0.412	260.99	22.967f	0.202	6.65	24.691f	21.22	406.56	46.826	
0.512	328.32	23.325f	0.253	6.77	24.742f	21.96	334.41	38.687	
0.612	396.63	23.574f	0.304	6.85	24.782f	22.65	285.59	32.692	
0.712	465.44	23.752f	0.355	6.91	24.781f	23.26	249.88	28.193	
0.812	534.88	23.886f	0.406	6.97	24.779f	23.89	223.29	24.846	
0.912	604.91	23.989f	0.458	7.03	24.777f	24.52	202.66	22.253	
1.012	675.53	24.071f	0.509	7.09	24.775f	25.16	186.25	20.192	
1.112	746.76	24.138f	0.561	7.15	24.773f	25.83	172.92	18.521	
1.212	818.59	24.194f	0.613	7.21	24.771f	26.49	161.82	17.132	
1.312	891.02	24.241f	0.664	7.27	24.770f	27.18	152.52	15.969	
1.412	964.05	24.281f	0.716	7.33	24.767f	27.88	144.57	14.978	
1.512	1,037.68	24.315f	0.768	7.39	24.766f	28.58	137.72	14.127	
1.612	1,111.92	24.345f	0.821	7.45	24.764f	29.31	131.80	13.391	
1.712	1,186.75	24.371f	0.873	7.51	24.762f	30.04	126.56	12.744	
1.812	1,262.19	24.395f	0.925	7.57	24.760f	30.79	121.97	12.177	
1.912	1,338.33	24.416f	0.978	7.62	24.791f	31.43	117.42	11.662	
2.012	1,415.08	24.439f	1.031	7.66	24.883f	31.84	112.52	11.180	
2.112	1,492.15	24.464f	1.083	7.69	24.976f	32.27	108.13	10.753	
2.212	1,569.13	24.490f	1.136	7.69	24.991f	32.37	103.16	10.342	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

Draft is from USK.

HYDROSTATIC PROPERTIES at 0.25 M. AFT TRIM



- ① Displacement 1=4 MT
 ② LCB (use top scale)
 ③ VCB (KB) 1=.004 M.
 ④ Immersion 1=.02 MT/cm
 ⑤ WPA 1=1.95 Sq.M.
 ⑥ LCF (use top scale)
 ⑦ Moment/Trim 1=.07 M.-MT/cm
 ⑧ KML 1=3 M.
 ⑨ KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

15-02-19 10:56:08 Cybermarine Technologies PTE. Ltd.
GHS 16.50

HYDROSTATIC PROPERTIES

Trim: Aft 0.500/50.000, No Heel, Fixed VCG = 0.000

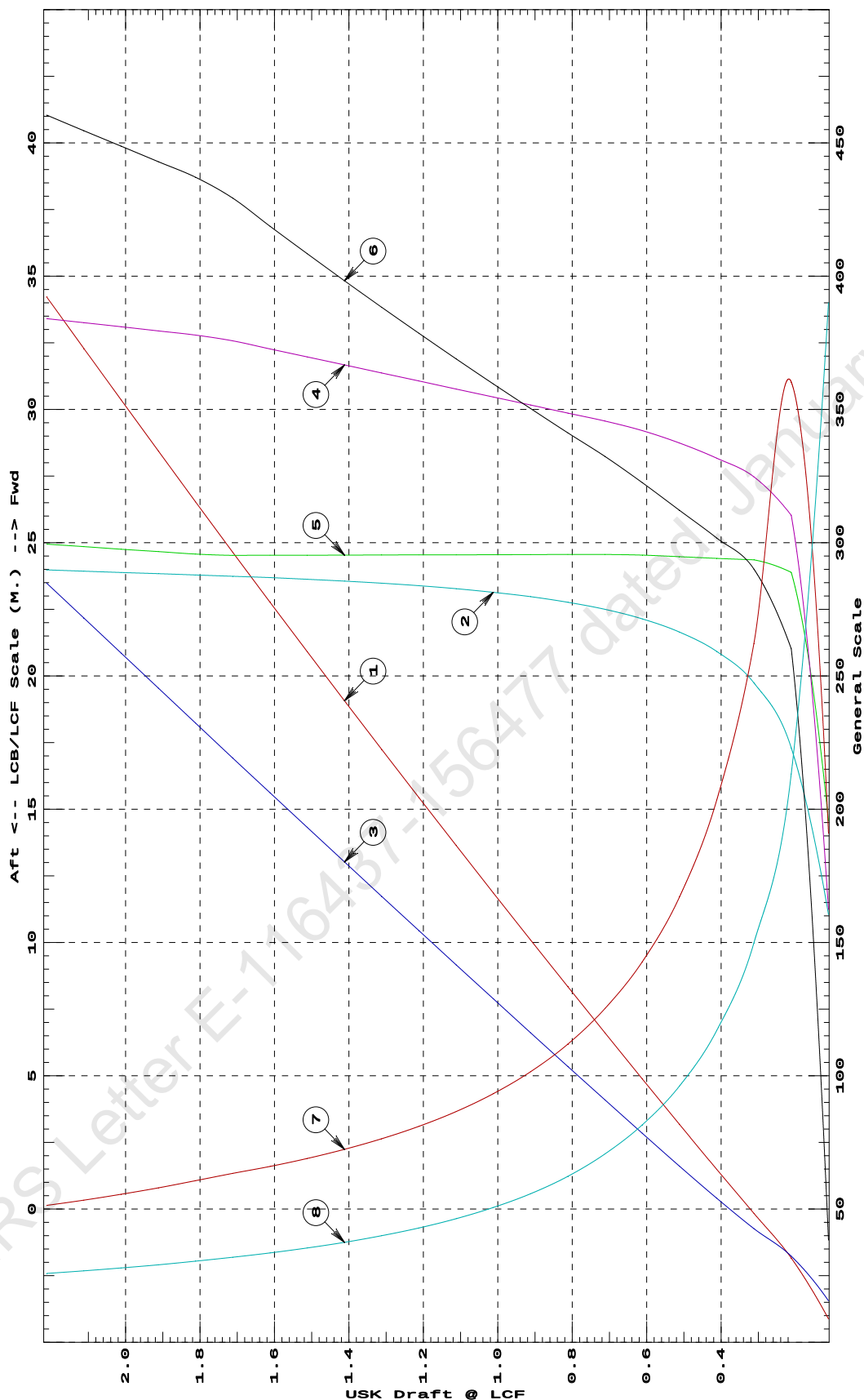
LCF	Displacement	Buoyancy-Ctr.		Weight/		Moment/			
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF----	cm trim----	KML-----	KMT-----	
0.112	35.14	11.050f	0.062	3.25	14.553f	2.68	381.82	155.997	
0.212	126.38	17.305f	0.129	6.21	23.886f	18.21	720.54	85.309	
0.312	193.59	19.742f	0.171	6.49	24.356f	20.31	524.44	59.946	
0.412	259.63	20.921f	0.216	6.63	24.413f	21.12	406.77	46.784	
0.512	326.97	21.646f	0.264	6.75	24.475f	21.85	334.11	38.459	
0.612	395.34	22.141f	0.314	6.84	24.535f	22.58	285.54	32.637	
0.712	464.31	22.499f	0.364	6.91	24.561f	23.27	250.53	28.272	
0.812	533.71	22.767f	0.414	6.97	24.557f	23.89	223.77	24.904	
0.912	603.73	22.974f	0.465	7.03	24.554f	24.52	203.09	22.303	
1.012	674.39	23.140f	0.515	7.09	24.549f	25.17	186.61	20.234	
1.112	745.60	23.274f	0.567	7.15	24.546f	25.83	173.19	18.551	
1.212	817.41	23.386f	0.618	7.21	24.543f	26.50	162.10	17.163	
1.312	889.82	23.480f	0.669	7.27	24.538f	27.18	152.73	15.994	
1.412	962.83	23.560f	0.721	7.33	24.535f	27.88	144.77	15.001	
1.512	1,036.44	23.629f	0.773	7.39	24.530f	28.59	137.92	14.149	
1.612	1,110.65	23.689f	0.825	7.45	24.527f	29.31	131.94	13.407	
1.712	1,185.47	23.742f	0.877	7.51	24.524f	30.05	126.73	12.762	
1.812	1,261.17	23.790f	0.929	7.56	24.573f	30.60	121.30	12.168	
1.912	1,337.60	23.837f	0.982	7.59	24.665f	31.00	115.89	11.627	
2.012	1,414.34	23.885f	1.034	7.62	24.757f	31.42	111.06	11.148	
2.112	1,491.42	23.932f	1.087	7.65	24.852f	31.82	106.67	10.716	
2.212	1,568.81	23.980f	1.139	7.68	24.947f	32.23	102.71	10.332	

Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.

Trim is per 50.00m.

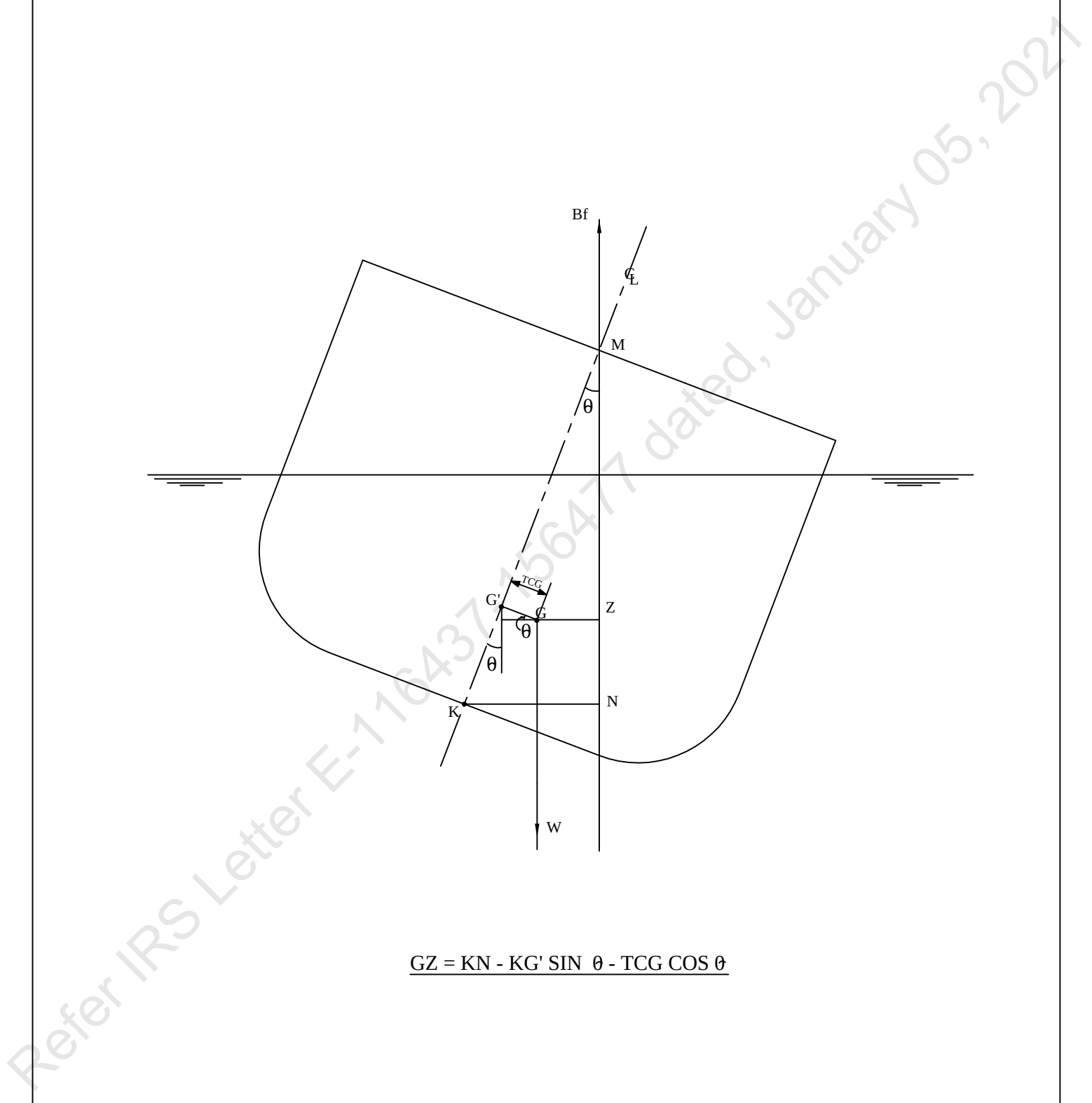
Draft is from USK.

HYDROSTATIC PROPERTIES at 0.5 M. AFT TRIM



- ① Displacement 1=4 MT
- ② LCB (use top scale)
- ③ VCB (KB) 1=.004 M.
- ④ Immersion 1=.02 MT/cm
- ⑤ WPA 1=1.95 Sq.M.
- ⑥ LCF (use top scale)
- ⑦ Moment/Trim 1=.07 M.-MT/cm
- ⑧ KML 1=2 M.
- ⑨ KMT 1=.4 M.

Specific Gravity = 1.025 Assumed KG = 0.00 M.
 Trim is per 50 M. "K" = BPL

[illegible]

The diagram shows a cross-section of a ship's hull. Key points and lines are labeled:

- B**: Center of Buoyancy, located on the vertical line of buoyancy.
- F**: Center of Flotation, located on the line of buoyancy.
- L**: Center of Lateral Resistance, located on the line of buoyancy.
- G**: Center of Gravity, located on the vertical line of gravity.
- G'**: A point on the line of buoyancy, representing the center of buoyancy at an angle θ .
- K**: A point on the line of buoyancy, representing the center of buoyancy at an angle θ .
- Z**: A point on the line of buoyancy, representing the center of buoyancy at an angle θ .
- N**: A point on the line of buoyancy, representing the center of buoyancy at an angle θ .
- M**: A point on the line of buoyancy, representing the center of buoyancy at an angle θ .
- Bf**: A vertical line representing the line of buoyancy.
- W**: A vertical line representing the line of gravity.
- TCG**: The Transverse Center of Gravity, shown as a horizontal distance from the line of buoyancy to the center of gravity.
- θ : The angle between the line of buoyancy and the line of gravity.

The formula for the distance from the center of buoyancy to the center of gravity is given by:

$$GZ = KN - KG' \sin \theta - TCG \cos \theta$$

Profile View

VOLUME USED FOR KN COMPUTATION



CROSS CURVE TABLES

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.
GHS 16.50

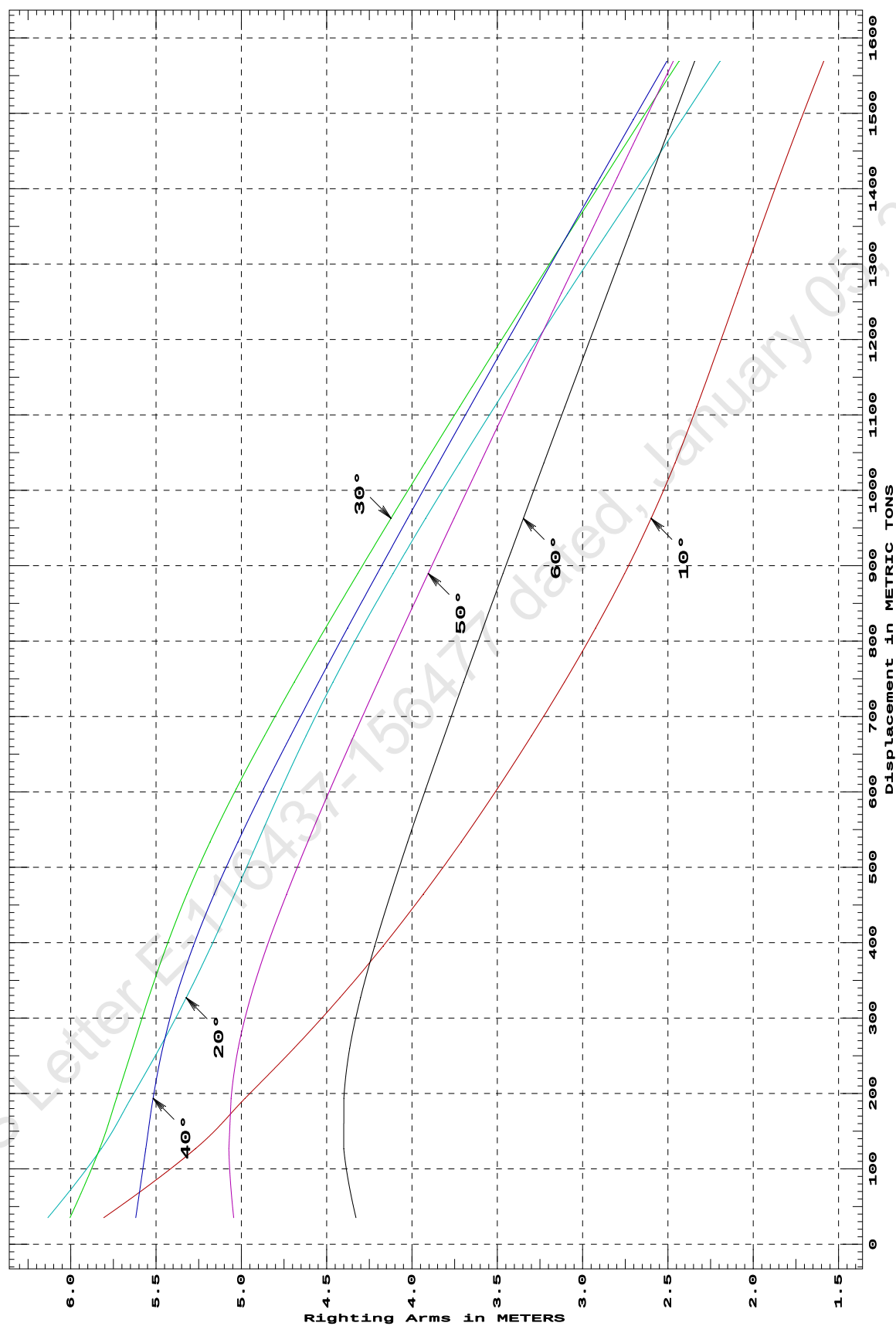
CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
35.31	5.807s	6.134s	6.004s	5.619s	5.045s	4.329s
126.67	5.270s	5.821s	5.830s	5.560s	5.073s	4.400s
193.60	4.978s	5.647s	5.732s	5.517s	5.060s	4.400s
259.63	4.693s	5.480s	5.639s	5.462s	5.019s	4.366s
326.98	4.423s	5.323s	5.543s	5.384s	4.947s	4.301s
395.34	4.170s	5.176s	5.439s	5.282s	4.849s	4.219s
464.24	3.934s	5.039s	5.319s	5.159s	4.735s	4.126s
533.64	3.712s	4.905s	5.183s	5.020s	4.611s	4.025s
603.65	3.501s	4.767s	5.030s	4.870s	4.479s	3.920s
674.25	3.300s	4.621s	4.865s	4.712s	4.342s	3.811s
745.46	3.107s	4.463s	4.690s	4.547s	4.200s	3.698s
817.28	2.924s	4.293s	4.507s	4.377s	4.054s	3.584s
889.68	2.754s	4.111s	4.318s	4.203s	3.905s	3.467s
962.69	2.599s	3.920s	4.123s	4.026s	3.754s	3.348s
1,036.38	2.458s	3.722s	3.924s	3.844s	3.600s	3.227s
1,110.58	2.331s	3.516s	3.721s	3.661s	3.444s	3.105s
1,185.40	2.211s	3.305s	3.514s	3.474s	3.286s	2.981s
1,261.19	2.093s	3.088s	3.303s	3.284s	3.125s	2.855s
1,337.62	1.973s	2.867s	3.089s	3.091s	2.962s	2.728s
1,414.37	1.850s	2.642s	2.873s	2.898s	2.799s	2.601s
1,491.44	1.721s	2.418s	2.656s	2.703s	2.634s	2.473s
1,568.83	1.588s	2.195s	2.436s	2.507s	2.469s	2.344s
Distances in METERS.---Specific Gravity = 1.025.-----						

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.
GHS 16.50

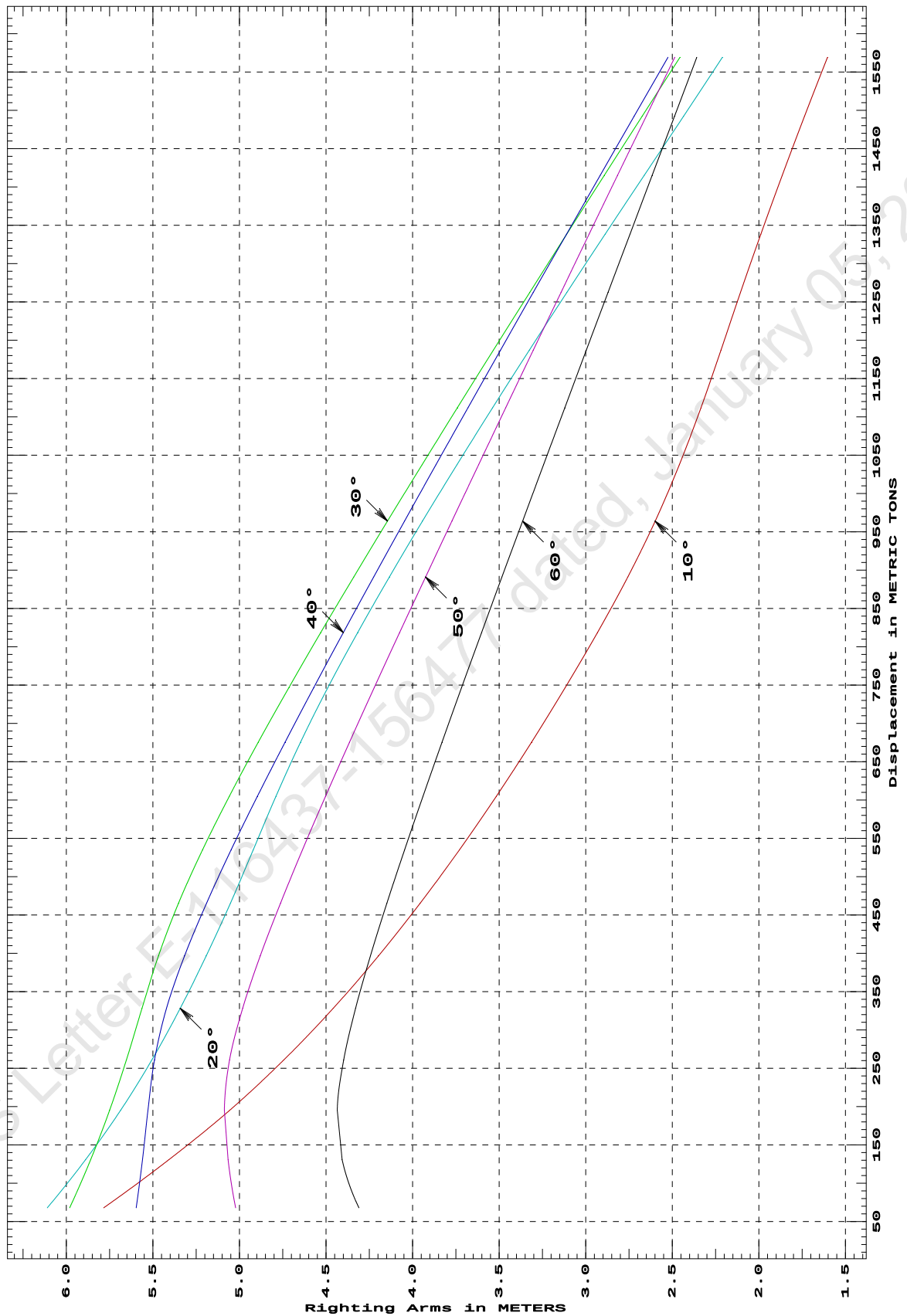
CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Fwd 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
67.99	5.785s	6.110s	5.981s	5.596s	5.022s	4.310s
130.85	5.407s	5.889s	5.859s	5.560s	5.066s	4.407s
195.25	5.056s	5.686s	5.750s	5.529s	5.087s	4.435s
261.01	4.744s	5.506s	5.653s	5.490s	5.057s	4.398s
328.33	4.459s	5.342s	5.563s	5.421s	4.981s	4.329s
396.62	4.197s	5.192s	5.468s	5.317s	4.880s	4.244s
465.44	3.956s	5.053s	5.352s	5.190s	4.763s	4.149s
534.86	3.730s	4.921s	5.214s	5.048s	4.637s	4.047s
604.89	3.516s	4.789s	5.058s	4.896s	4.503s	3.940s
675.51	3.313s	4.647s	4.891s	4.737s	4.364s	3.830s
746.74	3.119s	4.489s	4.714s	4.571s	4.221s	3.717s
818.57	2.932s	4.317s	4.530s	4.400s	4.075s	3.601s
891.00	2.757s	4.135s	4.340s	4.225s	3.925s	3.484s
964.03	2.599s	3.943s	4.145s	4.046s	3.773s	3.365s
1,037.66	2.459s	3.744s	3.945s	3.865s	3.619s	3.244s
1,111.90	2.334s	3.538s	3.742s	3.680s	3.462s	3.121s
1,186.73	2.219s	3.326s	3.534s	3.493s	3.303s	2.997s
1,262.18	2.107s	3.110s	3.324s	3.303s	3.143s	2.871s
1,338.33	1.989s	2.888s	3.110s	3.111s	2.980s	2.744s
1,415.08	1.865s	2.663s	2.893s	2.917s	2.816s	2.616s
1,492.21	1.736s	2.435s	2.675s	2.721s	2.651s	2.487s
1,569.14	1.604s	2.210s	2.456s	2.525s	2.486s	2.358s
Distances in METERS.---Specific Gravity = 1.025.-----						

CROSS CURVES OF STABILITY - Stbd Heel
at 0.25 M. FWD TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.
GHS 16.50

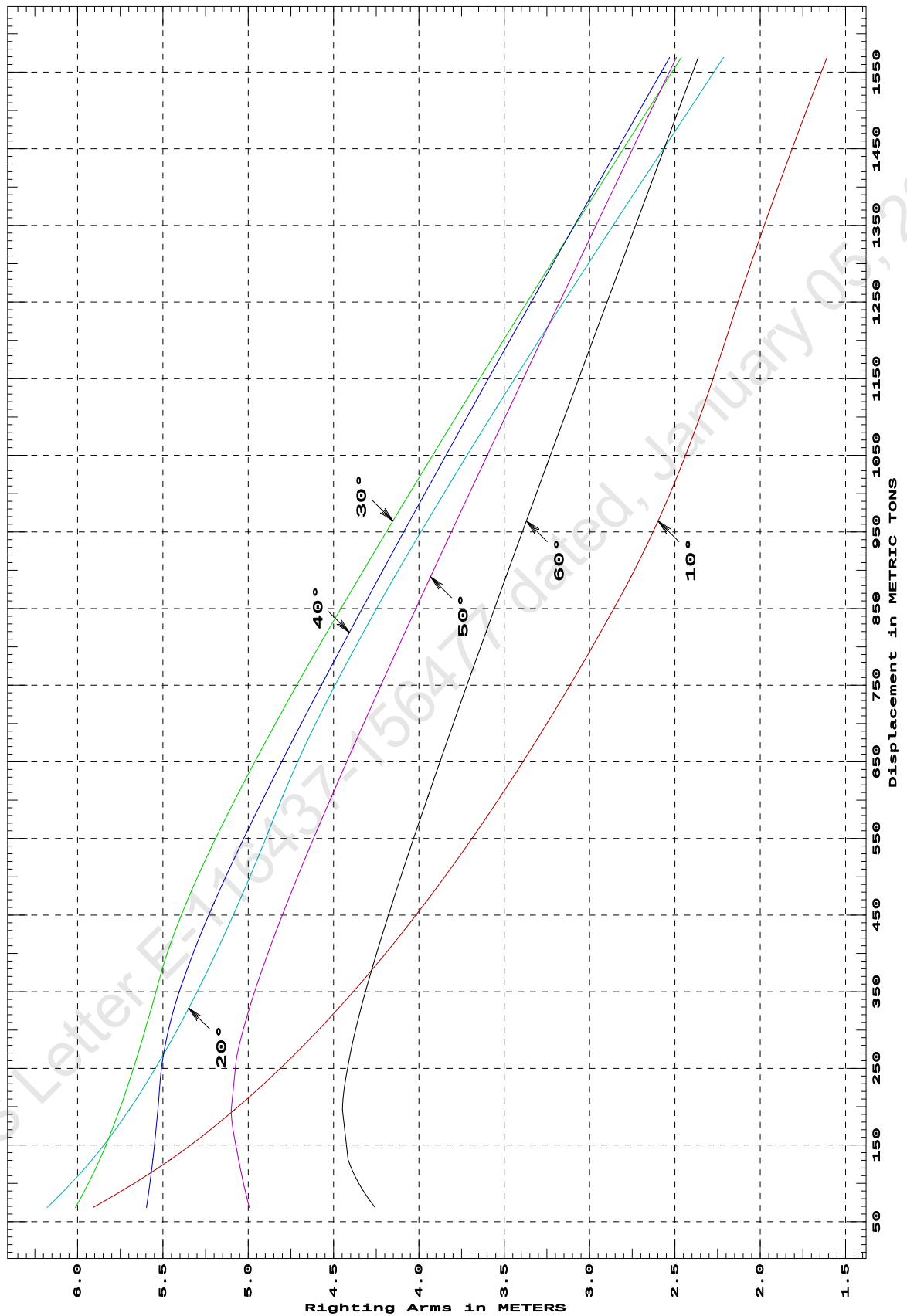
CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: zero at zero heel (trim righting arm held at zero)

Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
68.53	5.910s	6.179s	6.013s	5.596s	4.993s	4.256s
131.38	5.453s	5.913s	5.870s	5.558s	5.057s	4.416s
195.72	5.082s	5.699s	5.755s	5.529s	5.100s	4.448s
261.52	4.760s	5.514s	5.657s	5.502s	5.071s	4.409s
328.77	4.471s	5.349s	5.568s	5.434s	4.993s	4.339s
396.99	4.206s	5.197s	5.480s	5.328s	4.890s	4.252s
465.83	3.963s	5.057s	5.363s	5.200s	4.772s	4.156s
535.26	3.735s	4.925s	5.223s	5.058s	4.645s	4.054s
605.29	3.521s	4.798s	5.067s	4.905s	4.511s	3.947s
675.92	3.317s	4.657s	4.899s	4.745s	4.372s	3.836s
747.15	3.123s	4.498s	4.723s	4.579s	4.228s	3.723s
818.99	2.936s	4.326s	4.539s	4.407s	4.082s	3.607s
891.43	2.757s	4.143s	4.347s	4.232s	3.932s	3.490s
964.46	2.599s	3.950s	4.152s	4.053s	3.780s	3.370s
1,038.11	2.459s	3.751s	3.952s	3.871s	3.625s	3.249s
1,112.35	2.334s	3.545s	3.748s	3.687s	3.468s	3.126s
1,187.19	2.222s	3.333s	3.541s	3.499s	3.309s	3.002s
1,262.64	2.112s	3.116s	3.330s	3.310s	3.149s	2.877s
1,338.68	1.994s	2.895s	3.117s	3.117s	2.986s	2.750s
1,415.33	1.871s	2.670s	2.900s	2.923s	2.822s	2.621s
1,492.22	1.742s	2.441s	2.682s	2.728s	2.657s	2.493s
1,569.15	1.609s	2.216s	2.463s	2.531s	2.492s	2.363s
Distances in METERS.---Specific Gravity = 1.025.-----						

CROSS CURVES OF STABILITY - Stbd Heel
at LEVEL TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.250/50.000 at zero heel (trim righting arm held at zero)

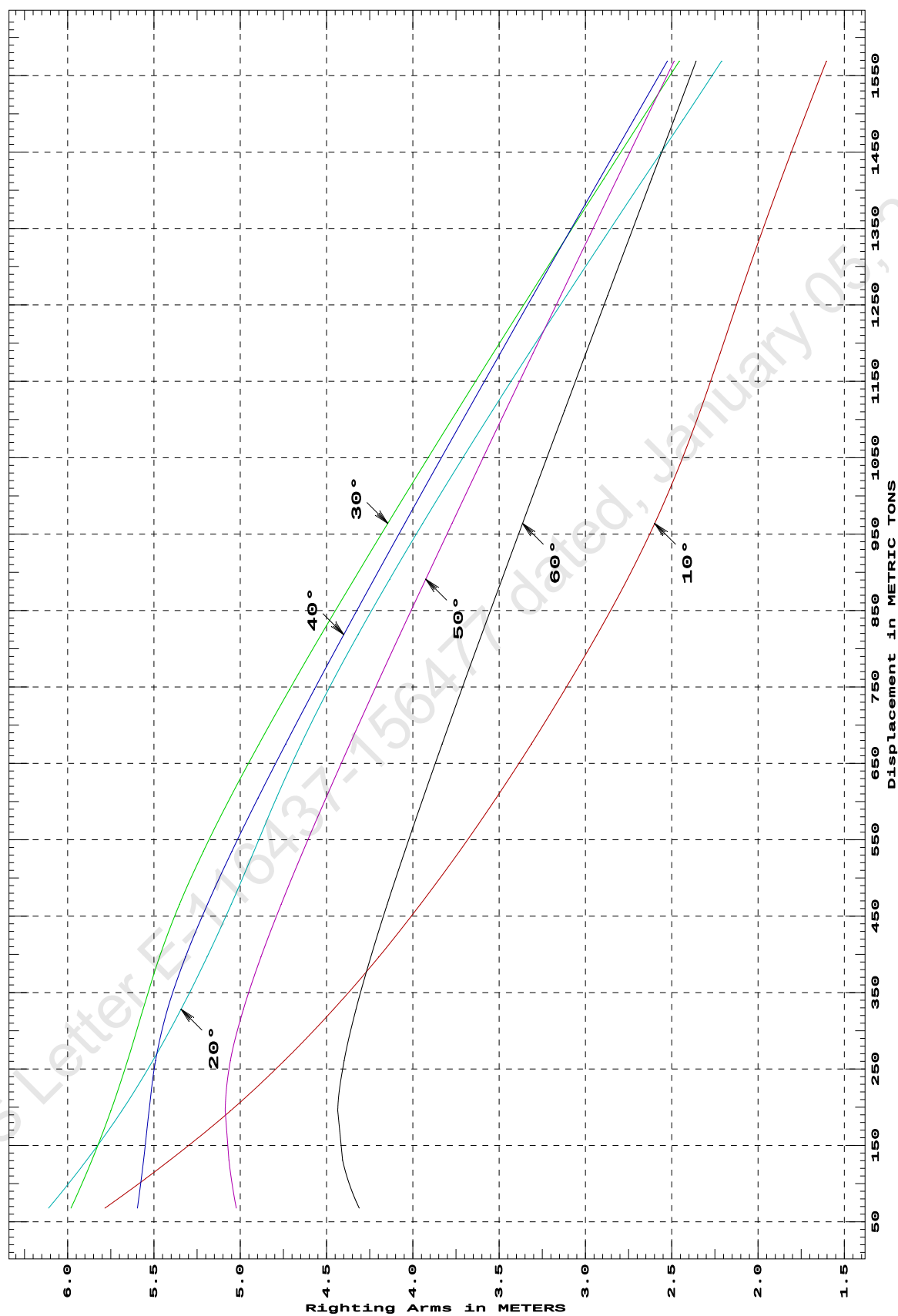
Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
67.99	5.784s	6.111s	5.981s	5.596s	5.023s	4.311s
130.84	5.407s	5.889s	5.860s	5.560s	5.066s	4.407s
195.22	5.057s	5.686s	5.750s	5.529s	5.087s	4.435s
260.99	4.744s	5.506s	5.653s	5.490s	5.057s	4.398s
328.32	4.459s	5.343s	5.563s	5.421s	4.981s	4.329s
396.63	4.197s	5.192s	5.468s	5.317s	4.880s	4.244s
465.44	3.956s	5.053s	5.352s	5.190s	4.763s	4.149s
534.88	3.730s	4.921s	5.214s	5.048s	4.637s	4.047s
604.91	3.516s	4.790s	5.058s	4.896s	4.503s	3.940s
675.53	3.313s	4.647s	4.891s	4.737s	4.364s	3.830s
746.76	3.119s	4.489s	4.714s	4.571s	4.221s	3.717s
818.59	2.932s	4.317s	4.530s	4.400s	4.075s	3.601s
891.02	2.757s	4.135s	4.340s	4.225s	3.925s	3.484s
964.05	2.599s	3.943s	4.145s	4.046s	3.773s	3.365s
1,037.68	2.459s	3.743s	3.945s	3.865s	3.619s	3.243s
1,111.92	2.334s	3.538s	3.742s	3.680s	3.462s	3.121s
1,186.75	2.219s	3.326s	3.534s	3.493s	3.303s	2.997s
1,262.19	2.107s	3.110s	3.324s	3.303s	3.143s	2.871s
1,338.33	1.989s	2.888s	3.110s	3.111s	2.980s	2.744s
1,415.08	1.865s	2.663s	2.893s	2.917s	2.816s	2.616s
1,492.15	1.736s	2.435s	2.675s	2.721s	2.651s	2.487s
1,569.13	1.604s	2.210s	2.456s	2.525s	2.486s	2.358s
Distances in METERS.---Specific Gravity = 1.025.-----						

15-02-19 10:56:39

Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY - Stbd Heel
at 0.25 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

15-02-19 10:56:39 Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY

Showing righting arms in heel at VCG = 0.00

Trim: Aft 0.500/50.000 at zero heel (trim righting arm held at zero)

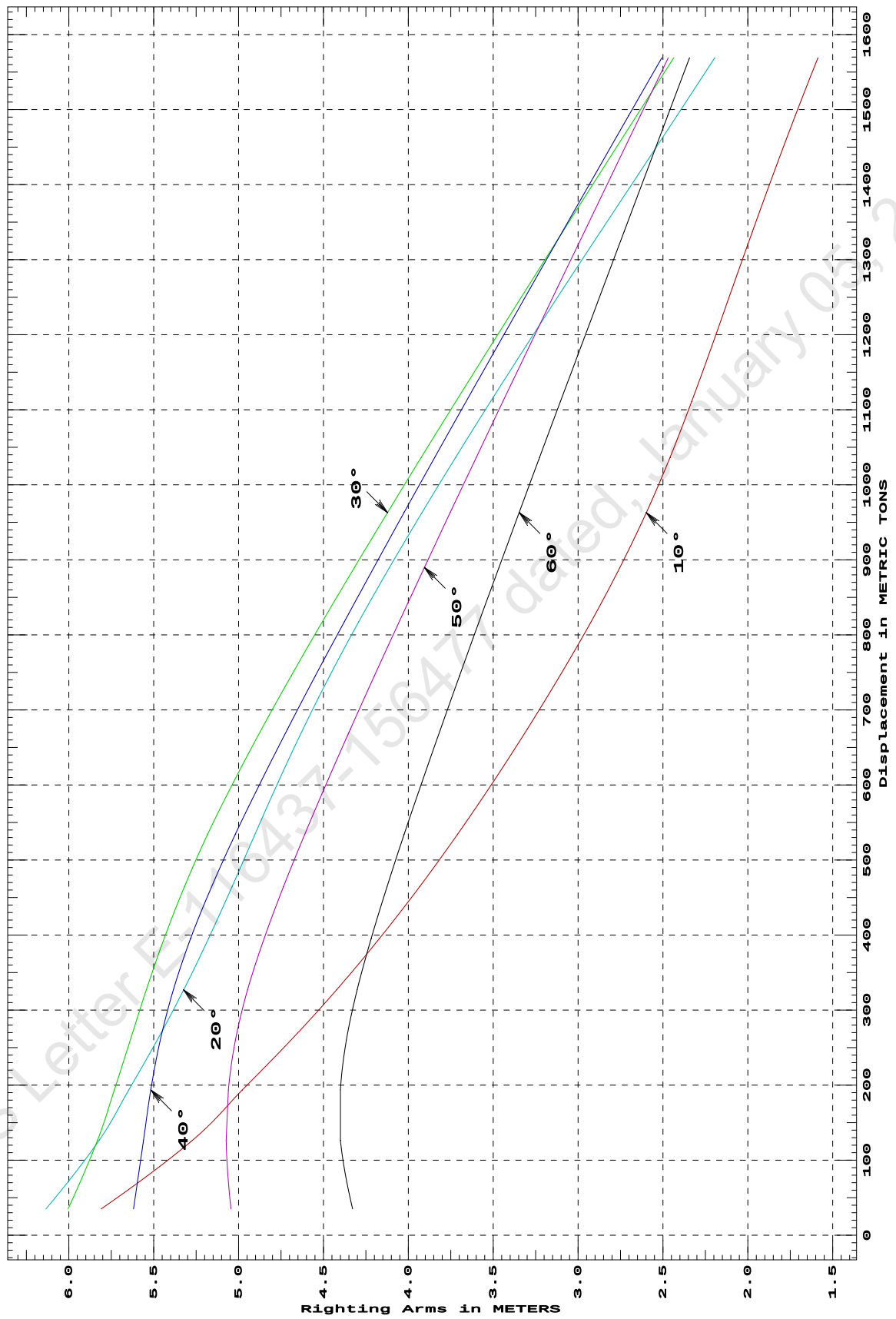
Displacement METRIC TONS	Heel Angles in Degrees					
	10.00s	20.00s	30.00s	40.00s	50.00s	60.00s
35.14	5.809s	6.135s	6.004s	5.619s	5.045s	4.328s
126.38	5.272s	5.822s	5.831s	5.560s	5.073s	4.400s
193.59	4.978s	5.647s	5.732s	5.517s	5.060s	4.400s
259.63	4.693s	5.481s	5.639s	5.462s	5.019s	4.366s
326.97	4.423s	5.324s	5.543s	5.384s	4.947s	4.301s
395.34	4.170s	5.177s	5.439s	5.282s	4.849s	4.219s
464.31	3.934s	5.039s	5.319s	5.159s	4.735s	4.125s
533.71	3.712s	4.905s	5.183s	5.020s	4.611s	4.025s
603.73	3.501s	4.767s	5.030s	4.870s	4.479s	3.920s
674.39	3.300s	4.621s	4.864s	4.711s	4.341s	3.810s
745.60	3.107s	4.463s	4.689s	4.547s	4.199s	3.698s
817.41	2.924s	4.292s	4.506s	4.377s	4.054s	3.583s
889.82	2.754s	4.111s	4.317s	4.203s	3.905s	3.466s
962.83	2.598s	3.920s	4.123s	4.025s	3.753s	3.347s
1,036.44	2.458s	3.721s	3.924s	3.844s	3.600s	3.227s
1,110.65	2.331s	3.516s	3.721s	3.660s	3.443s	3.105s
1,185.47	2.211s	3.305s	3.514s	3.474s	3.285s	2.981s
1,261.17	2.093s	3.088s	3.303s	3.284s	3.125s	2.855s
1,337.60	1.973s	2.867s	3.089s	3.091s	2.962s	2.728s
1,414.34	1.850s	2.643s	2.873s	2.898s	2.799s	2.601s
1,491.42	1.721s	2.418s	2.656s	2.703s	2.634s	2.473s
1,568.81	1.588s	2.195s	2.437s	2.507s	2.469s	2.344s
Distances in METERS.---Specific Gravity = 1.025.-----						

15-02-19 10:56:39

Cybermarine Technologies PTE. Ltd.

GHS 16.50

CROSS CURVES OF STABILITY - Stbd Heel
at 0.5 M. AFT TRIM (initial)



Specific Gravity = 1.025 Assumed KG = 0.00 M.
"K" = BPL

DEADWEIGHT TABLE

Draft M	Disp T	Dwt T
0.612	396.99	21.89
0.712	465.83	90.73
0.812	535.26	160.16
0.912	605.29	230.19
1.012	675.92	300.82
1.112	747.15	372.05
1.212	818.99	443.89
1.312	891.43	516.33
1.412	964.46	589.36
1.512	1038.11	663.01
1.612	1112.35	737.25
1.712	1187.19	812.09
1.812	1262.64	887.54
1.912	1338.68	963.58
2.012	1415.33	1040.23
2.112	1492.22	1117.12
2.212	1569.15	1194.05

Full Load Displacement

15-02-19 10:57:40 Cybermarine Technologies PTE. Ltd.
GHS 16.50

MAXIMUM VCG vs. DISPLACEMENT

Heeling moment is present from: wind

Trim = Fwd 0.500/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
300.00	15.372	94%	0d	12d	
500.00	14.274	47%	0d	9d	
750.00	12.752	16%	0d	6d	
1,000.00	10.916	0%	0d	5d	
1,250.00	8.412	0%	2d	3d	
1,500.00	5.807	0%	6d	2d	
#1,750.00	3.468	0%	15d	1d	

Heeling moment is present from: wind

Trim = Fwd 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
300.00	15.445	95%	0d	14d	
500.00	14.325	51%	0d	10d	
750.00	12.852	14%	0d	7d	
1,000.00	11.011	0%	0d	5d	
1,250.00	8.477	0%	2d	4d	
1,500.00	5.833	0%	6d	3d	
#1,750.00	3.494	0%	15d	2d	

Heeling moment is present from: wind

Trim = zero at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
300.00	15.471	95%	0d	17d	
500.00	14.340	51%	0d	11d	
750.00	12.878	13%	0d	8d	
1,000.00	11.044	0%	0d	6d	
1,250.00	8.477	0%	2d	4d	
1,500.00	5.832	0%	6d	3d	
#1,750.00	3.467	0%	LARGE	2d	

Heeling moment is present from: wind

Trim = Aft 0.250/50.000 at zero heel (trim righting arm held at zero)

Displacement		--- Margins ---			
METRIC TONS	Max VCG	LIM1	LIM2	LIM3	
300.00	15.445	95%	0d	14d	
500.00	14.325	51%	0d	10d	
750.00	12.852	14%	0d	7d	
1,000.00	11.011	0%	0d	5d	
1,250.00	8.462	0%	2d	4d	
1,500.00	5.832	0%	6d	3d	
#1,750.00	3.493	0%	LARGE	2d	

#: MAX. DISPLACEMENT ALLOWED IS 1569.15 T ONLY.

NOTE: MAXIMUM VCG DOES NOT COVER OPERATION/LIFTING OF WEIGHTS AND IS VALID FOR UNMANNED TOWING OPERATION ONLY, WITH CRANE IN STOWED POSITION.

15-02-19 10:57:40 Cybermarine Technologies PTE. Ltd.
GHS 16.50

Heeling moment is present from: wind
Trim = Aft 0.500/50.000 at zero heel (trim righting arm held at zero)
Displacement --- Margins ---
METRIC TONS Max VCG LIM1 LIM2 LIM3

300.00	15.373	94%	0d	12d
500.00	14.274	47%	0d	9d
750.00	12.752	16%	0d	6d
1,000.00	10.917	0%	0d	5d
1,250.00	8.412	0%	2d	3d
1,500.00	5.805	0%	6d	2d
# 1,750.00	3.469	0%	15d	1d

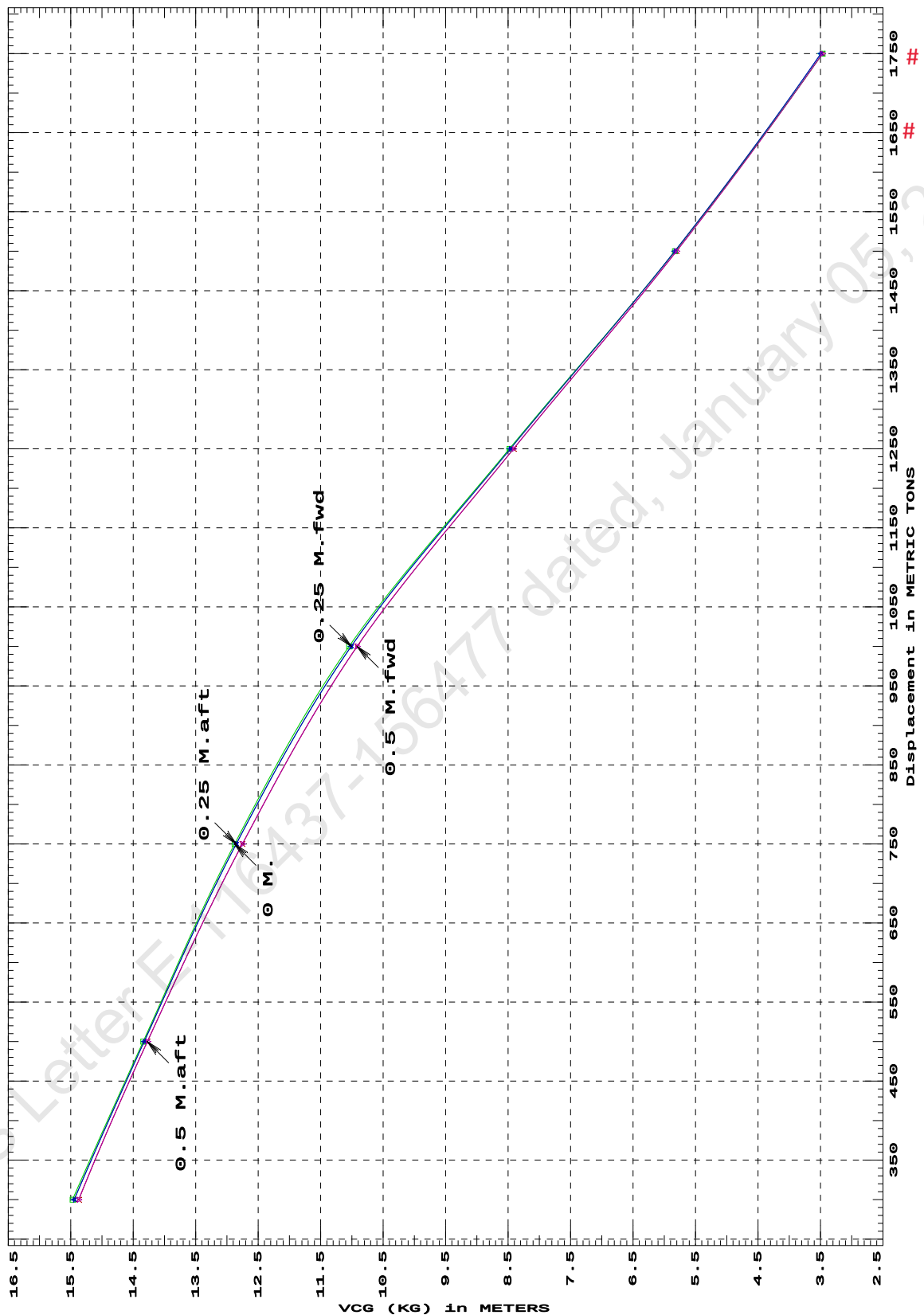
Distances in METERS.---Specific Gravity = 1.025.---d = degrees.

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2-----Min/Max
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad
(2) Angle from Equilibrium to RAzero or Flood > 20.00 deg
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg

#: MAX. DISPLACEMENT ALLOWED IS 1569.15 T ONLY.

NOTE: MAXIMUM VCG DOES NOT COVER OPERATION/LIFTING OF WEIGHTS AND IS VALID FOR UNMANNED TOWING OPERATION ONLY, WITH CRANE IN STOWED POSITION.

IS CODE 2008 PART B CHAPTER-2 SEC.2 MAXIMUM VCG (KG)
at various trims (initial)

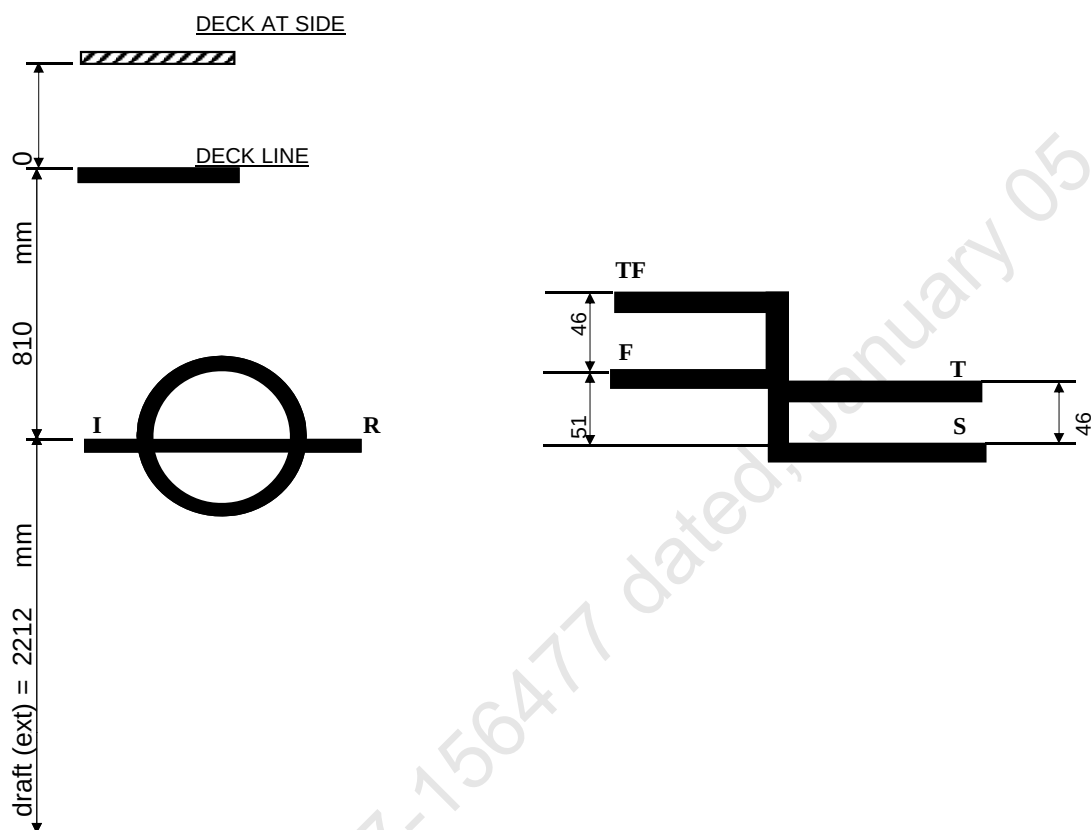


Specific Gravity = 1.025
Trim is per 50 M.
"K" = BPL

NOTE: MAXIMUM VCG DOES NOT COVER OPERATIONS/LIFTING OF WEIGHTS AND IS VALID FOR UNMANNED TOWING OPERATION ONLY, WITH CRANE IN STOWED POSITION.

#: MAX. DISPLACEMENT ALLOWED IS 1569.15 T ONLY.

TABLE OF FREEBOARDS



LOADLINE	FREEBOARD (mm)	DRAFT (ext) KL (m)	DISPLACEMENT (ext) T	DEADWEIGHT T
SUMMER	810	2.212	1569.15	1194.05
FRESH WATER	759	2.263	1568.79	1193.69
TROPICAL	764	2.258	1604.13	1229.03
TROPICAL FRESH WATER	713	2.309	1603.68	1228.58

Deck Plate Thickness = 10 mm

Keel Plate Thickness = 12 mm

LOADING CONDITIONS

#:SEE ACTUAL LOADING CONDITION

SUMMARY OF CONDITIONS #					
CONDITION NO.	1	2	3	4	5
CONDITION	Condition 00 : Lightship Condition	Condition 1 : Lightship + Crane + Ballast	Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Aft	Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Stbd	Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo Crane Lifting 45.1T @ 12m Radius Fwd
TOTAL TANK LOAD (T)	0.000	120.570	449.170	449.170	449.170
DWT CONST.(T)	0.000	0.000	0.000	0.000	0.000
DEADWEIGHT (T)	0.000	0.000	0.000	0.000	0.000
LIGHT WEIGHT (T)	375.100	375.100	375.100	375.100	375.100
DISPLACEMENT (T)	375.100	645.520	1569.150	1569.160	1569.150
FWD DRAFT (M)	0.580	0.899	2.140	2.226	2.313
AFT DRAFT (M)	0.580	1.039	2.284	2.197	2.111
MEAN DRAFT (M)	0.580	0.969	2.212	2.212	2.212
TRIM (M)	0.000	0.140a	0.145a	0.029f	0.203f
L.C.G. (FROM AP) (M)	25.000f	24.469f	24.713f	25.058f	25.403f
L.C.B. (FROM AP) (M)	25.000f	24.461f	24.705f	25.059f	25.414f
M.C.T. (T-CM)	22.230	24.430	31.130	30.930	31.130
KG (M)	3.000	3.522	3.968	3.968	3.968
GM _T (M)	31.429	17.294	6.131	6.127	6.131
MAX. GZ (M)	4.193	3.434	0.894	0.573	0.893

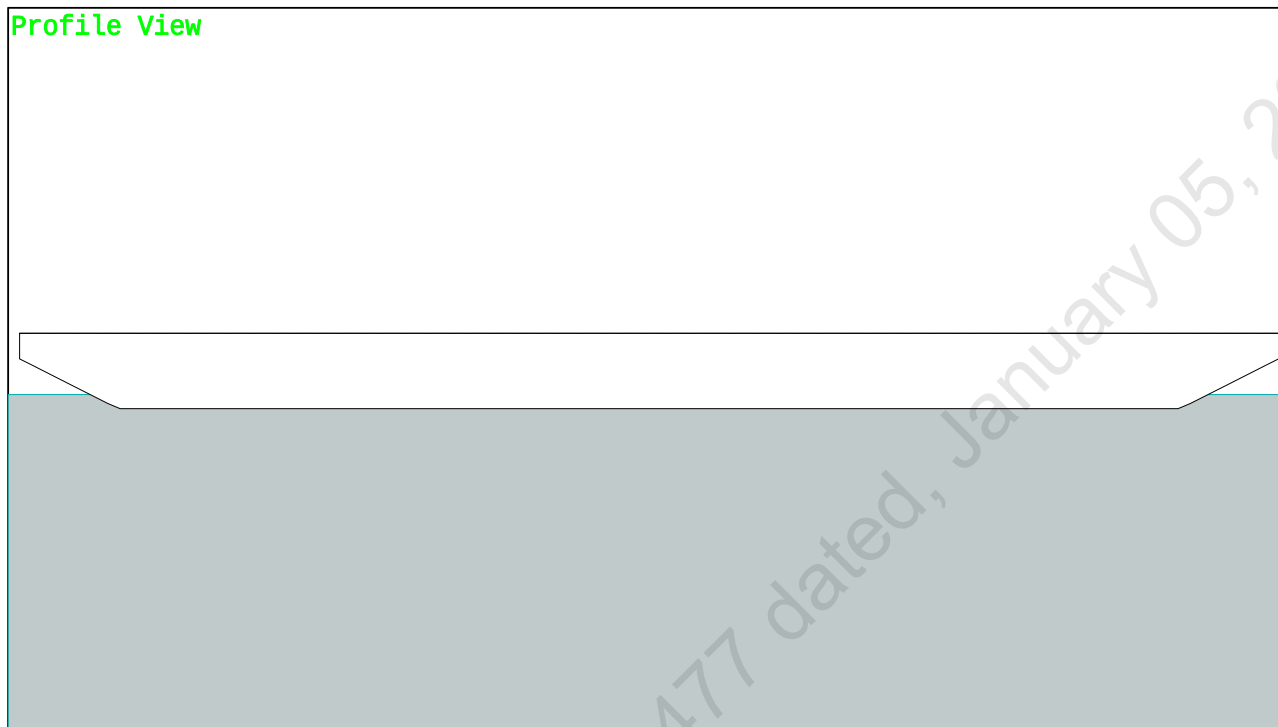
#:SEE ACTUAL LOADING CONDITION

SUMMARY OF CONDITIONS #					
CONDITION NO.	6	7	8	9	10
CONDITION	Condition 5 : Lightship + Crane + Ballast + Deck Cargo Towing Condition with Max. draft 2.212m	Condition 6 : Lightship + Max. Deck Cargo Max. deck cargo loading condition	Condition 7 : Lightship + Load at Aft + Ballast Crane Lifting 36.6T @ 18m Radius Aft	Condition 8 : Lightship + Load at Stbd + Ballast Crane Lifting 36.6T @ 18m Radius Stbd	Condition 9 : Lightship + Load at Fwd + Ballast Crane Lifting 36.60T @ 18m Radius Fwd
TOTAL TANK LOAD (T)	445.160	0.000	337.180	337.180	337.180
DWT CONST.(T)	0.000	0.000	0.000	0.000	0.000
DEADWEIGHT (T)	0.000	0.000	0.000	0.000	0.000
LIGHT WEIGHT (T)	375.100	375.100	375.100	375.100	375.100
DISPLACEMENT (T)	1575.980 1569.15	1569.150	998.880	998.890	998.880
FWD DRAFT (M)	2.165	2.212	1.318	1.443	1.601
AFT DRAFT (M)	2.277	2.212	1.598	1.465	1.315
MEAN DRAFT (M)	2.221	2.212	1.458	1.454	1.458
TRIM (M)	0.113a	0.000	0.280a	0.022a	0.286f
L.C.G. (FROM AP) (M)	24.775f	25.000f	24.242f	24.941f	25.775f
L.C.B. (FROM AP) (M)	24.771f	25.000f	24.213f	24.939f	25.805f
M.C.T. (T-CM)	31.420	31.320	27.010	27.100	27.010
KG (M)	3.078	3.380	5.986	5.986	5.986
GM _T (M)	7.025	6.963	8.338	8.459	8.338
MAX. GZ (M)	1.103	1.101	1.850	1.243	1.849

Condition 00 : Lightship
Lightship condition

Condition Graphic - Draft: 0.580 Trim: zero Heel: zero

Profile View



Plan View



01-01-21 20:57:40 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 00 : Lightship
Lightship condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 0.580 @ 50.00f, 0.580 @ 25.00f, 0.580 @ 0.00
Trim: zero, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WEIGHT	375.10	25.000f	0.000	3.000	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	375.10	25.000f	0.000	0.284

	Righting Arms:		0.000	0.000	
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	666.6	25.000f	0.000	299.07 34.145
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		6.83	22.23	296.35	31.429
Distances in METERS.-----		Moments in m.-MT.			

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

DRAFT AT FP (M) = 0.580
DRAFT AT MS (M) = 0.580
DRAFT AT AP (M) = 0.580
DRAFT AT FWD MARK (M) = 0.580
DRAFT AT AFT MARK (M) = 0.580

01-01-21 20:57:40 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 00 : Lightship
Lightship condition

HYDROSTATIC PROPERTIES
No Trim, No Heel, VCG = 3.000

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft---Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
0.580 375.10 25.000f 0.284 6.83 25.000f 22.23 296.35 31.429
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.000f TCG = 0.000 VCG = 3.000

Origin Depth---	Trim---	Degrees of Heel---	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. 50% DeckImm Area---Angle
0.568	0.00	0.00	375.00	0.000 -0.025	0.0000 14.707
0.568	0.00	0.04s	375.08	0.000 0.000	-0.0000 14.662
0.548	0.00	5.04s	375.24	0.000 2.595	0.1132 9.662
0.374	0.00	10.04s	375.07	0.000 3.748	0.4003 4.662
0.100	0.00	14.71s	375.10	0.000 4.095	0.7242 0.000
0.077	0.00	15.04s	375.10	0.000 4.108	0.7484 -0.338
-0.286	0.00	20.04s	375.10	0.000 4.193	1.1126 -5.338
-0.690	0.00	25.04s	375.31	0.000 4.134	1.4770 -10.338
-1.121	0.00	30.04s	375.15	0.000 3.983	1.8318 -15.338
-1.556	0.00	35.04s	374.93	0.000 3.737	2.1694 -20.338
-1.978	0.00	40.04s	374.95	0.000 3.409	2.4818 -25.338
-2.384	0.00	45.04s	375.14	0.000 3.025	2.7629 -30.338
-2.773	0.00	50.04s	374.97	0.000 2.598	3.0086 -35.338
-3.140	0.00	55.04s	375.13	0.000 2.138	3.2155 -40.338
-3.483	0.00	60.04s	375.21	0.000 1.654	3.3811 -45.338

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

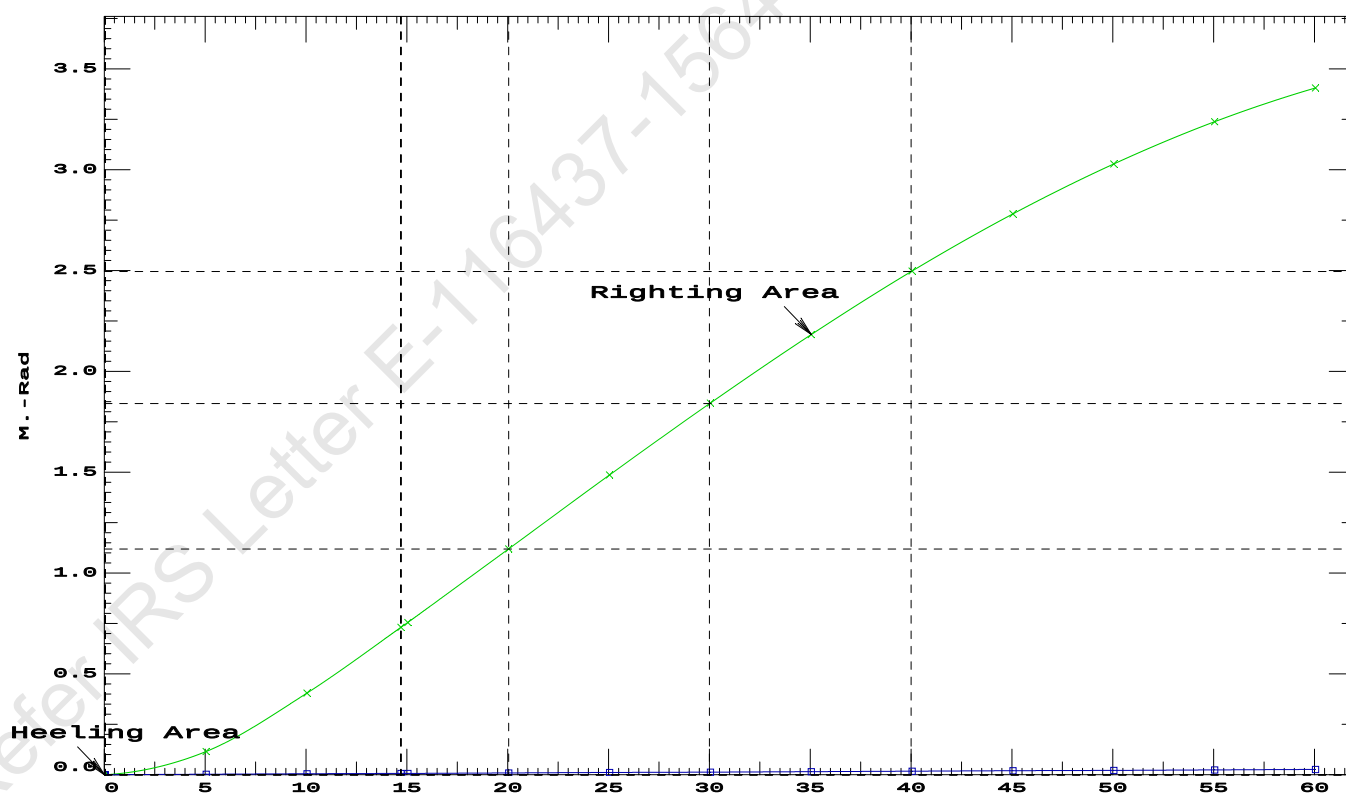
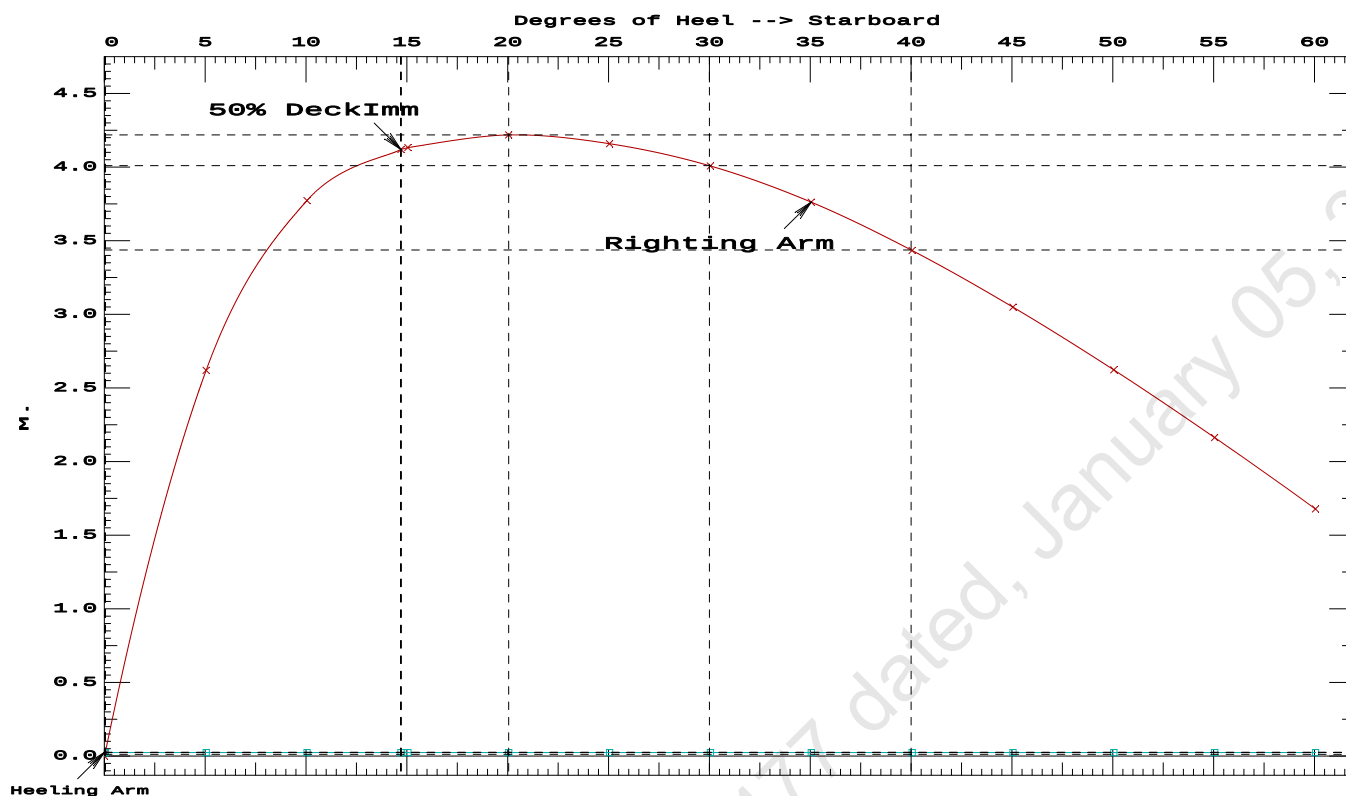
Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 9.31 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 1.1126 P
(2) Angle from Equilibrium to RZero or Flood > 20.00 deg LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 14.66 P

01-01-21 20:57:40 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 00 : Lightship
Lightship condition



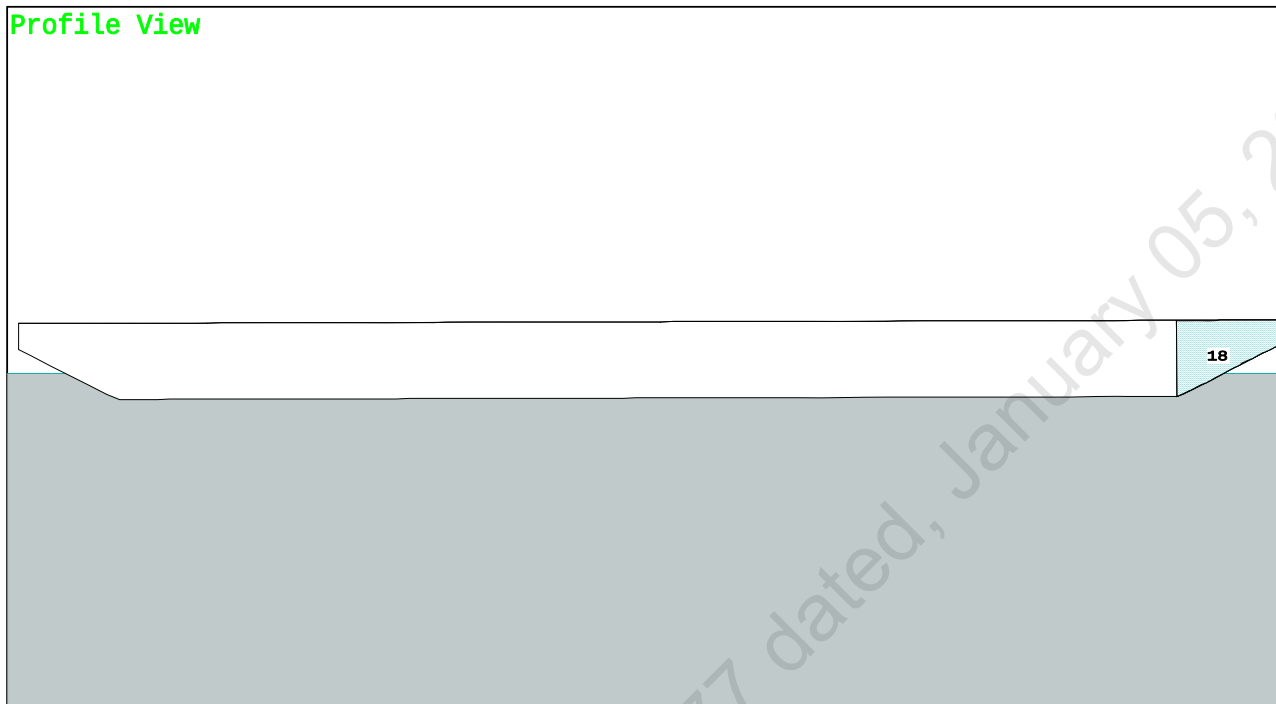
01-01-21 20:57:44 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 1 : Lightship + Crane + Ballast

Condition Graphic - Draft: 0.899 @ 50.000f, 1.039 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

16 WBTk6.P 17 WBTk6.C 18 WBTk6.S

01-01-21 20:57:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 1 : Lightship + Crane + Ballast

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 0.899 @ 50.00f, 0.969 @ 25.00f, 1.039 @ 0.00

Trim: Aft 0.140/50.000, Heel: zero

Part-----	Weight(MT)	LCG	TCG	VCG	
LIGHT SHIP	375.10	25.000f	0.000	3.000	
Crane	150.00	4.500f	0.000	6.120	
Total Fixed----->	525.10	19.144f	0.000	3.891	
Load-----	SpGr-----	Weight(MT)	LCG	TCG	VCG
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p
WBTK6.C	1.000	1.025	40.60	47.664f	0.000
WBTK6.S	1.000	1.025	39.99	47.660f	4.967s
Total Tanks----->			120.57	47.661f	0.000
Total Weight----->			645.67	24.469f	0.000
		Displ(MT)	LCB	TCB	VCB
HULL	1.025	645.52	24.461f	0.000	0.486

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA	LCF	TCF	BML
Total Waterplane---->	1.025	689.5	24.874f	0.000	192.28
		MT/cm	m.	MT/cm	GML
		7.07		24.43	189.24
Distances in METERS.-----		Moments in m.-MT.			
					17.294

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

117.6 Cu.M. (7%) SALT WATER

150.00 MT Crane

DRAFT AT FP (M) = 0.899
 DRAFT AT MS (M) = 0.969
 DRAFT AT AP (M) = 1.039
 DRAFT AT FWD MARK (M) = 0.910
 DRAFT AT AFT MARK (M) = 1.028

01-01-21 20:57:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 1 : Lightship + Crane + Ballast

HYDROSTATIC PROPERTIES

Trim: Aft 0.140/50.000, No Heel, VCG = 3.522

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
0.969 645.52 24.461f 0.486 7.07 24.874f 24.43 189.24 17.294
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 125.5 m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.469f TCG = 0.000 VCG = 3.522

Free Surface Adjustment: 0.194

Adjusted CG: LCG = 24.470f TCG = 0.000 VCG = 3.716

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. 50% DeckImm Area----Angle
1.027	0.16a	0.00	645.68	0.000 -0.012	0.0000 8.547
1.027	0.16a	0.04s	645.67	0.000 0.000	-0.0000 8.509
1.016	0.16a	5.04s	645.67	0.000 1.505	0.0657 3.509
0.987	0.17a	8.55s	645.67	0.000 2.463	0.1876 0.000
0.958	0.18a	10.04s	645.67	0.000 2.750	0.2554 -1.491
0.774	0.21a	15.04s	645.67	0.000 3.271	0.5223 -6.491
0.509	0.25a	20.04s	645.67	0.000 3.433	0.8174 -11.491
0.494	0.25a	20.31s	645.67	0.000 3.434	0.8339 -11.766
0.237	0.31a	25.04s	645.67	0.000 3.337	1.1147 -16.491
-0.035	0.37a	30.04s	645.67	0.000 3.097	1.3964 -21.491
-0.307	0.43a	35.04s	645.67	0.000 2.778	1.6534 -26.491
-0.576	0.49a	40.04s	645.67	0.000 2.407	1.8800 -31.491
-0.840	0.55a	45.04s	645.67	0.000 2.001	2.0726 -36.491
-1.098	0.60a	50.04s	645.67	0.000 1.567	2.2285 -41.491
-1.347	0.65a	55.04s	645.67	0.000 1.114	2.3456 -46.491
-1.585	0.69a	60.04s	645.67	0.000 0.648	2.4226 -51.491

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

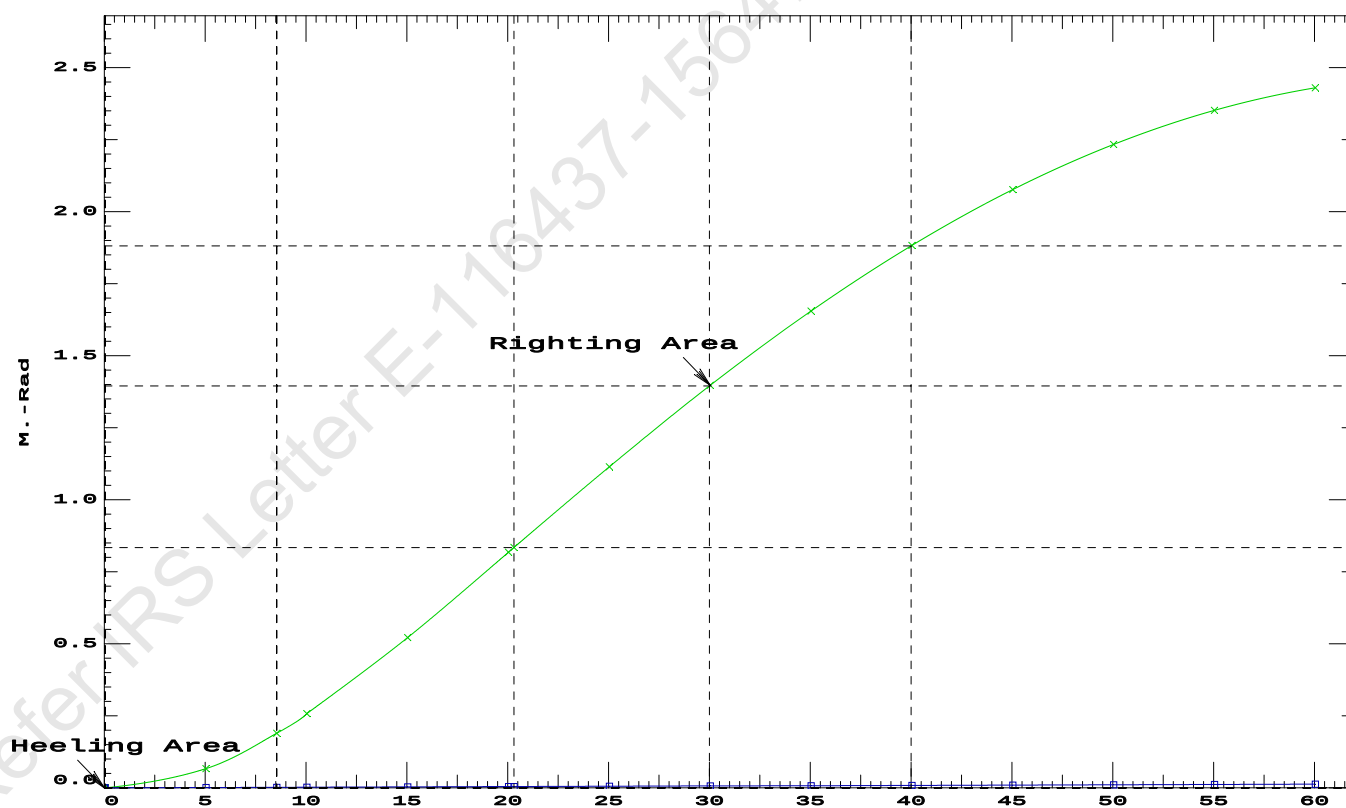
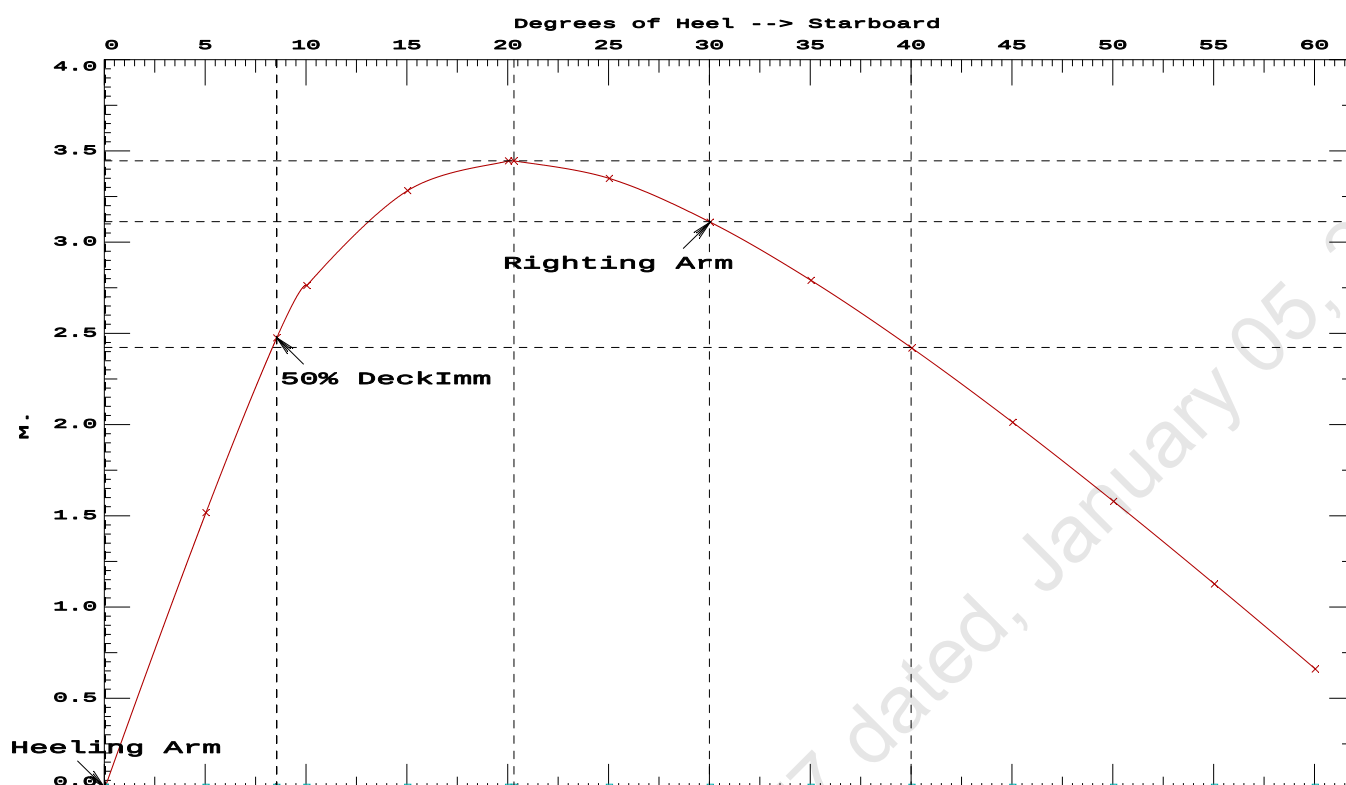
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 125.5 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 7.85 (constant)

LIM-----INTACT CRITERIA-IS CODE 2008PART B ----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.8339 P
(2) Angle from Equilibrium to RZero or Flood > 20.00 deg LARGE
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 8.51 P

01-01-21 20:57:44 Cybermarine Technologies PTE. Ltd.

GHS 17.22

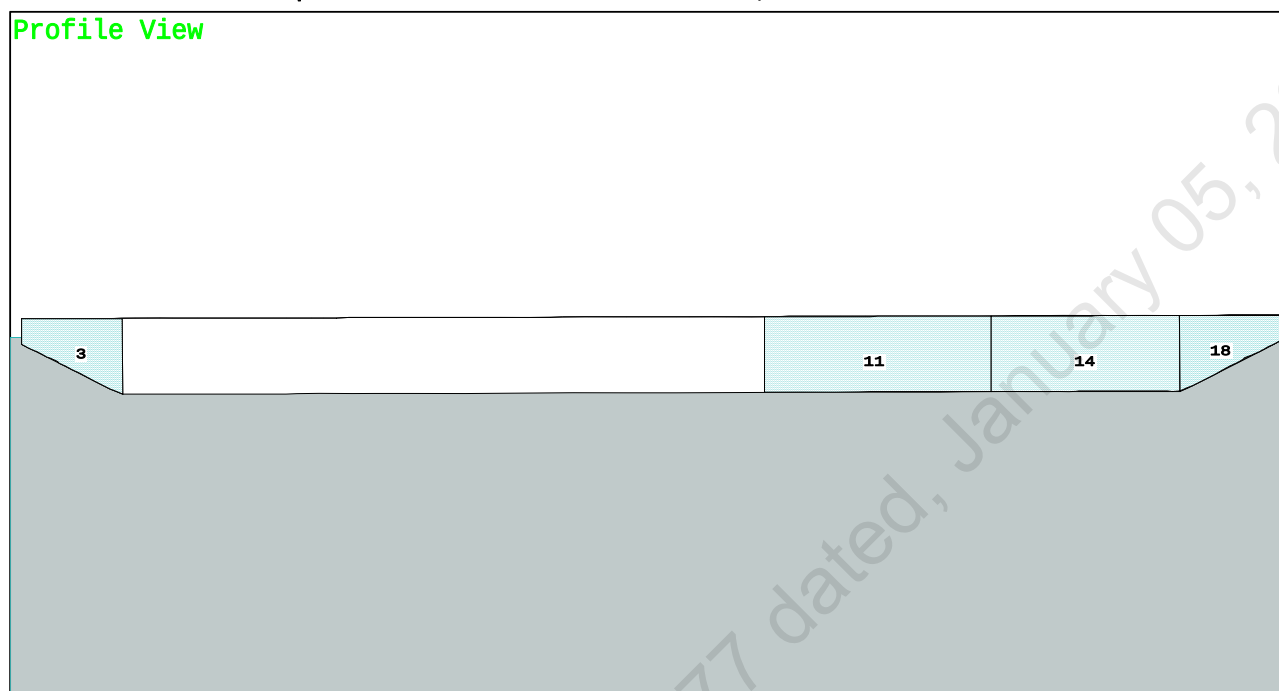
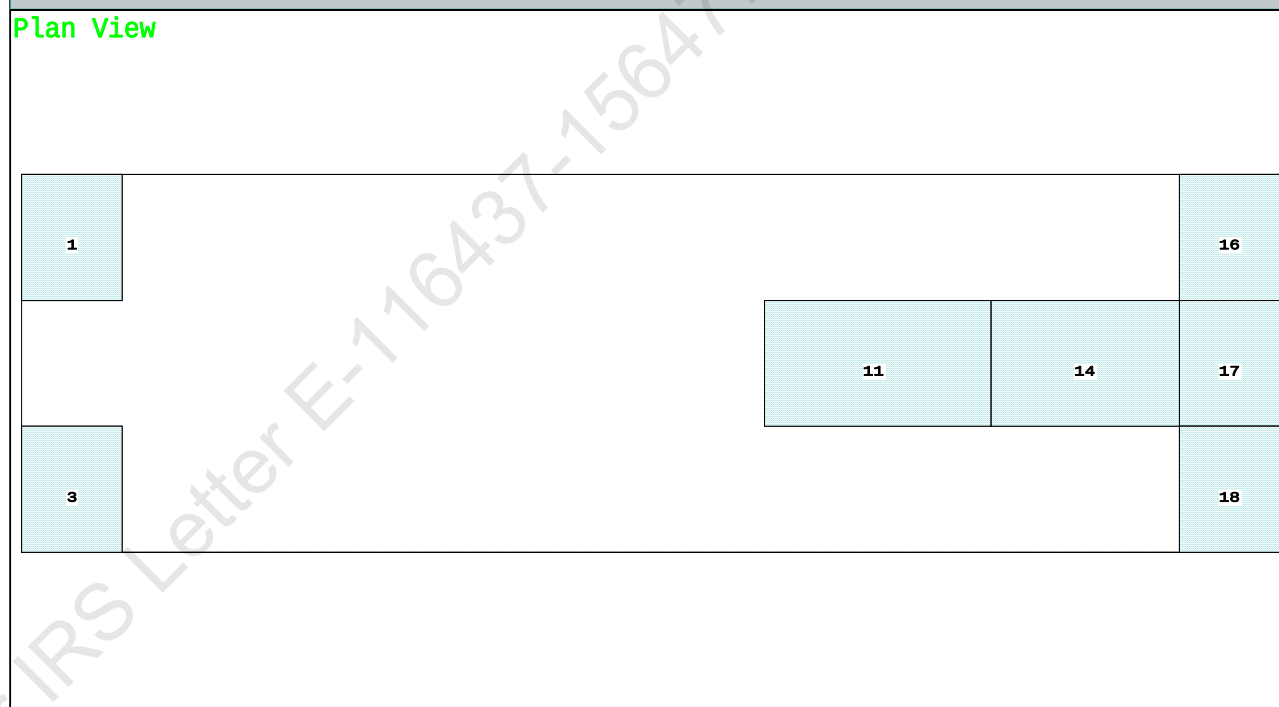
Condition 1 : Lightship + Crane + Ballast

01-01-21 20:57:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Aft

Condition Graphic - Draft: 2.140 @ 50.000f, 2.284 @ 0.000 Heel: zero

Profile View**Plan View****Tanks**

1 WBTk1.P 3 WBTk1.S 11 WBTk4.C 14 WBTk5.C 16 WBTk6.P 17 WBTk6.C 18 WBTk6.S

01-01-21 20:57:51 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.140 @ 50.00f, 2.212 @ 25.00f, 2.284 @ 0.00

Trim: Aft 0.145/50.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	7.500a	0.000	33.320	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,119.98	20.946f	0.000	4.884	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTk6.S	1.000	1.025	39.99	47.660f	4.967s	1.915	41.84*
Total Tanks----->			449.17	34.106f	0.000	1.684	381.88
Total Weight----->			1,569.15	24.713f	0.000	3.968	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,569.15	24.705f	0.000	1.135	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	750.6	24.994f	0.000	102.04	8.965
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			7.69		31.13	99.21	6.131
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

438.2 Cu.M. (26%) SALT WATER

150.00 MT Crane

45.10 MT Crane Load

549.78 MT Deck Cargo

DRAFT AT FP (M) = 2.140
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.284
DRAFT AT FWD MARK (M) = 2.151
DRAFT AT AFT MARK (M) = 2.273

HYDROSTATIC PROPERTIES

Trim: Aft 0.145/50.000, No Heel, VCG = 3.968

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT-----
2.212	1,569.15	24.705f	1.135	7.69	24.994f	31.13	99.21	6.131
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 381.9 m.-MT

01-01-21 20:57:51 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.713f TCG = 0.000 VCG = 3.968

Free Surface Adjustment: 0.243

Adjusted CG: LCG = 24.714f TCG = 0.000 VCG = 4.212

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
2.272	0.17a	0.00	1,569.2	0.000	-0.010	0.0000	2.805
2.272	0.17a	0.09p	1,569.2	0.000	0.000	-0.0000	2.713
2.270	0.17a	2.81p	1,569.2	0.000	0.291	0.0069	0.000
2.263	0.17a	5.09p	1,569.2	0.000	0.535	0.0234	-2.287
2.321	0.23a	10.09p	1,569.2	0.000	0.868	0.0861	-7.287
2.408	0.26a	12.87p	1,569.2	0.000	0.894	0.1293	-10.067
2.494	0.29a	15.09p	1,569.2	0.000	0.877	0.1636	-12.287
2.715	0.35a	20.09p	1,569.2	0.000	0.760	0.2365	-17.287
2.951	0.43a	25.09p	1,569.2	0.000	0.574	0.2952	-22.287
3.176	0.52a	30.09p	1,569.2	0.000	0.340	0.3354	-27.287
3.377	0.60a	35.09p	1,569.2	0.000	0.080	0.3539	-32.287
3.432	0.63a	36.57p	1,569.2	0.000	0.000	0.3549	-33.764
3.553	0.68a	40.09p	1,569.2	0.000	-0.193	0.3490	-37.287
3.703	0.76a	45.09p	1,569.2	0.000	-0.472	0.3200	-42.287
3.824	0.83a	50.09p	1,569.2	0.000	-0.752	0.2666	-47.287
3.916	0.90a	55.09p	1,569.2	0.000	-1.029	0.1889	-52.287
3.977	0.95a	60.09p	1,569.2	0.000	-1.300	0.0872	-57.287

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

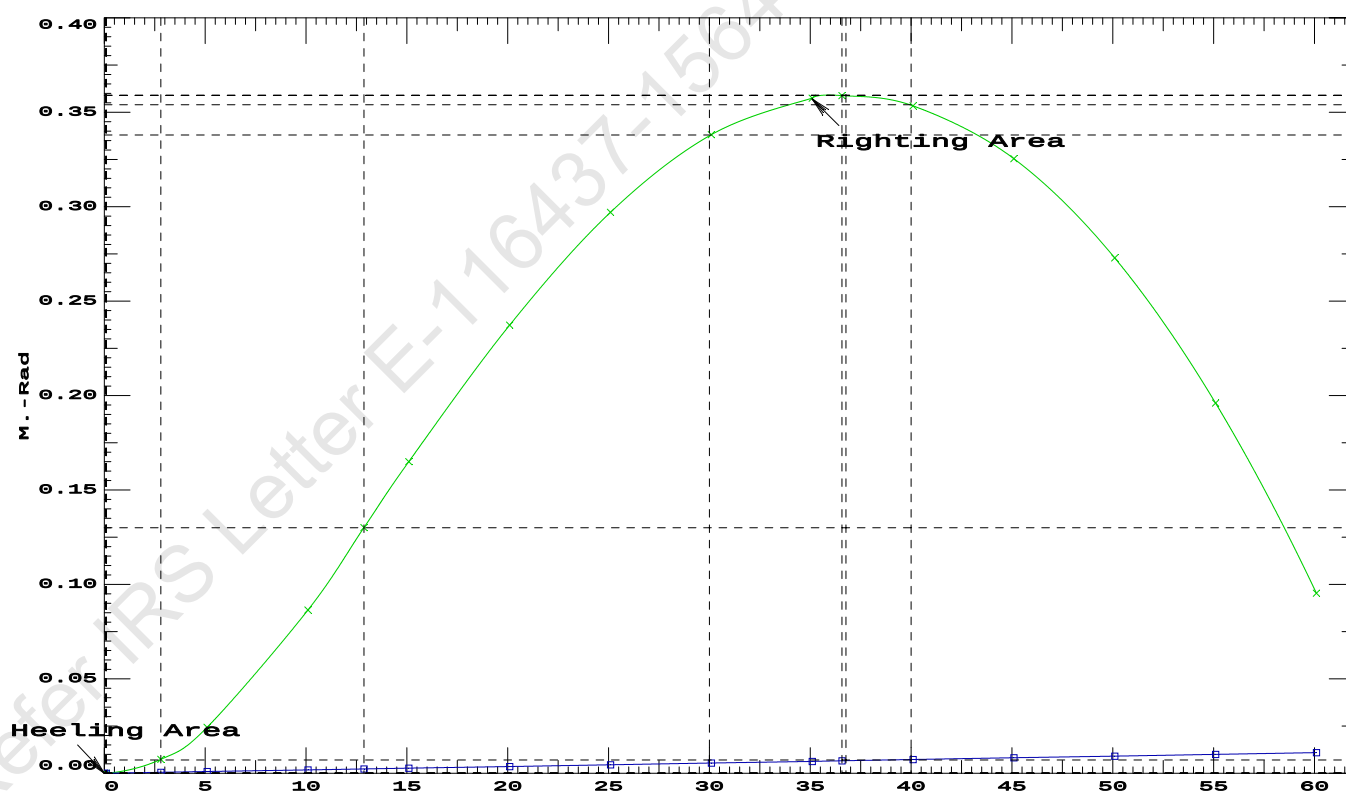
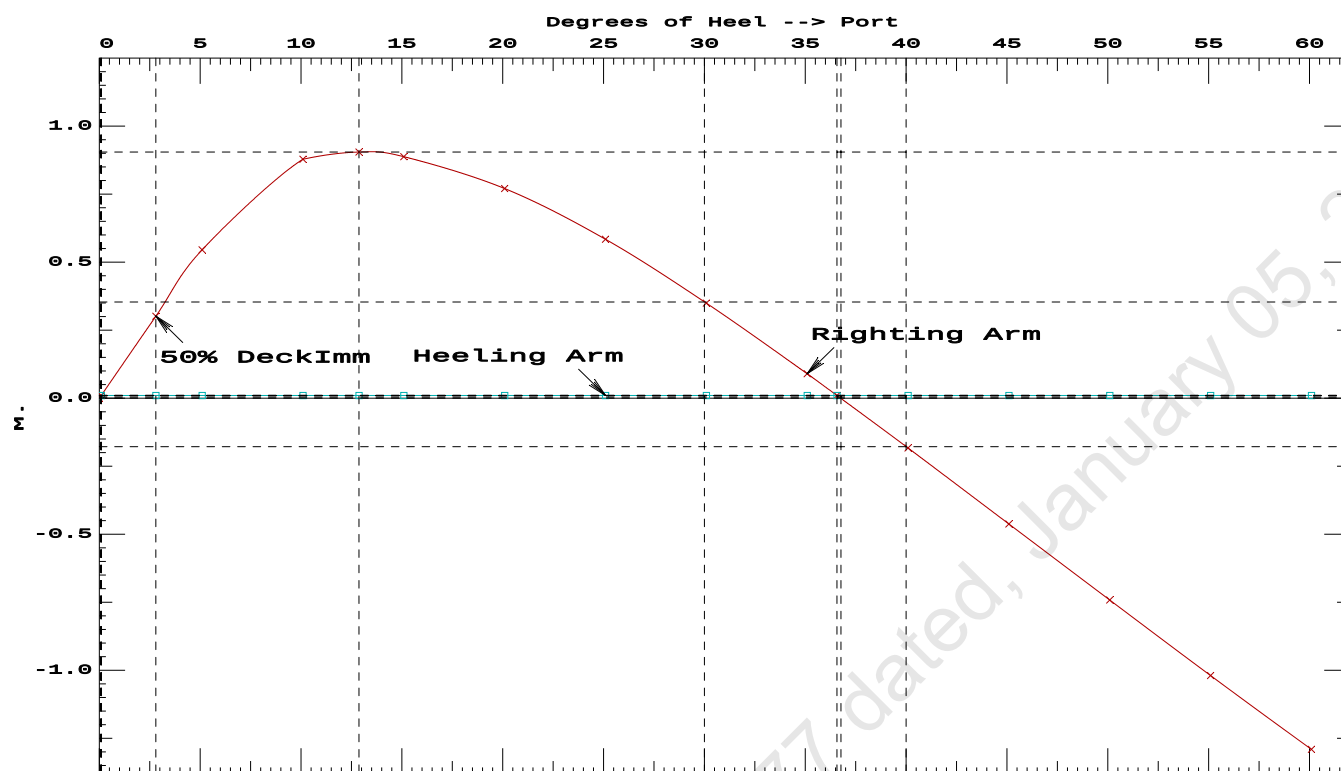
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 381.9 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Port heeling moment = 16.26 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2-----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1293 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	36.48 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.71 P

01-01-21 20:57:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 45.1T @ 12m Radius Aft

01-01-21 20:57:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.713f TCG = 0.000 VCG = 3.968

Free Surface Adjustment: 0.243

Adjusted CG: LCG = 24.714f TCG = 0.000 VCG = 4.212

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	in Heel---	Res. Area
2.272	0.17a	0.01p	1,569.2	0.000	0.000	0.0000
2.264	0.17a	5.01p	1,569.2	0.000	0.535	0.0234
2.319	0.22a	10.01p	1,569.2	0.000	0.875	0.0863
2.408	0.26a	12.87p	1,569.2	0.000	0.904	0.1314
2.491	0.29a	15.01p	1,569.2	0.000	0.888	0.1647
2.711	0.35a	20.01p	1,569.2	0.000	0.772	0.2386
2.947	0.43a	25.01p	1,569.2	0.000	0.587	0.2984
3.172	0.51a	30.01p	1,569.2	0.000	0.353	0.3398
3.374	0.60a	35.01p	1,569.2	0.000	0.094	0.3594
3.438	0.63a	36.74p	1,569.2	0.000	0.000	0.3609
3.551	0.68a	40.01p	1,569.2	0.000	-0.179	0.3558
3.701	0.76a	45.01p	1,569.2	0.000	-0.458	0.3281
3.822	0.83a	50.01p	1,569.2	0.000	-0.738	0.2759
3.914	0.90a	55.01p	1,569.2	0.000	-1.015	0.1995
3.976	0.95a	60.01p	1,569.2	0.000	-1.286	0.0990

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 381.9 m.-MT was applied to artificially modify the CG.

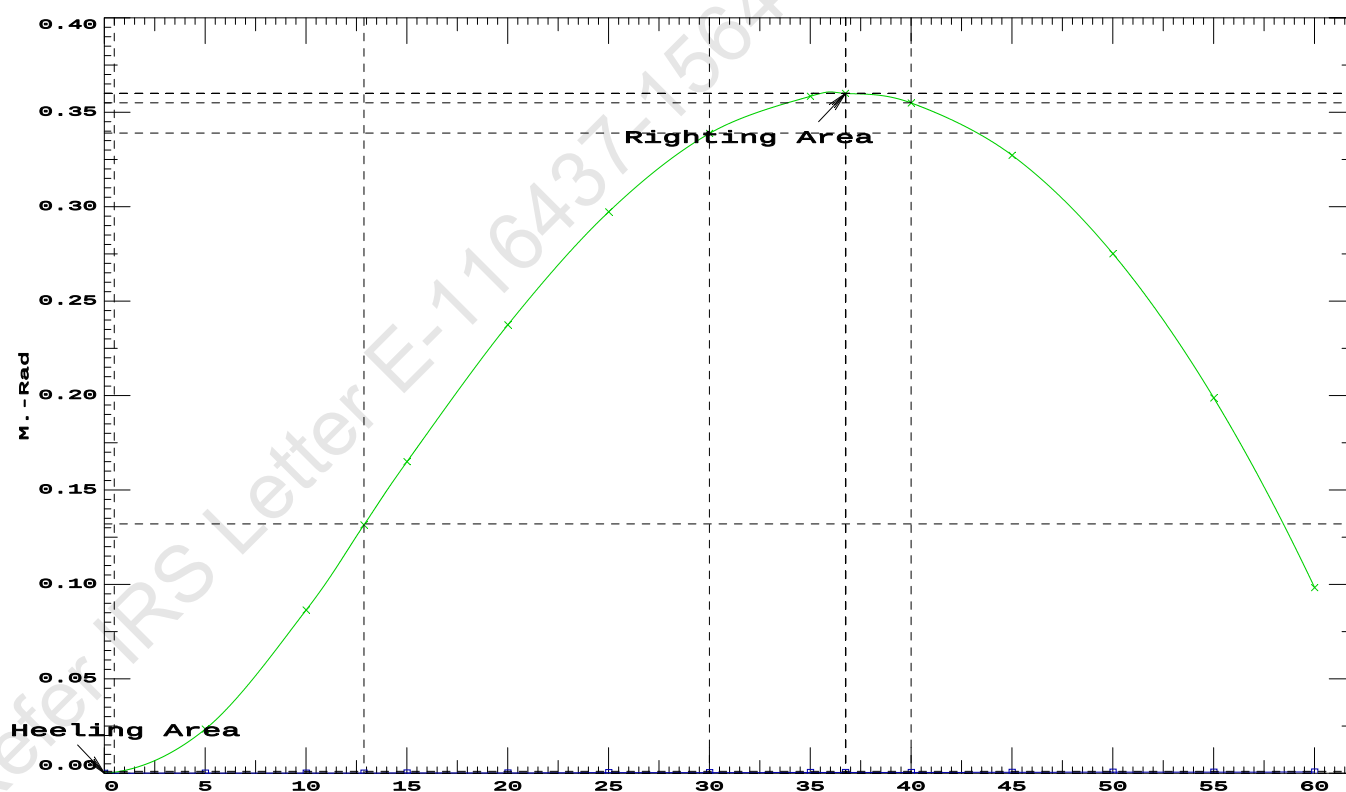
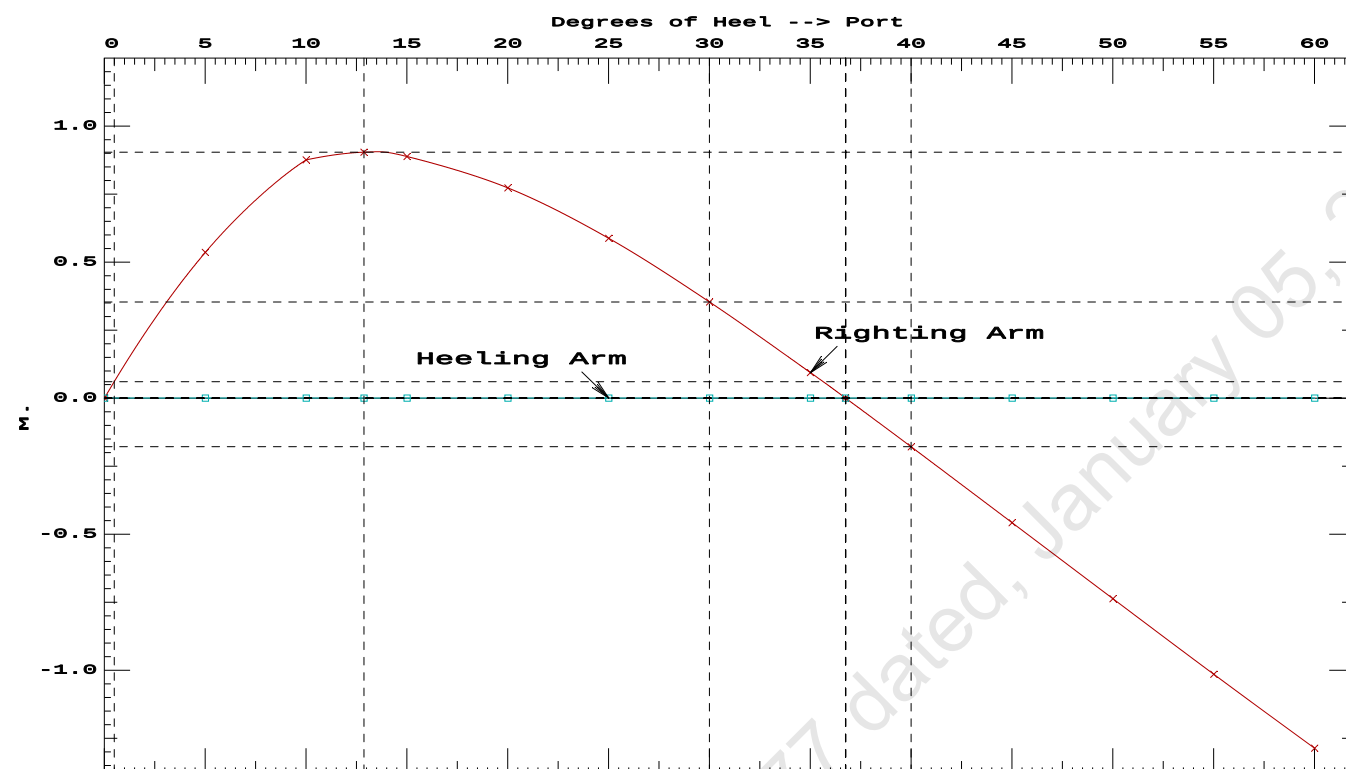
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
 Port heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	141796 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	884.143 P

01-01-21 20:57:51 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 2 : Lightship + Load at Aft + Ballast + Deck CargoCrane Lifting 45.1T @ 12m Radius Aft

01-01-21 20:57:57 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 2 : Lightship + Load at Aft + Ballast + Deck Cargo**Crane Lifting 45.10T @ 12m Radius Aft**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.140 @ 50.00f, 2.212 @ 25.00f, 2.284 @ 0.00

Trim: Aft 0.145/50.000, Heel: Port 0.01 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	7.500a	0.000	33.320	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,119.98	20.946f	0.000	4.884	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTk6.S	1.000	1.025	39.99	47.660f	4.967s	1.915	41.84*
Total Tanks----->			449.17	34.106f	0.000	1.684	381.88
Total Weight----->			1,569.15	24.713f	0.000	3.968	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,569.15	24.705f	0.001p	1.135	

	Righting Arms:			0.000	-0.001s		
	External Arms:			0.000	-0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		750.6	24.994f	0.000	102.04	8.965
			MT/cm-----	m.-	MT/cm-----	GML-----	GMT-----
			7.69		31.13	99.21	6.131
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = -0.01

01-01-21 20:57:57 Cybermarine Technologies PTE. Ltd.
GHS 17.22

After hook load drop							
WEIGHT and DISPLACEMENT and WATERPLANE STATUS							
USK draft: 2.315 @ 50.00f, 2.153 @ 25.00f, 1.992 @ 0.00							
Trim: Fwd 0.323/50.000, Heel: 0.00 deg.							
Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,074.88	22.139f	0.000	3.691	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTk6.S	1.000	1.025	39.99	47.660f	4.967s	1.915	41.84*
Total Tanks----->			449.17	34.106f	0.000	1.684	381.88
Total Weight----->			1,524.05	25.666f	0.000	3.100	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,524.05	25.679f	0.001	1.106	

	Righting Arms:			0.000	-0.001s		
Part-----			SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane---->	1.025			746.9	25.130f	0.016p	103.52
				MT/cm-----	m.-	MT/cm-----	GML-----
				7.66		30.95	101.53
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

01-01-21 20:57:57 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.691

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.979	0.37f	0.00	1,524.1	0.000	-0.000	0.0000
1.979	0.37f	0.07s	1,524.1	0.000	0.010	0.0000
1.966	0.38f	5.00s	1,524.1	0.000	0.652	0.0285
1.937	0.48f	10.00s	1,524.1	0.000	1.140	0.1078
2.005	0.62f	15.00s	1,524.1	0.000	1.285	0.2162
2.050	0.68f	17.14s	1,523.9	0.000	1.295	0.2644
2.118	0.76f	20.00s	1,524.1	0.000	1.277	0.3288
2.221	0.94f	25.00s	1,524.1	0.000	1.181	0.4369
2.302	1.13f	30.00s	1,524.1	0.000	1.029	0.5338

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

01-01-21 20:57:57 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.691

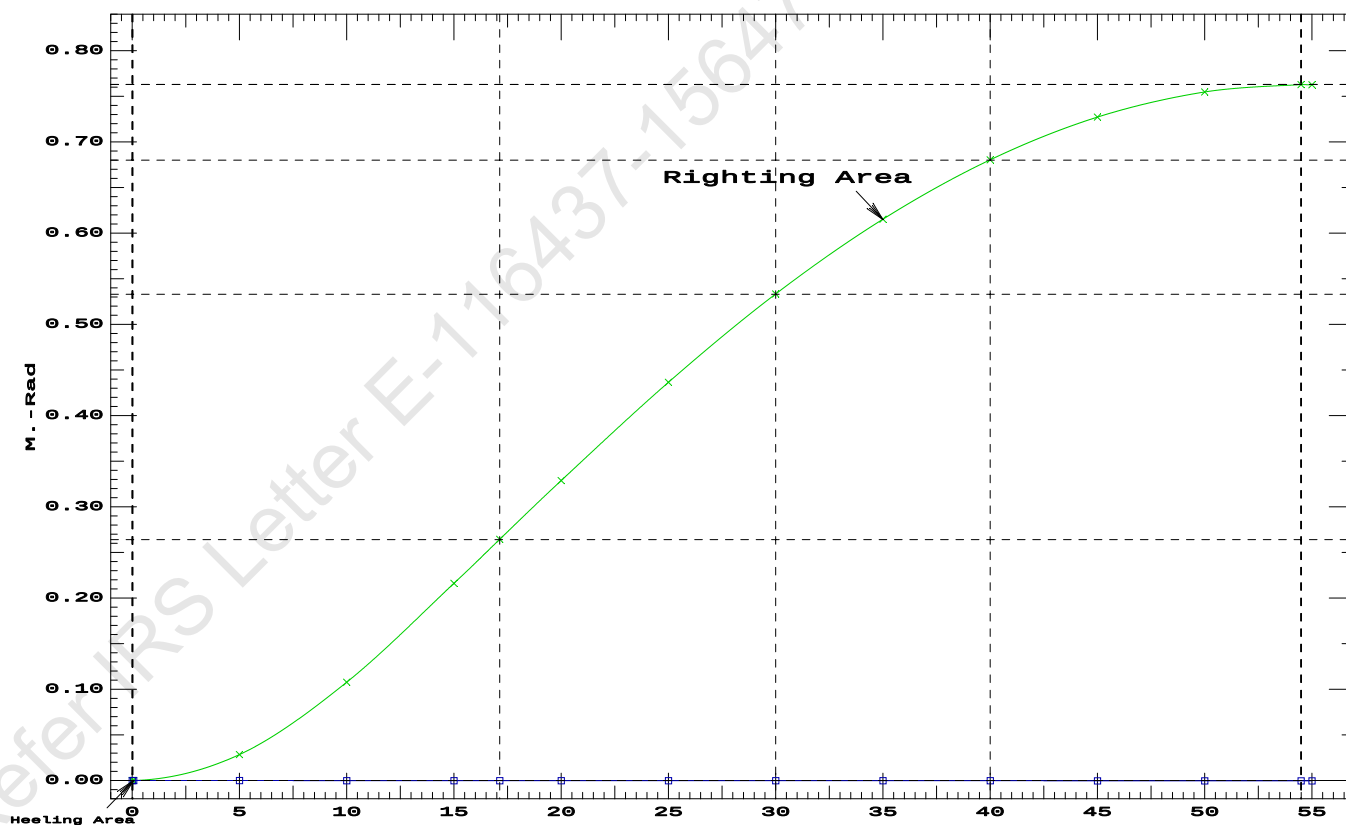
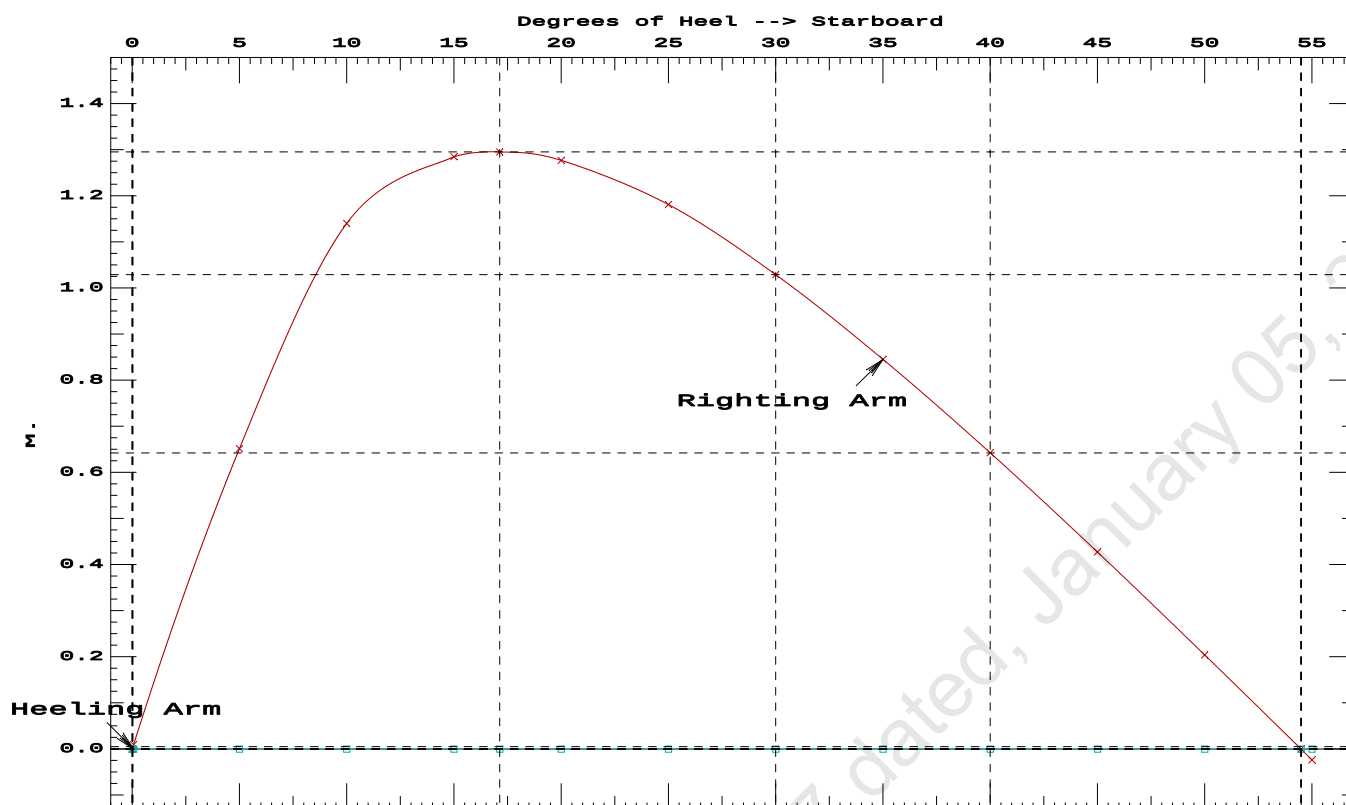
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.979	0.37f	0.00	1,524.1	0.000	-0.000	0.0000
1.979	0.37f	0.07s	1,524.1	0.000	0.010	0.0000
1.966	0.38f	5.00s	1,524.1	0.000	0.652	0.0285
1.937	0.48f	10.00s	1,524.1	0.000	1.140	0.1078
2.005	0.62f	15.00s	1,524.1	0.000	1.285	0.2162
2.050	0.68f	17.14s	1,523.9	0.000	1.295	0.2644
2.118	0.76f	20.00s	1,524.1	0.000	1.277	0.3288
2.221	0.94f	25.00s	1,524.1	0.000	1.181	0.4369
2.302	1.13f	30.00s	1,524.1	0.000	1.029	0.5338
2.364	1.31f	35.00s	1,524.1	0.000	0.846	0.6158
2.406	1.49f	40.00s	1,524.1	0.000	0.643	0.6809
2.428	1.66f	45.00s	1,524.1	0.000	0.428	0.7277
2.431	1.82f	50.00s	1,524.1	0.000	0.204	0.7553
2.418	1.95f	54.49s	1,524.1	0.000	0.000	0.7633
2.416	1.96f	55.00s	1,524.1	0.000	-0.023	0.7632

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAzero > 20.00 deg 54.42 P

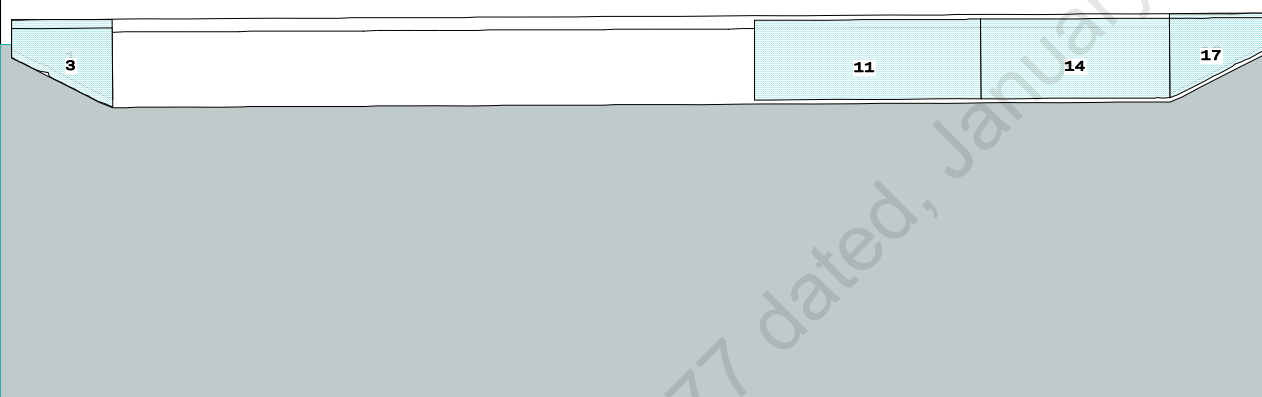
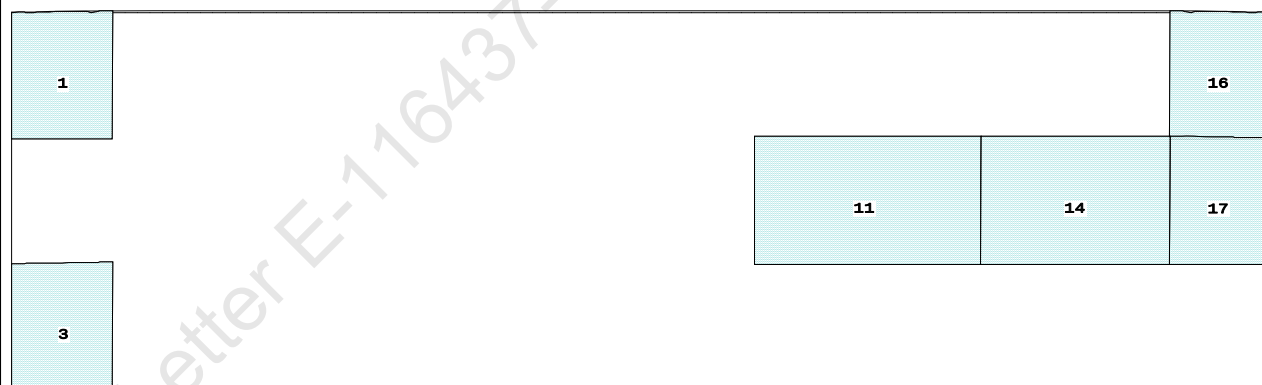


02-01-21 09:30:02 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd

CG - Draft: 2.028 @ 50.000f, 2.291 @ 0.000 Heel: stbd 1.96 deg.

Profile View**Plan View****Tanks**

1 WBTk1.P 3 WBTk1.S 11 WBTk4.C 14 WBTk5.C 16 WBTk6.P 17 WBTk6.C

02-01-21 09:30:02 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.028 @ 50.00f, 2.160 @ 25.00f, 2.291 @ 0.00

Trim: Aft 0.263/50.000, Heel: Stbd 1.96 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	4.500f	12.000s	33.320	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,119.98	21.429f	0.483s	4.884	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTK1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTK1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTK6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks----->			409.19	32.782f	0.485p	1.662	340.04
Total Weight----->			1,529.17	24.467f	0.224s	4.022	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,529.17	24.451f	0.323s	1.114	

	Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		749.2	24.939f	0.017s	104.00	9.202
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			7.68		30.92	101.09	6.293
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

399.2 Cu.M. (24%) SALT WATER

150.00 MT Crane
549.78 MT Deck Cargo

45.10 MT Crane Load

DRAFT AT FP (M) = 2.028
DRAFT AT MS (M) = 2.160
DRAFT AT AP (M) = 2.291
DRAFT AT FWD MARK (M) = 2.049
DRAFT AT AFT MARK (M) = 2.270

HYDROSTATIC PROPERTIES

Trim: Aft 0.263/50.000, Heel: Stbd 1.96 deg., VCG = 4.022

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
2.160 1,529.17 24.451f 1.114 7.68 24.939f 30.92 101.09 6.293
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 340.0 m.-MT

02-01-21 09:30:02 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.467f TCG = 0.224s VCG = 4.022

Free Surface Adjustment: 0.222

Adjusted CG: LCG = 24.468f TCG = 0.216s VCG = 4.244

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
2.279	0.30a	0.00	1,529.2	0.000	-0.227	0.0000	2.775
2.278	0.30a	2.05s	1,529.2	0.000	0.000	-0.0041	0.721
2.277	0.30a	2.77s	1,529.2	0.000	0.079	-0.0036	0.000
2.271	0.32a	7.05s	1,529.2	0.000	0.539	0.0196	-4.279
2.392	0.44a	12.05s	1,529.2	0.000	0.756	0.0789	-9.279
2.453	0.48a	13.67s	1,529.2	0.000	0.765	0.1005	-10.900
2.597	0.56a	17.05s	1,529.2	0.000	0.731	0.1451	-14.279
2.835	0.68a	22.05s	1,529.2	0.000	0.594	0.2040	-19.279
3.081	0.83a	27.05s	1,529.2	0.000	0.389	0.2474	-24.279
3.305	0.99a	32.05s	1,529.2	0.000	0.145	0.2710	-29.279
3.420	1.07a	34.83s	1,529.2	0.000	0.000	0.2745	-32.059
3.507	1.14a	37.05s	1,529.2	0.000	-0.119	0.2722	-34.279
3.682	1.29a	42.05s	1,529.2	0.000	-0.391	0.2501	-39.279
3.829	1.42a	47.05s	1,529.2	0.000	-0.667	0.2039	-44.279
3.947	1.54a	52.05s	1,529.2	0.000	-0.942	0.1336	-49.279
4.033	1.65a	57.05s	1,529.2	0.000	-1.213	0.0396	-54.279
4.087	1.74a	62.05s	1,529.2	0.000	-1.475	-0.0778	-59.279

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

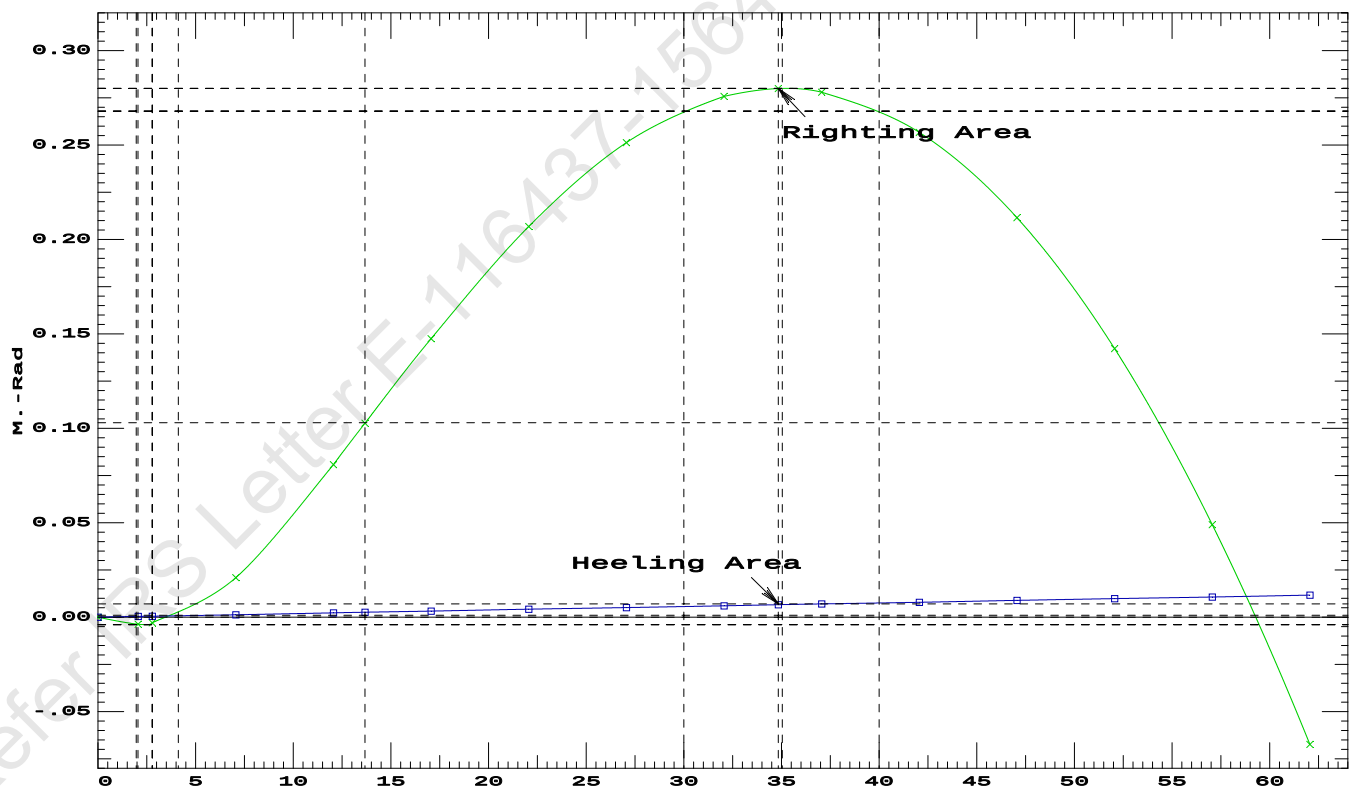
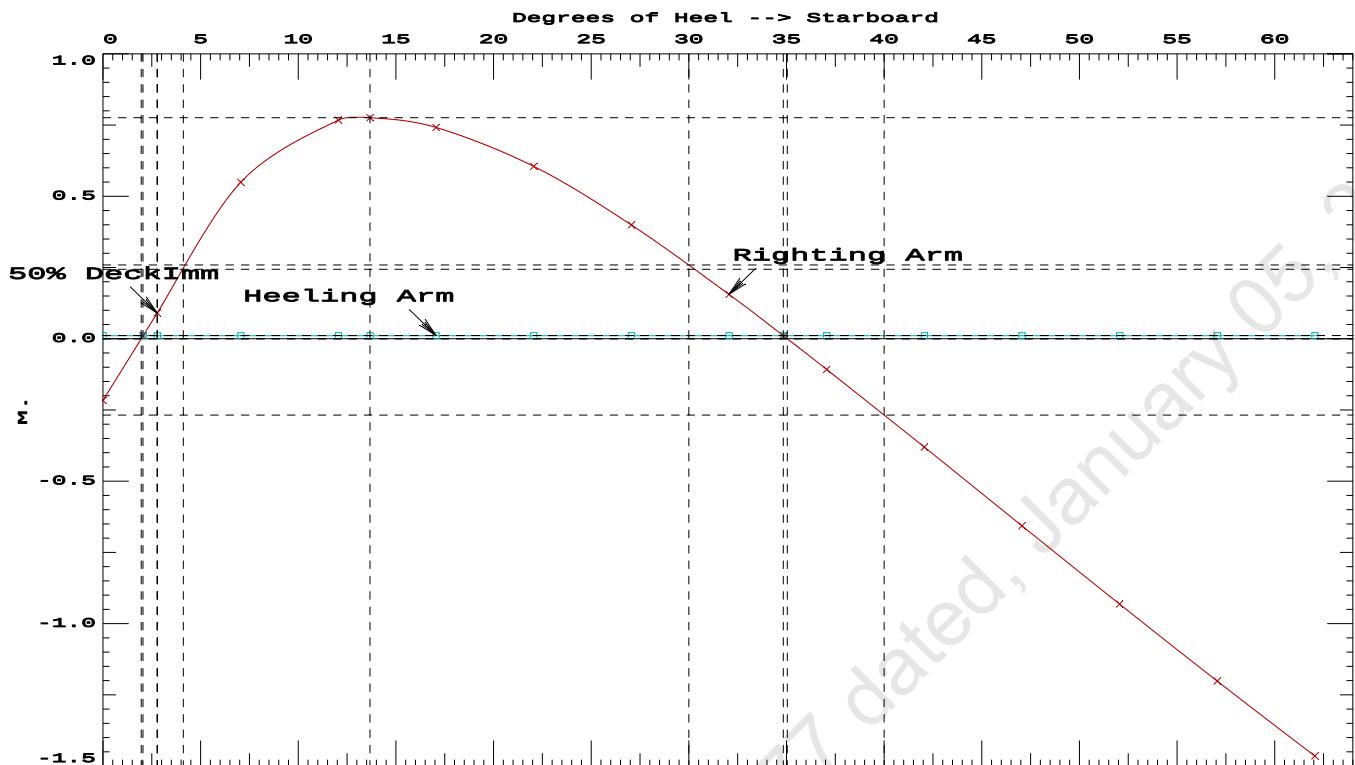
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 340.0 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
 Stbd heeling moment = 16.50 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2-----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1005 P
(2) Angle from Equilibrium to RAzero or Flood	>	20.00 deg	32.78 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	0.72 P

02-01-21 09:30:02 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd



02-01-21 09:30:02 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.467f TCG = 0.224s VCG = 4.022

Free Surface Adjustment: 0.222

Adjusted CG: LCG = 24.468f TCG = 0.216s VCG = 4.244

Origin Depth---	Trim---Heel---	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. Area
2.278	0.30a 1.96s	1,529.2	0.000 0.000	0.0000
2.270	0.32a 6.96s	1,529.2	0.000 0.541	0.0236
2.389	0.44a 11.96s	1,529.2	0.000 0.766	0.0835
2.453	0.48a 13.68s	1,529.2	0.000 0.775	0.1067
2.593	0.55a 16.96s	1,529.2	0.000 0.743	0.1506
2.831	0.68a 21.96s	1,529.2	0.000 0.608	0.2107
3.076	0.83a 26.96s	1,529.2	0.000 0.404	0.2553
3.301	0.99a 31.96s	1,529.2	0.000 0.160	0.2802
3.428	1.08a 35.03s	1,529.2	0.000 0.000	0.2845
3.503	1.14a 36.96s	1,529.2	0.000 -0.103	0.2828
3.679	1.28a 41.96s	1,529.2	0.000 -0.376	0.2620
3.827	1.42a 46.96s	1,529.2	0.000 -0.652	0.2172
3.945	1.54a 51.96s	1,529.2	0.000 -0.927	0.1483
4.032	1.65a 56.96s	1,529.2	0.000 -1.197	0.0556
4.086	1.74a 61.96s	1,529.2	0.000 -1.460	-0.0604

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 340.0 m.-MT was applied to artificially modify the CG.

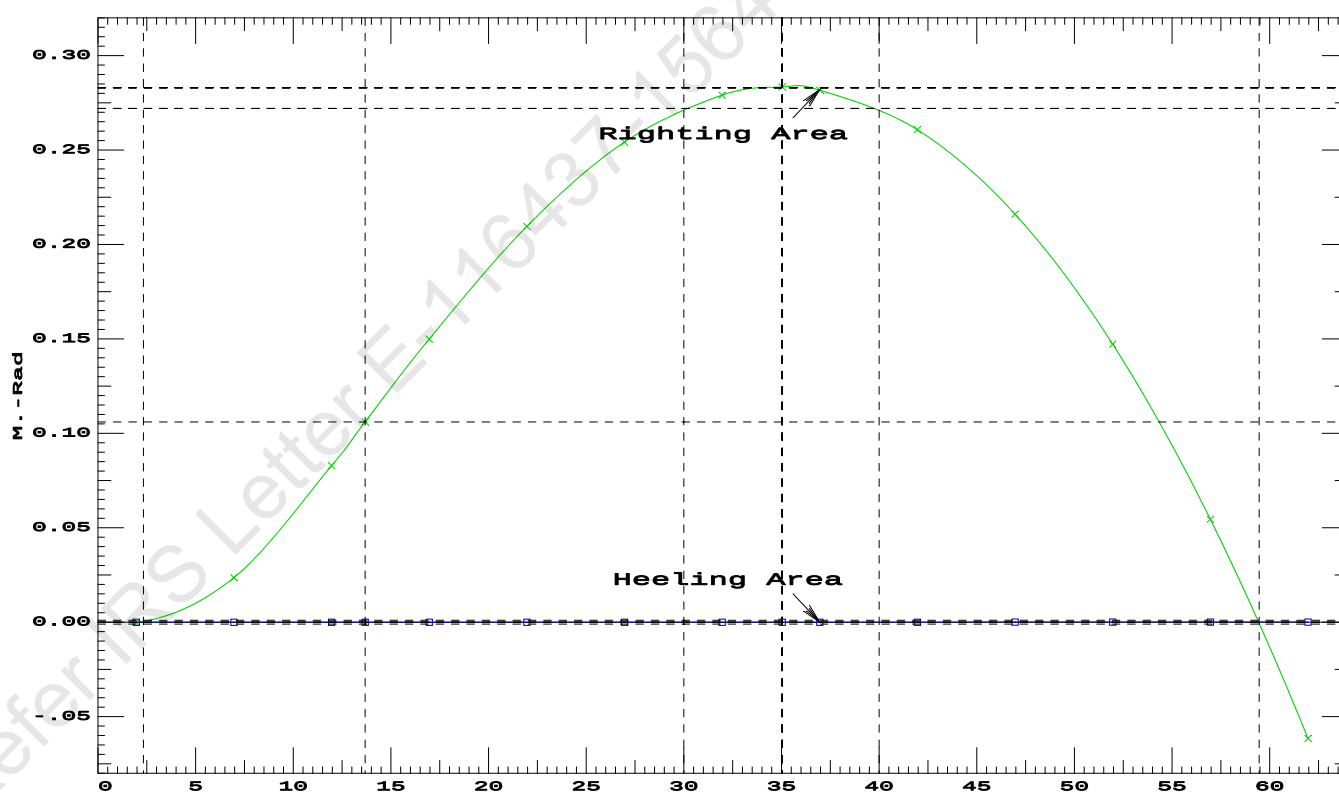
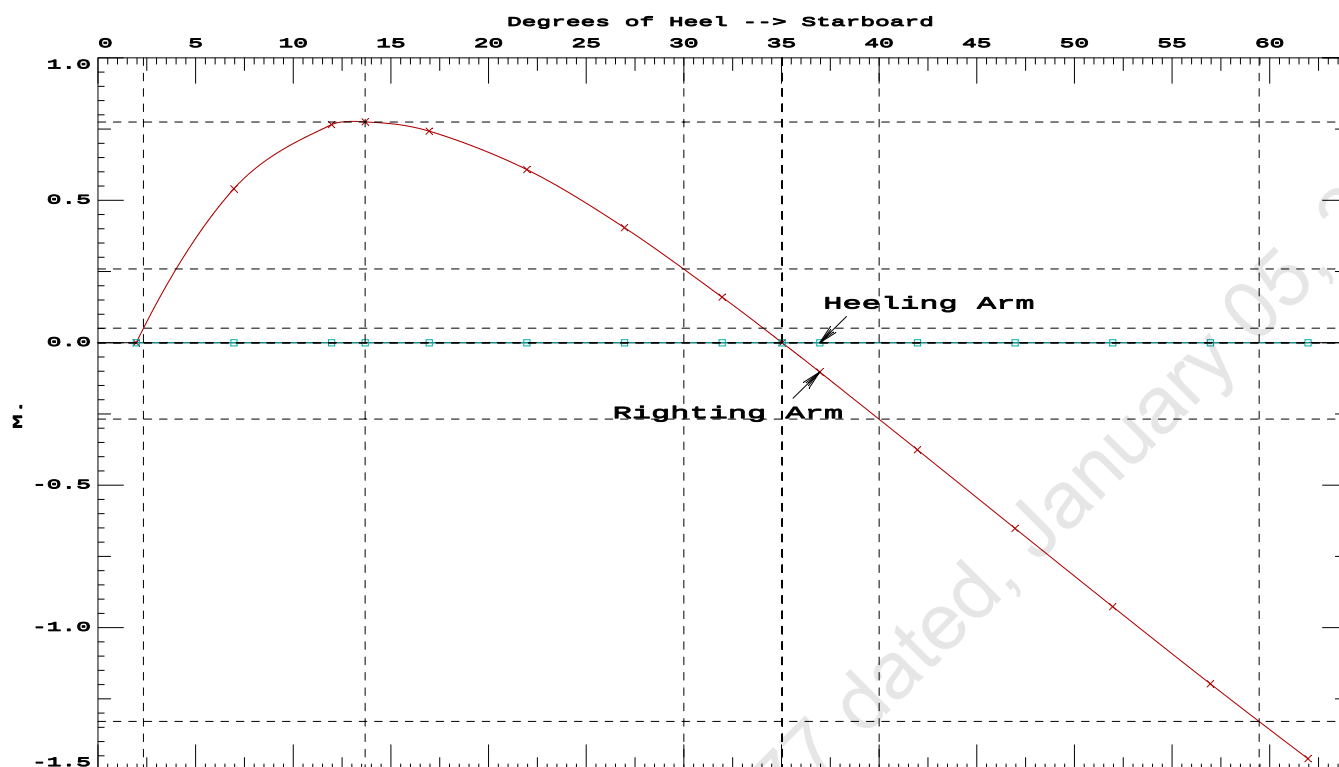
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
 Stbd heeling moment = 0.10

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	1.96 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	1185765 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	7538.71 P

02-01-21 09:30:02 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 3 : Lightship + Load at Port + Ballast + Deck CargoCrane Lifting 45.1T @ 12m Radius Stbd

02-01-21 09:23:53 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 3 : Lightship + Load at Port + Ballast + Deck Cargo**Crane Lifting 45.10T @ 12m Radius Port**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.028 @ 50.00f, 2.160 @ 25.00f, 2.291 @ 0.00

Trim: Aft 0.263/50.000, Heel: Stbd 1.96 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	4.500f	12.000s	33.320	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,119.98	21.429f	0.483s	4.884	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks----->			409.19	32.782f	0.485p	1.662	340.04
Total Weight----->			1,529.17	24.467f	0.224s	4.022	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,529.17	24.451f	0.324s	1.114	

	Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	749.2	24.939f	0.017s	104.00	9.202
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			7.68		30.92	101.09	6.293
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 1.96

02-01-21 09:23:53 Cybermarine Technologies PTE. Ltd.
GHS 17.22

After hook load drop							
WEIGHT and DISPLACEMENT and WATERPLANE STATUS							
USK draft: 2.119 @ 50.00f, 2.101 @ 25.00f, 2.084 @ 0.00							
Trim: Fwd 0.035/50.000, Heel: Port 1.00 deg.							
Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,074.88	22.139f	0.000	3.691	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
Total Tanks----->			409.19	32.782f	0.485p	1.662	340.04
Total Weight----->			1,484.07	25.074f	0.134p	3.132	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,484.11	25.074f	0.170p	1.078	

	Righting Arms:			0.001	0.000p		
Part-----	SpGr-----		WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		749.8	25.018f	0.009p	107.53	9.483
			MT/cm-----	m-----	MT/cm-----	GML-----	GMT-----
			7.69		31.31	105.48	7.429
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 4867.93; t3: 30.00; t: 15.00
limmarg: 1793.17; t3: 15.00; t: 7.50
limmarg: 379.69; t3: 7.50; t: 3.75
limmarg: -13.67; t3: 3.75; t: 1.88
limmarg: 149.36; t3: 5.68; t: 0.94
limmarg: 59.41; t3: 4.74; t: 0.47
limmarg: 21.94; t3: 4.27; t: 0.23
limmarg: 5.44; t3: 4.04; t: 0.12
limmarg: -2.70; t3: 3.92; t: 0.06
limmarg: -0.02; t3: 3.96; t: 0.03
limmarg: 4.75; t3: 4.03; t: 0.01

Theta3 is 4.03 degrees, as demonstrated below.

02-01-21 09:23:53 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.691

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.070	0.04f	1.96s	1,484.1	0.000	-0.398	0.0000
2.072	0.04f	1.00p	1,484.1	0.000	0.000	-0.0103
2.065	0.04f	4.03p	1,484.1	0.000	0.405	0.0005
2.057	0.04f	6.00p	1,484.1	0.000	0.667	0.0189
2.075	0.05f	11.00p	1,484.1	0.000	1.136	0.0986
2.194	0.07f	16.00p	1,484.0	0.000	1.271	0.2061
2.250	0.07f	17.84p	1,484.0	0.000	1.278	0.2471
2.352	0.08f	21.00p	1,484.1	0.000	1.257	0.3173
2.500	0.10f	26.00p	1,484.1	0.000	1.149	0.4231
2.630	0.12f	31.00p	1,484.1	0.000	0.988	0.5167

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs -2 deg to abs 4 > 1.000 1.047 P

02-01-21 09:23:53 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.691

Origin	Degrees of	Displacement	Residual Arms	Res.
Depth---Trim---Heel---	Weight(MT)---	in Trim--in Heel---	Area	
2.070	0.04f	1.96s	1,484.1	0.000 -0.398 0.0000
2.072	0.04f	1.00p	1,484.1	0.000 0.000 -0.0103
2.065	0.04f	4.03p	1,484.1	0.000 0.405 0.0005
2.057	0.04f	6.00p	1,484.1	0.000 0.667 0.0189
2.075	0.05f	11.00p	1,484.1	0.000 1.136 0.0986
2.194	0.07f	16.00p	1,484.0	0.000 1.271 0.2061
2.250	0.07f	17.84p	1,484.0	0.000 1.278 0.2471
2.352	0.08f	21.00p	1,484.1	0.000 1.257 0.3173
2.500	0.10f	26.00p	1,484.1	0.000 1.149 0.4231
2.630	0.12f	31.00p	1,484.1	0.000 0.988 0.5167
2.739	0.14f	36.00p	1,484.1	0.000 0.798 0.5948
2.827	0.16f	41.00p	1,484.1	0.000 0.590 0.6555
2.893	0.18f	46.00p	1,484.1	0.000 0.370 0.6975
2.938	0.20f	51.00p	1,484.1	0.000 0.144 0.7199
2.954	0.21f	54.11p	1,484.1	0.000 0.000 0.7239
2.960	0.21f	56.00p	1,484.1	0.000 -0.087 0.7224

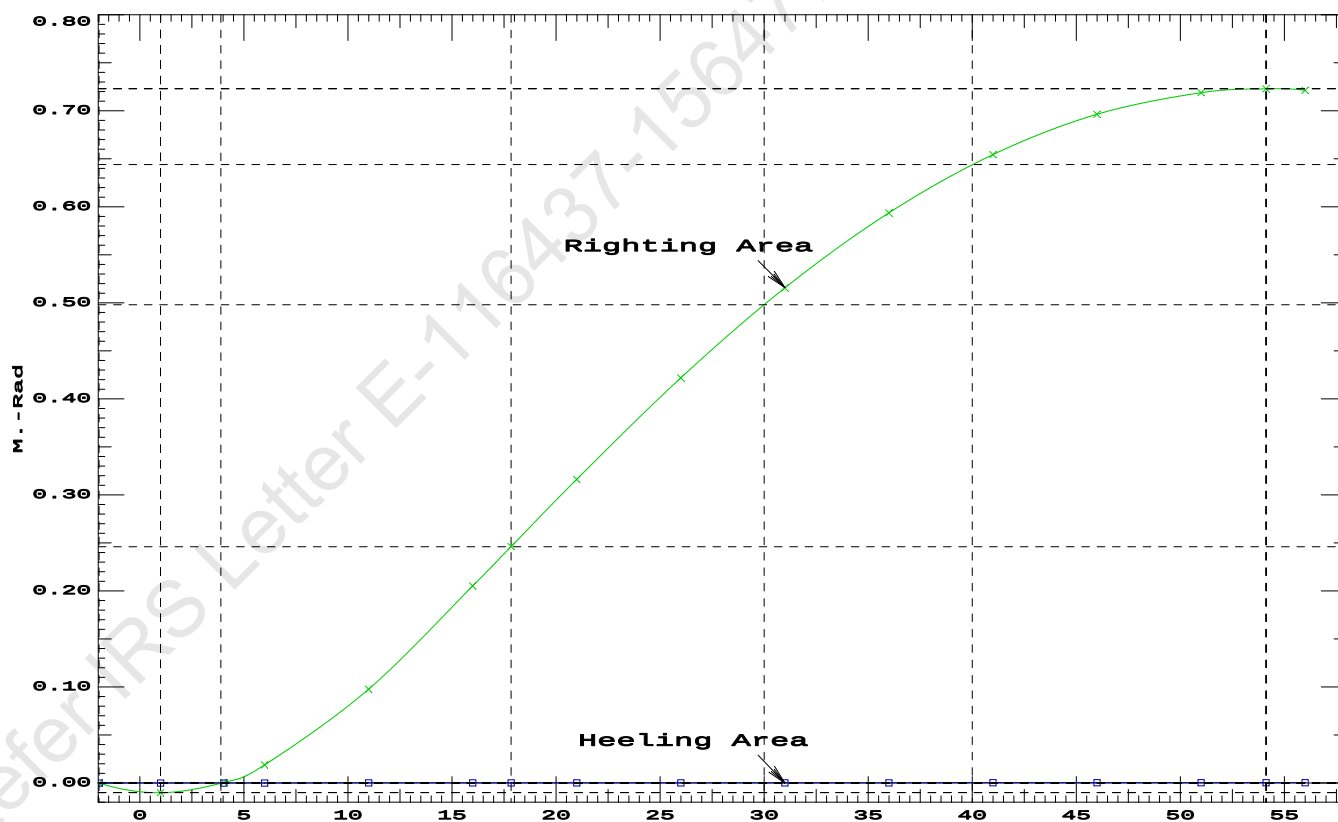
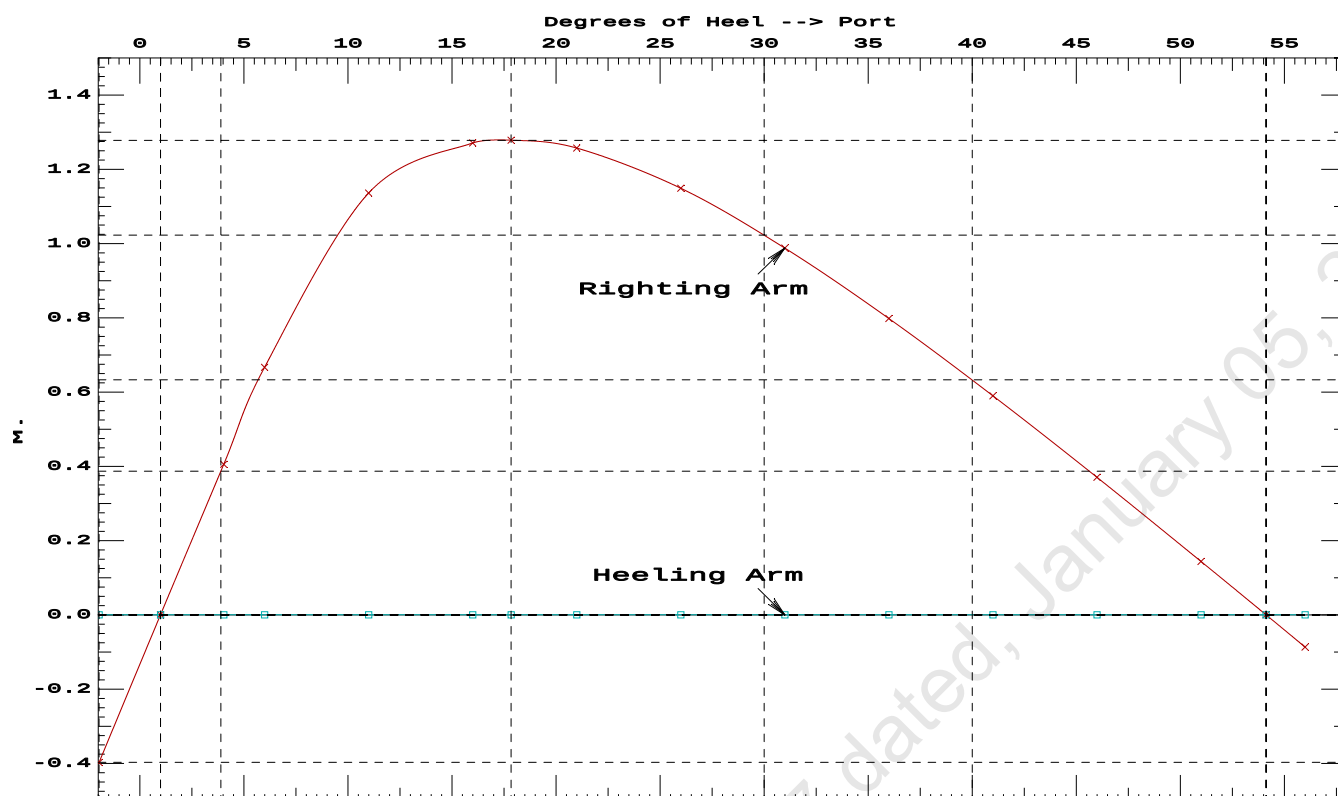
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs -2 deg to RZero > 1.000 71.547 P
(2) Angle from abs 4 deg to RZero > 20.00 deg 50.08 P

02-01-21 09:23:53 Cybermarine Technologies PTE. Ltd.
GHS 17.22

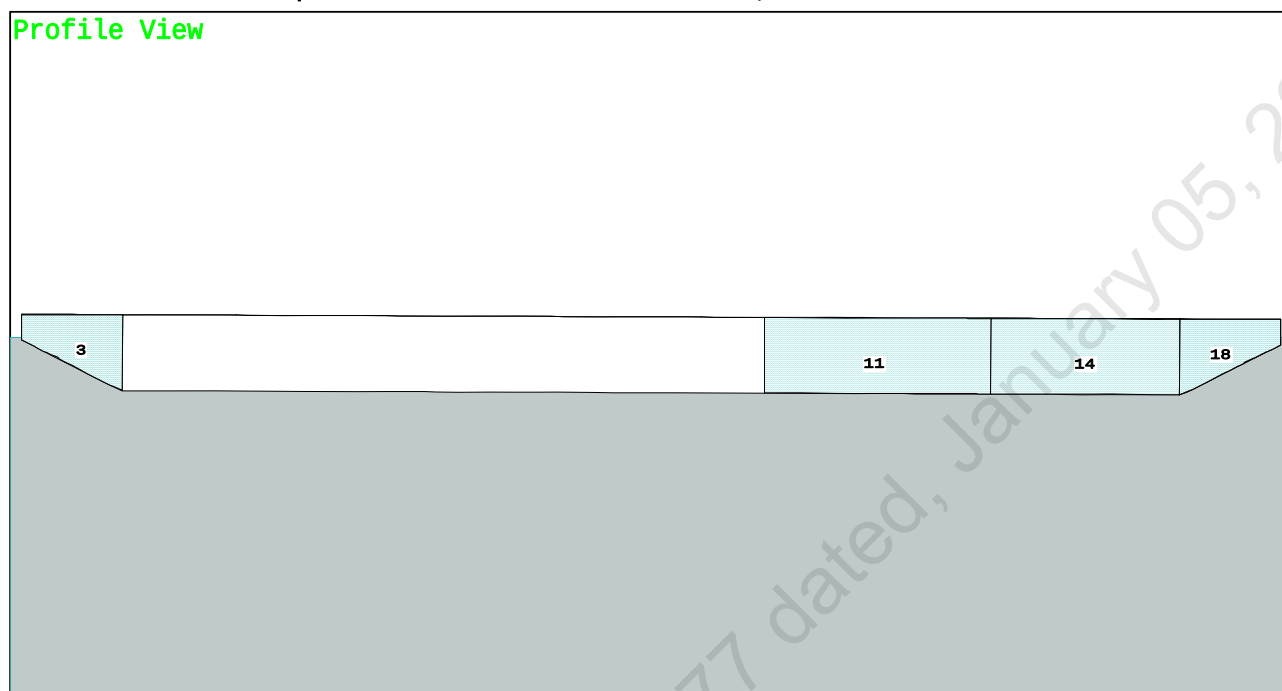
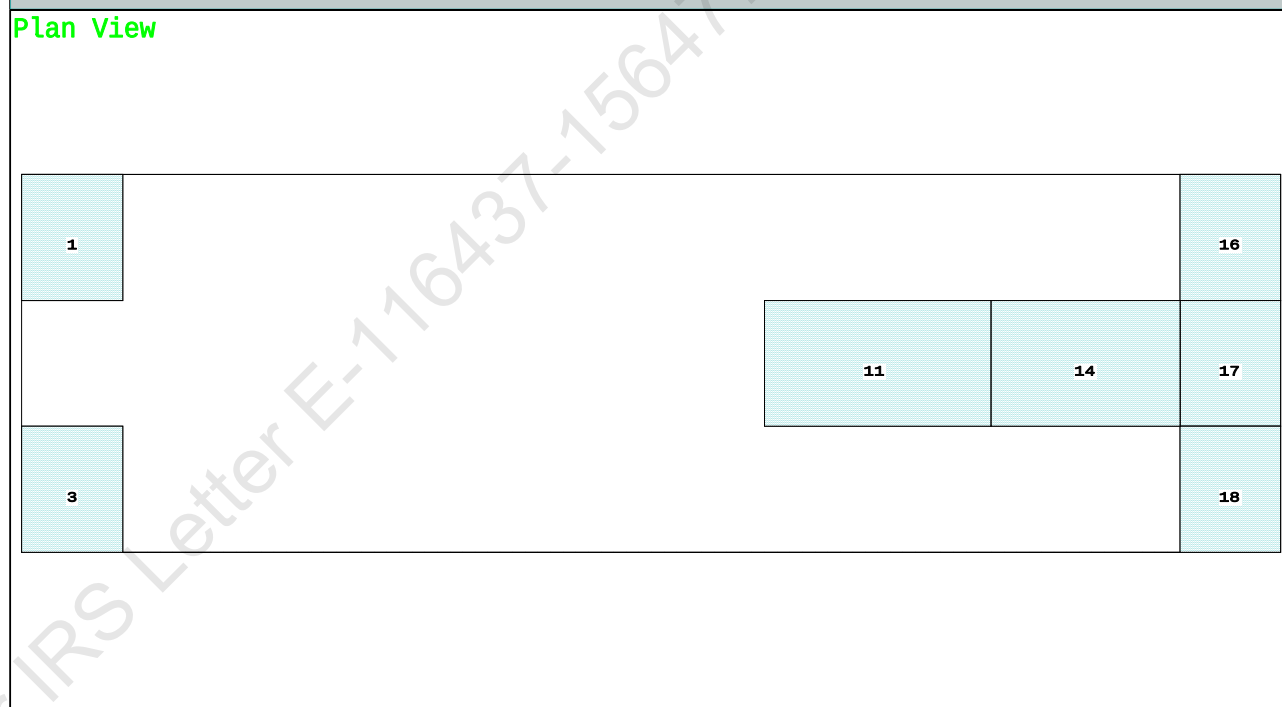


01-01-21 20:58:15 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Fwd

Condition Graphic - Draft: 2.313 @ 50.000f, 2.111 @ 0.000 Heel: zero

Profile View**Plan View****Tanks**

1 WBTk1.P 3 WBTk1.S 11 WBTk4.C 14 WBTk5.C 16 WBTk6.P 17 WBTk6.C 18 WBTk6.S

01-01-21 20:58:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.313 @ 50.00f, 2.212 @ 25.00f, 2.111 @ 0.00

Trim: Fwd 0.203/50.000, Heel: zero

Part-----			Weight(MT)-----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	16.500f	0.000	33.320	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,119.98	21.912f	0.000	4.884	
	Load-----	SpGr-----	Weight(MT)-----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTk6.S	1.000	1.025	39.99	47.660f	4.967s	1.915	41.84*
Total Tanks----->			449.17	34.106f	0.000	1.684	381.88
Total Weight----->			1,569.15	25.403f	0.000	3.968	
			Displ(MT)-----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,569.15	25.414f	0.000	1.135	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->		1.025	750.6	25.008f	0.000	102.03	8.964
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			7.69		31.13	99.20	6.131
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

438.2 Cu.M. (26%) SALT WATER

150.00 MT Crane
549.78 MT Deck Cargo

45.10 MT Crane Load

DRAFT AT FP (M) = 2.313
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.111
DRAFT AT FWD MARK (M) = 2.297
DRAFT AT AFT MARK (M) = 2.127

HYDROSTATIC PROPERTIES

Trim: Fwd 0.203/50.000, No Heel, VCG = 3.968

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)-----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
2.212	1,569.15	25.414f	1.135	7.69	25.008f	31.13	99.20	6.131
Distances in METERS.-----						Specific Gravity = 1.025.-----		
						Moment in m.-MT.		
						Trim is per 50.00m.		

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 381.9 m.-MT

01-01-21 20:58:15 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.403f TCG = 0.000 VCG = 3.968

Free Surface Adjustment: 0.243

Adjusted CG: LCG = 25.402f TCG = 0.000 VCG = 4.212

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
2.099	0.23f	0.00	1,569.0	0.000	-0.010	0.0000	2.693
2.099	0.23f	0.09p	1,569.0	0.000	0.000	-0.0000	2.599
2.095	0.23f	2.69p	1,569.2	0.000	0.279	0.0063	0.000
2.087	0.24f	5.09p	1,569.2	0.000	0.535	0.0234	-2.401
2.085	0.32f	10.09p	1,569.3	0.000	0.866	0.0860	-7.401
2.135	0.37f	12.87p	1,569.0	0.000	0.893	0.1290	-10.174
2.192	0.40f	15.09p	1,569.2	0.000	0.875	0.1633	-12.401
2.346	0.49f	20.09p	1,569.2	0.000	0.758	0.2360	-17.401
2.503	0.60f	25.09p	1,569.2	0.000	0.571	0.2945	-22.401
2.636	0.72f	30.09p	1,569.2	0.000	0.337	0.3345	-27.401
2.747	0.84f	35.09p	1,569.2	0.000	0.077	0.3528	-32.401
2.775	0.88f	36.52p	1,569.2	0.000	0.000	0.3537	-33.832
2.837	0.96f	40.09p	1,569.2	0.000	-0.196	0.3477	-37.401
2.905	1.07f	45.09p	1,569.2	0.000	-0.474	0.3185	-42.401
2.951	1.17f	50.09p	1,569.2	0.000	-0.754	0.2649	-47.401
2.975	1.26f	55.09p	1,569.2	0.000	-1.031	0.1870	-52.401
2.978	1.34f	60.09p	1,569.2	0.000	-1.302	0.0852	-57.401

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

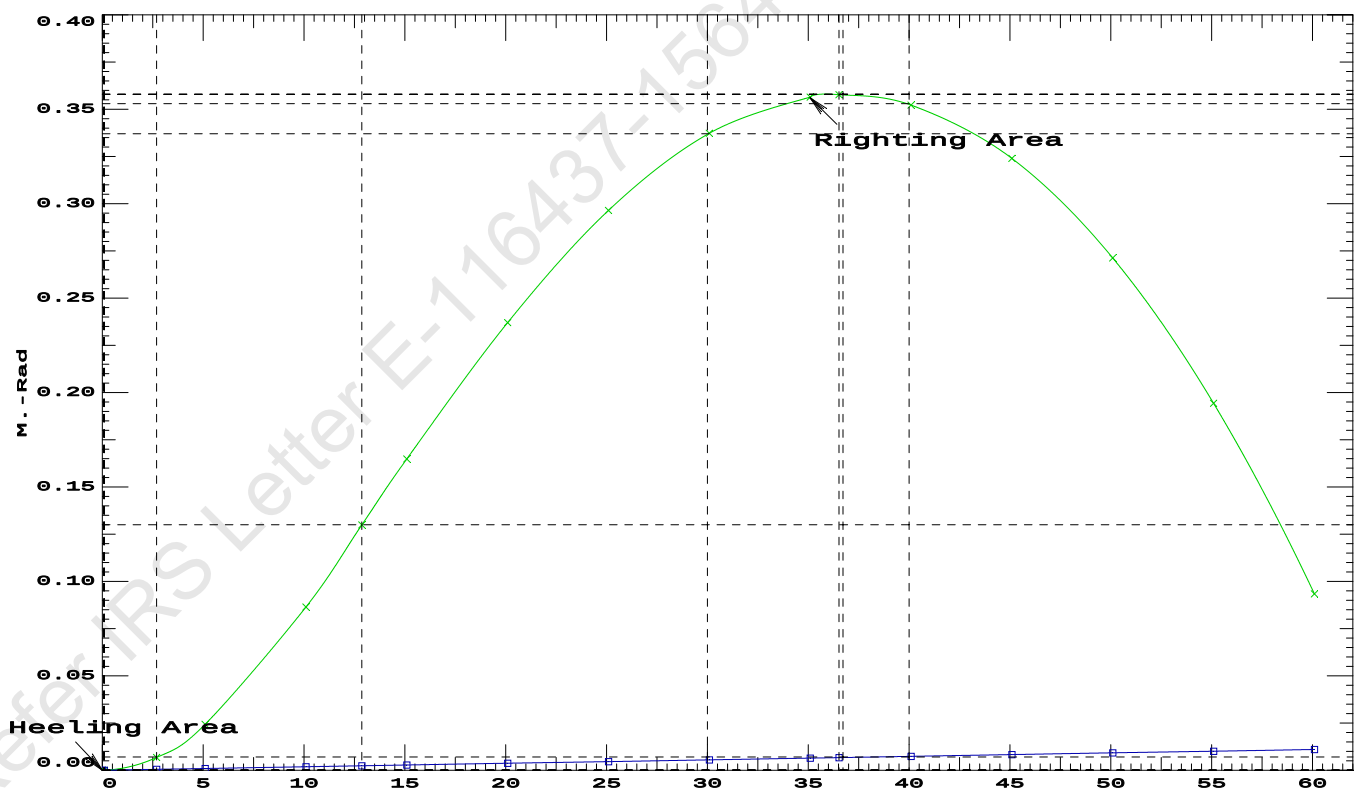
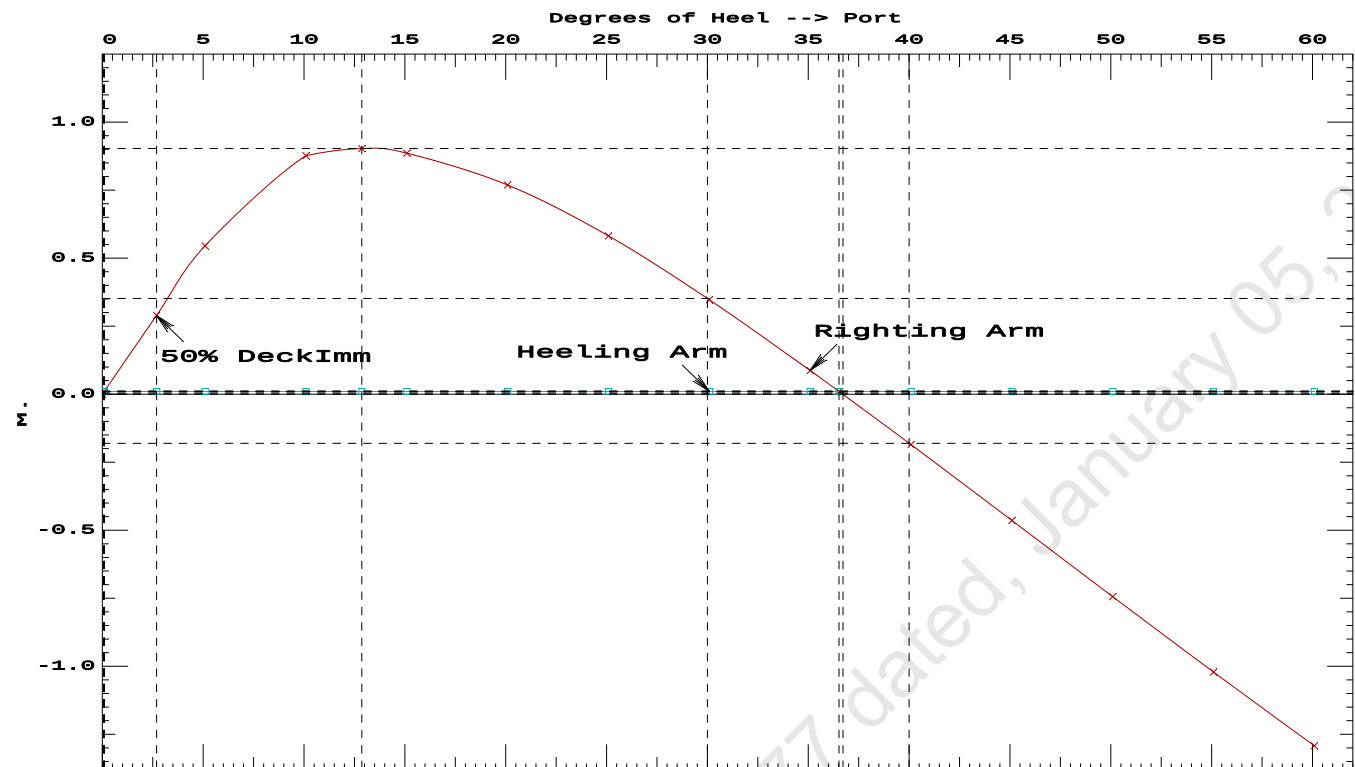
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 381.9 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
 Port heeling moment = 16.49 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2-----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1290 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	36.43 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	2.60 P

01-01-21 20:58:15 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Fwd



01-01-21 20:58:15 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo
Crane Lifting 45.1T @ 12m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.403f TCG = 0.000 VCG = 3.968

Free Surface Adjustment: 0.243

Adjusted CG: LCG = 25.402f TCG = 0.000 VCG = 4.212

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.098	0.23f	0.01p	1,569.2	0.000	0.000	0.0000
2.087	0.24f	5.01p	1,569.2	0.000	0.535	0.0234
2.084	0.32f	10.01p	1,569.3	0.000	0.873	0.0862
2.136	0.37f	12.87p	1,569.2	0.000	0.902	0.1311
2.190	0.40f	15.01p	1,569.2	0.000	0.886	0.1644
2.343	0.49f	20.01p	1,569.2	0.000	0.771	0.2381
2.500	0.60f	25.01p	1,569.2	0.000	0.585	0.2978
2.634	0.72f	30.01p	1,569.2	0.000	0.351	0.3389
2.746	0.84f	35.01p	1,569.2	0.000	0.091	0.3584
2.779	0.88f	36.70p	1,569.2	0.000	0.000	0.3598
2.836	0.96f	40.01p	1,569.2	0.000	-0.181	0.3546
2.904	1.07f	45.01p	1,569.2	0.000	-0.460	0.3267
2.950	1.17f	50.01p	1,569.2	0.000	-0.739	0.2743
2.975	1.26f	55.01p	1,569.2	0.000	-1.016	0.1977
2.978	1.33f	60.01p	1,569.2	0.000	-1.288	0.0971

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 381.9 m.-MT was applied to artificially modify the CG.

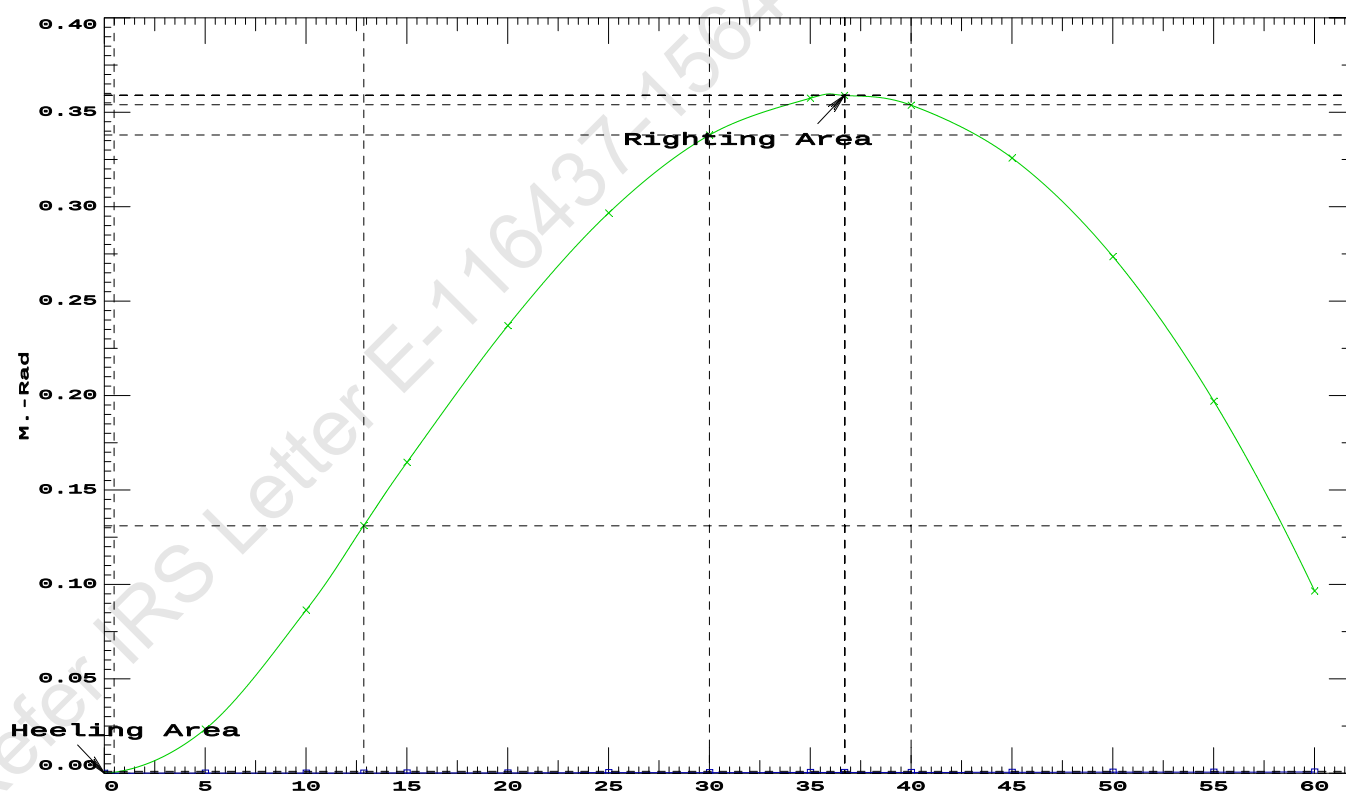
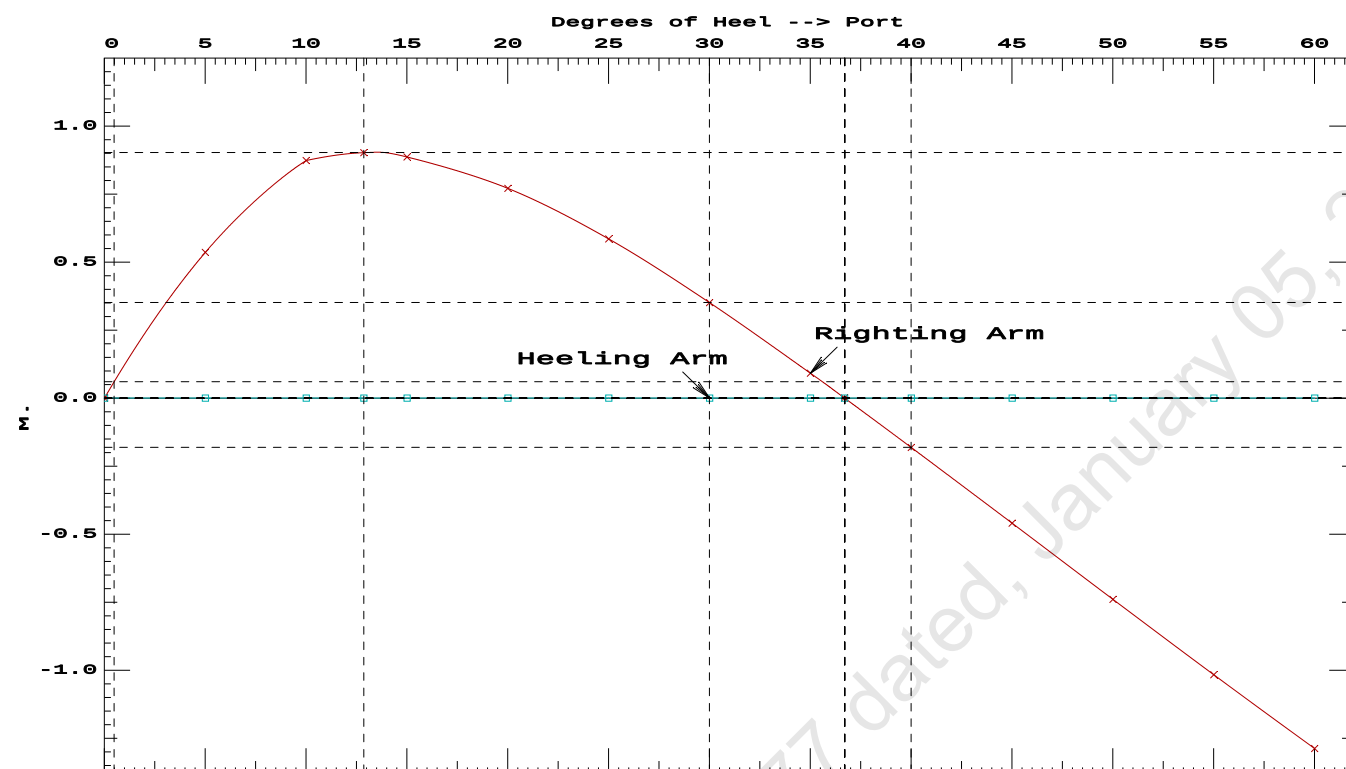
Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
 Port heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT-----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	141533 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	882.505 P

01-01-21 20:58:15 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 4 : Lightship + Load at Fwd + Ballast + Deck CargoCrane Lifting 45.1T @ 12m Radius Fwd

01-01-21 20:58:22 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 4 : Lightship + Load at Fwd + Ballast + Deck Cargo**Crane Lifting 45.10T @ 12m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.313 @ 50.00f, 2.212 @ 25.00f, 2.111 @ 0.00

Trim: Fwd 0.203/50.000, Heel: Port 0.01 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Crane Load			45.10	16.500f	0.000	33.320	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,119.98	21.912f	0.000	4.884	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTk6.S	1.000	1.025	39.99	47.660f	4.967s	1.915	41.84*
Total Tanks----->			449.17	34.106f	0.000	1.684	381.88
Total Weight----->			1,569.15	25.403f	0.000	3.968	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,569.15	25.414f	0.001p	1.135	

	Righting Arms:			0.000	-0.001s		
	External Arms:			0.000	-0.001		
	Residual Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		750.6	25.008f	0.000	102.03	8.964
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			7.69		31.13	99.20	6.131
Distances in METERS.-----							
-----Moments in m.-MT.							

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = -0.01

01-01-21 20:58:22 Cybermarine Technologies PTE. Ltd.
GHS 17.22

After hook load drop							
WEIGHT and DISPLACEMENT and WATERPLANE STATUS							
USK draft: 2.315 @ 50.00f, 2.153 @ 25.00f, 1.992 @ 0.00							
Trim: Fwd 0.323/50.000, Heel: 0.00 deg.							
Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Deck Cargo			549.78	25.000f	0.000	3.500	
Total Fixed----->			1,074.88	22.139f	0.000	3.691	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk1.P	1.000	1.025	39.99	2.340f	4.967p	1.915	41.85*
WBTk1.S	1.000	1.025	39.99	2.340f	4.967s	1.915	41.85*
WBTk4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk6.P	1.000	1.025	39.99	47.660f	4.967p	1.915	41.84*
WBTk6.C	1.000	1.025	40.60	47.664f	0.000	1.904	41.85*
WBTk6.S	1.000	1.025	39.99	47.660f	4.967s	1.915	41.84*
Total Tanks----->			449.17	34.106f	0.000	1.684	381.88
Total Weight----->			1,524.05	25.666f	0.000	3.100	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL	1.025		1,524.06	25.679f	0.001	1.106	

Righting Arms:				0.000	-0.001s		
Part-----	SpGr-----		WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		746.9	25.130f	0.016p	103.52	9.192
			MT/cm-----	m.-	MT/cm-----	GML-----	GMT-----
			7.66		30.95	101.53	7.198
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: INFINITY; t3: 30.0; t: 15.00
limmarg: INFINITY; t3: 15.0; t: 7.50
limmarg: INFINITY; t3: 7.5; t: 3.75
limmarg: INFINITY; t3: 3.8; t: 1.88
limmarg: INFINITY; t3: 1.9; t: 0.94
limmarg: INFINITY; t3: 1.0; t: 0.47
limmarg: INFINITY; t3: 0.5; t: 0.23
limmarg: INFINITY; t3: 0.3; t: 0.12
limmarg: INFINITY; t3: 0.2; t: 0.06
limmarg: INFINITY; t3: 0.1; t: 0.03
limmarg: INFINITY; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

01-01-21 20:58:22 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.691

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. Area
1.979	0.37f	0.00	1,524.1	0.000	-0.000	0.0000
1.979	0.37f	0.07s	1,524.1	0.000	0.010	0.0000
1.966	0.38f	5.00s	1,524.1	0.000	0.652	0.0285
1.937	0.48f	10.00s	1,524.1	0.000	1.140	0.1078
2.005	0.62f	15.00s	1,524.1	0.000	1.285	0.2162
2.050	0.68f	17.14s	1,523.9	0.000	1.295	0.2644
2.118	0.76f	20.00s	1,524.1	0.000	1.277	0.3288
2.221	0.94f	25.00s	1,524.1	0.000	1.181	0.4369
2.302	1.13f	30.00s	1,524.1	0.000	1.029	0.5338

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 LARGE

01-01-21 20:58:22 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 22.139f TCG = 0.000 VCG = 3.691

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.979	0.37f	0.00	1,524.1	0.000	-0.000	0.0000
1.979	0.37f	0.07s	1,524.1	0.000	0.010	0.0000
1.966	0.38f	5.00s	1,524.1	0.000	0.652	0.0285
1.937	0.48f	10.00s	1,524.1	0.000	1.140	0.1078
2.005	0.62f	15.00s	1,524.1	0.000	1.285	0.2162
2.050	0.68f	17.14s	1,523.9	0.000	1.295	0.2644
2.118	0.76f	20.00s	1,524.1	0.000	1.277	0.3288
2.221	0.94f	25.00s	1,524.1	0.000	1.181	0.4369
2.302	1.13f	30.00s	1,524.1	0.000	1.029	0.5338
2.364	1.31f	35.00s	1,524.1	0.000	0.846	0.6158
2.406	1.49f	40.00s	1,524.1	0.000	0.643	0.6809
2.428	1.66f	45.00s	1,524.1	0.000	0.428	0.7277
2.431	1.82f	50.00s	1,524.1	0.000	0.204	0.7553
2.418	1.95f	54.49s	1,524.0	0.000	0.000	0.7633
2.416	1.96f	55.00s	1,524.1	0.000	-0.023	0.7632

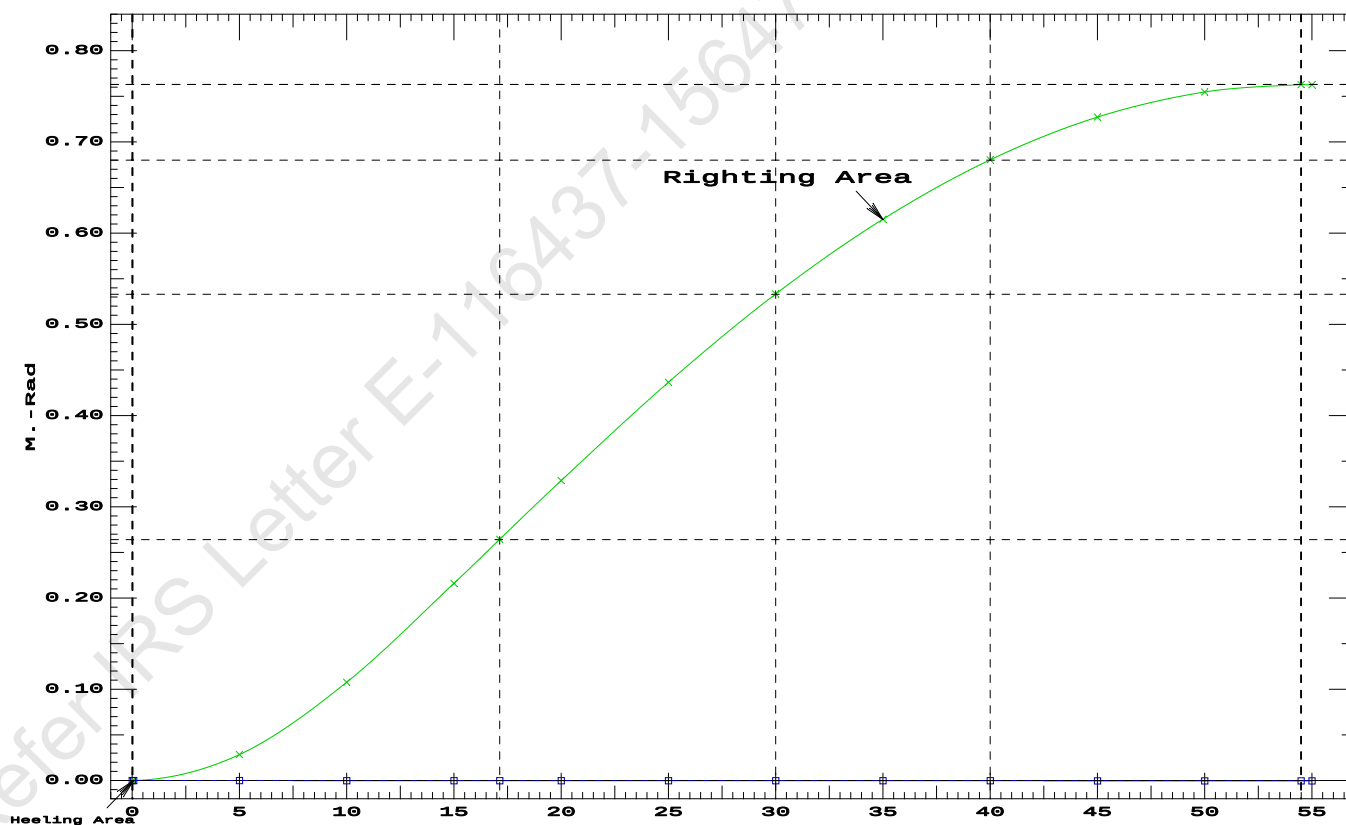
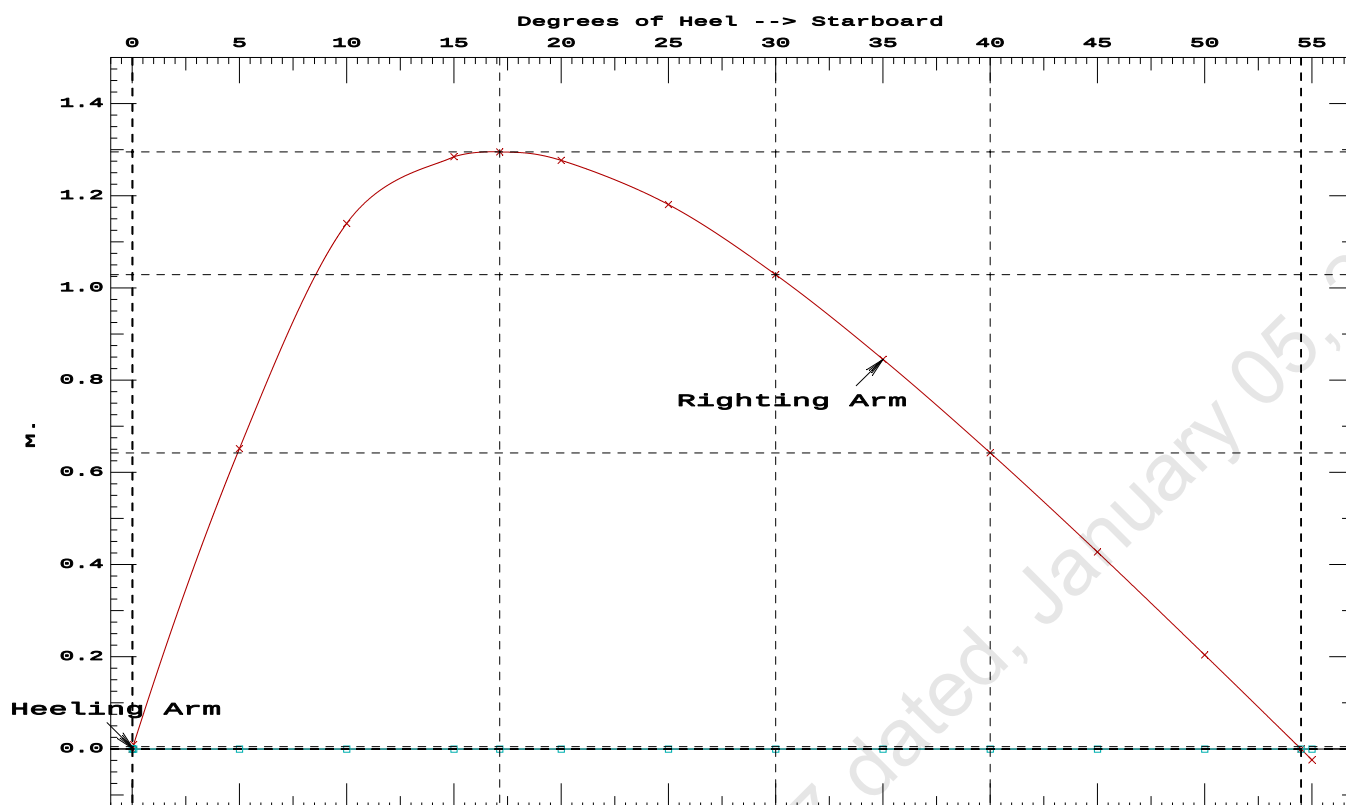
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1074.88 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAzero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAzero > 20.00 deg 54.42 P

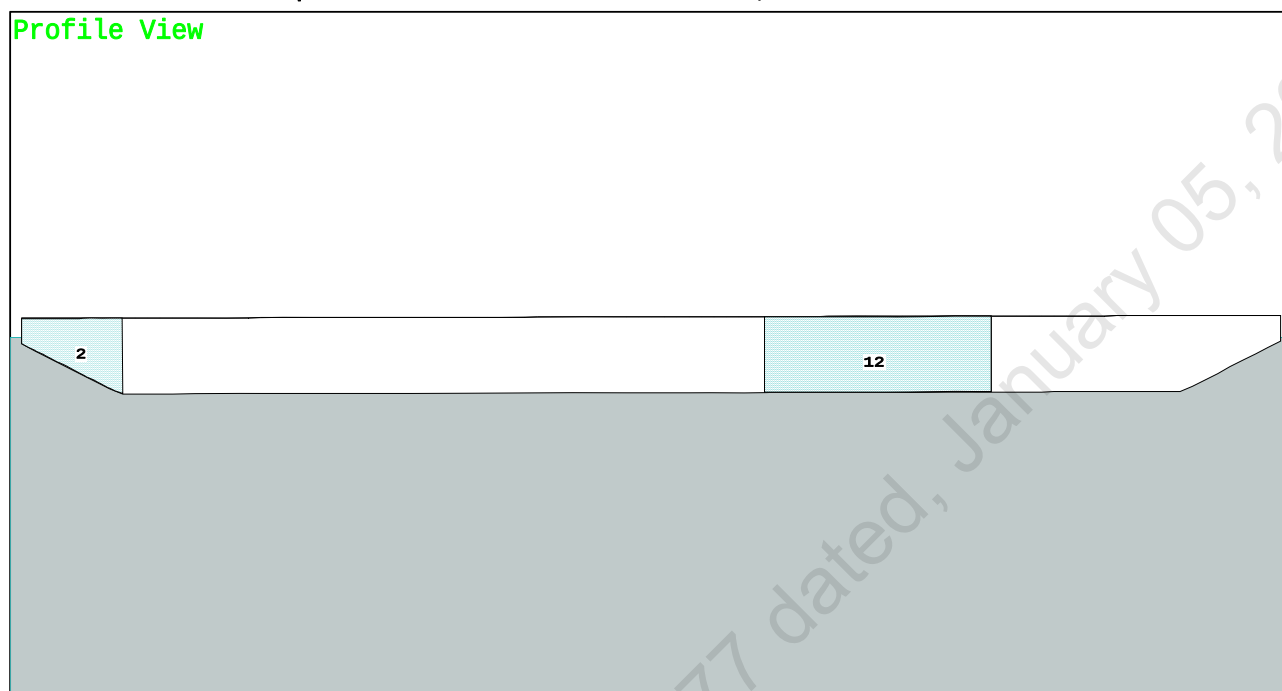
01-01-21 20:58:22 Cybermarine Technologies PTE. Ltd.
GHS 17.22



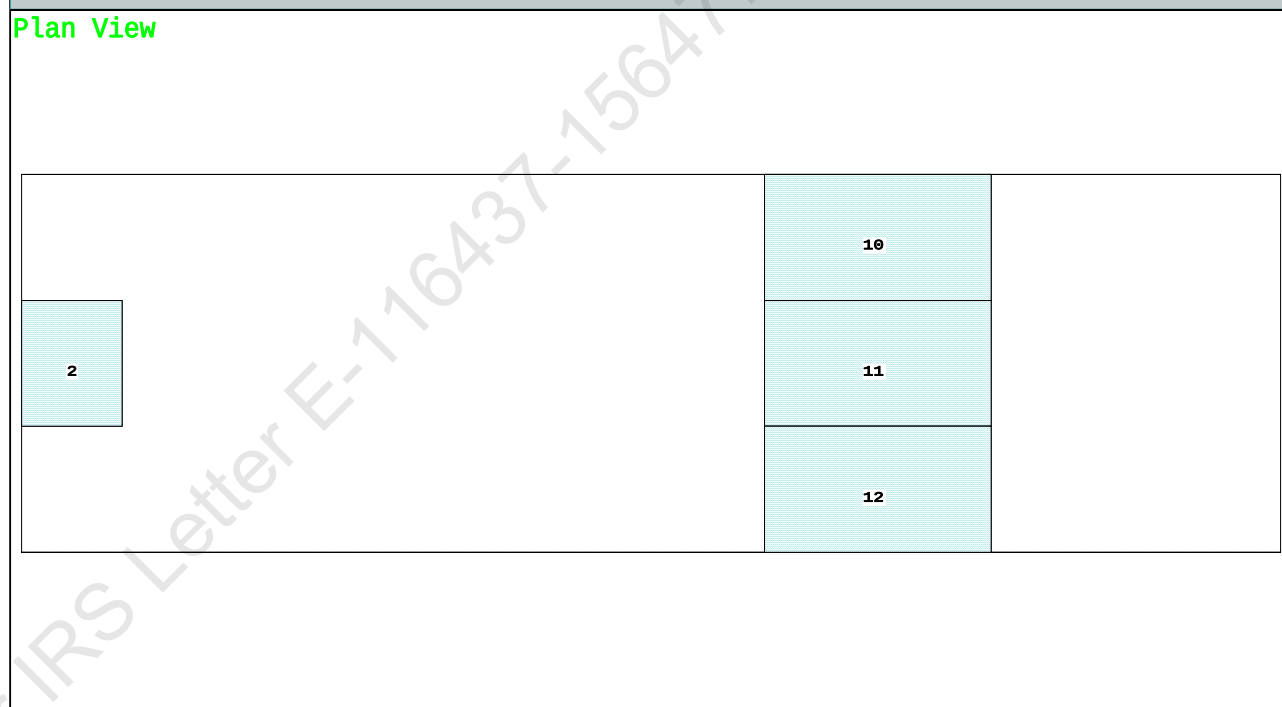
GHS 17.22

Towing Condition with Max. draft 2.212m

Profile View



Plan View



2 WBTK1.C 10 WBTK4.P 11 WBTK4.C 12 WBTK4.S

01-01-21 20:59:24 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 5 : Lightship + Crane + Ballast + Deck CargoTowing Condition with Max. draft 2.212m

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.165 @ 50.00f, 2.221 @ 25.00f, 2.277 @ 0.00

Trim: Aft 0.113/50.000, Heel: zero

Part-----			Weight(MT)	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			150.00	4.500f	0.000	6.120	
Deck Cargo			605.72	25.000f	0.000	3.500	
Total Fixed----->			1,130.82	22.281f	0.000	3.682	
	Load-----	SpGr-----	Weight(MT)	LCG-----	TCG-----	VCG-----	FSM-----
WBTK1.C	1.000	1.025	40.60	2.336f	0.000	1.904	41.85*
WBTK4.P	1.000	1.025	134.48	34.000f	4.980p	1.511	94.17*
WBTK4.C	1.000	1.025	135.61	34.000f	0.000	1.500	94.17*
WBTK4.S	1.000	1.025	134.48	34.000f	4.980s	1.511	94.17*
Total Tanks----->			445.16	31.112f	0.000	1.544	324.37
Total Weight----->			1,575.98	24.775f	0.000	3.078	
			Displ(MT)	LCB-----	TCB-----	VCB-----	
HULL		1.025	# 1,575.98	24.771f	0.000	1.139	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		750.6	24.995f	0.000	101.61	8.964
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			7.69		31.42	99.67	7.025
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

434.3 Cu.M. (26%) SALT WATER

150.00 MT Crane

605.72 MT Deck Cargo

DRAFT AT FP (M) = 2.165
 DRAFT AT MS (M) = ~~2.221~~ 2.212 M
 DRAFT AT AP (M) = 2.277
 DRAFT AT FWD MARK (M) = 2.174
 DRAFT AT AFT MARK (M) = 2.268

#: MAX. DISPLACEMENT ALLOWED IS 1569.15 T ONLY
 MAX DRAFT EXTREME ALLOWED IS 2.212 M.

01-01-21 20:59:24 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 5 : Lightship + Crane + Ballast + Deck Cargo
Towing Condition with Max. draft 2.212m

HYDROSTATIC PROPERTIES

Trim: Aft 0.113/50.000, No Heel, VCG = 3.078

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
2.221 1,575.98 24.771f 1.139 7.69 24.995f 31.42 99.67 7.025
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 324.4 m.-MT

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.775f TCG = 0.000 VCG = 3.078

Free Surface Adjustment: 0.206

Adjusted CG: LCG = 24.776f TCG = 0.000 VCG = 3.284

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. 50% DeckImm Area----Angle
2.265	0.13a	0.00	1,576.0	0.000 -0.010	0.0000 2.833
2.265	0.13a	0.08s	1,576.0	0.000 0.000	-0.0000 2.752
2.263	0.13a	2.83s	1,576.0	0.000 0.338	0.0081 0.000
2.256	0.13a	5.08s	1,576.0	0.000 0.613	0.0268 -2.248
2.310	0.18a	10.08s	1,576.0	0.000 1.019	0.0995 -7.248
2.479	0.22a	15.08s	1,576.0	0.000 1.102	0.1944 -12.248
2.495	0.23a	15.45s	1,576.0	0.000 1.103	0.2016 -12.621
2.696	0.27a	20.08s	1,576.0	0.000 1.060	0.2896 -17.248
2.928	0.33a	25.08s	1,576.0	0.000 0.949	0.3778 -22.248
3.146	0.39a	30.08s	1,576.0	0.000 0.787	0.4538 -27.248
3.342	0.46a	35.08s	1,576.0	0.000 0.596	0.5144 -32.248
3.513	0.52a	40.08s	1,576.0	0.000 0.388	0.5575 -37.248
3.657	0.58a	45.08s	1,576.0	0.000 0.171	0.5819 -42.248
3.749	0.62a	48.90s	1,576.0	0.000 0.000	0.5876 -46.064
3.774	0.64a	50.08s	1,576.0	0.000 -0.053	0.5871 -47.248
3.861	0.69a	55.08s	1,576.0	0.000 -0.280	0.5726 -52.248
3.920	0.73a	60.08s	1,576.0	0.000 -0.506	0.5383 -57.248

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

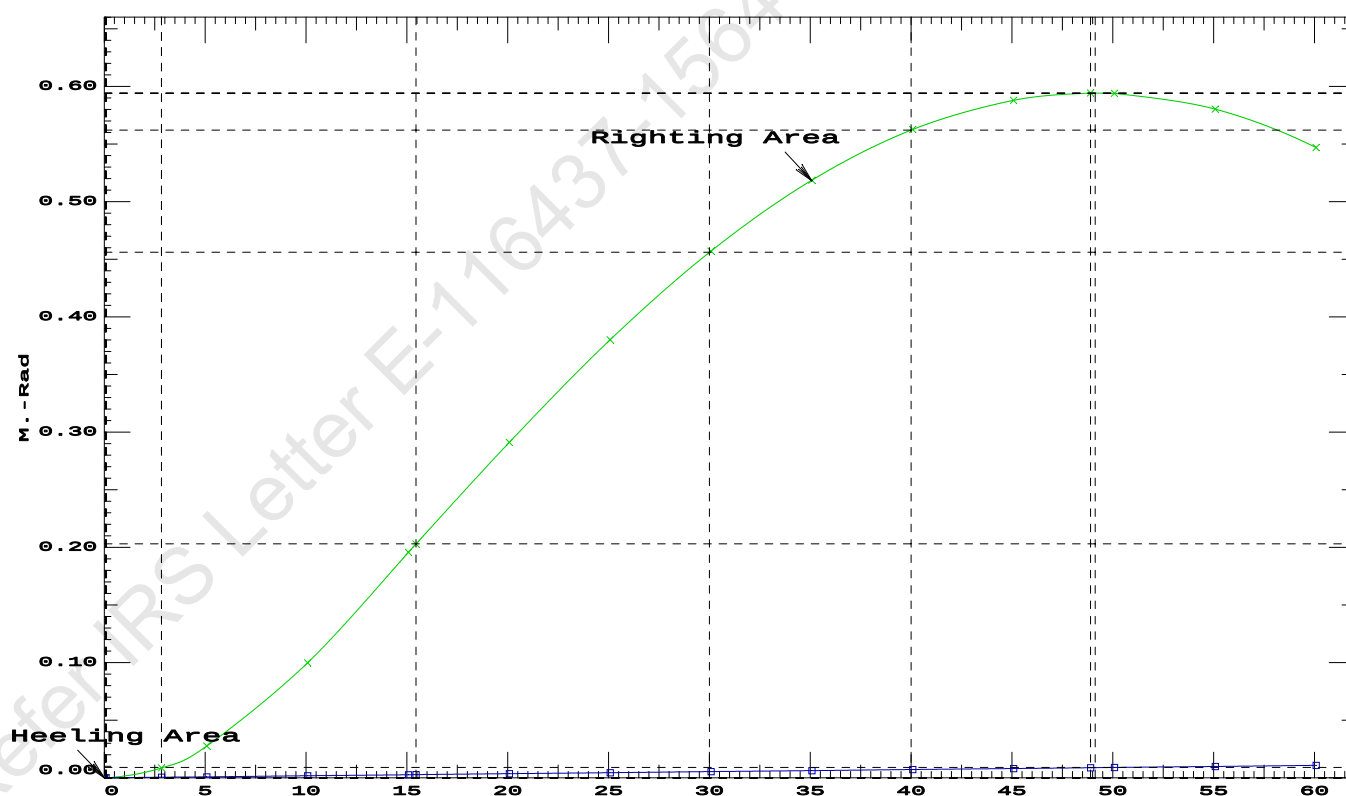
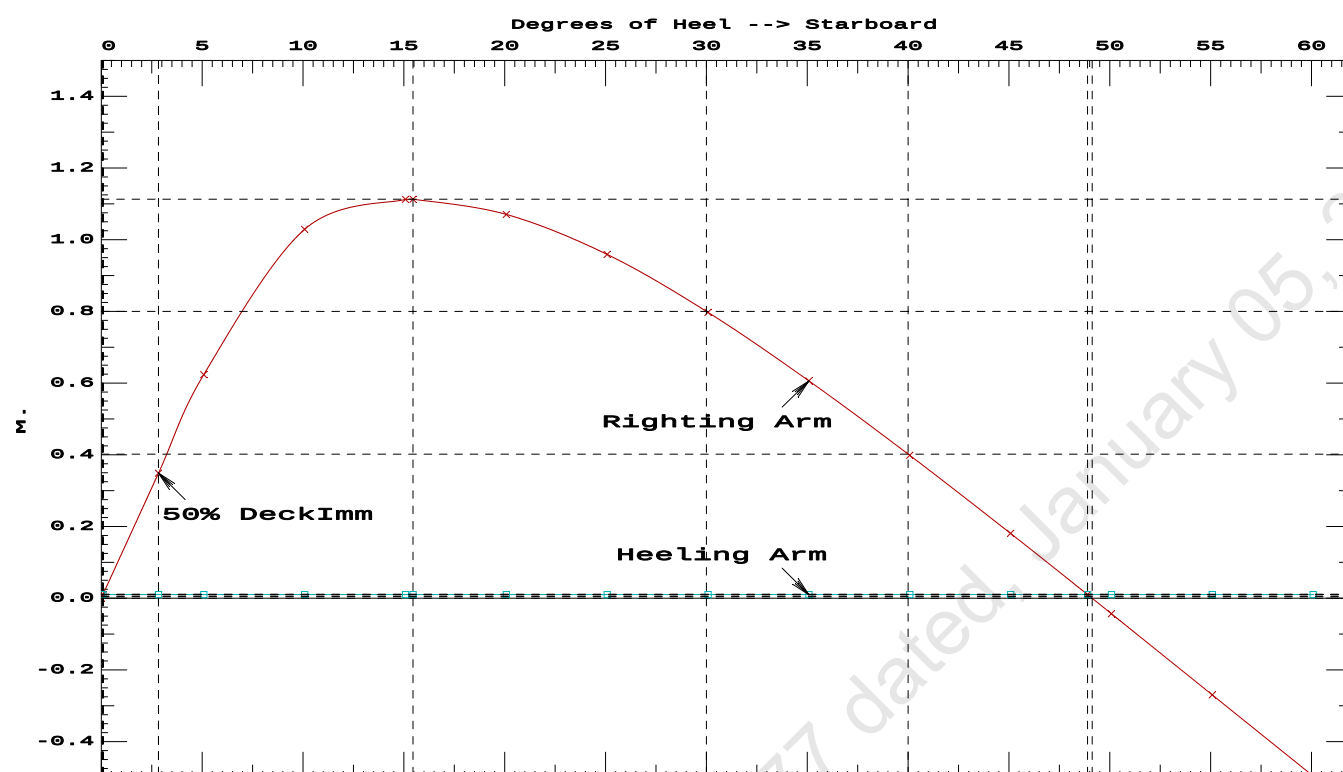
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 324.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 16.23 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.2016 P
(2) Angle from Equilibrium to RZero or Flood > 20.00 deg 48.82 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 2.75 P

01-01-21 20:59:24 Cybermarine Technologies PTE. Ltd.

GHS 17.22

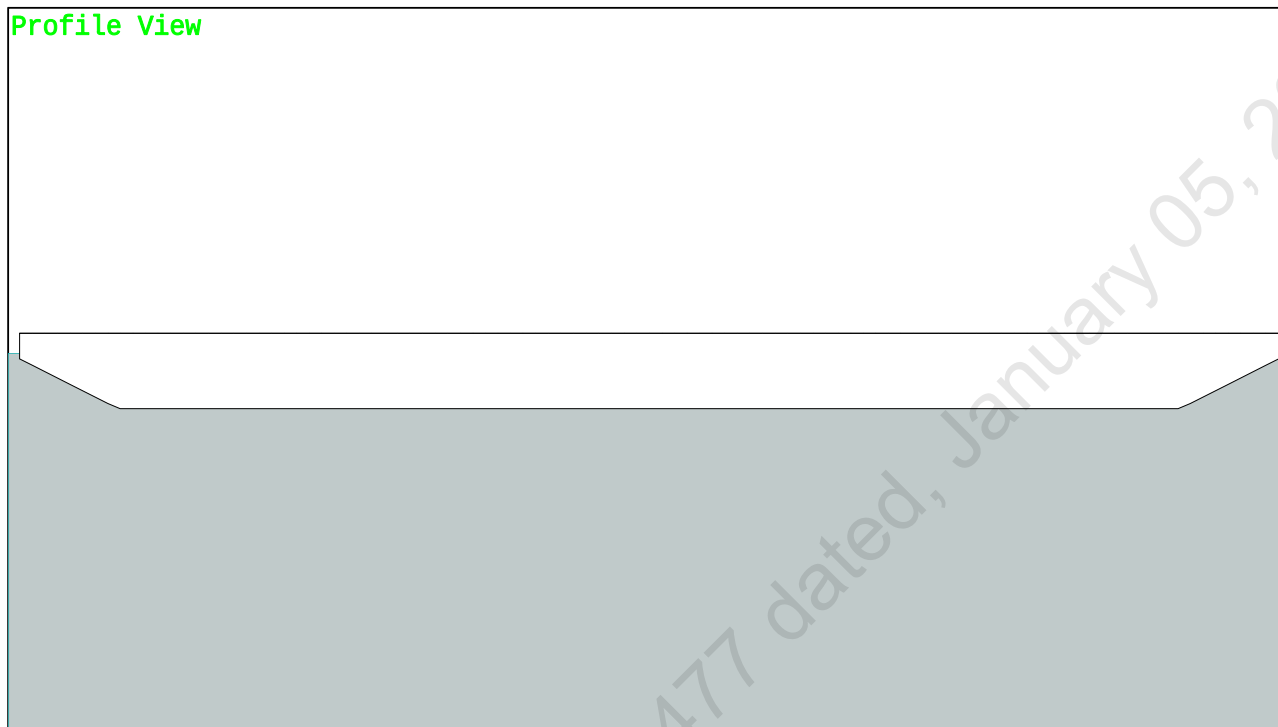
Condition 5 : Lightship + Crane + Ballast + Deck Cargo**Towing Condition with Max. draft 2.212m**

01-01-21 21:06:17 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 6 : Lightship + Max.Deck Cargo
Max. deck cargo loading condition

Condition Graphic - Draft: 2.212 Trim: zero Heel: zero

Profile View



Plan View



01-01-21 21:06:17 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 6 : Lightship + Max.Deck Cargo
Max. deck cargo loading condition

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
LIGHT SHIP	375.10	25.000f	0.000	3.000	
Deck Cargo	1,194.05	25.000f	0.000	3.500	
Total Weight----->	1,569.15	25.000f	0.000	3.380	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
Total Tanks----->		0.00			0.00
		Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1.025	1,569.15	25.000f	0.000	1.134

Righting Arms:		0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----BMT
Total Waterplane---->	1.025	750.6	25.000f	0.000	102.05 9.209
		MT/cm-----	m.-MT/cm-----	GML-----	GMT
		7.69	31.32	99.81	6.963
Distances in METERS.-----			Moments in m.-MT.		

SUMMARY OF LOADING

0.0 Cu.M. (0%) SALT WATER

1,194.05 MT Deck Cargo

DRAFT AT FP (M) = 2.212
DRAFT AT MS (M) = 2.212
DRAFT AT AP (M) = 2.212
DRAFT AT FWD MARK (M) = 2.212
DRAFT AT AFT MARK (M) = 2.212

01-01-21 21:06:17 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 6 : Lightship + Max.Deck Cargo
Max. deck cargo loading condition

HYDROSTATIC PROPERTIES
No Trim, No Heel, VCG = 3.380

LCF Displacement Buoyancy-Ctr. Weight/ Moment/
Draft----Weight(MT)----LCB-----VCB-----cm-----LCF---cm trim----GML-----GMT
2.212 1,569.15 25.000f 1.134 7.69 25.000f 31.32 99.81 6.963
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.
Trim is per 50.00m.

Draft is from USK.

RESIDUAL RIGHTING ARMS vs HEEL ANGLE
LCG = 25.000f TCG = 0.000 VCG = 3.380

Origin Depth---	Trim----	Degrees of Heel----	Displacement Weight(MT)---	Residual Arms in Trim--in Heel----	Res. 50% DeckImm Area----Angle
2.200	0.00	0.00	1,569.1	0.000 -0.005	0.0000 3.085
2.200	0.00	0.04s	1,569.2	0.000 0.000	-0.0000 3.042
2.196	0.00	3.09s	1,569.0	0.000 0.371	0.0098 -0.000
2.190	0.00	5.04s	1,569.1	0.000 0.608	0.0266 -1.958
2.221	0.00	10.04s	1,569.1	0.000 1.018	0.0989 -6.958
2.366	0.00	15.04s	1,569.2	0.000 1.100	0.1938 -11.958
2.369	0.00	15.13s	1,569.2	0.000 1.101	0.1954 -12.042
2.558	0.00	20.04s	1,569.1	0.000 1.053	0.2887 -16.958
2.762	0.00	25.04s	1,569.1	0.000 0.935	0.3760 -21.958
2.949	0.00	30.04s	1,569.2	0.000 0.765	0.4506 -26.958
3.114	0.00	35.04s	1,569.2	0.000 0.567	0.5089 -31.958
3.254	0.00	40.04s	1,569.2	0.000 0.351	0.5491 -36.958
3.370	0.00	45.04s	1,569.1	0.000 0.126	0.5700 -41.958
3.423	0.00	47.78s	1,569.2	0.000 0.000	0.5730 -44.693
3.460	0.00	50.04s	1,569.1	0.000 -0.105	0.5709 -46.958
3.524	0.00	55.04s	1,569.1	0.000 -0.338	0.5516 -51.958
3.561	0.00	60.04s	1,569.1	0.000 -0.571	0.5119 -56.958

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

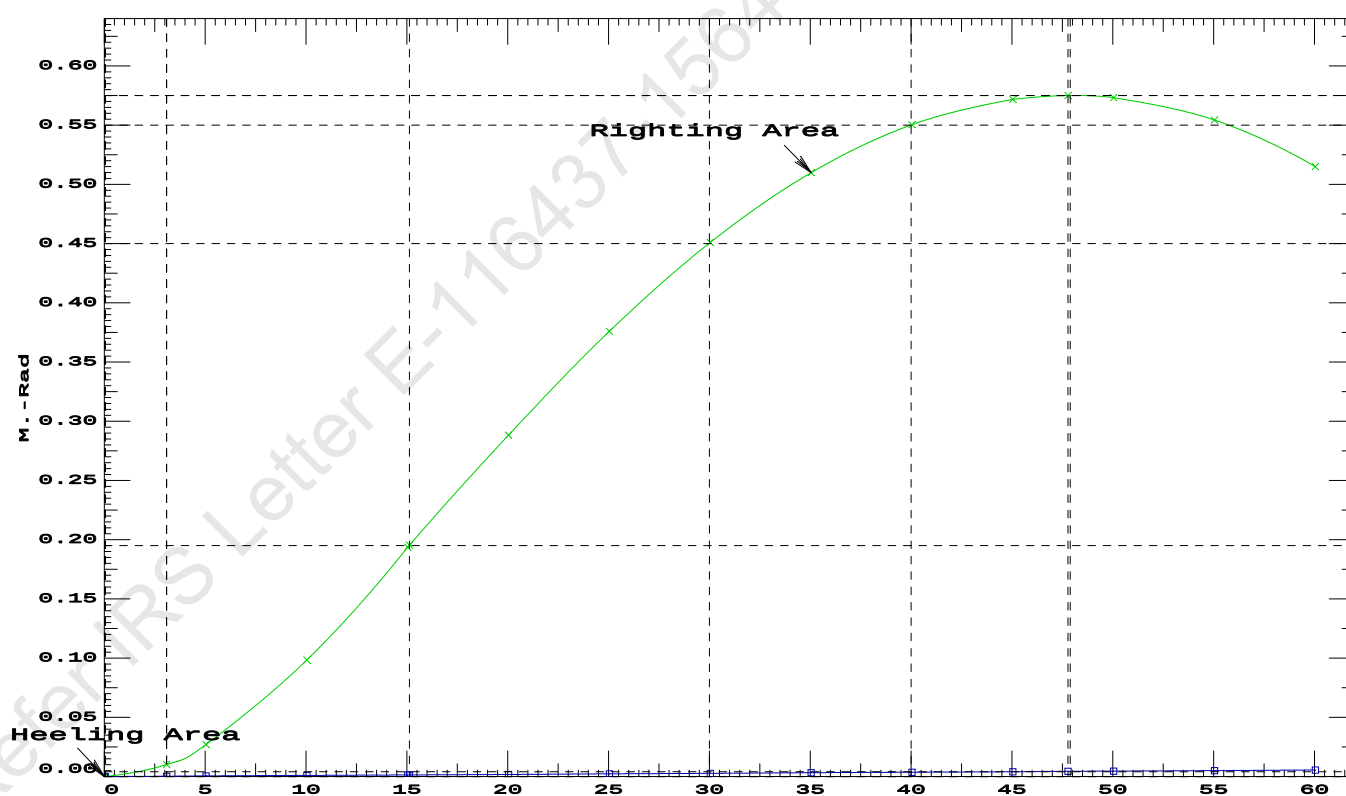
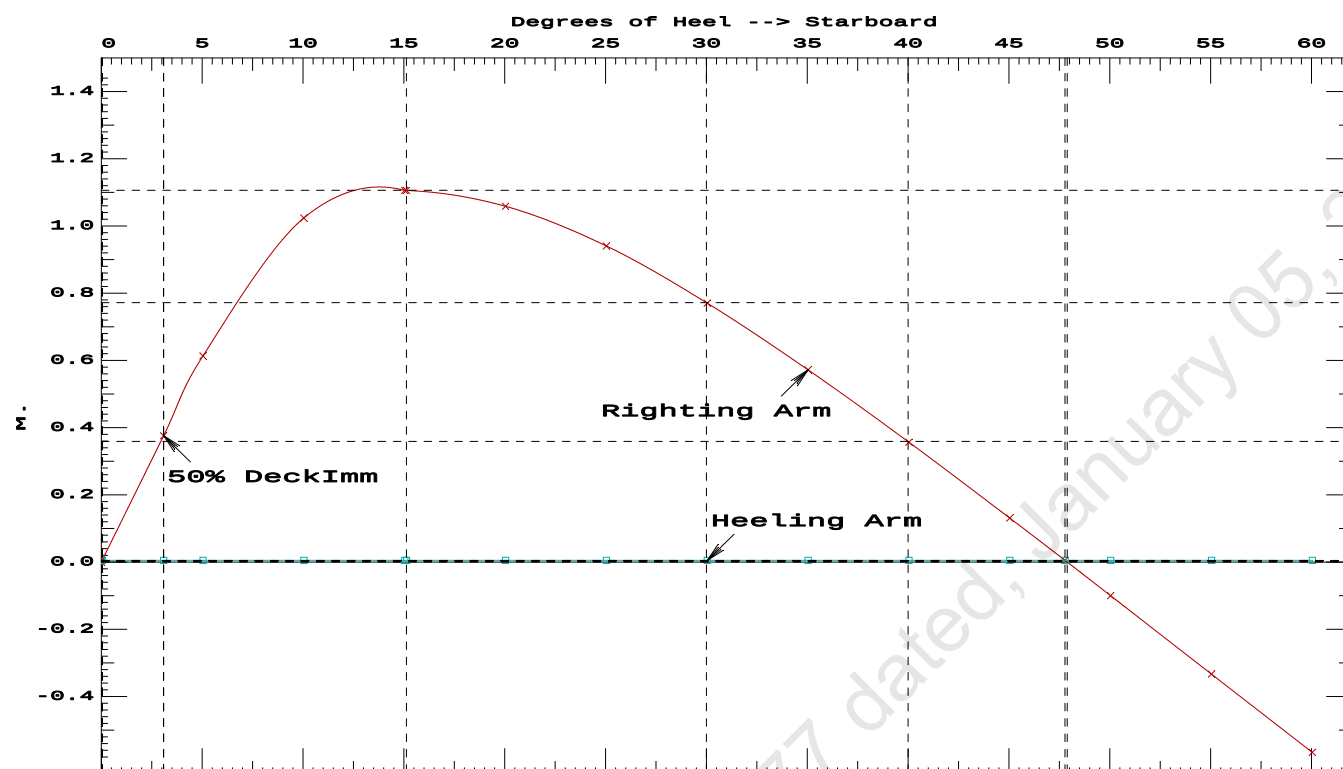
Note: No tank loads are present.

Note: The Residual Righting Arms shown above are in excess of the
wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 8.20 (constant)

LIM-----IS CODE 2008 PART B CHAPTER-2 SEC.2----Min/Max-----Attained
(1) Residual Area from abs 0 deg to MaxRA > 0.0800 m.-Rad 0.1954 P
(2) Angle from Equilibrium to RAZero or Flood > 20.00 deg 47.73 P
(3) Angle from Equilibrium to 50% Dk Imm Angle > 0.00 deg 3.04 P

01-01-21 21:06:17 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 6 : Lightship + Max.Deck Cargo
Max. deck cargo loading condition



01-01-21 20:59:39 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

Condition Graphic - Draft: 1.318 @ 50.000f, 1.598 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 WBT5.P 14 WBT5.C 15 WBT5.S

01-01-21 20:59:39 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.318 @ 50.00f, 1.458 @ 25.00f, 1.598 @ 0.00

Trim: Aft 0.280/50.000, Heel: zero

Part-----			Weight(MT)	LCG----	TCG-----	VCG	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	14.580a	0.000	76.930	
Total Fixed----->			661.70	15.066f	0.000	8.268	
	Load-----	SpGr-----	Weight(MT)	LCG----	TCG-----	VCG-----	FSM-----
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks----->			337.18	42.250f	0.000	1.507	235.45
Total Weight----->			998.88	24.242f	0.000	5.986	
			Displ(MT)	LCB-----	TCB-----	VCB	
HULL		1.025	998.88	24.213f	0.000	0.741	

	Righting Arms:			0.000	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	718.2	24.738f	0.000	140.46	13.582
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			7.36		27.01	135.22	8.338
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

329.0 Cu.M. (20%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.318
DRAFT AT MS (M) = 1.458
DRAFT AT AP (M) = 1.598
DRAFT AT FWD MARK (M) = 1.340
DRAFT AT AFT MARK (M) = 1.576

HYDROSTATIC PROPERTIES

Trim: Aft 0.280/50.000, No Heel, VCG = 5.986

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT
1.460	998.88	24.213f	0.741	7.36	24.738f	27.01	135.22	8.338
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 235.4 m.-MT

01-01-21 20:59:39 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.242f TCG = 0.000 VCG = 5.986

Free Surface Adjustment: 0.236

Adjusted CG: LCG = 24.243f TCG = 0.000 VCG = 6.222

Origin Depth	Degrees of Trim	Heel	Displacement Weight(MT)	Residual Arms in Trim	in Heel	Res. 50% DeckImm Area	Angle
1.586	0.32a	0.00	998.88	0.000	-0.014	0.0000	5.461
1.586	0.32a	0.09s	998.88	0.000	0.000	-0.0000	5.369
1.574	0.32a	5.09s	998.88	0.000	0.733	0.0320	0.369
1.572	0.32a	5.46s	998.88	0.000	0.787	0.0369	0.000
1.541	0.33a	10.09s	998.88	0.000	1.449	0.1273	-4.631
1.506	0.43a	15.09s	998.88	0.000	1.848	0.2735	-9.631
1.504	0.44a	15.46s	998.88	0.000	1.850	0.2853	-9.997
1.475	0.58a	20.09s	998.88	0.000	1.700	0.4320	-14.631
1.436	0.73a	25.09s	998.88	0.000	1.350	0.5665	-19.631
1.388	0.88a	30.09s	998.88	0.000	0.915	0.6659	-24.631
1.332	1.03a	35.09s	998.88	0.000	0.437	0.7252	-29.631
1.276	1.16a	39.46s	998.88	0.000	0.000	0.7420	-33.997
1.267	1.17a	40.09s	998.88	0.000	-0.064	0.7416	-34.631
1.193	1.31a	45.09s	998.88	0.000	-0.576	0.7138	-39.631
1.111	1.44a	50.09s	998.88	0.000	-1.091	0.6410	-44.631
1.019	1.56a	55.09s	998.88	0.000	-1.602	0.5235	-49.631
0.917	1.65a	60.09s	998.88	0.000	-2.104	0.3618	-54.631

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

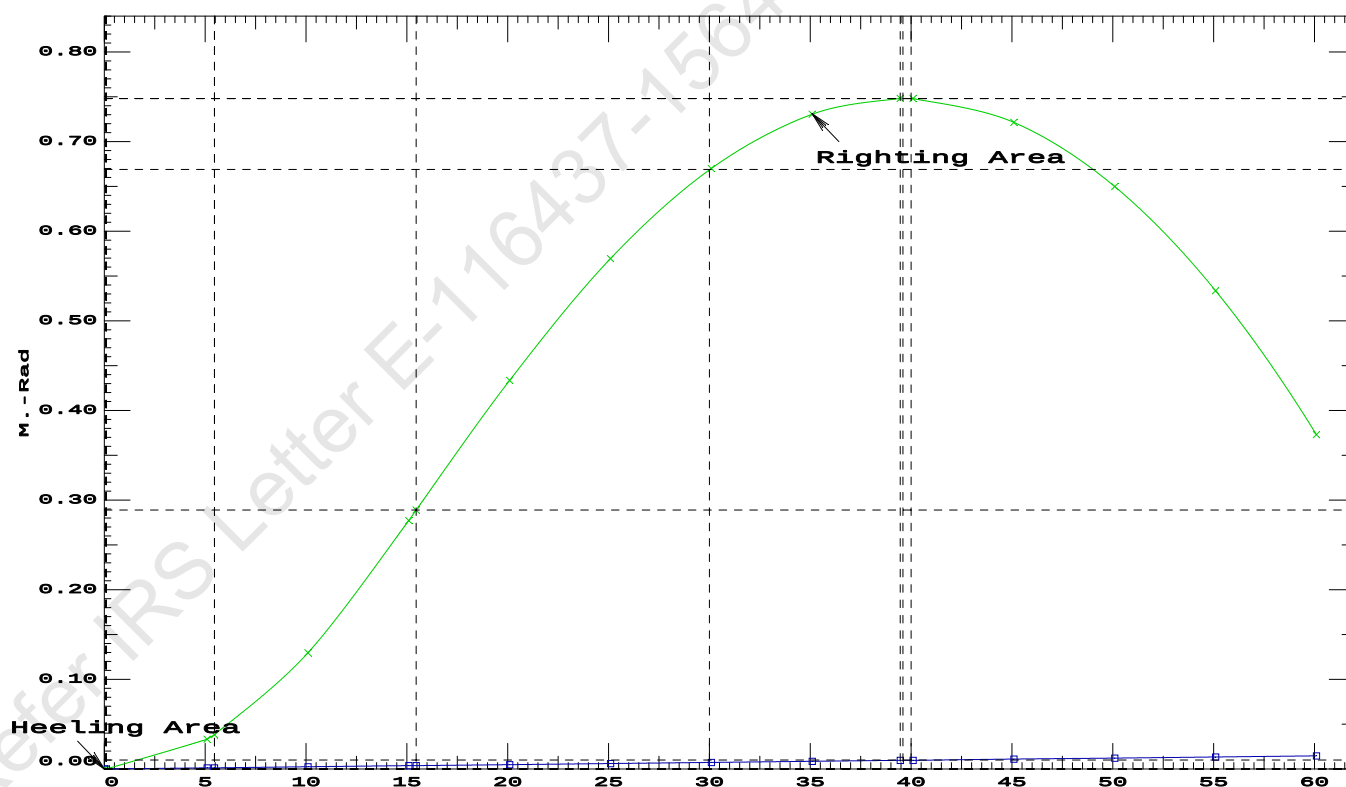
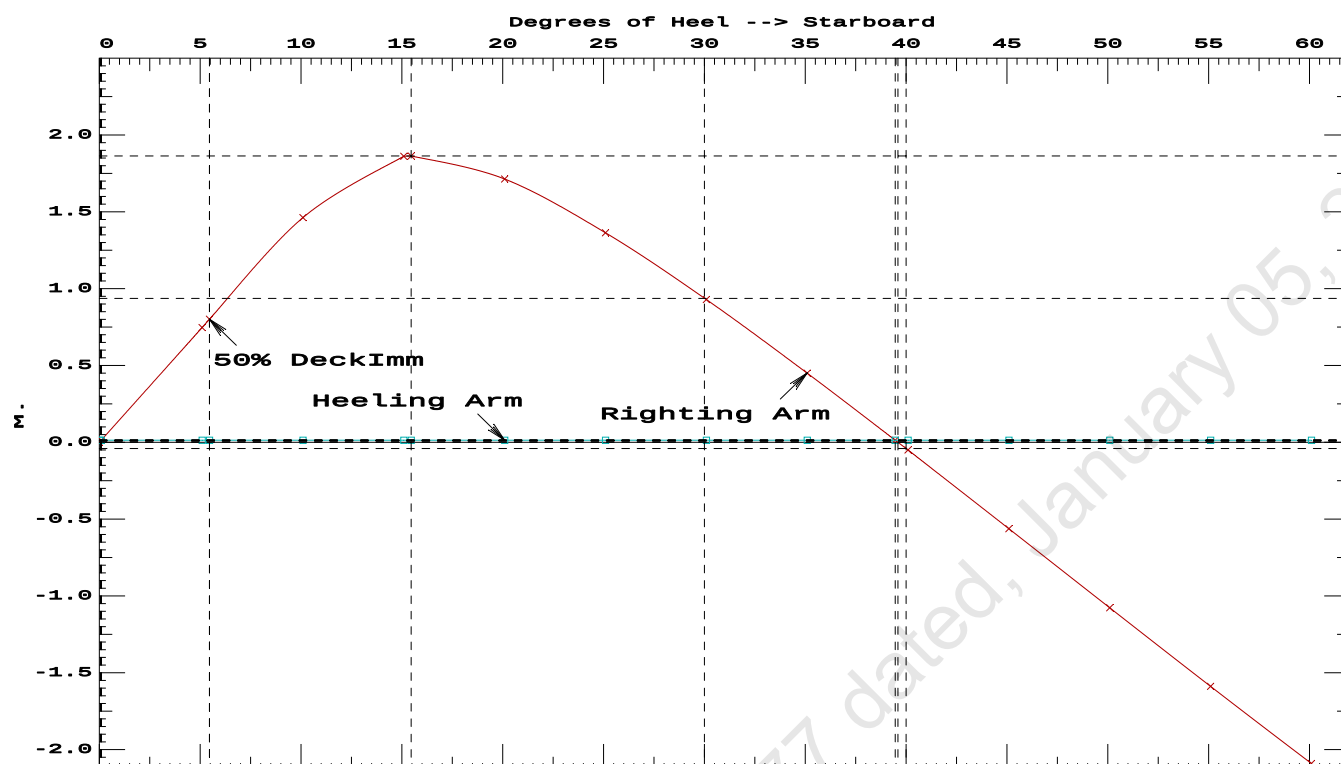
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 13.90 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2-----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2853 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	39.37 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	5.37 P

01-01-21 20:59:39 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft



01-01-21 20:59:39 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.242f TCG = 0.000 VCG = 5.986
Free Surface Adjustment: 0.236
Adjusted CG: LCG = 24.243f TCG = 0.000 VCG = 6.222

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. Area
1.586	0.32a	0.01p	998.88	0.000 0.000	0.0000
1.574	0.32a	5.01p	998.88	0.000 0.733	0.0320
1.542	0.33a	10.01p	998.88	0.000 1.450	0.1273
1.507	0.43a	15.01p	998.88	0.000 1.860	0.2739
1.504	0.44a	15.48p	998.88	0.000 1.862	0.2892
1.476	0.58a	20.01p	998.88	0.000 1.717	0.4336
1.437	0.73a	25.01p	998.88	0.000 1.370	0.5706
1.389	0.88a	30.01p	998.88	0.000 0.935	0.6718
1.333	1.03a	35.01p	998.88	0.000 0.458	0.7329
1.274	1.16a	39.58p	998.88	0.000 0.000	0.7514
1.268	1.17a	40.01p	998.88	0.000 -0.043	0.7512
1.195	1.31a	45.01p	998.88	0.000 -0.555	0.7252
1.112	1.44a	50.01p	998.88	0.000 -1.069	0.6544
1.020	1.55a	55.01p	998.88	0.000 -1.581	0.5387
0.919	1.65a	60.01p	998.88	0.000 -2.083	0.3788

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

Note: Angle of MaxRA refers to the absolute Righting Arm curve.

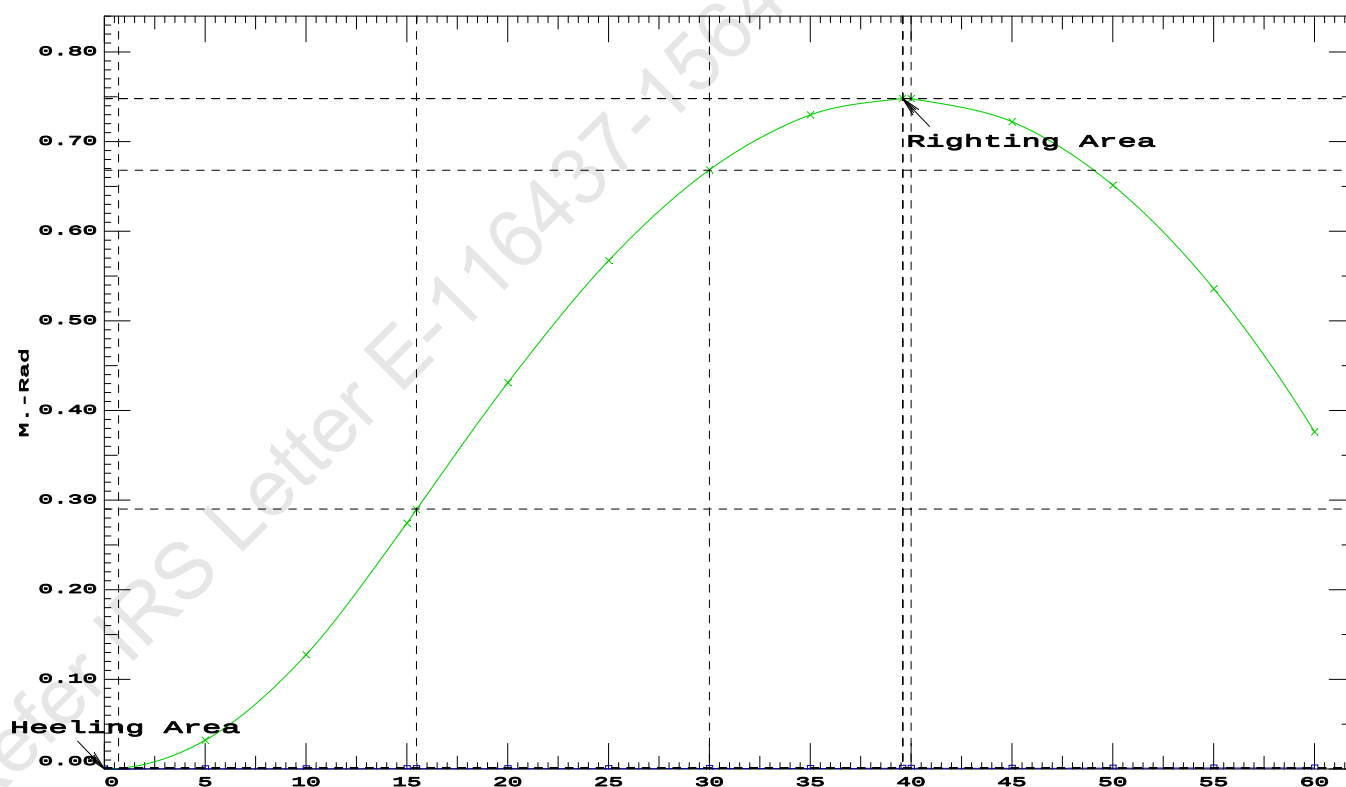
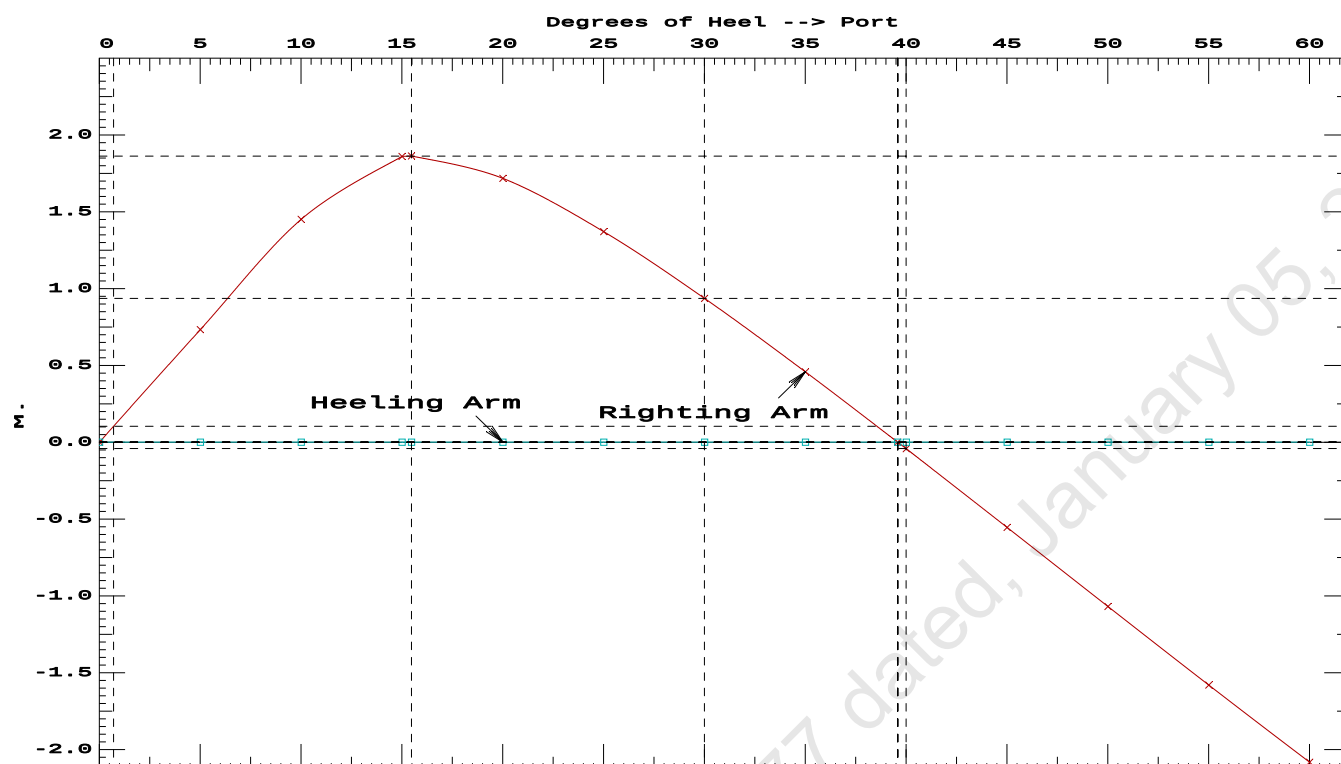
LIM-----STABILITY CRITERIA FOR CRANE OPERAT-----Min/Max-----Attained

(1) Absolute Angle at Equilibrium	<	15.00 deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	186013 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	1087.53 P

01-01-21 20:59:39 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 7 : Lightship + Load at Aft + Ballast
Crane Lifting 36.6T @ 18m Radius Aft



01-01-21 20:59:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 7 : Lightship + Load at Aft + BallastCrane Lifting 36.60T @ 18m Radius Aft

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.318 @ 50.00f, 1.458 @ 25.00f, 1.598 @ 0.00

Trim: Aft 0.280/50.000, Heel: Port 0.01 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP	375.10	25.000f	0.000	3.000	
Crane	250.00	4.500f	0.000	6.120	
Crane Load	36.60	14.580a	0.000	76.930	
Total Fixed----->	661.70	15.066f	0.000	8.268	
Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p
WBTK5.C	1.000	1.025	113.02	42.250f	0.000
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s
Total Tanks----->			337.18	42.250f	0.000
Total Weight----->			998.88	24.242f	0.000
			Displ(MT)----	LCB-----	TCB-----
HULL	1.025		998.75	24.213f	0.002p

Righting Arms:			0.000	-0.001s	
External Arms:			0.000	-0.001p	
Residual Righting Arms:			0.000	0.000	
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----
Total Waterplane---->	1.025		717.2	24.743f	0.009p
			MT/cm-----	m.-MT/cm-----	GML-----
			7.35	26.89	134.63
Distances in METERS.-----					
Moments in m.-MT.					

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = -0.01

01-01-21 20:59:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.535 @ 50.00f, 1.408 @ 25.00f, 1.281 @ 0.00
Trim: Fwd 0.254/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
LIGHT SHIP	375.10	25.000f	0.000	3.000		
Crane	250.00	4.500f	0.000	6.120		
Total Fixed----->	625.10	16.801f	0.000	4.248		
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td>	LCG-----	TCG-----	VCG-----	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511
Total Tanks----->			337.18	42.250f	0.000	1.507
Total Weight----->			962.28	25.719f	0.000	3.288
		Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		962.28	25.732f	0.001	0.715

Righting Arms:			0.000	0.001s		
External Arms:			0.000	0.001		
Residual Righting Arms:			0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane----->	1.025	715.1	25.239f	0.002s	143.88	14.030
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		7.33	27.20	141.31	11.458	
Distances in METERS.-----						Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 844875200.00; t3: 30.0; t: 15.00
limmarg: 296443808.00; t3: 15.0; t: 7.50
limmarg: 77062752.00; t3: 7.5; t: 3.75
limmarg: 19249968.00; t3: 3.8; t: 1.88
limmarg: 5044931.00; t3: 1.9; t: 0.94
limmarg: 1261490.38; t3: 1.0; t: 0.47
limmarg: 384287.44; t3: 0.5; t: 0.23
limmarg: 98800.98; t3: 0.3; t: 0.12
limmarg: 43435.99; t3: 0.2; t: 0.06
limmarg: 26024.30; t3: 0.1; t: 0.03
limmarg: 6198.45; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

01-01-21 20:59:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 16.801f TCG = 0.000 VCG = 4.248

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.270	0.29f	0.01p	962.28	0.000	-0.002	0.0000
1.269	0.29f	0.00	962.28	0.000	0.000	-0.0000
1.270	0.29f	0.07s	962.29	0.000	0.014	0.0000
1.259	0.29f	5.00s	962.26	0.000	1.027	0.0448
1.218	0.30f	10.00s	962.28	0.000	2.031	0.1785
1.090	0.38f	15.00s	962.28	0.000	2.710	0.3877
0.942	0.49f	19.32s	962.29	0.000	2.824	0.5987
0.918	0.51f	20.00s	962.28	0.000	2.822	0.6326
0.737	0.64f	25.00s	962.28	0.000	2.713	0.8766
0.549	0.76f	30.00s	962.28	0.000	2.505	1.1050

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 625.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 62.984 P

01-01-21 20:59:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 16.801f TCG = 0.000 VCG = 4.248

Origin Depth---	Degrees of Trim----	Displacement Heel----	Residual Arms Weight(MT)----	Res. in Trim--	Res. in Heel---	Area -->
1.270	0.29f	0.01p	962.28	0.000	-0.002	0.0000
1.269	0.29f	0.00	962.28	0.000	0.000	-0.0000
1.270	0.29f	0.07s	962.29	0.000	0.014	0.0000
1.259	0.29f	5.00s	962.26	0.000	1.027	0.0448
1.218	0.30f	10.00s	962.28	0.000	2.031	0.1785
1.090	0.38f	15.00s	962.28	0.000	2.710	0.3877
0.942	0.49f	19.32s	962.29	0.000	2.824	0.5987
0.918	0.51f	20.00s	962.28	0.000	2.822	0.6326
0.737	0.64f	25.00s	962.28	0.000	2.713	0.8766
0.549	0.76f	30.00s	962.28	0.000	2.505	1.1050
0.355	0.89f	35.00s	962.28	0.000	2.239	1.3124
0.156	1.01f	40.00s	962.28	0.000	1.936	1.4948
-0.044	1.13f	45.00s	962.28	0.000	1.607	1.6496
-0.246	1.24f	50.00s	962.28	0.000	1.257	1.7747
-0.445	1.34f	55.00s	962.28	0.000	0.894	1.8687
-0.642	1.43f	60.00s	962.28	0.000	0.520	1.9305
-0.832	1.51f	65.00s	962.28	0.000	0.140	1.9593
-0.900	1.53f	66.83s	962.28	0.000	0.000	1.9615
-1.016	1.57f	70.00s	962.28	0.000	-0.243	1.9548

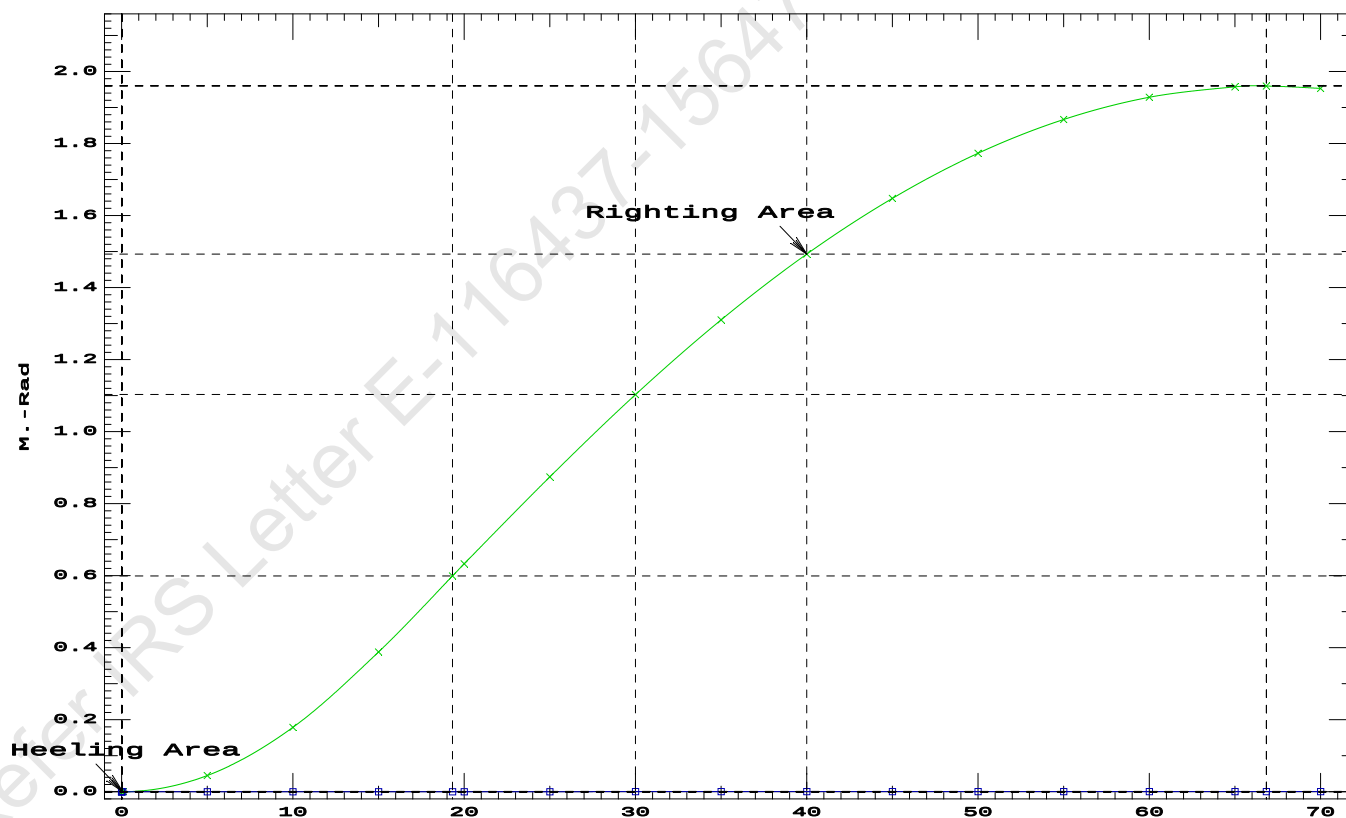
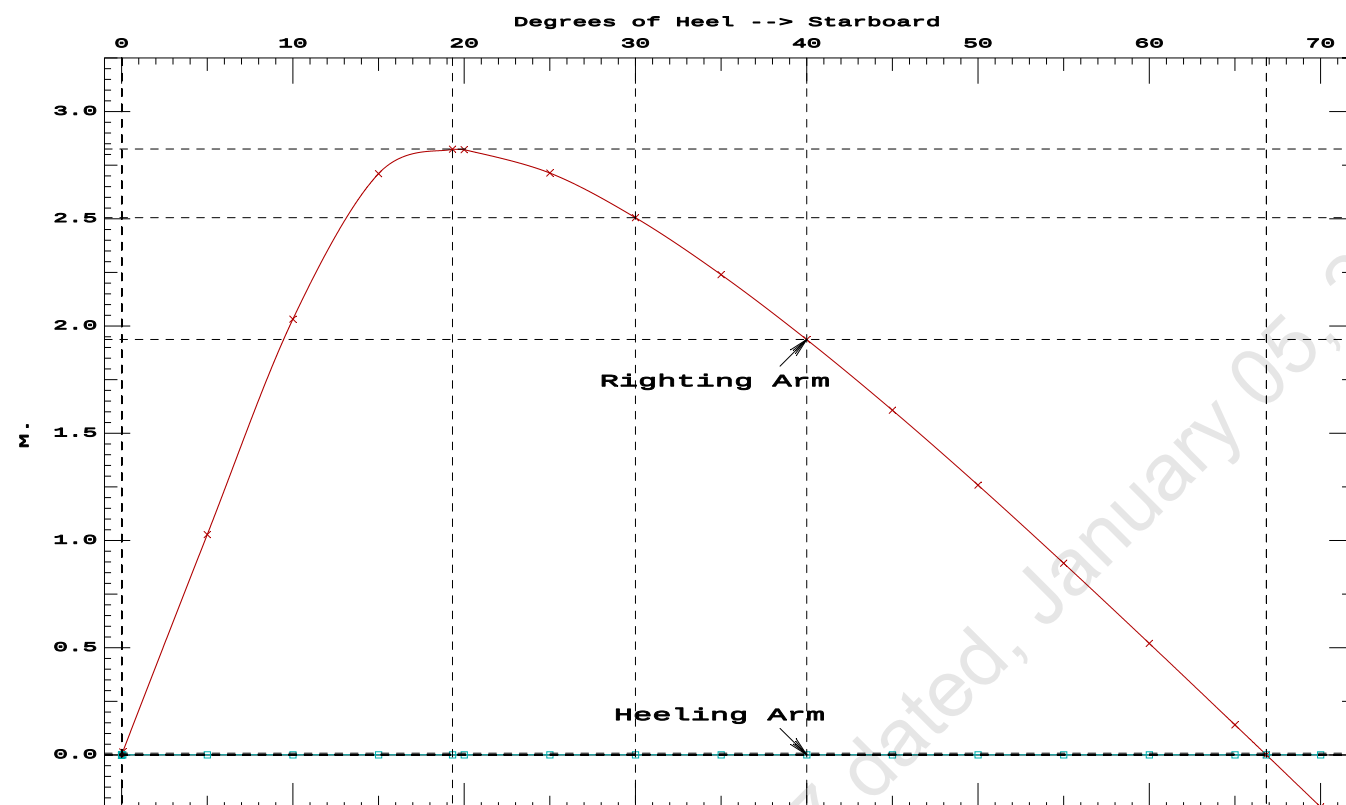
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 625.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 LARGE
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 66.76 P

01-01-21 20:59:44 Cybermarine Technologies PTE. Ltd.
GHS 17.22

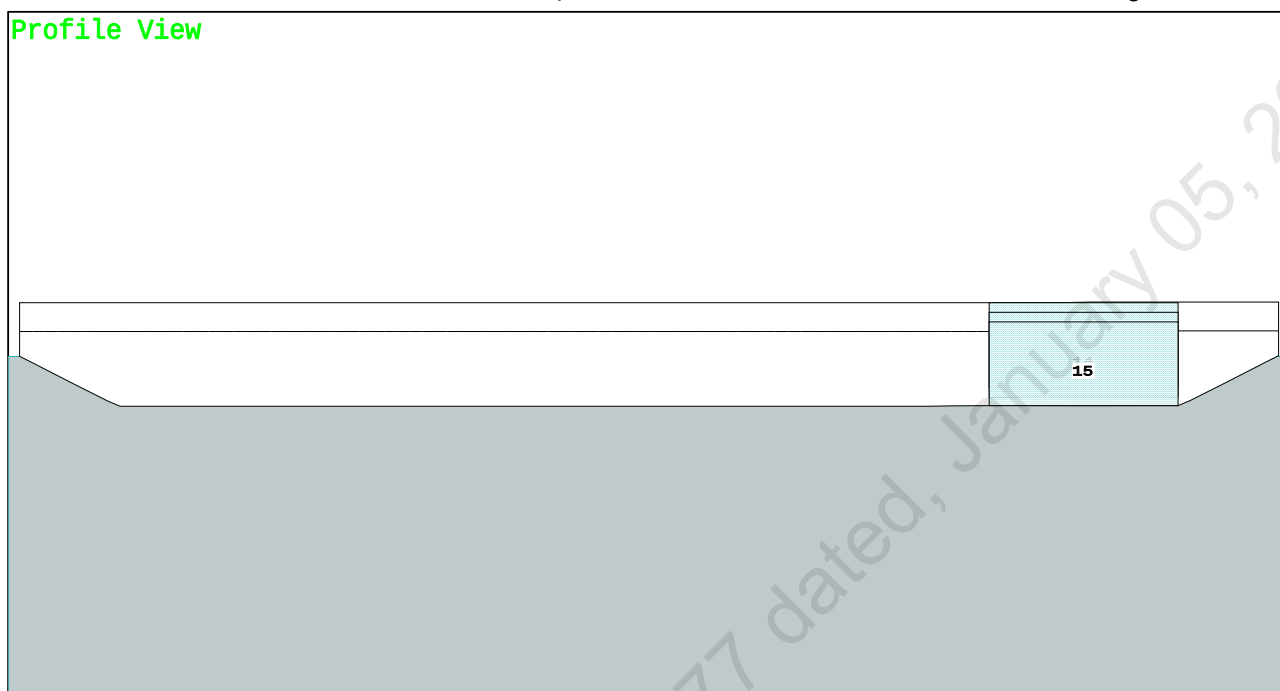
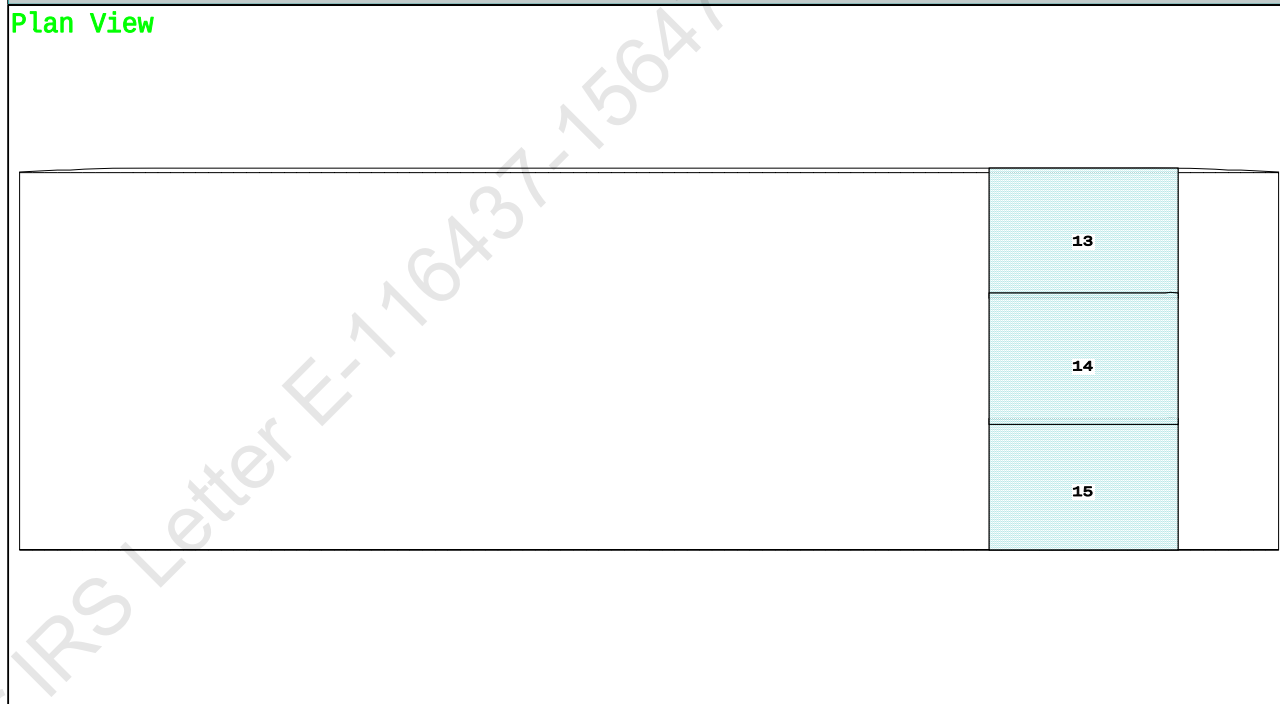


01-01-21 20:59:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

CG - Draft: 1.443 @ 50.000f, 1.465 @ 0.000 Heel: stbd 4.37 deg.

Profile View**Plan View****Tanks**

13 WBTk5.P 14 WBTk5.C 15 WBTk5.S

01-01-21 20:59:48 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.443 @ 50.00f, 1.454 @ 25.00f, 1.465 @ 0.00

Trim: Aft 0.022/50.000, Heel: Stbd 4.37 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	4.500f	18.000s	76.930	
Total Fixed----->			661.70	16.121f	0.996s	8.268	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks----->			337.18	42.250f	0.000	1.507	235.45
Total Weight----->			998.88	24.941f	0.660s	5.986	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	998.89	24.939f	1.057s	0.780	

	Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	719.7	24.979f	0.122s	140.85	13.681
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			7.38		27.10	135.63	8.459
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

329.0 Cu.M. (20%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.443
DRAFT AT MS (M) = 1.454
DRAFT AT AP (M) = 1.465
DRAFT AT FWD MARK (M) = 1.445
DRAFT AT AFT MARK (M) = 1.463

HYDROSTATIC PROPERTIES

Trim: Aft 0.022/50.000, Heel: Stbd 4.37 deg., VCG = 5.986

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)----	LCB-----	VCB-----	cm-----	LCF-----	cm trim-----	GML-----	GMT-----
1.454	998.89	24.939f	0.780	7.38	24.979f	27.10	135.63	8.459
Distances in METERS.-----				Specific Gravity = 1.025.-----				
				Moment in m.-MT.				
				Trim is per 50.00m.				

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 235.4 m.-MT

01-01-21 20:59:48 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.941f TCG = 0.660s VCG = 5.986

Free Surface Adjustment: 0.236

Adjusted CG: LCG = 24.941f TCG = 0.641s VCG = 6.221

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
1.458	0.02a	0.00	998.88	0.000	-0.655	0.0000	5.992
1.448	0.02a	4.46s	998.88	0.000	0.000	-0.0255	1.534
1.441	0.02a	5.99s	998.88	0.000	0.227	-0.0225	0.000
1.416	0.03a	9.46s	998.88	0.000	0.731	0.0066	-3.466
1.344	0.03a	14.46s	998.88	0.000	1.228	0.0932	-8.466
1.322	0.03a	15.75s	998.88	0.000	1.243	0.1211	-9.760
1.255	0.04a	19.46s	998.88	0.000	1.140	0.1998	-13.466
1.156	0.05a	24.46s	998.88	0.000	0.828	0.2866	-18.466
1.048	0.07a	29.46s	998.88	0.000	0.426	0.3420	-23.466
0.937	0.08a	34.25s	998.88	0.000	0.000	0.3601	-28.261
0.932	0.08a	34.46s	998.88	0.000	-0.019	0.3601	-28.466
0.810	0.09a	39.46s	998.88	0.000	-0.484	0.3383	-33.466
0.681	0.10a	44.46s	998.88	0.000	-0.958	0.2754	-38.466
0.548	0.11a	49.46s	998.88	0.000	-1.433	0.1710	-43.466
0.410	0.12a	54.46s	998.88	0.000	-1.902	0.0255	-48.466
0.268	0.12a	59.46s	998.88	0.000	-2.360	-0.1606	-53.466
0.125	0.13a	64.46s	998.88	0.000	-2.802	-0.3859	-58.466

Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

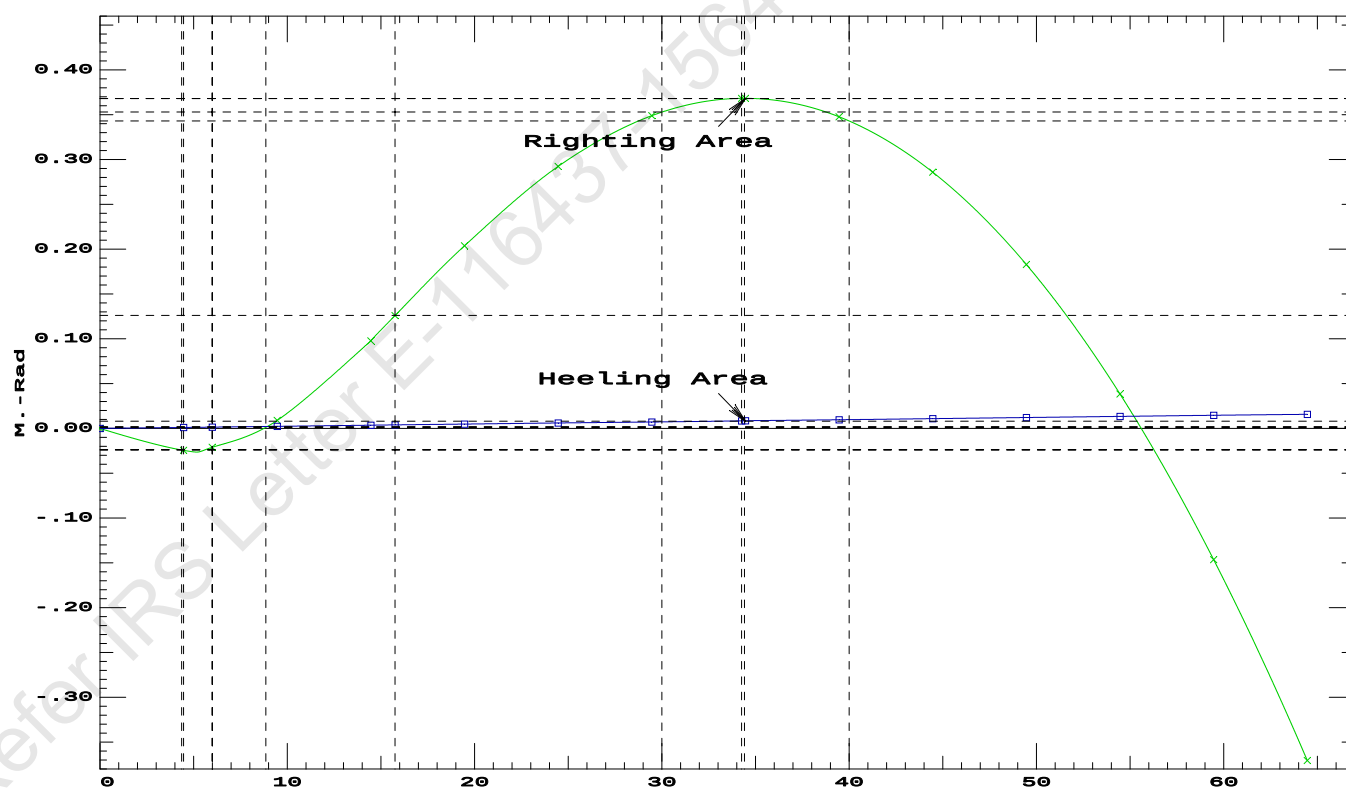
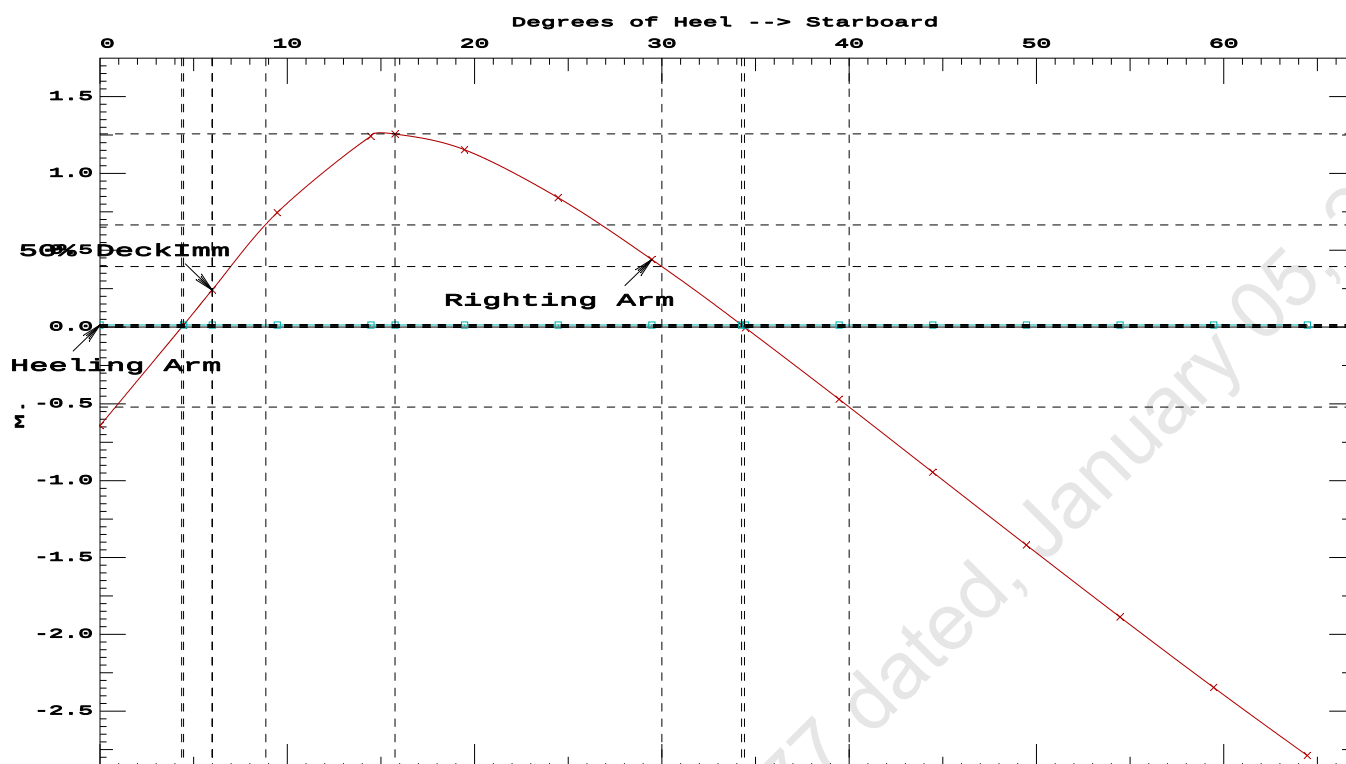
Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 14.05 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2-----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.1211 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	29.80 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	1.53 P

01-01-21 20:59:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd



01-01-21 20:59:48 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 24.941f TCG = 0.660s VCG = 5.986

Free Surface Adjustment: 0.236

Adjusted CG: LCG = 24.941f TCG = 0.642s VCG = 6.221

Origin Depth---	Degrees of Trim----	Displacement Heel----	Weight(MT)---	Residual Arms in Trim--	Res. in Heel---	Area
1.449	0.02a	4.36s	998.88	0.000	0.000	0.0000
1.457	0.03a	0.64p	998.88	0.000	0.735	0.0321
1.443	0.02a	5.64p	998.88	0.000	1.465	0.1281
1.403	0.03a	10.64p	998.88	0.000	2.166	0.2867
1.329	0.03a	15.38p	998.88	0.000	2.493	0.4817
1.324	0.03a	15.64p	998.88	0.000	2.492	0.4930
1.232	0.05a	20.64p	998.88	0.000	2.294	0.7056
1.131	0.06a	25.64p	998.88	0.000	1.910	0.8904
1.022	0.07a	30.64p	998.88	0.000	1.442	1.0372
0.905	0.08a	35.64p	998.88	0.000	0.930	1.1410
0.781	0.09a	40.64p	998.88	0.000	0.392	1.1988
0.689	0.10a	44.20p	998.88	0.000	0.000	1.2110
0.652	0.10a	45.64p	998.88	0.000	-0.160	1.2090
0.517	0.11a	50.64p	998.88	0.000	-0.717	1.1709
0.379	0.12a	55.64p	998.88	0.000	-1.273	1.0841

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.10

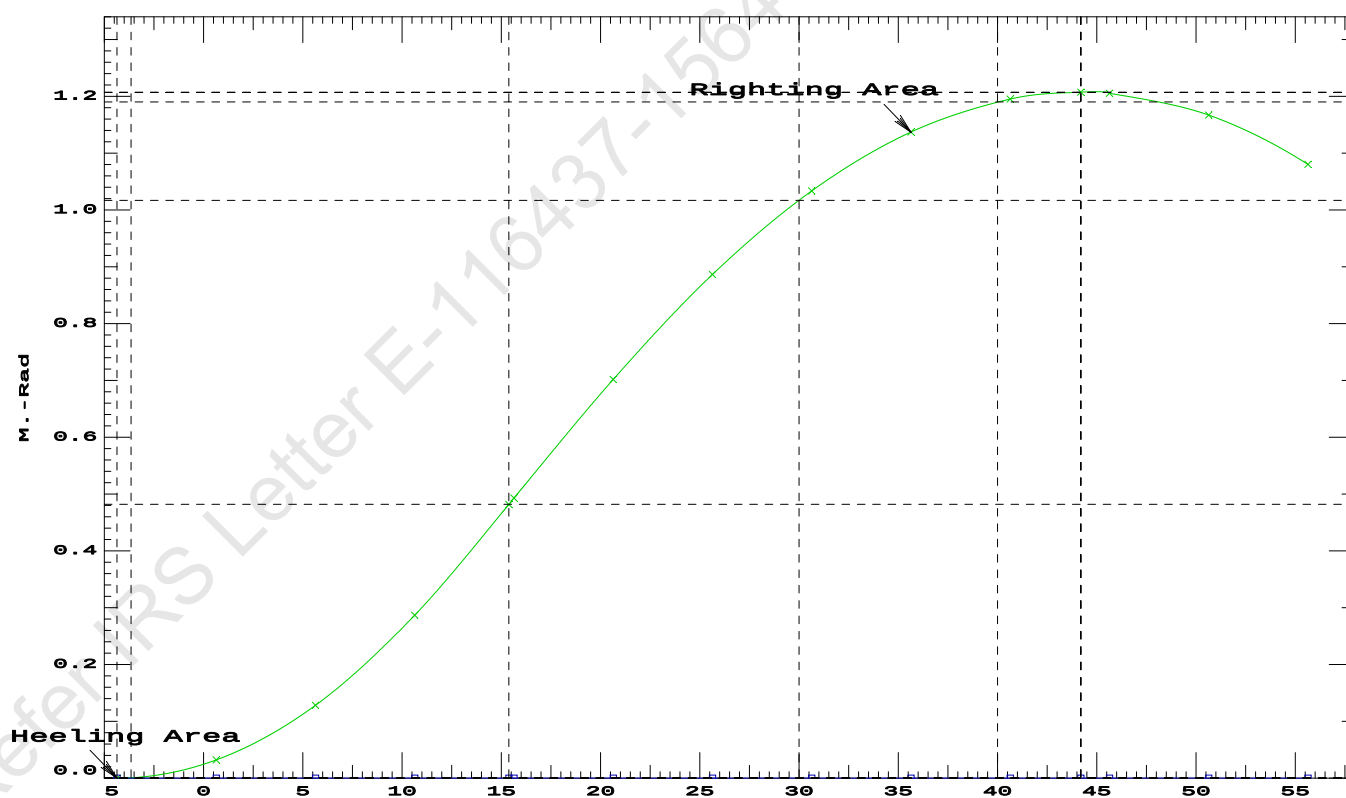
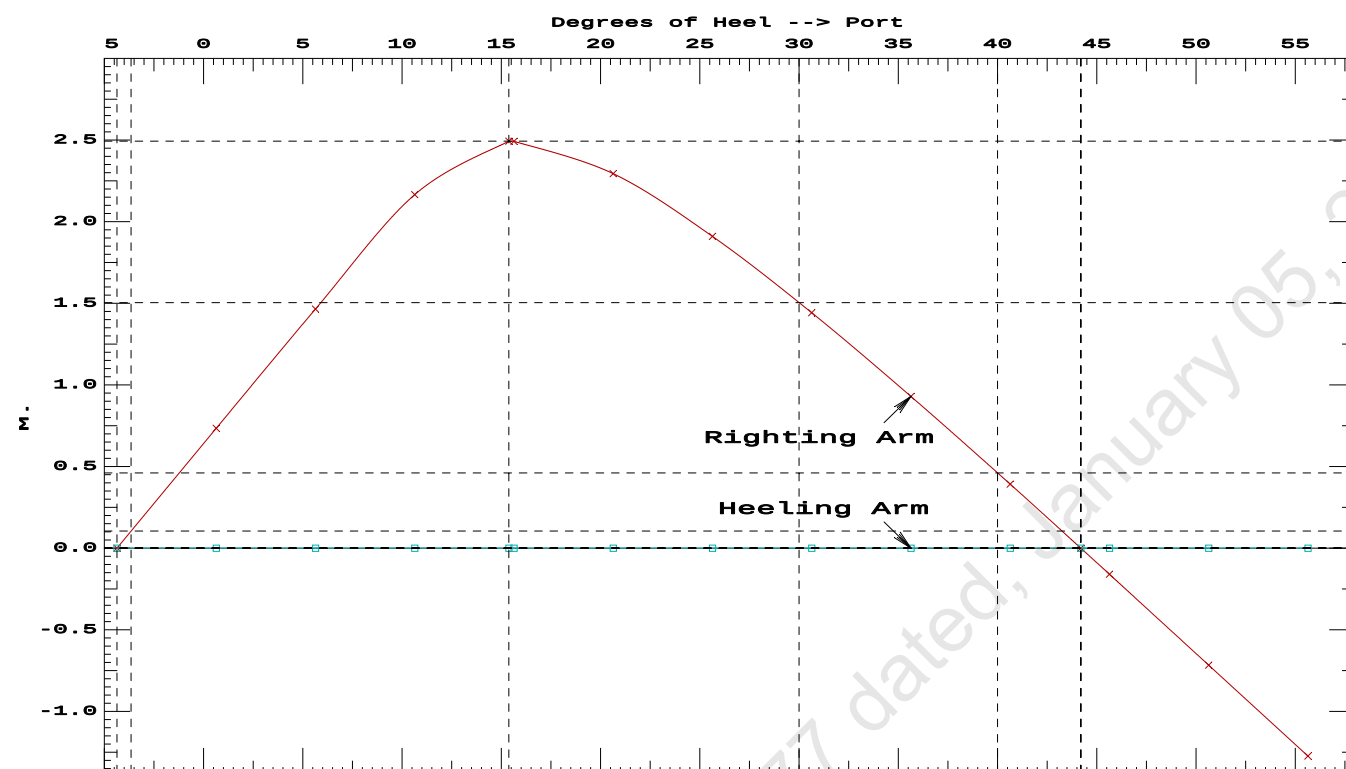
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	4.36 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	2490101 P
(3) Abs Ratio from Equilibrium to RAZero or Flood	>	1.667	14273.6 P

01-01-21 20:59:48 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 8 : Lightship + Load at Stbd + Ballast
Crane Lifting 36.6T @ 18m Radius Stbd



01-01-21 20:59:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 8 : Lightship + Load at Stbd + Ballast**Crane Lifting 36.6T @ 18m Radius Stbd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 1.443 @ 50.00f, 1.454 @ 25.00f, 1.465 @ 0.00

Trim: Aft 0.022/50.000, Heel: Stbd 4.36 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	4.500f	18.000s	76.630	
Total Fixed----->			661.70	16.121f	0.996s	8.251	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks----->			337.18	42.250f	0.000	1.507	235.45
Total Weight----->			998.88	24.941f	0.660s	5.975	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	998.89	24.939f	1.056s	0.780	

	Righting Arms:			0.000	0.000s		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->		1.025	719.7	24.979f	0.122s	140.85	13.680
			MT/cm-----	m.-MT/cm-----	MT/cm-----	GML-----	GMT-----
			7.38		27.10	135.64	8.470
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = 4.36

01-01-21 20:59:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.535 @ 50.00f, 1.408 @ 25.00f, 1.281 @ 0.00
Trim: Fwd 0.254/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----	
LIGHT SHIP	375.10	25.000f	0.000	3.000		
Crane	250.00	4.500f	0.000	6.120		
Total Fixed----->	625.10	16.801f	0.000	4.248		
Load-----	SpGr-----	Weight(MT)---- <td>LCG-----</td> <td>TCG-----</td> <td>VCG-----</td>	LCG-----	TCG-----	VCG-----	
WBTK5.P	1.000	1.025	112.08	42.251f	4.980p	1.511
WBTK5.C	1.000	1.025	113.02	42.250f	0.000	1.500
WBTK5.S	1.000	1.025	112.08	42.251f	4.980s	1.511
Total Tanks----->			337.18	42.250f	0.000	1.507
Total Weight----->			962.28	25.719f	0.000	3.288
		Displ(MT)---- <td>LCB-----</td> <td>TCB-----</td> <td>VCB-----</td> <td></td>	LCB-----	TCB-----	VCB-----	
HULL	1.025		962.21	25.732f	0.001	0.715

Righting Arms:			0.000	-0.001s		
External Arms:			0.000	-0.001		
Residual Righting Arms:			0.000	0.000		
Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	715.1	25.239f	0.002p	143.89	14.031
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		7.33	27.20	141.32	11.459	
Distances in METERS.-----						Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 3138.93; t3: 30.00; t: 15.00
limmarg: 1036.43; t3: 15.00; t: 7.50
limmarg: 195.42; t3: 7.50; t: 3.75
limmarg: -26.20; t3: 3.75; t: 1.88
limmarg: 69.41; t3: 5.68; t: 0.94
limmarg: 17.94; t3: 4.74; t: 0.47
limmarg: -4.31; t3: 4.27; t: 0.23
limmarg: 7.71; t3: 4.53; t: 0.12
limmarg: 2.07; t3: 4.41; t: 0.06
limmarg: -0.69; t3: 4.35; t: 0.03
limmarg: 3.00; t3: 4.43; t: 0.01

Theta3 is 4.43 degrees, as demonstrated below.

01-01-21 20:59:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 16.801f TCG = 0.000 VCG = 4.248

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.261	0.29f	4.36s	962.26	0.000	-0.896	0.0000
1.269	0.29f	0.00	962.26	0.000	0.000	-0.0341
1.261	0.29f	4.43p	962.28	0.000	0.909	0.0010
1.258	0.29f	5.00p	962.28	0.000	1.027	0.0107
1.218	0.30f	10.00p	962.28	0.000	2.031	0.1443
1.090	0.38f	15.00p	962.28	0.000	2.710	0.3535
0.942	0.49f	19.32p	962.28	0.000	2.824	0.5645
0.918	0.51f	20.00p	962.28	0.000	2.822	0.5984
0.737	0.64f	25.00p	962.28	0.000	2.713	0.8424
0.549	0.76f	30.00p	962.28	0.000	2.505	1.0708

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 625.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Res. Area Ratio from abs -4.4 deg to abs 4.4 > 1.000 1.030 P

01-01-21 20:59:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 16.801f TCG = 0.000 VCG = 4.248

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
1.261	0.29f	4.36s	962.26	0.000	-0.896	0.0000
1.269	0.29f	0.00	962.26	0.000	0.000	-0.0341
1.261	0.29f	4.43p	962.28	0.000	0.909	0.0010
1.258	0.29f	5.00p	962.28	0.000	1.027	0.0107
1.218	0.30f	10.00p	962.28	0.000	2.031	0.1443
1.090	0.38f	15.00p	962.28	0.000	2.710	0.3535
0.942	0.49f	19.32p	962.28	0.000	2.824	0.5645
0.918	0.51f	20.00p	962.28	0.000	2.822	0.5984
0.737	0.64f	25.00p	962.28	0.000	2.713	0.8424
0.549	0.76f	30.00p	962.28	0.000	2.505	1.0708
0.355	0.89f	35.00p	962.28	0.000	2.239	1.2782
0.156	1.01f	40.00p	962.28	0.000	1.936	1.4607
-0.044	1.13f	45.00p	962.28	0.000	1.607	1.6155
-0.246	1.24f	50.00p	962.28	0.000	1.257	1.7406
-0.445	1.34f	55.00p	962.28	0.000	0.894	1.8345
-0.642	1.43f	60.00p	962.28	0.000	0.520	1.8963
-0.832	1.51f	65.00p	962.28	0.000	0.140	1.9251
-0.900	1.53f	66.83p	962.28	0.000	0.000	1.9274
-1.016	1.57f	70.00p	962.28	0.000	-0.243	1.9206

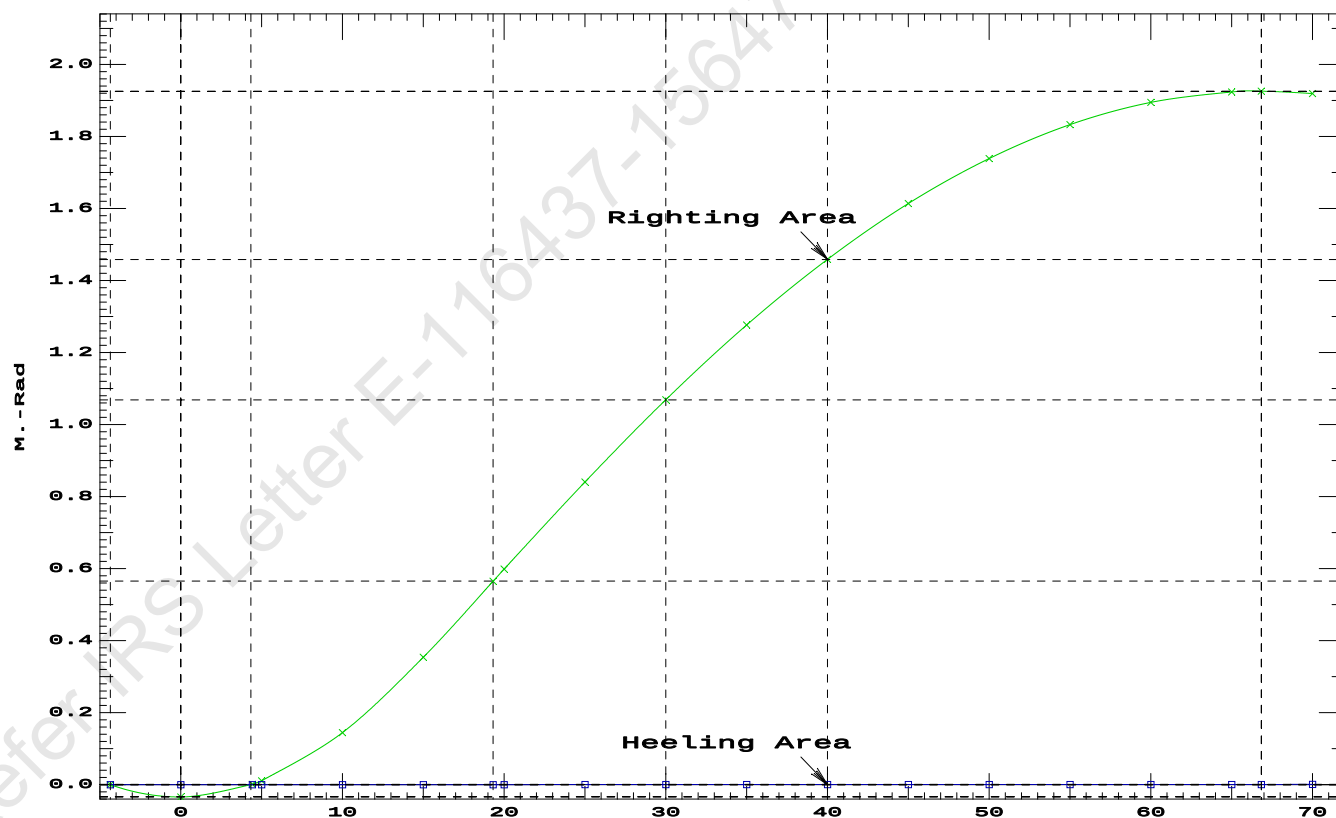
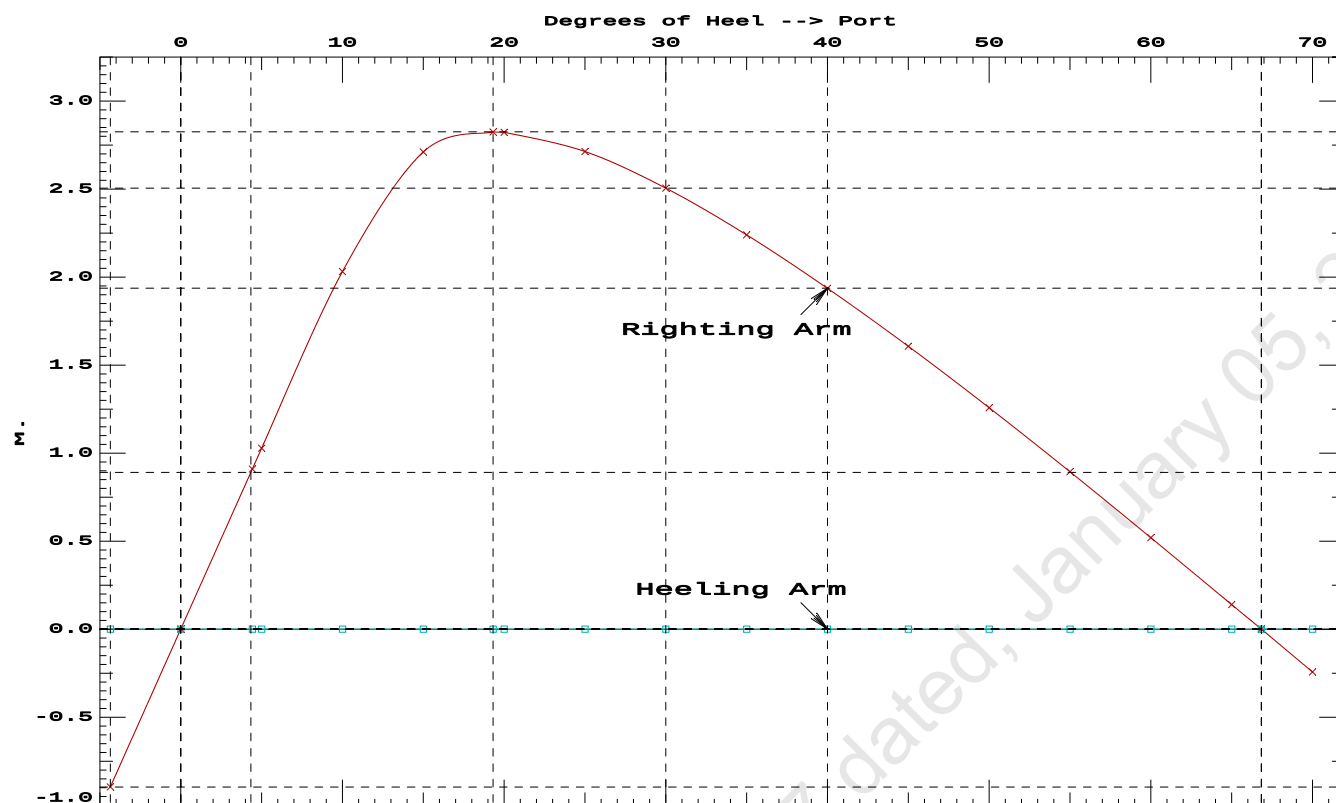
Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 625.10 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Res. Area Ratio from abs -4.4 deg to RAZero > 1.000 57.495 P
(2) Angle from abs 4.4 deg to RAZero > 20.00 deg 62.40 P

01-01-21 20:59:55 Cybermarine Technologies PTE. Ltd.
GHS 17.22



01-01-21 20:59:59 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

Condition Graphic - Draft: 1.601 @ 50.000f, 1.315 @ 0.000 Heel: zero

Profile View



Plan View



Tanks

13 WBT5.P 14 WBT5.C 15 WBT5.S

01-01-21 20:59:59 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 1.601 @ 50.00f, 1.458 @ 25.00f, 1.315 @ 0.00
Trim: Fwd 0.286/50.000, Heel: zero

Part-----	Weight(MT)	LCG	TCG	VCG		
LIGHT SHIP	375.10	25.000f	0.000	3.000		
Crane	250.00	4.500f	0.000	6.120		
Crane Load	36.60	27.250f	0.000	76.930		
Total Fixed----->	661.70	17.379f	0.000	8.268		
Load-----	SpGr	Weight(MT)	LCG	TCG	VCG	FSM
WBTk5.P	1.000	1.025	112.08	42.251f	4.980p	1.511 78.49*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500 78.48*
WBTk5.S	1.000	1.025	112.08	42.251f	4.980s	1.511 78.49*
Total Tanks----->		337.18	42.250f	0.000	1.507	235.45
Total Weight----->		998.88	25.775f	0.000	5.986	
		Displ(MT)	LCB	TCB	VCB	
HULL	1.025	998.88	25.805f	0.000	0.742	

Righting Arms:			0.000	0.000		
Part-----	SpGr	WPA	LCF	TCF	BML	BMT
Total Waterplane---->	1.025	718.2	25.268f	0.000	140.46	13.582
		MT/cm	m	MT/cm	GML	GMT
		7.36		27.01	135.22	8.338
Distances in METERS.-----			Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

SUMMARY OF LOADING

329.0 Cu.M. (20%) SALT WATER

250.00 MT Crane

36.60 MT Crane Load

DRAFT AT FP (M) = 1.601
DRAFT AT MS (M) = 1.458
DRAFT AT AP (M) = 1.315
DRAFT AT FWD MARK (M) = 1.578
DRAFT AT AFT MARK (M) = 1.338

HYDROSTATIC PROPERTIES

Trim: Fwd 0.286/50.000, No Heel, VCG = 5.986

LCF	Displacement	Buoyancy-Ctr.	Weight/	Moment/				
Draft----	Weight(MT)	LCB	VCB	cm	LCF	cm trim	GML	GMT
1.460	998.88	25.805f	0.742	7.36	25.268f	27.01	135.22	8.338
Distances in METERS.-----			Specific Gravity = 1.025.-----					
			Moment in m.-MT.					
			Trim is per 50.00m.					

Draft is from USK.

Formal Free Surface included.

Note: GMT includes the formal free surface moment 235.4 m.-MT

01-01-21 20:59:59 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.775f TCG = 0.000 VCG = 5.986

Free Surface Adjustment: 0.236

Adjusted CG: LCG = 25.773f TCG = 0.000 VCG = 6.222

Origin Depth	Degrees of Trim	Degrees of Heel	Displacement Weight(MT)	Residual Arms in Trim	Residual Arms in Heel	Res. 50% DeckImm Area	Res. 50% DeckImm Angle
1.303	0.33f	0.00	998.85	0.000	-0.014	0.0000	5.448
1.303	0.33f	0.09s	998.88	0.000	0.000	-0.0000	5.354
1.291	0.33f	5.09s	998.88	0.000	0.733	0.0320	0.354
1.289	0.33f	5.45s	998.88	0.000	0.785	0.0366	0.000
1.250	0.34f	10.09s	998.88	0.000	1.448	0.1273	-4.646
1.127	0.44f	15.09s	998.88	0.000	1.847	0.2735	-9.646
1.116	0.45f	15.46s	998.88	0.000	1.849	0.2852	-10.010
0.965	0.59f	20.09s	998.88	0.000	1.699	0.4318	-14.646
0.792	0.75f	25.09s	998.88	0.000	1.349	0.5663	-19.646
0.611	0.90f	30.09s	998.88	0.000	0.914	0.6656	-24.646
0.423	1.05f	35.09s	998.88	0.000	0.436	0.7248	-29.646
0.256	1.18f	39.45s	998.88	0.000	0.000	0.7415	-34.002
0.231	1.20f	40.09s	998.88	0.000	-0.065	0.7411	-34.646
0.035	1.34f	45.09s	998.88	0.000	-0.577	0.7132	-39.646
-0.160	1.47f	50.09s	998.88	0.000	-1.092	0.6404	-44.646
-0.354	1.59f	55.09s	998.88	0.000	-1.603	0.5228	-49.646
-0.543	1.69f	60.09s	998.88	0.000	-2.105	0.3609	-54.646
Distances in METERS.-----Specific Gravity = 1.025.-----Area in m.-Rad.							

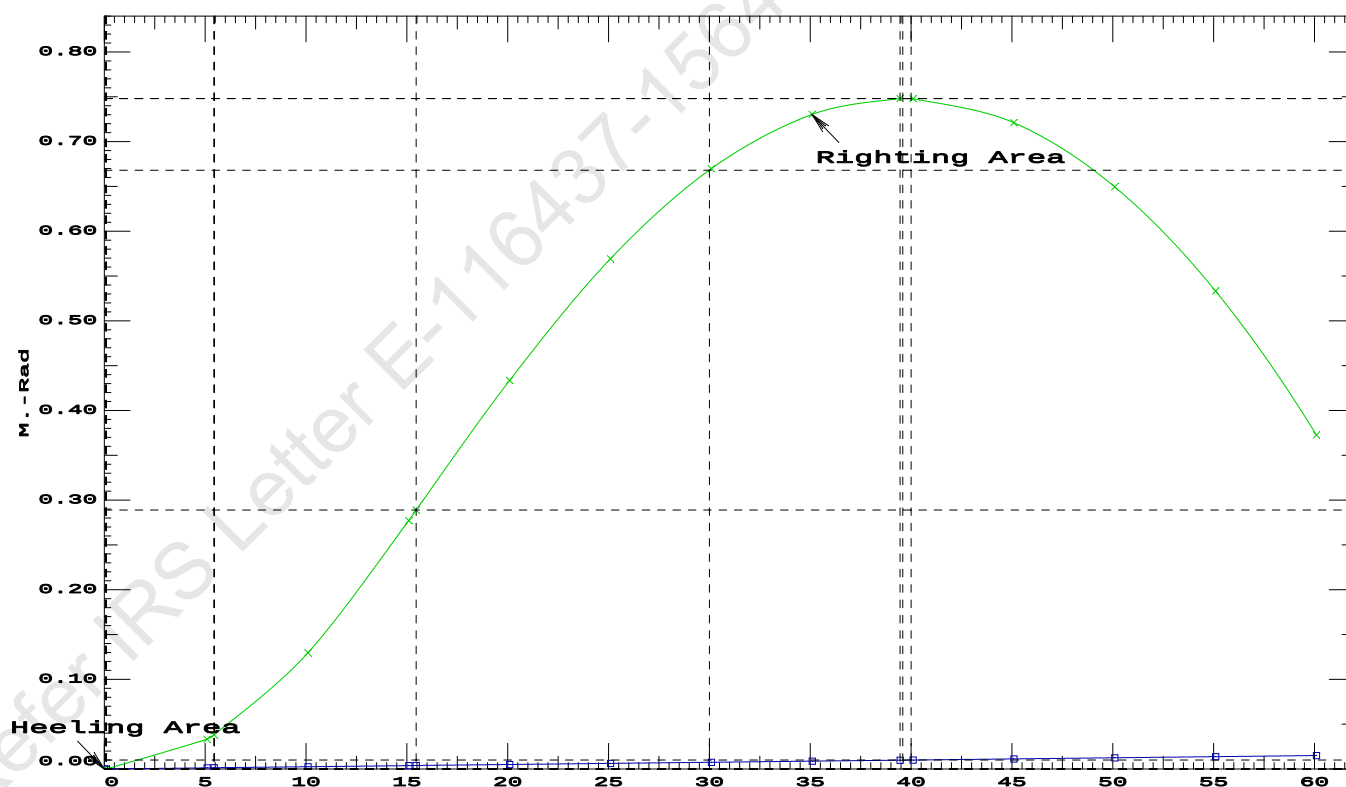
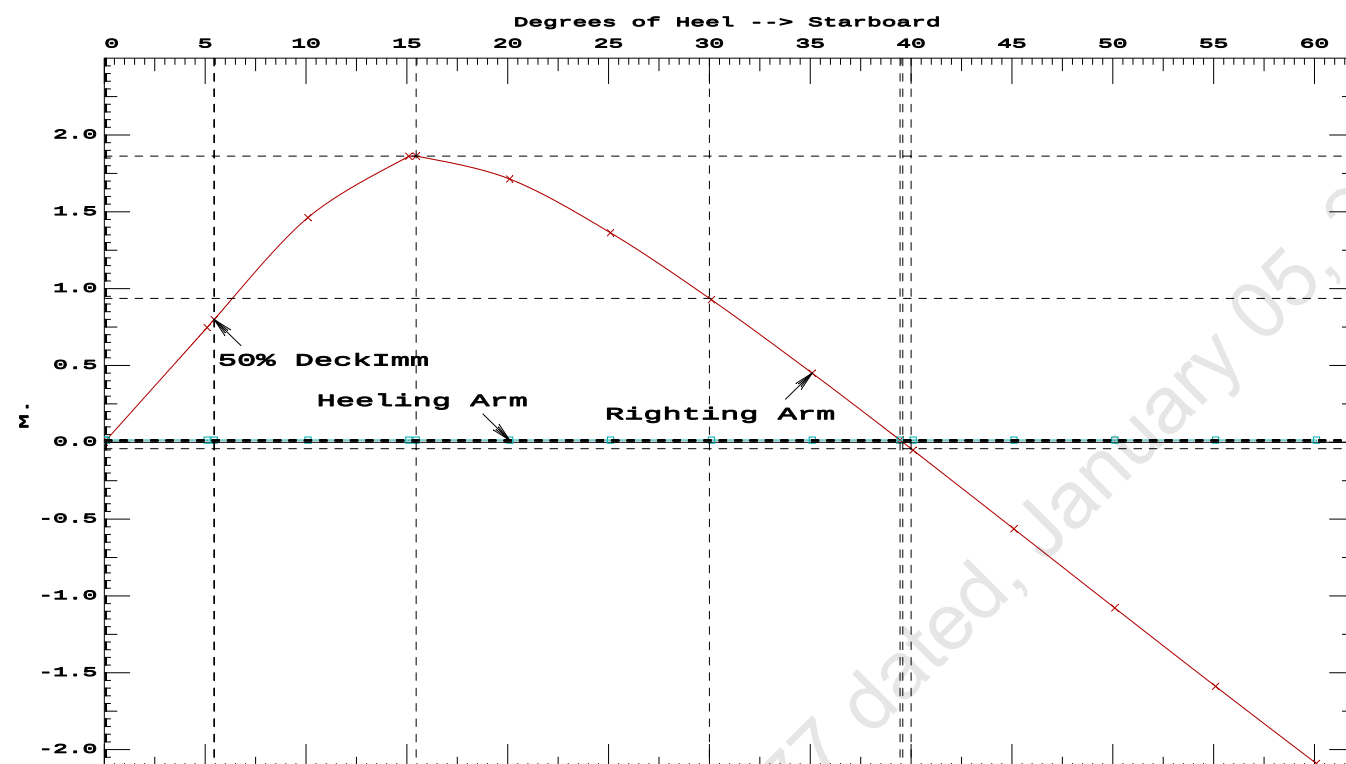
Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the wind heeling arms derived from these moments (in m.-MT):
Stbd heeling moment = 14.26 (constant)

LIM-----	IS CODE 2008 PART B CHAPTER-2 SEC.2-----	Min/Max-----	Attained
(1) Residual Area from abs 0 deg to MaxRA	>	0.0800 m.-Rad	0.2852 P
(2) Angle from Equilibrium to RAZero or Flood	>	20.00 deg	39.36 P
(3) Angle from Equilibrium to 50% Dk Imm Angle	>	0.00 deg	5.35 P

01-01-21 20:59:59 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd



01-01-21 20:59:59 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Total CG: LCG = 25.775f TCG = 0.000 VCG = 5.986

Free Surface Adjustment: 0.236

Adjusted CG: LCG = 25.773f TCG = 0.000 VCG = 6.222

Origin Depth---	Degrees of Trim----	Heel----	Displacement Weight(MT)---	Residual Arms in Trim--in Heel---	Res. Area
1.302	0.33f	0.01p	998.75	0.000 0.000	0.0000
1.291	0.33f	5.01p	998.84	0.000 0.733	0.0320
1.251	0.34f	10.01p	998.88	0.000 1.450	0.1273
1.130	0.44f	15.01p	998.88	0.000 1.859	0.2739
1.115	0.45f	15.48p	998.88	0.000 1.862	0.2892
0.968	0.59f	20.01p	998.88	0.000 1.717	0.4335
0.796	0.74f	25.01p	998.88	0.000 1.369	0.5705
0.615	0.90f	30.01p	998.88	0.000 0.935	0.6717
0.427	1.05f	35.01p	998.88	0.000 0.457	0.7327
0.251	1.19f	39.58p	998.88	0.000 0.000	0.7511
0.234	1.20f	40.01p	998.88	0.000 -0.043	0.7510
0.039	1.34f	45.01p	998.88	0.000 -0.555	0.7249
-0.157	1.47f	50.01p	998.88	0.000 -1.070	0.6541
-0.351	1.59f	55.01p	998.88	0.000 -1.581	0.5384
-0.539	1.69f	60.01p	998.88	0.000 -2.083	0.3784

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Weight and Center of Gravity used for the righting arms above include tank loads. However, the tank load centers were NOT ALLOWED TO SHIFT with heel and trim changes. Rather, a constant Free Surface Moment of 235.4 m.-MT was applied to artificially modify the CG.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Port heeling moment = 1.00

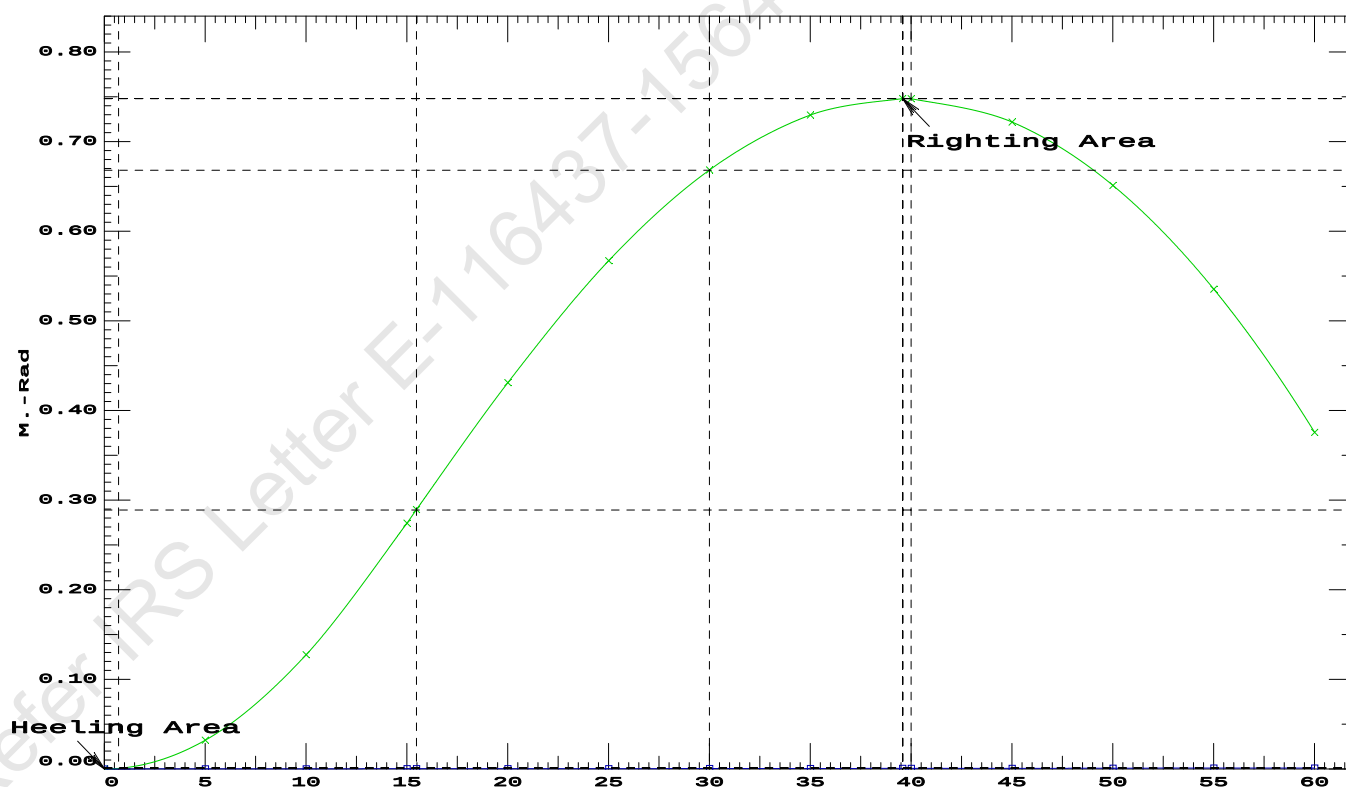
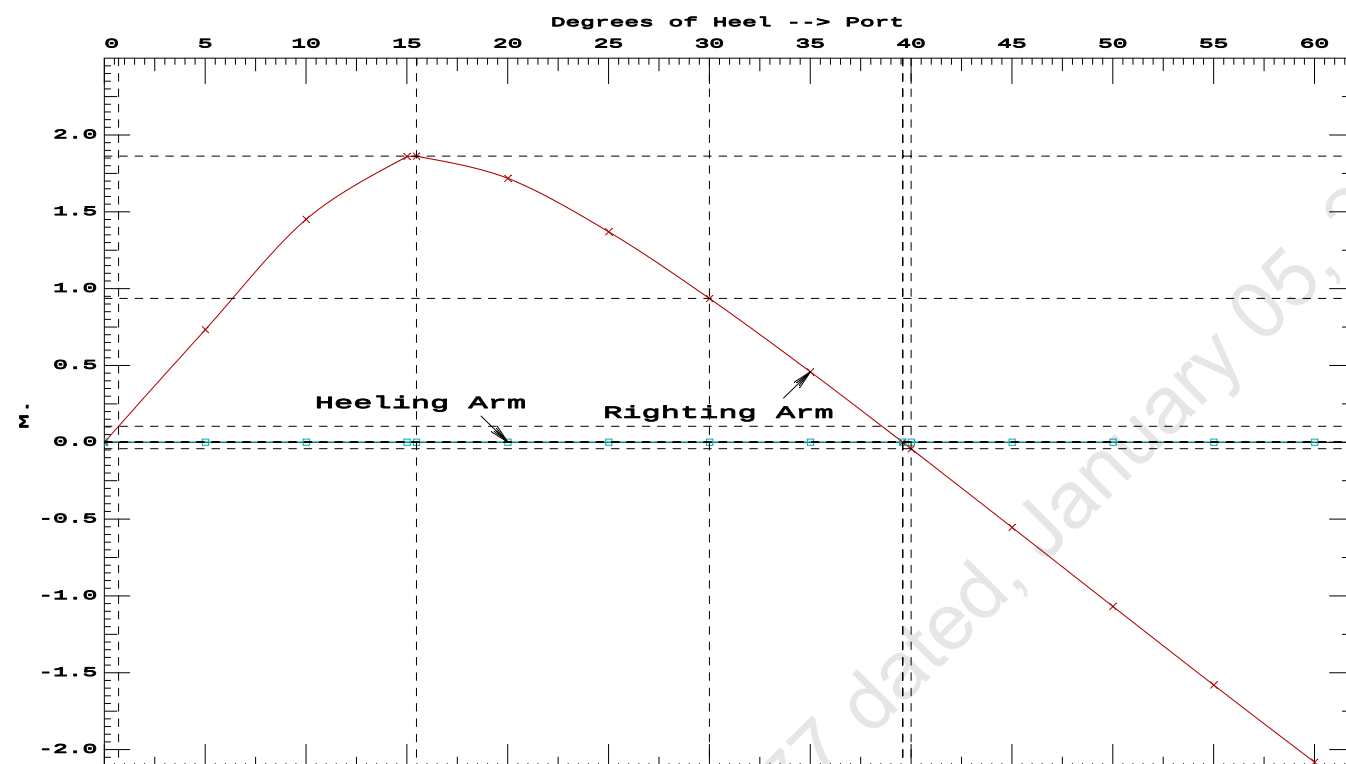
Note: Angle of MaxRA refers to the absolute Righting Arm curve.

LIM-----	STABILITY CRITERIA FOR CRANE OPERAT----	Min/Max-----	Attained
(1) Absolute Angle at Equilibrium	<	15.00 deg	0.01 P
(2) Rise in Abs. RA from Equilibrium to MaxRA	>	66.7 %	185939 P
(3) Abs Ratio from Equilibrium to RAzero or Flood	>	1.667	1087.30 P

01-01-21 20:59:59 Cybermarine Technologies PTE. Ltd.

GHS 17.22

Condition 9 : Lightship + Load at Fwd + Ballast
Crane Lifting 36.60T @ 18m Radius Fwd



01-01-21 21:00:04 Cybermarine Technologies PTE. Ltd.
GHS 17.22

Condition 9 : Lightship + Load at Fwd + Ballast**Crane Lifting 36.60T @ 18m Radius Fwd**

Crane in operation with load in place

WEIGHT and DISPLACEMENT and WATERPLANE STATUS

USK draft: 2.320 @ 50.00f, 2.194 @ 25.00f, 2.069 @ 0.00

Trim: Fwd 0.251/50.000, Heel: Port 0.01 deg.

Part-----			Weight(MT)----	LCG-----	TCG-----	VCG-----	
LIGHT SHIP			375.10	25.000f	0.000	3.000	
Crane			250.00	4.500f	0.000	6.120	
Crane Load			36.60	27.250f	0.000	76.930	
Deck Cargo			556.61	25.000f	0.000	3.500	
Total Fixed----->			1,218.31	20.861f	0.000	6.090	
	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks----->			337.18	42.250f	0.000	1.507	235.45
Total Weight----->			1,555.49	25.498f	0.000	5.096	
			Displ(MT)----	LCB-----	TCB-----	VCB-----	
HULL		1.025	1,555.49	25.517f	0.001p	1.126	

	Righting Arms:			0.001	-0.001s		
	External Arms:			0.000	-0.001		
	Residual Righting Arms:			0.001	0.000		
Part-----		SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025		750.5	25.010f	0.000	102.91	9.135
			MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
			7.69		30.78	98.94	5.165
Distances in METERS.-----				Moments in m.-MT.			

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

Angle Theta1 = -0.01

01-01-21 21:00:04 Cybermarine Technologies PTE. Ltd.
GHS 17.22

After hook load drop
WEIGHT and DISPLACEMENT and WATERPLANE STATUS
USK draft: 2.257 @ 50.00f, 2.147 @ 25.00f, 2.036 @ 0.00
Trim: Fwd 0.221/50.000, Heel: 0.00 deg.

Part-----	Weight(MT)----	LCG-----	TCG-----	VCG-----
LIGHT SHIP	375.10	25.000f	0.000	3.000
Crane	250.00	4.500f	0.000	6.120
Deck Cargo	556.61	25.000f	0.000	3.500
Total Fixed----->	1,181.71	20.663f	0.000	3.896

	Load-----	SpGr-----	Weight(MT)----	LCG-----	TCG-----	VCG-----	FSM-----
WBTk5.P	1.000	1.025	112.08	42.251f	4.980p	1.511	78.49*
WBTk5.C	1.000	1.025	113.02	42.250f	0.000	1.500	78.48*
WBTk5.S	1.000	1.025	112.08	42.251f	4.980s	1.511	78.49*
Total Tanks----->			337.18	42.250f	0.000	1.507	235.45
Total Weight----->			1,518.89	25.455f	0.000	3.365	

	Displ(MT)----	LCB-----	TCB-----	VCB-----
HULL	1,518.89	25.466f	0.001	1.101

Righting Arms: 0.000 0.000s

Part-----	SpGr-----	WPA-----	LCF-----	TCF-----	BML-----	BMT-----
Total Waterplane---->	1.025	750.4	25.011f	0.000	105.35	9.352
		MT/cm-----	m.-MT/cm-----	GML-----	GMT-----	
		7.69	31.32	103.09	7.088	

Distances in METERS.-----Moments in m.-MT.

Note: FSM values marked with an asterisk (*) are formal values which are not the same as the true values in the present condition.

limmarg: 495515936.00; t3: 30.0; t: 15.00
limmarg: 205068288.00; t3: 15.0; t: 7.50
limmarg: 60344208.00; t3: 7.5; t: 3.75
limmarg: 15182684.00; t3: 3.8; t: 1.88
limmarg: 3986225.00; t3: 1.9; t: 0.94
limmarg: 996077.56; t3: 1.0; t: 0.47
limmarg: 303104.50; t3: 0.5; t: 0.23
limmarg: 78016.20; t3: 0.3; t: 0.12
limmarg: 34285.12; t3: 0.2; t: 0.06
limmarg: 20528.70; t3: 0.1; t: 0.03
limmarg: 4865.55; t3: 0.1; t: 0.01

Theta3 is 0.07 degrees, as demonstrated below.

01-01-21 21:00:04 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.663f TCG = 0.000 VCG = 3.896

Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.024	0.25f	0.01p	1,518.9	0.000	-0.001	0.0000
2.024	0.25f	0.00s	1,518.9	0.000	0.000	-0.0000
2.024	0.25f	0.07s	1,518.9	0.000	0.009	0.0000
2.012	0.26f	5.00s	1,518.9	0.000	0.631	0.0275
1.995	0.33f	10.00s	1,519.0	0.000	1.107	0.1045
2.079	0.42f	15.00s	1,518.9	0.000	1.234	0.2092
2.109	0.45f	16.22s	1,518.9	0.000	1.237	0.2355
2.211	0.52f	20.00s	1,518.9	0.000	1.206	0.3166
2.336	0.64f	25.00s	1,518.9	0.000	1.090	0.4176
2.440	0.77f	30.00s	1,518.9	0.000	0.917	0.5056

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1181.71 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM-----STABILITY CRITERION-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to abs 0.1 > 1.000 49.656 P

01-01-21 21:00:04 Cybermarine Technologies PTE. Ltd.
GHS 17.22

RESIDUAL RIGHTING ARMS vs HEEL ANGLE

Fixed CG: LCG = 20.663f TCG = 0.000 VCG = 3.896

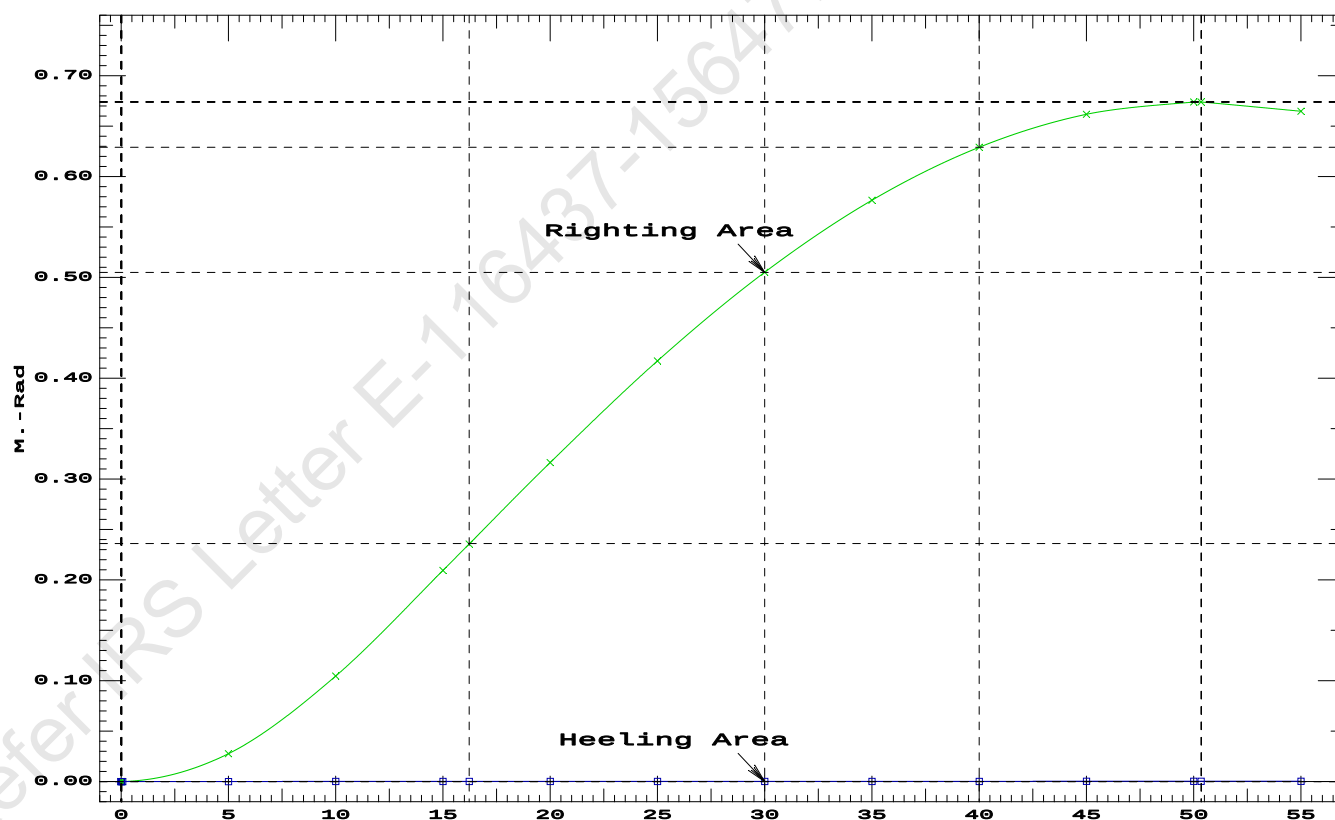
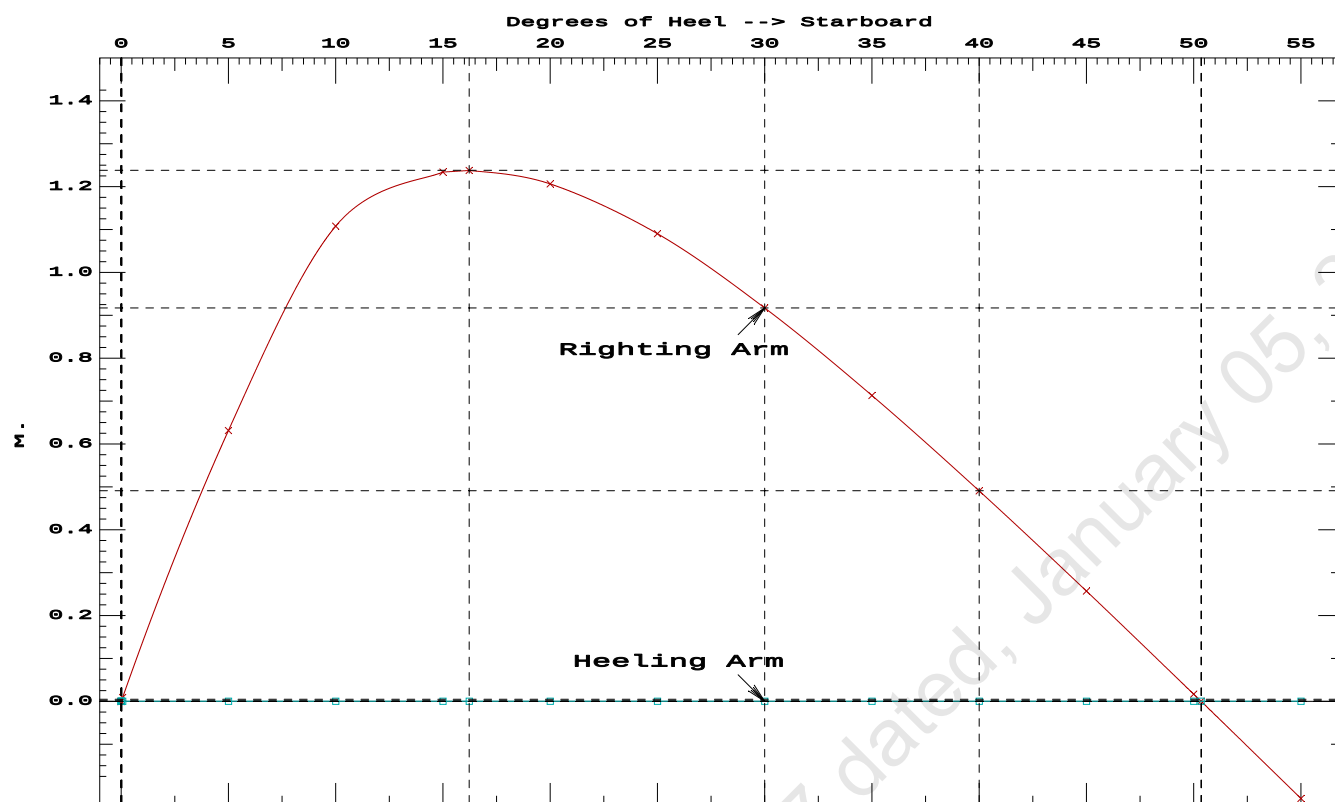
Origin Depth	Degrees of Trim	Displacement Heel	Weight(MT)	Residual Arms in Trim	Res. in Heel	Area
2.024	0.25f	0.01p	1,518.9	0.000	-0.001	0.0000
2.024	0.25f	0.00s	1,518.9	0.000	0.000	-0.0000
2.024	0.25f	0.07s	1,518.9	0.000	0.009	0.0000
2.012	0.26f	5.00s	1,518.9	0.000	0.631	0.0275
1.995	0.33f	10.00s	1,519.0	0.000	1.107	0.1045
2.079	0.42f	15.00s	1,518.9	0.000	1.234	0.2092
2.109	0.45f	16.22s	1,518.9	0.000	1.237	0.2355
2.211	0.52f	20.00s	1,518.9	0.000	1.206	0.3166
2.336	0.64f	25.00s	1,518.9	0.000	1.090	0.4176
2.440	0.77f	30.00s	1,518.9	0.000	0.917	0.5056
2.523	0.90f	35.00s	1,518.9	0.000	0.713	0.5769
2.587	1.02f	40.00s	1,518.9	0.000	0.490	0.6295
2.630	1.14f	45.00s	1,518.9	0.000	0.257	0.6622
2.653	1.25f	50.00s	1,518.9	0.000	0.017	0.6742
2.653	1.25f	50.34s	1,518.9	0.000	0.000	0.6742
2.655	1.35f	55.00s	1,518.9	0.000	-0.227	0.6650

Distances in METERS.---Specific Gravity = 1.025.---Area in m.-Rad.

Note: The Center of Gravity shown above is for the Fixed Weight of 1181.71 MT. As the tank load centers shift with heel and trim, the total Center of Gravity varies. The righting arms shown above include the effect of the C.G. variation.

Note: The Residual Righting Arms shown above are in excess of the overturning arms derived from these moments (in m.-MT):
Stbd heeling moment = 0.50

LIM-----RINA PT E, CH 19, SEC 2: 2.2.3-----Min/Max-----Attained
(1) Residual Area Ratio from abs 0 deg to RAZero > 1.000 6608572 P
(2) Angle from abs 0.1 deg to RAZero > 20.00 deg 50.27 P



LIGHTSHIP PARTICULARS

Lightship Weight	=	375.10	T
VCG	=	3.000	M above Baseline
LCG	=	25.000	M fwd of AP
TCG	=	0.000	M off centerline

(Reference: Lightship particulars are taken from approved Draft Survey Report of CHISTI-Y/015.)

03-04-19 10:01:31 Cybermarine Technologies PTE. Ltd.
GHS 16.50

LATERAL PLANE STATUS

USK draft: 0.500 @ 50.00f, 0.500 @ 25.00f, 0.500 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	21.6	25.000f	-0.248	121.7	25.000f	1.289
CRANE.C				35.2	7.500f	4.712
DECKCARGO.C				40.0	25.000f	3.512
Total Lateral Plane->	21.6	25.000f	-0.248	196.9	21.871f	2.352
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.600 @ 50.00f, 0.600 @ 25.00f, 0.600 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	26.0	25.000f	-0.297	117.2	25.000f	1.236
CRANE.C				35.2	7.500f	4.612
DECKCARGO.C				40.0	25.000f	3.412
Total Lateral Plane->	26.0	25.000f	-0.297	192.4	21.799f	2.306
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.700 @ 50.00f, 0.700 @ 25.00f, 0.700 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	30.5	25.000f	-0.346	112.8	25.000f	1.183
CRANE.C				35.2	7.500f	4.512
DECKCARGO.C				40.0	25.000f	3.312
Total Lateral Plane->	30.5	25.000f	-0.346	188.0	21.723f	2.259
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.800 @ 50.00f, 0.800 @ 25.00f, 0.800 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	35.0	25.000f	-0.395	108.3	25.000f	1.130
CRANE.C				35.2	7.500f	4.412
DECKCARGO.C				40.0	25.000f	3.212
Total Lateral Plane->	35.0	25.000f	-0.395	183.5	21.642f	2.214
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 0.900 @ 50.00f, 0.900 @ 25.00f, 0.900 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	39.6	25.000f	-0.443	103.7	25.000f	1.077
CRANE.C				35.2	7.500f	4.312
DECKCARGO.C				40.0	25.000f	3.112
Total Lateral Plane->	39.6	25.000f	-0.443	178.9	21.557f	2.169
Distances in METERS.-----						

03-04-19 10:01:31 Cybermarine Technologies PTE. Ltd.
GHS 16.50

LATERAL PLANE STATUS

USK draft: 1.000 @ 50.00f, 1.000 @ 25.00f, 1.000 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	44.1	25.000f	-0.492	99.1	25.000f	1.025
CRANE.C				35.2	7.500f	4.212
DECKCARGO.C				40.0	25.000f	3.012
Total Lateral Plane->	44.1	25.000f	-0.492	174.3	21.467f	2.124
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.100 @ 50.00f, 1.100 @ 25.00f, 1.100 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	48.8	25.000f	-0.540	94.5	25.000f	0.972
CRANE.C				35.2	7.500f	4.112
DECKCARGO.C				40.0	25.000f	2.912
Total Lateral Plane->	48.8	25.000f	-0.540	169.7	21.370f	2.081
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.200 @ 50.00f, 1.200 @ 25.00f, 1.200 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	53.4	25.000f	-0.589	89.8	25.000f	0.920
CRANE.C				35.2	7.500f	4.012
DECKCARGO.C				40.0	25.000f	2.812
Total Lateral Plane->	53.4	25.000f	-0.589	165.0	21.267f	2.038
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.300 @ 50.00f, 1.300 @ 25.00f, 1.300 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	58.2	25.000f	-0.637	85.1	25.000f	0.868
CRANE.C				35.2	7.500f	3.912
DECKCARGO.C				40.0	25.000f	2.712
Total Lateral Plane->	58.2	25.000f	-0.637	160.3	21.158f	1.997
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.400 @ 50.00f, 1.400 @ 25.00f, 1.400 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	62.9	25.000f	-0.685	80.4	25.000f	0.817
CRANE.C				35.2	7.500f	3.812
DECKCARGO.C				40.0	25.000f	2.612
Total Lateral Plane->	62.9	25.000f	-0.685	155.6	21.040f	1.956
Distances in METERS.-----						

03-04-19 10:01:31 Cybermarine Technologies PTE. Ltd.
GHS 16.50

LATERAL PLANE STATUS

USK draft: 1.500 @ 50.00f, 1.500 @ 25.00f, 1.500 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	67.7	25.000f	-0.733	75.6	25.000f	0.765
CRANE.C				35.2	7.500f	3.712
DECKCARGO.C				40.0	25.000f	2.512
Total Lateral Plane->	67.7	25.000f	-0.733	150.8	20.915f	1.916
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.600 @ 50.00f, 1.600 @ 25.00f, 1.600 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	72.5	25.000f	-0.781	70.8	25.000f	0.714
CRANE.C				35.2	7.500f	3.612
DECKCARGO.C				40.0	25.000f	2.412
Total Lateral Plane->	72.5	25.000f	-0.781	146.0	20.780f	1.878
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.700 @ 50.00f, 1.700 @ 25.00f, 1.700 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	77.4	25.000f	-0.829	65.9	25.000f	0.663
CRANE.C				35.2	7.500f	3.512
DECKCARGO.C				40.0	25.000f	2.312
Total Lateral Plane->	77.4	25.000f	-0.829	141.1	20.634f	1.841
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.800 @ 50.00f, 1.800 @ 25.00f, 1.800 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	82.3	25.000f	-0.876	61.0	25.000f	0.612
CRANE.C				35.2	7.500f	3.412
DECKCARGO.C				40.0	25.000f	2.212
Total Lateral Plane->	82.3	25.000f	-0.876	136.2	20.477f	1.806
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 1.900 @ 50.00f, 1.900 @ 25.00f, 1.900 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	87.2	25.000f	-0.924	56.0	25.000f	0.561
CRANE.C				35.2	7.500f	3.312
DECKCARGO.C				40.0	25.000f	2.112
Total Lateral Plane->	87.2	25.000f	-0.924	131.2	20.307f	1.772
Distances in METERS.-----						

03-04-19 10:01:31 Cybermarine Technologies PTE. Ltd.
GHS 16.50

LATERAL PLANE STATUS

USK draft: 2.000 @ 50.00f, 2.000 @ 25.00f, 2.000 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	92.2	25.000f	-0.971	51.1	25.000f	0.511
CRANE.C				35.2	7.500f	3.212
DECKCARGO.C				40.0	25.000f	2.012
Total Lateral Plane->	92.2	25.000f	-0.971	126.3	20.123f	1.739
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.100 @ 50.00f, 2.100 @ 25.00f, 2.100 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	97.2	25.000f	-1.019	46.1	25.000f	0.461
CRANE.C				35.2	7.500f	3.112
DECKCARGO.C				40.0	25.000f	1.912
Total Lateral Plane->	97.2	25.000f	-1.019	121.3	19.922f	1.709
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.212 @ 50.00f, 2.212 @ 25.00f, 2.212 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	102.8	25.000f	-1.072	40.5	25.000f	0.405
CRANE.C				35.2	7.500f	3.000
DECKCARGO.C				40.0	25.000f	1.800
Total Lateral Plane->	102.8	25.000f	-1.072	115.7	19.676f	1.677
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.300 @ 50.00f, 2.300 @ 25.00f, 2.300 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	107.2	25.000f	-1.114	36.1	25.000f	0.361
CRANE.C				35.2	7.500f	2.912
DECKCARGO.C				40.0	25.000f	1.712
Total Lateral Plane->	107.2	25.000f	-1.114	111.3	19.465f	1.653
Distances in METERS.-----						

LATERAL PLANE STATUS

USK draft: 2.400 @ 50.00f, 2.400 @ 25.00f, 2.400 @ 0.00

Trim: zero, Heel: zero

Part-----	LPA-----	LCP-----	HCP-----	LPA-----	LCP-----	HCP-----
HULL	112.2	25.000f	-1.162	31.1	25.000f	0.311
CRANE.C				35.2	7.500f	2.812
DECKCARGO.C				40.0	25.000f	1.612
Total Lateral Plane->	112.2	25.000f	-1.162	106.3	19.205f	1.629
Distances in METERS.-----						



MEMORANDUM OF FREEBOARDS

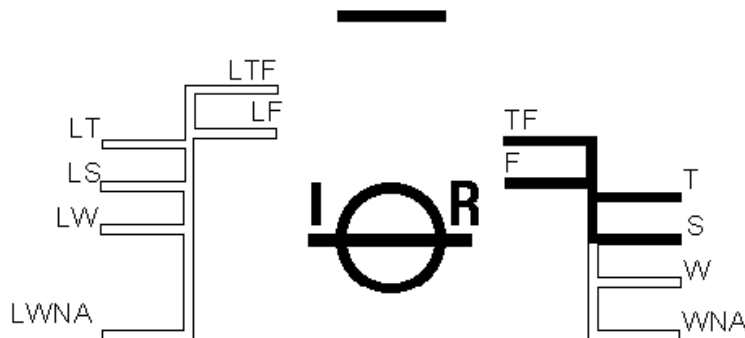
23 DEC 2020

Chisti Marine Engg. Works Yard No. CHISTI/015 K.B.26	Type B freeboard Freeboard Length : 48.00 m Summer WL Extreme Draught : 2.212 m - load line amidships
--	--

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of *
~~Wood~~ / *steel Deck *at side / *continued to side. (*Delete words not applicable).

Freeboards from Deck Line		Seasonal, Fresh Water, or Timber allowances		
Tropical Fresh Water	713	mm.	97	mm. above summer
Fresh Water	759	mm.	51	mm. above summer
Tropical	764	mm.	46	mm. above summer
Summer	810	mm.	Upper edge of line through centre	
Winter	Not	mm.	Assigned	mm. below summer
Winter North Atlantic	Not	mm.	Assigned	mm. below summer
Timber Tropical fresh water	—	mm.	—	mm. above timber summer
Timber fresh water	—	mm.	—	mm. above timber summer
Timber Tropical	—	m m.	—	mm. above timber summer
Timber Summer	—	mm.	—	mm. above centre of ring
Timber Winter	—	mm.	—	mm. below timber summer
Timber Winter North Atlantic	—	mm.	—	mm. below timber summer

Valid only for unmanned operation while under tow along Indian Coast, during the course of which the vessel does not go more than 20 nautical miles from the nearest land, and may cross gulfs or similar features recognised by the Administration.



At Fr.52
 ←



IRCLASS
Indian Register of Shipping
CIN: U61100MH1975NPL018244

52 A, Adi Shankaracharya Marg, Opp. Powai Lake,
Powai, Mumbai - 400 072, India
Phone : +91 22 30519400 / +91 22 7119 9400
Fax : +91 22 2570 3611
E-mail : ho@irclass.org ★ Website : www.irclass.org

Our ref: E-116313-156248

31-December-2020

Company : Cybermarine Knowledge System Pvt. Ltd.

Dear Sir/Madam,

Your document (detailed below) has been reviewed.

Project Details	
Project number	NC002582
Project name	Chisti Marine Engineering Works - CHISTI/015
Project Manager	Dhruvesh Singh
Project In-Charge	Veerendra P

Document Details	
Plan No.	CM20-V-111-0206
Plan Title	Draft Survey Report
Rev.No.	0
Upload.No.	1
Submitted Date	30-December-2020
Review Status	Approved
Parent Plan Title	

Notes

1. The review status Approved indicates:

The following Lightship particulars are approved:

Lightship Weight: 375.10 t

Lightship L.C.G: 25.00 m forward of AP (Fr.0)

Lightship V.C.G: 3.00 m above base line

Lightship T.C.G: 0.000 m off centreline

The Final trim and stability booklet prepared based on the approved lightship particulars should be submitted for approval. Copy of the approved lightship survey report should form part of the booklet.

Thanking you,
Yours Sincerely,

For INDIAN REGISTER OF SHIPPING

Renganathan S

Sr. Principal Surveyor

This is an electronically published document and does not require signature.

DRAFT SURVEY REPORT OF KB-26 (YARD NO. CHISTI/015)



LIGHTSHIP PARTICULARS:

SEE LETTER E-116313-156248

Lightship Displacement -	375.10 T
Lightship L.C.G -	25.000 m from AP (Fr.0)
Lightship V.C.G -	3.000 m above BL
Lightship T.C.G -	0.000 m off-centerline

Rev.	Description	Date	Prepared	Review		
<u>Issued for</u> : <u>Approval</u>						
Client: K.B. SHIPPING & CO. (JAMNAGAR) Builder: CHISTI MARINE ENGINEERING WORKS		Project: PONTOON (YARD NO. CHISTI/015)				
Doc #		Document Title:		Prepared		
CM20-V-111-0206		DRAFT SURVEY REPORT		OM		
Scale	Size			Sheet	Date	Checked
NTS	A4			1 of 14	30-12-2020	Review 1
				Review 2		
 www.cybermarine.net info@cybermarine.net						
10, Bukit Batok Crescent, #12-01, The Spire Building, Singapore-658079.		501, Technocity IT Premises Co-op. Soc. Ltd, Plot No. X-5/3, Mahape, Maharashtra- 400710, India.		22503, Katy Freeway #7, Katy, Texas, USA 77450.		

GENERAL PARTICULARS

Name of the Vessel	KB-26
Client	K.B.SHIPPING & CO., JAMNAGAR
Builder	CHISTI MARINE ENGINEERING WORKS
Yard No.	CHISTI/015
Type of Vessel	PONTOON
Class Notation	⚡SU-PONTOON, "INDIAN COASTAL SERVICE" DESCRIPTIVE NOTE "FOR CRANE OPERATION" "DECK STRENGTHENED FOR {10T/M ² }

Main Particulars

LOA	50.000 M
LBP	50.000 M
Breadth (Mld.)	15.000 M
Depth (Mld.)	3.000 M
Draft (Mld.)	2.200 M

References

AP	at Fr.0
FP	at Fr.100
Midship	at Fr.50

Frame spacing

Throughout	500 mm
------------	--------

Location of Test	Sikka,Jamnagar
Date	29-Dec-20
Weather	Calm
Wind	Light
Specific Gravity	1.025
Restraining Line	Loose
Witnessed by	Mr.Chiranjeev Malviya (IRS Surveyor)

(w.r.t keel line)

Location	Draft Port (m)	Draft Stbd (m)	Mean Draft (m)
Fwd (Fr.92)	0.580	0.580	0.580
Mid (Fr.50)	0.580	0.580	0.580
Aft (Fr.8)	0.580	0.580	0.580

Correction for Drafts

To correct for observed drafts at marks and obtain Drafts at perps.

LBP	-	50.000	m
-----	---	--------	---

Fwd	-	0.580	m
Aft	-	0.580	m

Correction to Drafts (wrt Keel line)

Location	At Fwd (Fr.92)	At Aft (Fr.8)	
Draft at Marks	0.580	0.580	m
Trim at Marks		0.000	m
Distance of Marks from PP	4.000	4.000	m
Distance between Marks		42.000	m
Draft correction factor		0.000	
Correction at PP	0.000	0.000	
Corrected Draft at PP	0.580	0.580	
Trim		0.000	m
Mean Draft		0.580	m
From Hydrostatics against above Mean Draft and Trim			
Displacement at density(T/M^3)=1.025 T/m^3		375.10	T
LCB		25.000	m
KMt		34.429	m
LCG		25.000	m
TCG		0.000	m

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

30-12-20 12:33:39
GHS 17.22

Cybermarine Technologies PTE. Ltd.
KB-26

HYDROSTATIC PROPERTIES

No Trim, No Heel, Fixed VCG = 0.000

Draft@	Displacement	Buoyancy-Ctr.	Weight/	Moment/
25.000f	---Weight(MT)---	LCB-----VCB-----	cm-----LCF---	cm trim----
0.580	375.10	25.000f 0.284	6.83 25.000f	22.46 299.36 34.429
Distances in METERS.-----Specific Gravity = 1.025.-----Moment in m.-MT.				
Trim is per 50.00m.				

Draft is from USK.

Weights to go off Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.00	0.000	0.00	0.000	0.00	0.000	0.00
TOTAL		0.000	0.000	0.000	0.000	0.000	0.000	0.000

Weights to go on Board

No	Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
-	-	0.000	0.000	0.00	0.000	0.00	0.000	0.00
TOTAL		0.000	0.000	0.000	0.000	0.000	0.000	0.000

Lightship Particulars

Item	WEIGHT T	LCG m	L-MOM T-m	VCG m	V-MOM T-m	TCG m	T-MOM T-m
Particulars as Draft Survey	375.10	25.000	9377.500	3.000	1125.300	0.000	0.000
Weights to go Off board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Weights to go on board	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Lightship Particulars	375.10	25.000	9377.50	3.000	1125.30	0.000	0.00

Lightship Particulars

Displacement	375.10	T
LCG	25.000	m From AP
TCG	0.000	m Off-Centreline
VCG	3.000	m above BL

NOTE

1. Hydrostatic Particulars computed by Computer Program at Sp.Gr. 1.025
2. LCB, LCG are defined from AP, Fwd of AP is +ve and Aft of AP is -ve
3. Trim by Fwd is -ve and Trim by Aft is +ve.
4. KB is measured from Base Line.
5. Draft is measured from Keel Line

LIGHT-SHIP SURVEY PROCEDURE AND REPORTING FORMAT

Part I



Pre requisites for Conducting Light-Ship Survey (To be provided /confirmed by Builder/Owner/Operator/Consultant prior commencement of Light-Ship Survey)

Name Of Builder/owner rep conducting trial	CHISTI MARINE ENG. WORKS
Hull Number/Vessel name	CHISTI-015 / K.B.26
Type of Vessel (Cargo/passenger)	PONTOON
Location of Vessel	SIKKA
Date of Proposed Light-Ship Survey	29/12/2020
Draft of Vessel and trim	0.58m - TRIM-NIL
Depth of Water	3.0 m
Anticipated tide on date of survey	5.2 m 29/12/2020
Weather prediction for proposed date	CLAM
Availability of following reports from builder/Owner/Owners rep	
Copy of Approved Inclining experiment report	☐ YES ☒ NO N/A
Details of Periodical Draft Survey report/any Draft Survey report conducted after the approval of Final TSB.	☐ YES ☒ NO N/A
Declaration of weights to go on	☐ YES ☒ NO N/A
Declaration of weights to go off - tank contents - Loose items	☐ YES ☒ NO N/A

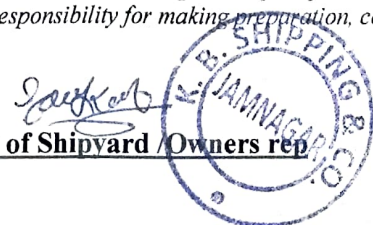


Facilities/Arrangement for conduct of Light-Ship Survey	
Availability of boat to take draft reading	☒ YES ☐ NO
Availability of draft mark plan/approved document like Prelim TSB/Final TSB for confirming location of draft marks	☒ YES ☐ NO
Safety Requirements- lighting, gas freeing, accessibility of areas requiring inspection as part of Survey	☒ YES ☐ NO
Availability of sounding tapes, hydrometer etc	☒ YES ☐ NO
Closing/isolating cross connected tanks	☐ YES ☐ NO N/A
Removal of gangway during draft reading/experiment	☒ YES ☐ NO
Permanent ballast used ? ; if any and its details	☐ YES ☒ NO N/A
If vessel has any list, reason for the list is established.	☐ YES ☒ NO N/A
A trial is conducted to see whether the vessel's draft corresponding to the loading condition is matching with the approved hydrostatics.	☐ YES ☒ NO N/A

NOTE : The responsibility of the IRS Surveyor is to verify that the survey is conducted according to accepted procedures and that all basic measurements and data are correctly taken and recorded. The responsibility for making preparation, conducting lightship survey and preparing report rest with owner/operator.

Confirm Readiness for test

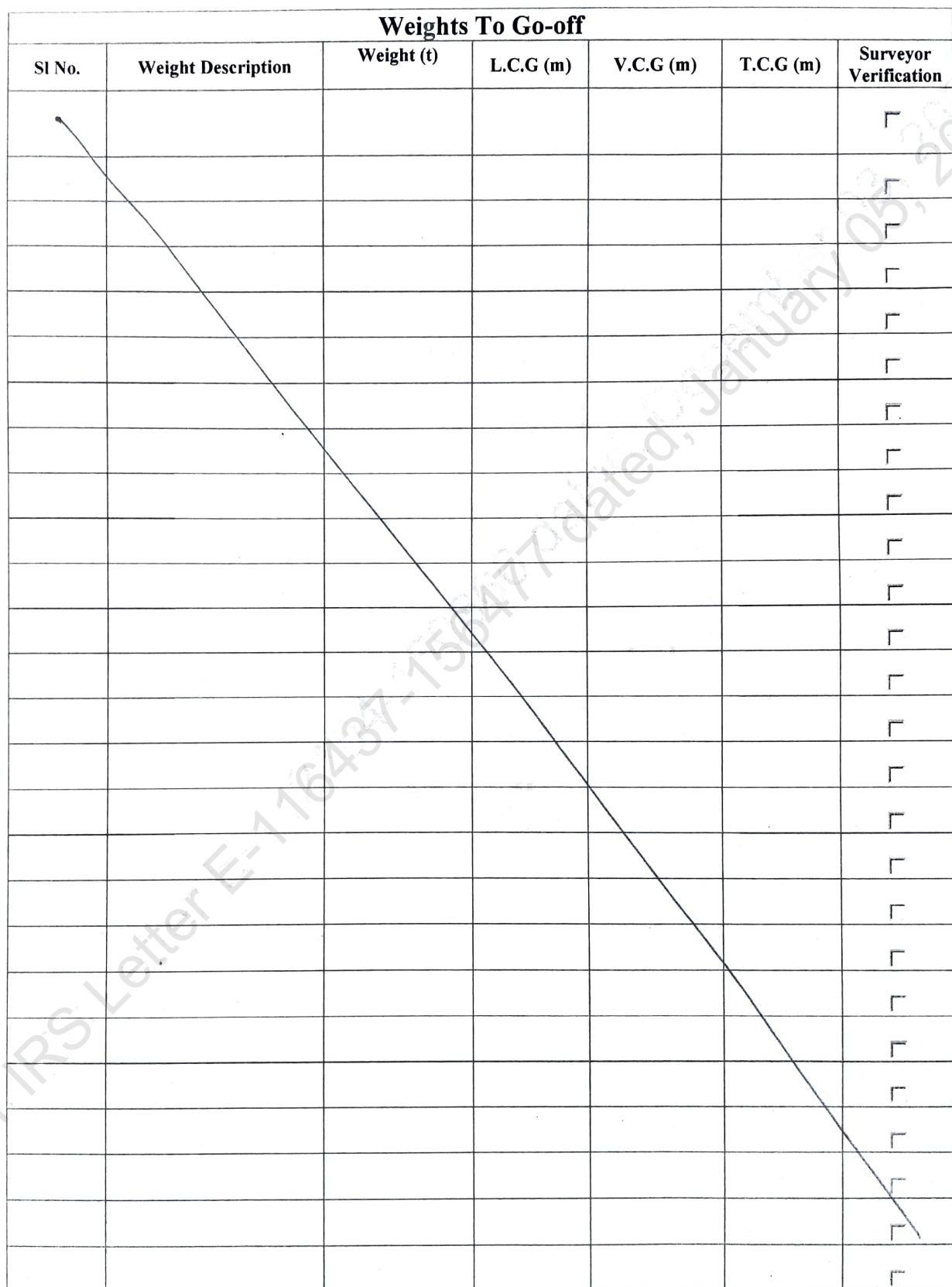
Name & Signature of Shipyard / Owners rep



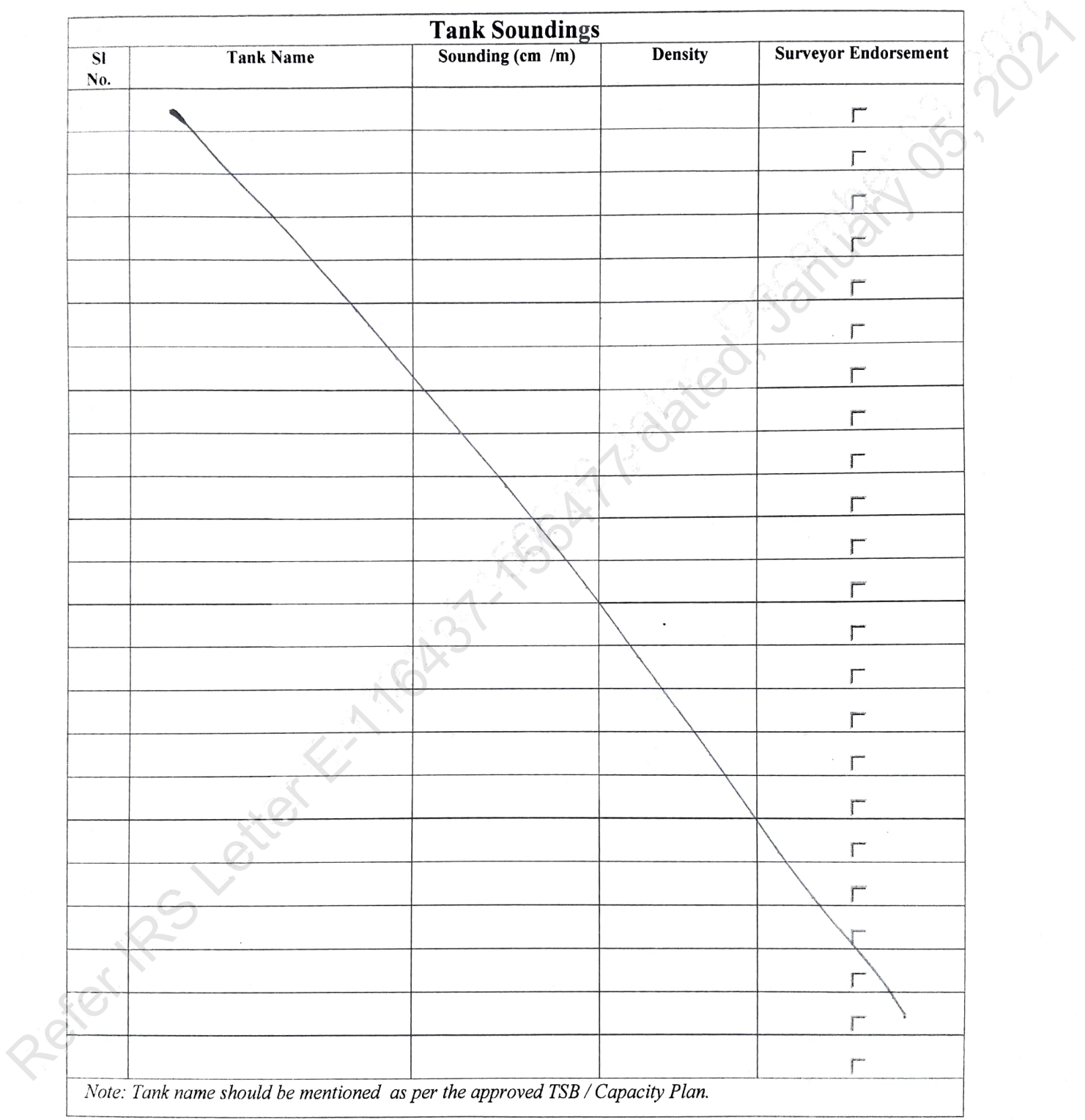


LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----



LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----



Note: Tank name should be mentioned as per the approved TSB / Capacity Plan.



Weights To Be Relocated

Sl No.	Weight Description	Weight (t)	L.C.G (m)		V.C.G (m)		T.C.G (m)		Surveyor Verification
			Initial	Final	Initial	Final	Initial	Final	
									┐
									┐
									┐
									┐
									┐
									┐
									┐
									┐
									┐
									┐
									┐
									┐

LCG: Longitudinal Center of Gravity from AP/Transom/Midship/Frame -----

LIGHTSHIP SURVEY PROCEDURE AND REPORTING FORMAT

Part II

REPORTING FORM FOR LIGHTSHIP SURVEY



Attending Surveyor should witness Light-Ship Survey only if the Part-I submitted by Owner/Builder/Operator/Consultant is found satisfactory

General Particulars

Name of Vessel	K.B-26
Builder/ Owner	Chisti Marine Eng. Works / K.B. Shipping
Yard Number/ IR Number	Chisti-015 / 65737
Type of Vessel Passenger Non Passenger	☒ Passenger ☒ Non Passenger Pontoon
Date(dd/mm/yyyy); Local Time of inspection/test	14:00 29/12/2020
Wind / Weather condition	5 knots / Calm
Location of Test	Sikka
Depth of water where vessel is moored, m	3.0 m
Method of reading Draft(By Boat-Y/N)	Boat

Location of Draft Marks	Frame Number	Draft Reading (Port side),m	Draft Reading (STBD Side),m
Aft	8	0.58 m	0.58 m
Midship	50	0.58 m	0.58 m
Forward	92	0.58 m	0.58 m

Note : Recommended to attach Photographs of draft; as record

Density of Water

Results of 3 samples from Forward, Aft and Mid of the vessel		
Sample 1: (ford)	1.025	t/m3
Sample 2: (mid)	1.025	t/m3
Sample 3: (aft)	1.025	t/m3



Verification of weights to go on / go off /re-location	SEE PART 1
Note: See Part 1 (the sheet for recording the above data) , verify and endorse.	

Special Comments on Light Ship Survey, if any.

Lightship Draft Survey Completed
Satisfactorily at Sikka.

Attending surveyor to forward both forms Part 1 and Part 2 together with completed attachments 1 and 2 to HO.

Name & designation of Attending Surveyor: -----


CHIRANJEEV MALVIYA
SURVEYOR
29/12/2020

